

ADS126x 32-Bit, Precision, 38-kSPS, Analog-to-Digital Converter (ADC) with Programmable Gain Amplifier (PGA) and Voltage Reference

1 Features

- Precision, 32-bit, $\Delta\Sigma$ ADC
- Auxiliary 24-bit, $\Delta\Sigma$ ADC (ADS1263)
- Data rates: 2.5 SPS to 38400 SPS
- Differential input, CMOS PGA
- 11 multifunction analog inputs
- High-accuracy architecture:
 - Offset drift: 1 nV/°C
 - Gain drift: 0.5 ppm/°C
 - Noise: 7 nV_{RMS} (2.5 SPS, gain = 32)
 - Linearity: 3 ppm
- 2.5-V internal voltage reference:
 - Temperature drift: 2 ppm/°C
- 50-Hz and 60-Hz rejection
- Single-cycle settled conversions
- Dual sensor excitation current sources
- Internal fault monitors
- Internal ADC test signal
- 8 general-purpose inputs/outputs

2 Applications

- Factory automation and control:
 - Analog input modules
 - Temperature controllers
 - Weigh modules
- Instrumentation:
 - Process analytics
 - Lab and field instrumentation
 - Weigh scales

3 Description

The ADS1262 and ADS1263 (ADS126x) are low-noise, low-drift, 38.4-kSPS, delta-sigma ($\Delta\Sigma$) ADCs with an integrated PGA, reference, and internal fault monitors. The ADS1263 integrates an auxiliary, 24-bit, $\Delta\Sigma$ ADC intended for background measurements. The sensor-ready ADCs provide complete, high-accuracy, one-chip measurement solutions for the most-demanding sensor applications, including weigh scales, strain-gauge sensors, thermocouples, and resistance temperature devices (RTD).

The ADCs are comprised of a low-noise, CMOS PGA (gains 1 to 32), a $\Delta\Sigma$ modulator, followed by a programmable digital filter. The flexible analog front-end (AFE) incorporates two sensor-excitation current sources suitable for direct RTD measurement.

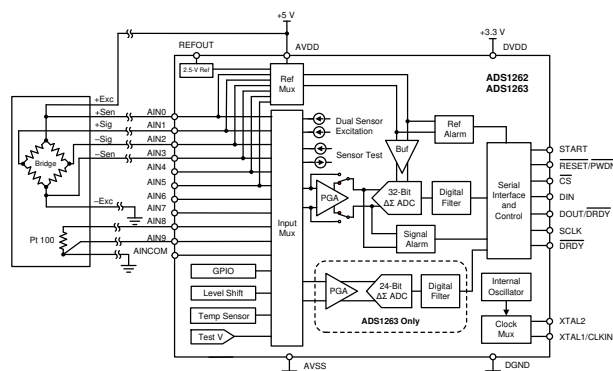
A single-cycle settling digital filter maximizes multiple-input conversion throughput, while providing 130-dB rejection of 50-Hz and 60-Hz line cycle interference.

The ADS1262 and ADS1263 are pin and functional compatible. These devices are available in a 28-pin TSSOP package and are fully specified over the –40°C to +125°C temperature range.

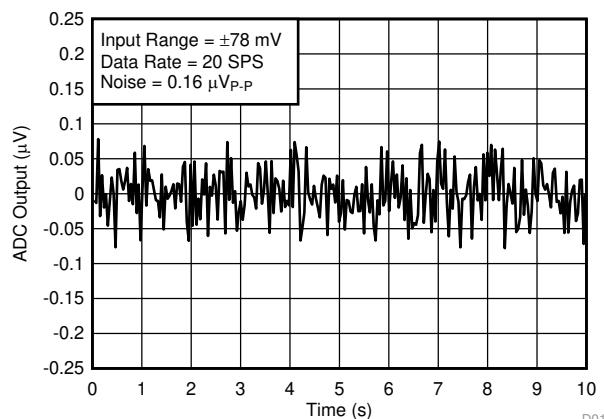
Device Information⁽¹⁾

PART NUMBER	PACKAGE	BODY SIZE (NOM)
ADS126x	TSSOP (28)	9.70 mm × 4.40 mm

- (1) For all available packages, see the package option addendum at the end of the data sheet.



Temperature Compensated Bridge Measurement



ADC Conversion Noise

D017



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4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision B (July 2015) to Revision C (May 2021)	Page
• Updated the numbering format for tables, figures, and cross-references throughout the document	1
• Added links to <i>Applications</i> section.....	1
• Changed <i>Functional Block Diagram</i> to correct XTAL1/CLKIN pin name.....	31
• Added discussion to <i>Internal Reference</i> section regarding 10 μ F (max) REFOUT capacitor ability to decrease reference noise.....	42
• Changed <i>IDAC Block Diagram</i> and text to include 2500 μ A setting.....	50
• Added last paragraph to <i>Pulse Conversion Mode</i> section to explain operation in chop mode.....	62
• Changed <i>EBh</i> to <i>AFh</i> in the example checksum computation in the <i>Checksum Mode</i> section.....	72
• Changed text in <i>GPIO Data Register</i> section regarding GPIO data read when programmed as an output... ..	103
• Changed calculation of V_{REFMIN} to use gain = 8 in <i>Detailed Design Procedure</i> application section.....	115
• Changed title of <i>What To Do and What Not To Do</i> from <i>Do's and Don'ts</i>	119

Changes from Revision A (May 2015) to Revision B (July 2015)	Page
• Changed ADS1263 from product preview to production data, and added text and specifications throughout data sheet to include the ADS1263 and ADC2.....	1
• Changed text throughout data sheet for clarity.....	1
• Added condition line to <i>Absolute Maximum Ratings</i> table.....	5
• Added <i>Crosstalk</i> section to <i>Electrical Characteristics</i> table.....	7
• Added Figure 7-32	12
• Added Figure 7-36	12
• Changed legend in Figure 7-45	12
• Added missing gain term in FSR definition of Equation 8	25

• Changed text in fourth paragraph of <i>Noise Performance</i> section to clarify conditions to achieve maximum ENOB.....	25
• Changed bit names from PGAH and PGAL to PGAH_ALM and PGAL_ALM, respectively, in <i>PGA Absolute Output-Voltage Monitor</i> section.....	40
• Changed Figure 9-12 to show correct name of bit 4.....	41
• Changed RMUX to RMUXP in second paragraph of <i>ADC Reference Voltage</i> section.....	41
• Changed text in last paragraph of <i>ADC Reference Voltage</i> section to show correct name of bit 4.....	41
• Changed text in <i>External Reference</i> section to clarify external reference inputs, polarity reversal switch, reference input current, and external reference buffer.....	42
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• Changed text in last paragraph of <i>Sensor-Excitation Current Sources (IDAC1 and IDAC2)</i> section to clarify settling time in IDAC rotation mode.....	50
• Changed text in <i>General-Purpose Input/Output (GPIO)</i> section regarding GPIO data readback when programmed as an output.....	52
• Changed Figure 9-27	52
• Changed TSIGP and TSIGN to TDACP and TDACN, respectively, in the last paragraph of the <i>Test DAC (TDAC)</i> section.....	53
• Changed text in <i>Test DAC (TDAC)</i> section allowing for any common-mode value instead of 0 V.....	53
• Added note (1) to Figure 9-30	57
• Changed $t_{h(DRSP)}$ value of 16 from max to min.....	61
• Added stop-start sequence text to restart conversions in <i>Continuous Conversion Mode</i> section.....	61
• Deleted software polling text from <i>Data Ready (DRDY)</i> section.....	67
• Added <i>Conversion Data Software Polling</i> section.....	67
• Added text to clarify data reset at conversion restart.....	68
• Added text to <i>Read Data Direct (ADC1)</i> section to clarify conversion restart.....	68
• Changed Figure 9-43 to show complete list of CRC bit settings.....	68
• Changed text in <i>Read Data by Command</i> section to clarify software polling.....	69
• Changed Figure 9-44 to show complete list of CRC bit settings.....	69
• Added text to <i>Offset Calibration Registers</i> section regarding offset calibration disabled in chop mode.....	76
• Added new step 1 to <i>Calibration Command Procedure</i> section.....	79
• Added text to <i>WREG Command</i> section regarding conversion restart.....	87
• Changed text in 2nd paragraph of <i>Register Map</i> section.....	88
• Changed <i>Group Update</i> column of Table 9-34	88
• Added software polling to Figure 10-16	120

5 Device Comparison

PRODUCT	INPUTS	AUXILIARY 24-BIT ADC
ADS1262	11	No
ADS1263	11	Yes

6 Pin Configuration and Functions

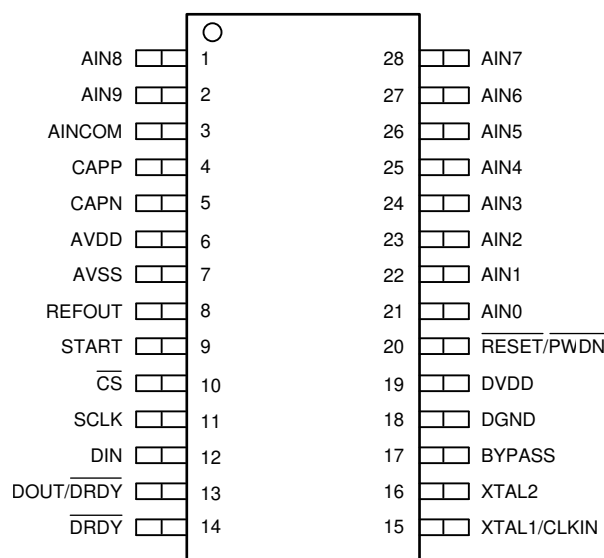


Figure 6-1. PW Package, 28-Pin TSSOP, Top View (Not To Scale)

Table 6-1. Pin Functions

PIN		I/O	DESCRIPTION
NO.	NAME		
1	AIN8	Analog input/output	Analog input 8, IDAC1, IDAC2, GPIO5
2	AIN9	Analog input/output	Analog input 9, IDAC1, IDAC2, GPIO6
3	AINCOM	Analog input/output	Analog input common, IDAC1, IDAC2, GPIO7, VBIAS
4	CAPP	Analog output	PGA output P: connect a 4.7-nF C0G dielectric capacitor from CAPP to CAPN
5	CAPN	Analog output	PGA output N: connect a 4.7-nF C0G dielectric capacitor from CAPP to CAPN
6	AVDD	Analog	Positive analog power supply
7	AVSS	Analog	Negative analog power supply
8	REFOUT	Analog Output	Internal reference voltage output, connect 1-μF capacitor to AVSS
9	START	Digital Input	Start conversion control
10	CS	Digital Input	Serial interface chip select (active low)
11	SCLK	Digital Input	Serial interface shift clock
12	DIN	Digital Input	Serial interface data input
13	DOUT/DRDY	Digital output	Serial interface data output and data ready indicator (active low)
14	DRDY	Digital output	Data ready indicator (active low)
15	XTAL1/CLKIN	Digital Input	1) Internal oscillator: Connect to DGND 2) External clock: Connect clock input 3) Crystal oscillator: Connect to crystal and crystal load capacitor
16	XTAL2	Digital Input	1) Internal oscillator: No connection (float) 2) External clock: No connection (float) 3) Crystal oscillator: Connect to crystal and crystal load capacitor
17	BYPASS	Analog Output	2-V sub-regulator external bypass; connect 1-μF capacitor to DGND
18	DGND	Digital	Digital ground

Table 6-1. Pin Functions (continued)

PIN		I/O	DESCRIPTION
NO.	NAME		
19	DVDD	Digital	Digital power supply
20	RESET/PWDN	Digital input	Reset (active low); hold low to power down the ADC
21	AIN0	Analog input/output	Analog input 0, REFP1, IDAC1, IDAC2
22	AIN1	Analog input/output	Analog input 1, REFN1, IDAC1, IDAC2
23	AIN2	Analog input/output	Analog input 2, REFP2, IDAC1, IDAC2
24	AIN3	Analog input/output	Analog input 3, REFN2, IDAC1, IDAC2, GPIO0
25	AIN4	Analog input/output	Analog input 4, REFP3, IDAC1, IDAC2, GPIO1
26	AIN5	Analog input/output	Analog input 5, REFN3, IDAC1, IDAC2, GPIO2
27	AIN6	Analog input/output	Analog input 6, IDAC1, IDAC2, GPIO3, TDACP
28	AIN7	Analog input/output	Analog input 7, IDAC1, IDAC2, GPIO4, TDACN

7 Specifications

7.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)⁽¹⁾

		MIN	MAX	UNIT
Voltage	AVDD to AVSS	−0.3	7	V
	AVSS to DGND	−3	0.3	V
	DVDD to DGND	−0.3	7	V
	Analog input	$V_{AVSS} - 0.3$	$V_{AVDD} + 0.3$	V
	Digital input	$V_{DGND} - 0.3$	$V_{DVDD} + 0.3$	V
Current	Input ⁽²⁾	−10	10	mA
Temperature	Junction, T_J	−50	150	°C
	Storage, T_{stg}	−60	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) Input pins are diode-clamped to the power supply rails. Limit the input current to 10 mA or less if the analog input voltage exceeds $V_{AVDD} + 0.3$ V or is below $V_{AVSS} - 0.3$ V, or if the digital input voltage exceeds $V_{DVDD} + 0.3$ V or is below $V_{DGND} - 0.3$ V.

7.2 ESD Ratings

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 ⁽¹⁾	±2000	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101 ⁽²⁾	±500	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.

(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

7.3 Recommended Operating Conditions

over operating ambient temperature range (unless otherwise noted)

			MIN	NOM	MAX	UNIT
POWER SUPPLY						
	Analog power supply	V_{AVDD} to V_{AVSS}	4.75	5	5.25	V
		V_{AVSS} to V_{DGND}	−2.6		0	
	Digital power supply	V_{DVDD} to V_{DGND}	2.7		5.25	V
ADC1 ANALOG INPUTS						
FSR	Full-scale differential input voltage range ⁽¹⁾		− V_{REF} / Gain		V_{REF} / Gain	V
V_{INP}, V_{INN}	Absolute input voltage ⁽²⁾	PGA enabled	See Equation 12			V
		PGA bypassed	$V_{AVSS} - 0.1$		$V_{AVDD} + 0.1$	
ADC2 ANALOG INPUTS (ADS1263)						
	Full-scale differential input voltage range		− V_{REF} / Gain		V_{REF} / Gain	V
	Absolute input voltage	Gain = 1, 2 and 4	$V_{AVSS} - 0.1$		$V_{AVDD} + 0.1$	V
		Gain = 8 to 128	See Equation 15			
VOLTAGE REFERENCE INPUTS						
V_{REF}	Differential reference voltage	$V_{REF} = V_{REFP} - V_{REFN}$	0.9		$V_{AVDD} - V_{AVSS} + 0.2$	V
V_{REFN}	Negative reference voltage		$V_{AVSS} - 0.1$		$V_{REFP} - 0.9$	V
V_{REFP}	Positive reference voltage		$V_{REFN} + 0.9$		$V_{AVDD} + 0.1$	V
CLOCK INPUT						
f_{CLK}	External clock frequency		1	7.3728	8	MHz
	External clock duty cycle		30%		70%	
	External crystal frequency		1	7.3728	8	MHz
GENERAL-PURPOSE INPUT/OUTPUT (GPIO)						
	Input voltage		V_{AVSS}		V_{AVDD}	V
DIGITAL INPUTS (other than GPIO)						
	Input voltage		V_{DGND}		V_{DVDD}	V
TEMPERATURE						
T_A	Operating ambient temperature		−40		125	°C

- (1) FSR is the ideal full-scale differential input voltage range, excluding noise, offset and gain errors. For ADC1, the maximum FSR is achieved with $V_{REF} = 5$ V and the PGA bypassed. If the PGA is enabled and $V_{REF} = 5$ V, the FSR is limited by the PGA input range. For ADC2, if $V_{REF} = 5$ V and gains = 8 to 128 then FSR is limited by the PGA input range.

- (2) V_{INP}, V_{INN} = Absolute Input Voltage. V_{IN} = Differential Input Voltage = $V_{INP} - V_{INN}$.

7.4 Thermal Information

THERMAL METRIC ⁽¹⁾		ADS126x	UNIT
		PW (TSSOP)	
		28 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	65.2	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	13.6	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	23.6	°C/W
ψ_{JT}	Junction-to-top characterization parameter	0.5	°C/W
ψ_{JB}	Junction-to-board characterization parameter	23.1	°C/W
$R_{\theta JC(bot)}$	Junction-to-case (bottom) thermal resistance	N/A	°C/W

- (1) For more information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics application report](#).

7.5 Electrical Characteristics

minimum and maximum specifications apply from $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$; typical specifications are at $T_A = 25^{\circ}\text{C}$; all specifications are at $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, $f_{CLK} = 7.3728\text{ MHz}$, ADC1 data rate = 20 SPS with PGA enabled and gain = 1, and ADC2 data rate = 10 SPS with gain = 1 (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
ADC1 ANALOG INPUTS							
	Absolute input current	Gain = 32		2		nA	
		PGA bypassed		150			
	Differential input current	Gain = 32		0.1		nA	
		PGA bypassed, V _{IN} = 5 V		150			
	Differential input impedance	PGA enabled		1		GΩ	
		PGA bypassed		40		MΩ	
	Channel-to-channel crosstalk	DC, V _{AVSS} ≤ V _{INX} ≤ V _{AVDD}		0.5		μV/V	
ADC1 PERFORMANCE							
	PGA gain			1, 2, 4, 8, 16, 32		V/V	
	Resolution			32		Bits	
DR	Data rate			2.5 38400		SPS	
	Noise performance			See Table 8-1			
INL	Integral nonlinearity	Gain = 1 to 32, PGA bypassed		3 12		ppm	
V _{OS}	Offset voltage	T _A = 25°C	Chop mode off	350 / Gain	800 / Gain	μV	
			Chop mode on	±0.1 / Gain	±0.5 / Gain		
		After calibration ⁽¹⁾		Noise / 4			
	Offset voltage drift	Chop mode off		30 / Gain + 10	100 / Gain + 50	nV/°C	
		Chop mode on		1	5		
GE	Gain error	T _A = 25°C, gain = 1 to 32		±50	±300	ppm	
		After calibration ⁽¹⁾		Noise / 4			
	Gain drift	Gain = 1 to 32, and PGA bypassed		0.5	4	ppm/°C	
NMRR	Normal-mode rejection ratio ⁽²⁾			See Table 9-6			
CMRR	Common-mode rejection ratio ⁽³⁾	f _{IN} = 60 Hz, data rate = 20 SPS		130		dB	
		f _{IN} = 60 Hz, data rate = 400 SPS		100	120		
PSRR	Power-supply rejection ratio ⁽⁴⁾	AVDD and AVSS		80	90	dB	
		DVDD		80	120		
ADC2 ANALOG INPUTS (ADS1263)							
	Absolute input current	Gain = 16		2		nA	
	Differential input current	Gain = 16		0.5		nA	
ADC2 PERFORMANCE (ADS1263)							
	Gain			1, 2, 4, 8, 16, 32, 64, 128		V/V	
	Resolution			24		Bits	
DR	Data rate			10, 100, 400, 800		SPS	
	Noise performance			See Table 8-3			
INL	Integral nonlinearity	Gain = 1 to 64		4	20	ppm	
		Gain = 128		7	30		
V _{OS}	Offset voltage	T _A = 25°C, gain = 1 to 128		±150	±500	μV	
	Offset voltage drift	Gain = 1 to 128		30	200	nV/°C	
GE	Gain error	T _A = 25°C, gain = 1 to 128		±500	±3000	ppm	
	Gain drift	Gain = 1 to 128		1	5	ppm/°C	
NMRR	Normal-mode rejection ratio			See Table 9-11			
CMRR	Common-mode rejection ratio	f _{IN} = 60 Hz, DR = 10 SPS		110		dB	
		f _{IN} = 60 Hz, DR = 400 SPS, gain = 8		75	90		
PSRR	Power-supply rejection ratio	AVDD and AVSS		75	90	dB	

7.5 Electrical Characteristics (continued)

minimum and maximum specifications apply from $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$; typical specifications are at $T_A = 25^{\circ}\text{C}$; all specifications are at $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, $f_{CLK} = 7.3728\text{ MHz}$, ADC1 data rate = 20 SPS with PGA enabled and gain = 1, and ADC2 data rate = 10 SPS with gain = 1 (unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP	MAX	UNIT	
CROSSTALK							
	Crosstalk	ADC1 to ADC2	20			μV/V	
		ADC2 to ADC1	1				
EXTERNAL VOLTAGE REFERENCE INPUTS							
	Reference input current ⁽⁵⁾	ADC1	150			nA	
		ADC2	1				
	Input current vs voltage	V _{REF} = 2 V to 4.8 V, ADC1	10			nA/V	
	Input current drift	ADC1	0.1			nA/°C	
	Input impedance	Differential, ADC1	50			MΩ	
	Low reference monitor	Threshold, ADC1	0.4			0.6	V
INTERNAL VOLTAGE REFERENCE							
	Reference voltage		2.5			V	
	Initial accuracy	T _A = 25°C	±0.1%			±0.2%	
	Reference voltage temperature drift	T _A = 0°C to +85°C	2			6	ppm/°C
		T _A = −40°C to +105°C	4			12	
	Reference voltage long term drift	T _A = 85°C, 1st 1000 hr	50			ppm	
	Thermal hysteresis	First 0°C to 85°C cycle	50			ppm	
	Output current		-10			10	mA
	Load regulation		40			μV/mA	
	Start-up time	Settling time to ±0.001% final value	50			ms	
TEMPERATURE SENSOR							
	Voltage	T _A = 25°C	122.4			mV	
	Temperature coefficient		420			μV/°C	
CURRENT SOURCES (IDAC1, IDAC2)							
	Currents		50, 100, 250, 500, 750, 1000, 1500, 2000, 2500, 3000			μA	
	Compliance range	All currents	V _{AVSS}			V _{AVDD} − 1.1	V
	Absolute error	All currents	±0.7%			±4%	
	Match error	IDAC1 current = IDAC2 current	±0.1%			±1%	ppm/°C
		IDAC1 current ≠ IDAC2 current	±1%				
	Temperature drift	Absolute	50			ppm/°C	
		Match	5				
LEVEL-SHIFT VOLTAGE							
	Voltage		(V _{AVDD} + V _{AVSS}) / 2			V	
	Output impedance		100			Ω	
SENSOR BIAS							
	Currents		±0.5, ±2, ±10, ±50, ±200			μA	
	Pull-up/pull-down resistor		10			MΩ	
TEST DAC (TDAC)							
	DAC reference voltage		V _{AVDD} − V _{AVSS}			V	
	Differential output voltage	18 binary weighted settings	−4			4	V
	Absolute output voltage	To V _{AVSS}	0.5			4.5	V
	Accuracy		±0.1%			±1.5%	
	Output impedance		See Table 9-8				

7.5 Electrical Characteristics (continued)

minimum and maximum specifications apply from $T_A = -40^{\circ}\text{C}$ to $+125^{\circ}\text{C}$; typical specifications are at $T_A = 25^{\circ}\text{C}$; all specifications are at $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, $f_{CLK} = 7.3728\text{ MHz}$, ADC1 data rate = 20 SPS with PGA enabled and gain = 1, and ADC2 data rate = 10 SPS with gain = 1 (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
PGA OVER-RANGE MONITOR							
	Differential alarm	Threshold		±105%			FSR
	Differential alarm accuracy			±1%		±3%	
	Absolute alarm thresholds	Low threshold		V _{AVSS} + 0.2			V
		High threshold		V _{AVDD} − 0.2			V
ADC CLOCK							
f _{CLK}	Internal oscillator frequency			7.3728			MHz
	Internal oscillator accuracy			±0.5%		±2%	
	External crystal start-up time	See Table 9-21 for recommended crystals		20			ms
GENERAL-PURPOSE INPUT/OUTPUTS (GPIO) ⁽⁶⁾							
V _{OH}	High-level output voltage	I _{OH} = 1 mA		0.8 · V _{AVDD}			V
V _{OL}	Low-level output voltage	I _{OL} = −1 mA		0.2 · V _{AVDD}			V
V _{IH}	High-level input voltage			0.7 · V _{AVDD} V _{AVDD}			V
V _{IL}	Low-level input voltage			V _{AVSS} 0.3 · V _{AVDD}			V
	Input hysteresis			0.5			V
DIGITAL INPUT/OUTPUT (Other Than GPIO)							
V _{OH}	High-level output voltage	I _{OH} = 1 mA		0.8 · V _{DVDD}			V
		I _{OH} = 8 mA		0.75 · V _{DVDD}			
V _{OL}	Low-level output voltage	I _{OL} = −1 mA		0.2 · V _{DVDD}			V
		I _{OL} = −8 mA		0.2 · V _{DVDD}			
V _{IH}	High-level input voltage			0.7 · V _{DVDD} V _{DVDD}			V
V _{IL}	Low-level input voltage			V _{DGND} 0.3 · V _{DVDD}			V
	Input hysteresis			0.1			V
	Input leakage			±10			μA
POWER SUPPLY							
I _{AVDD} I _{AVSS}	Analog supply current	Active mode, voltage reference off	ADS1262	4			mA
		Active mode, voltage reference on	ADS1262	4.2 6.5			
		Active mode, voltage reference on	ADS1263	4.3 6.5			
		Power-down mode		2 15			μA
I _{DVDD}	Digital supply current	Active mode	ADS1262 ADS1263	1 1.25			mA
		Power-down mode ⁽⁷⁾		25 50			μA
P _D	Power dissipation	Active mode, voltage reference on	ADS1262	24 37			mW
		Active mode, voltage reference on	ADS1263	25 37			
		Power-down mode		90 240			μW

- (1) Offset and gain calibration accuracy on the order of ADC conversion noise / 4. Conversion noise depends on data rate and PGA gain.
- (2) Normal-mode rejection ratio depends on the digital filter setting.
- (3) Common-mode rejection ratio is specified at data rate 20 SPS and 400 SPS.
- (4) Power-supply rejection ratio is specified at dc.
- (5) Specified with $V_{AVSS} \leq V_{REFN}$ and $V_{REFP} \leq V_{AVDD}$. For reference input voltage exceeding V_{AVDD} or V_{AVSS} , the ADC1 reference input current = 10 nA/ mV.
- (6) GPIO input and output voltages are referenced to V_{AVSS} .
- (7) External CLK input stopped. All other digital inputs maintained at V_{DVDD} or V_{DGND} .

7.6 Timing Requirements: Serial Interface

		MIN	MAX	UNIT
$t_{d(CSSC)}$	$\overline{CS}\downarrow$ before first SCLK $\uparrow$: delay time ⁽¹⁾	50		ns
$t_{d(DRSC)}$	$\overline{DRDY}\downarrow$ or $DOUT/\overline{DRDY}\downarrow$ before first SCLK $\uparrow$: delay time	0		ns
$t_{su(DI)}$	Valid DIN to SCLK $\downarrow$: setup time	35		ns
$t_{h(DI)}$	SCLK $\downarrow$ to valid DIN: hold time	25		ns
$t_c(SC)$	SCLK period ⁽²⁾	125	10^6	ns
$t_{w(SCH)}, t_{w(SCL)}$	SCLK high pulse duration or SCLK low pulse duration	40		ns
$t_{d(SCCS)}$	Last SCLK $\downarrow$ to CS $\uparrow$: delay time	40		ns
$t_{w(CSH)}$	$\overline{CS}$ high pulse duration	30		ns

(1) $\overline{CS}$ can be tied low.

(2) If serial interface time-out mode enabled, minimum SCLK frequency = 1 kHz. If serial interface time-out mode disabled (default), there is no minimum SCLK frequency.

7.7 Switching Characteristics: Serial Interface

over operating the ambient temperature range and DVDD = 2.7 V to 5.25 V (unless otherwise noted)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
$t_{w(DRH)}$	$\overline{DRDY}$ high pulse duration		16		$1/f_{CLK}$
$t_{p(CSDO)}$	$\overline{CS}\downarrow$ to $DOUT/\overline{DRDY}$ driven: propagation delay time	0		40	ns
$t_{p(SCDO)}$	SCLK $\uparrow$ to valid $DOUT/\overline{DRDY}$: propagation delay time			60	ns
$t_{h(SCDO)}$	SCLK $\uparrow$ to invalid $DOUT/\overline{DRDY}$: hold time	0			ns
$t_{p(CSDOZ)}$	$\overline{CS}\uparrow$ to $DOUT/\overline{DRDY}$ high impedance: propagation delay time			40	ns

7.8 Timing Diagrams

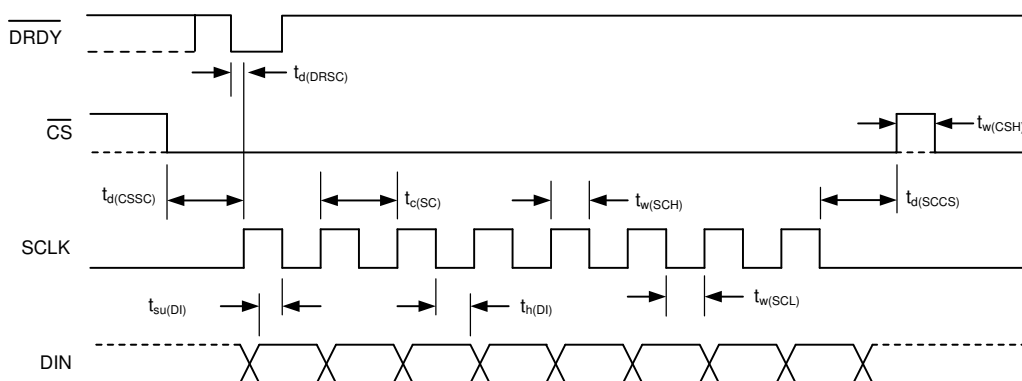


Figure 7-1. Serial Interface Timing Requirements

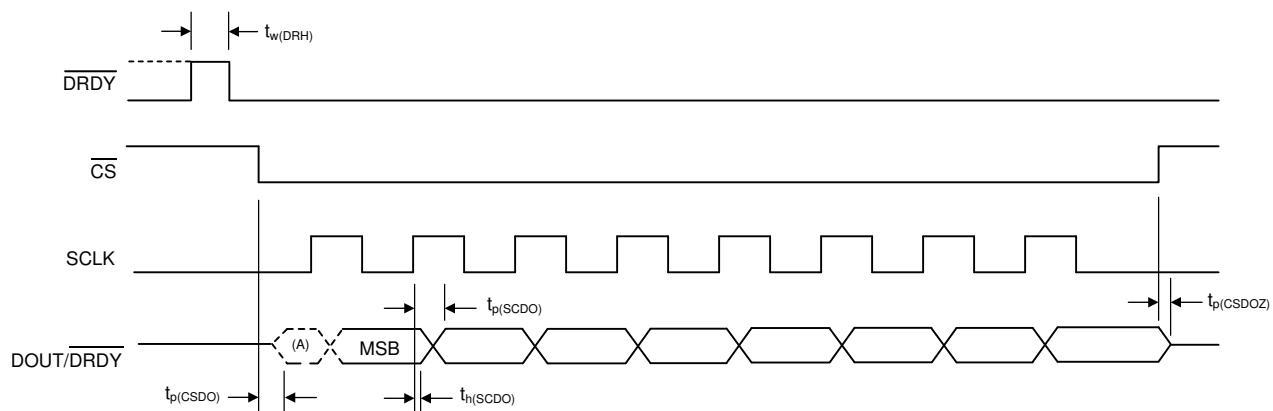


Figure 7-2. Serial Interface Switching Characteristics

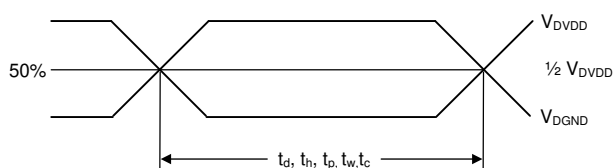
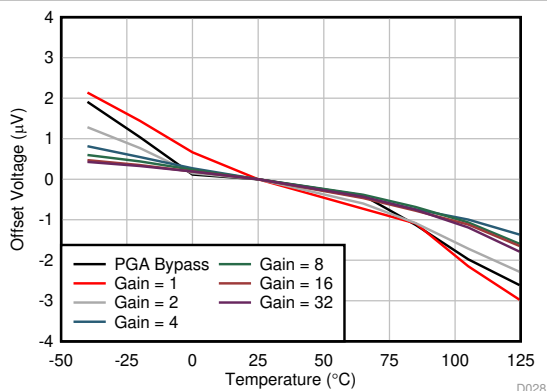


Figure 7-3. Timing Reference

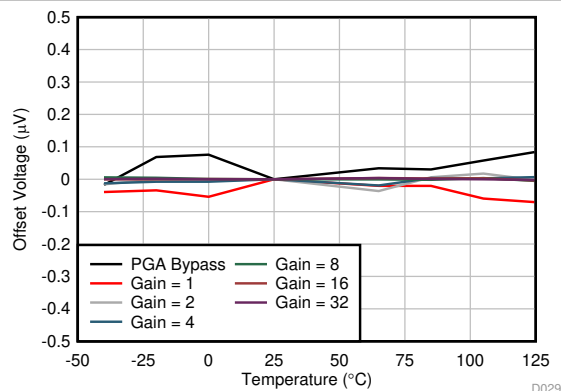
7.9 Typical Characteristics

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1



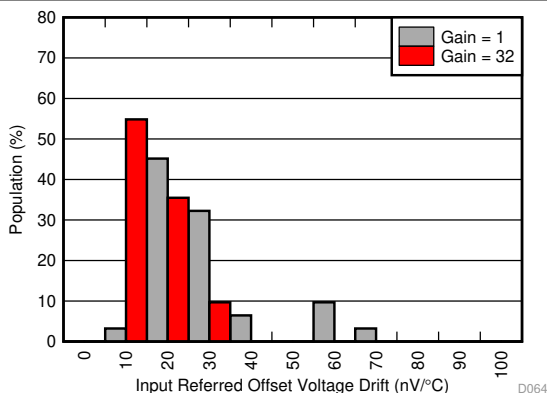
After offset calibration, shorted inputs

Figure 7-4. ADC1 Offset Voltage vs Temperature



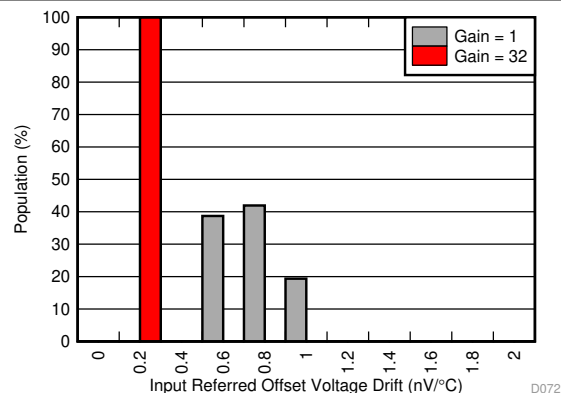
Chop mode on, after offset calibration, shorted inputs

Figure 7-5. ADC1 Offset Voltage vs Temperature



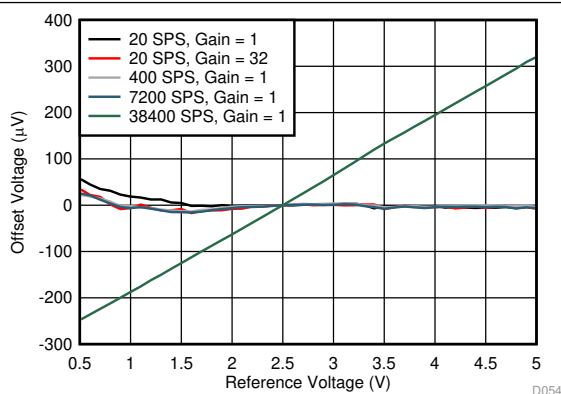
Shorted inputs, 30 units

Figure 7-6. ADC1 Offset Voltage vs Temperature Distribution



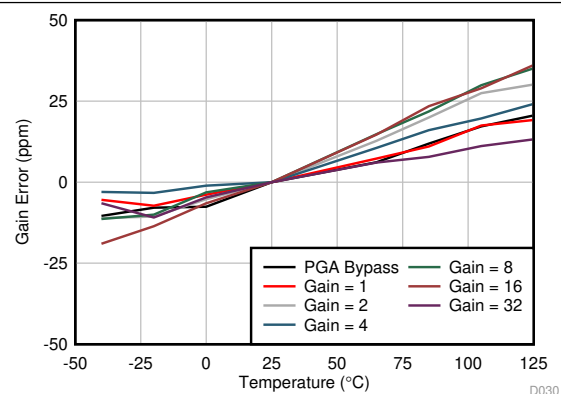
Chop mode on, shorted inputs, 30 units

Figure 7-7. ADC1 Offset Voltage vs Temperature Distribution



Shorted inputs

Figure 7-8. ADC1 Offset Voltage vs Reference Voltage



After gain calibration

Figure 7-9. ADC1 Gain Error vs Temperature

7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1

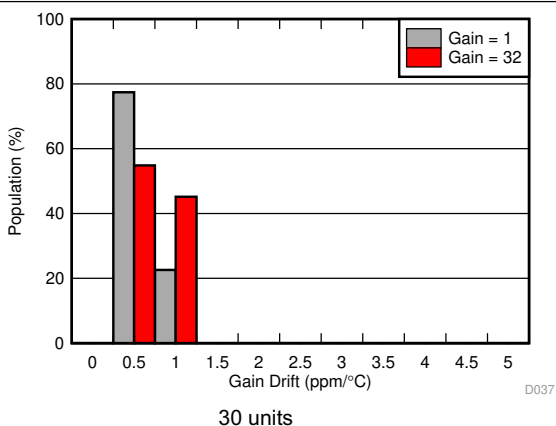


Figure 7-10. ADC1 Gain vs Temperature Distribution

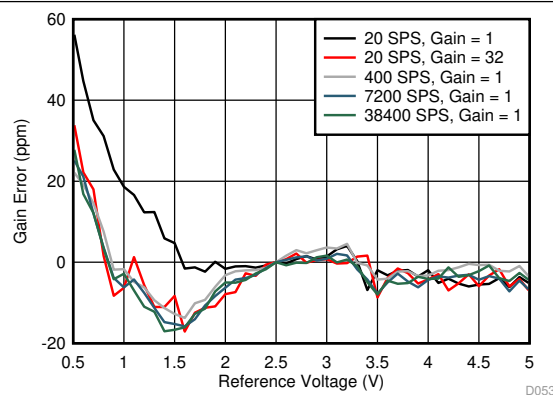


Figure 7-11. ADC1 Gain Error vs Reference Voltage

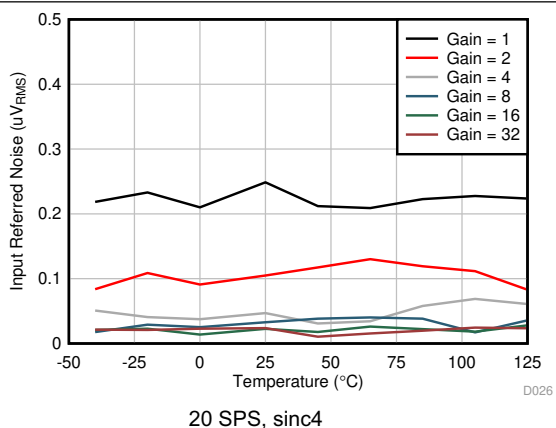


Figure 7-12. ADC1 Noise vs Temperature

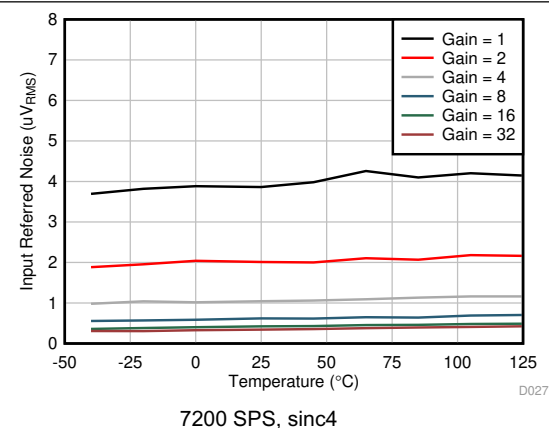


Figure 7-13. ADC1 Noise vs Temperature

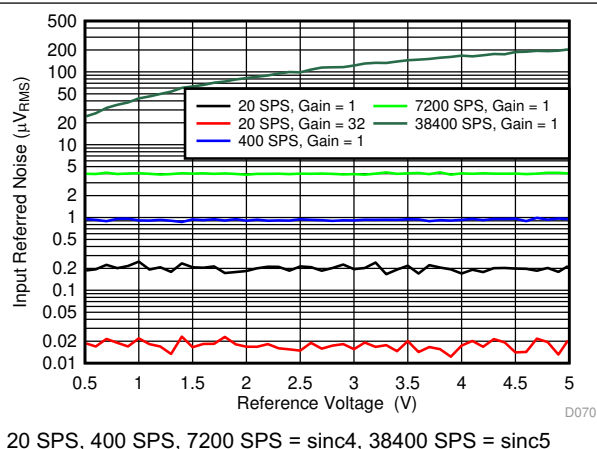


Figure 7-14. ADC1 Noise vs Reference Voltage

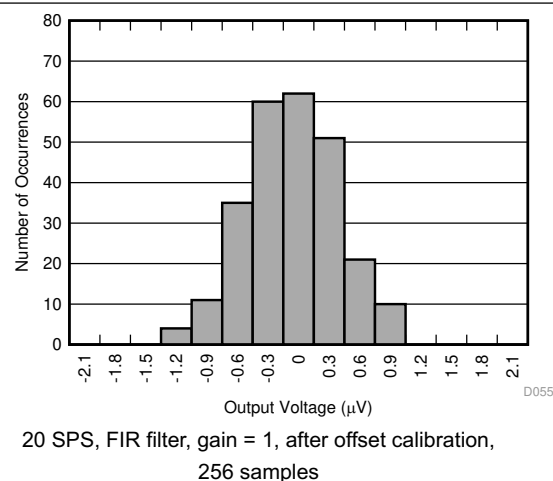
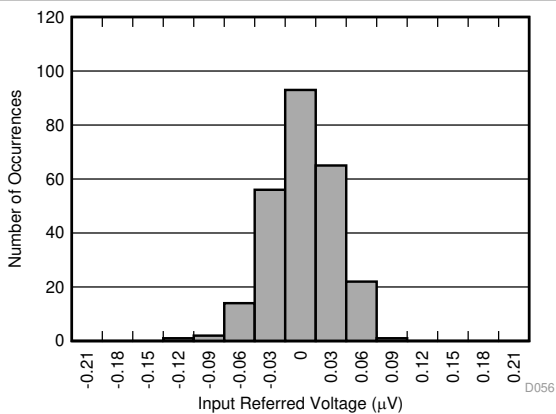


Figure 7-15. ADC1 Output Reading Distribution

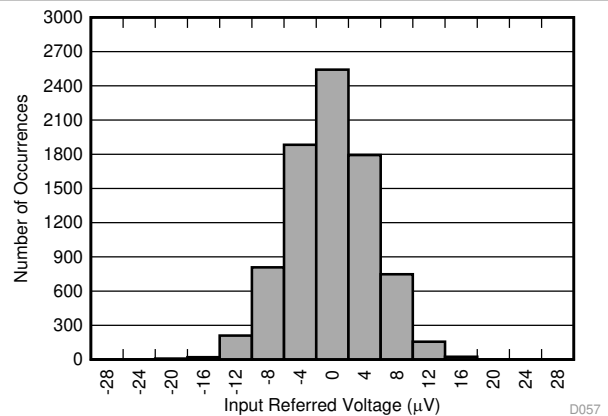
7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1



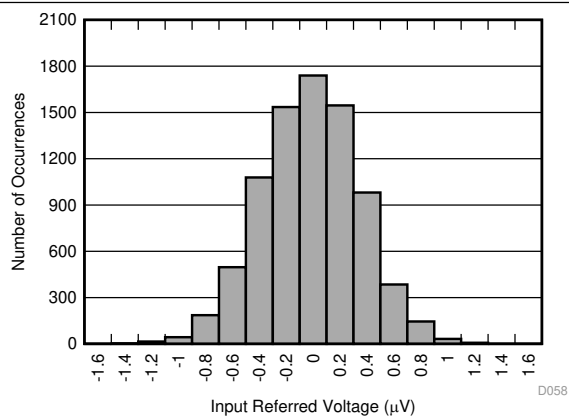
20 SPS, FIR filter, gain = 32, after offset calibration,
256 samples

Figure 7-16. ADC1 Output Reading Distribution



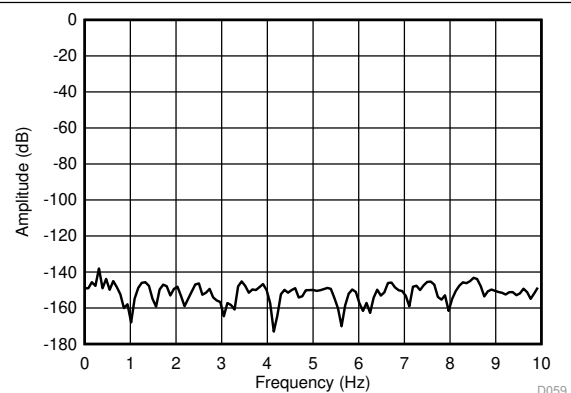
7200 SPS, sinc4 filter, gain = 1, after offset calibration,
8192 samples

Figure 7-17. ADC1 Output Reading Distribution



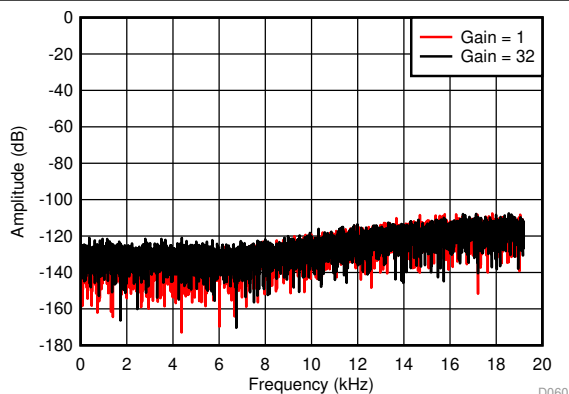
7200 SPS, sinc4 filter, gain = 32, after offset calibration,
8192 samples

Figure 7-18. ADC1 Output Reading Distribution



20 SPS, gain = 1, 256 points

Figure 7-19. ADC1 Output Spectrum



38400 SPS, 8192 points

Figure 7-20. ADC1 Output Spectrum

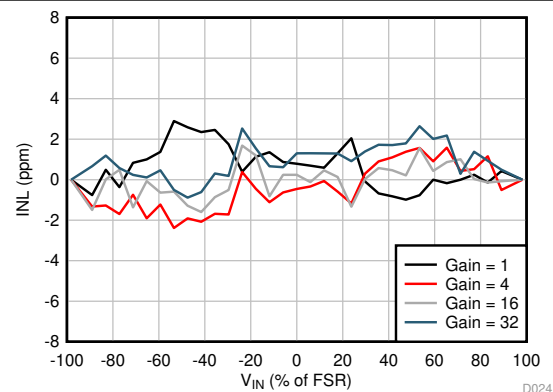


Figure 7-21. ADC1 INL vs V_{IN}

7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1

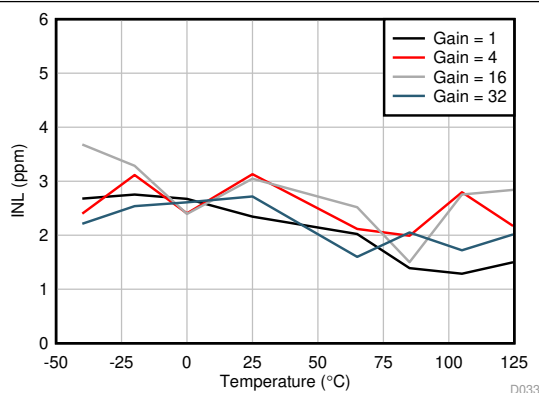


Figure 7-22. ADC1 INL vs Temperature

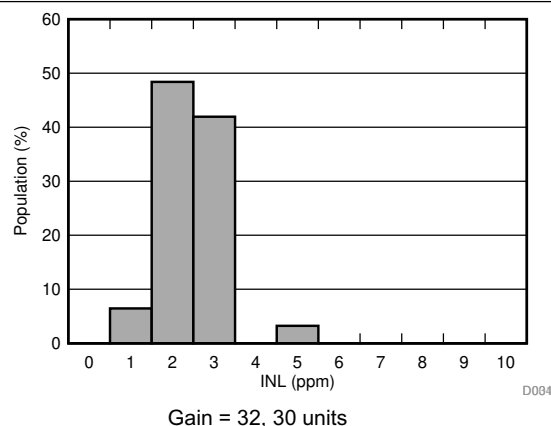


Figure 7-23. ADC1 INL Distribution

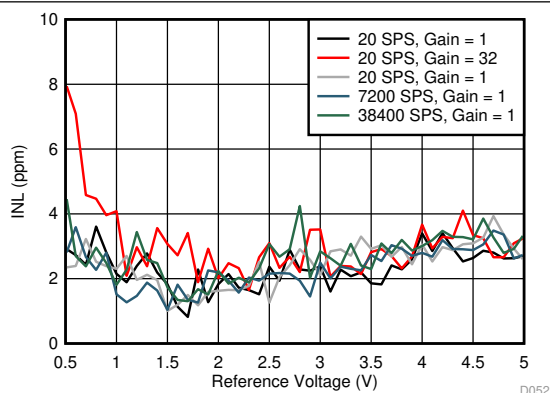


Figure 7-24. ADC1 INL vs Reference Voltage

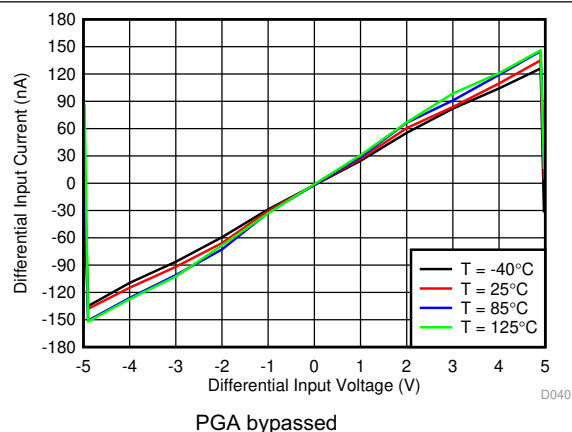


Figure 7-25. ADC1 Differential Input Current

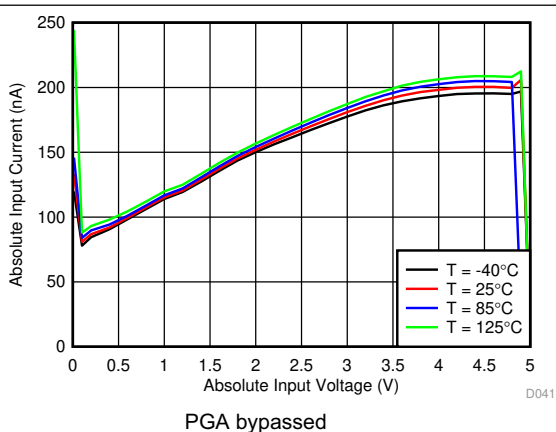


Figure 7-26. ADC1 Absolute Input Current

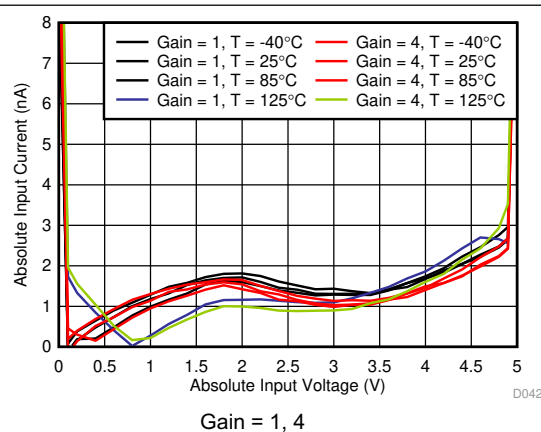
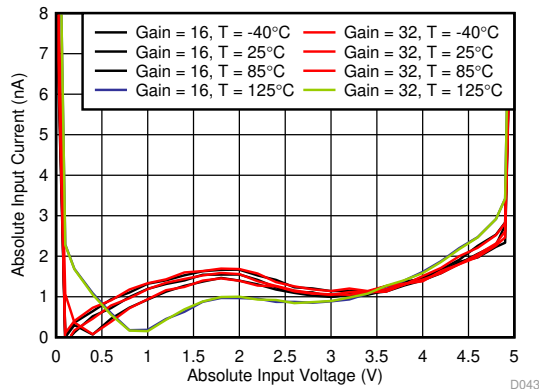


Figure 7-27. ADC1 Absolute Input Current

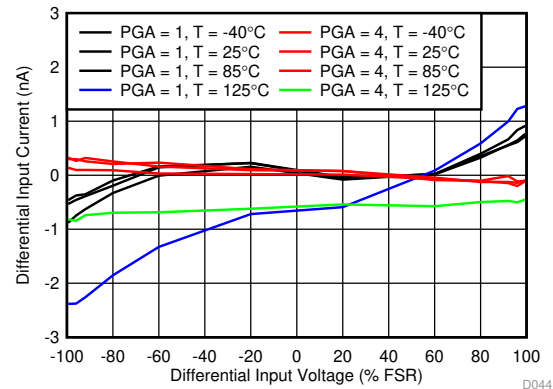
7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1



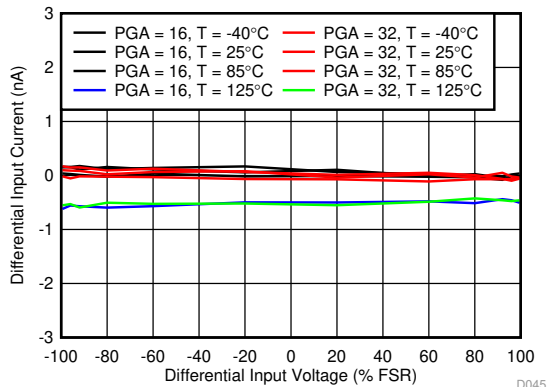
Gain = 16, 32

Figure 7-28. ADC1 Absolute Input Current



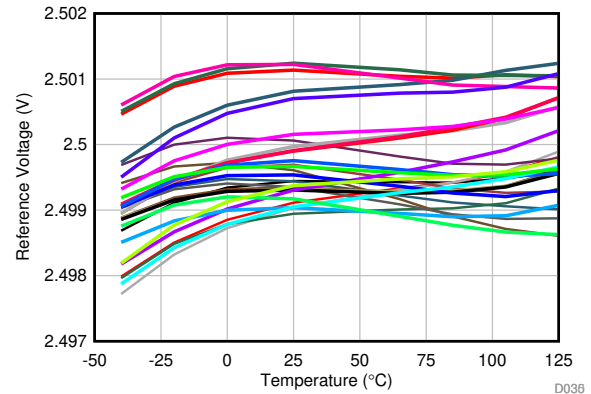
Gain = 1, 4

Figure 7-29. ADC1 Differential Input Current



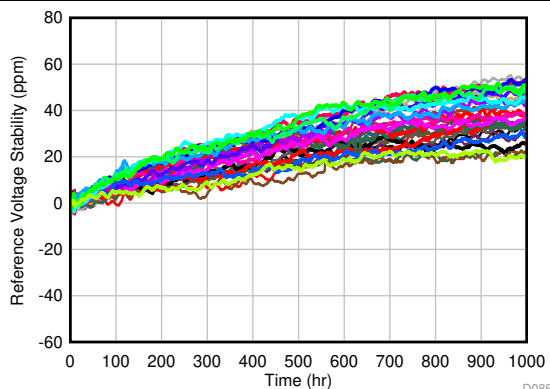
Gain = 16, 32

Figure 7-30. ADC1 Differential Input Current



30 units

Figure 7-31. Voltage Reference vs Temperature



$T_A = 85^\circ\text{C}$, 30 units

Figure 7-32. Voltage Reference Long-Term Drift

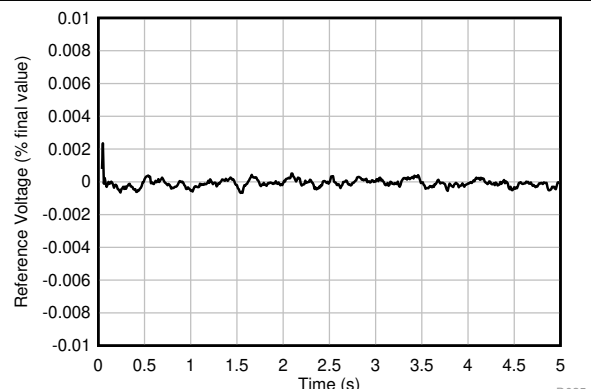


Figure 7-33. Voltage Reference Start-Up Time

7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1

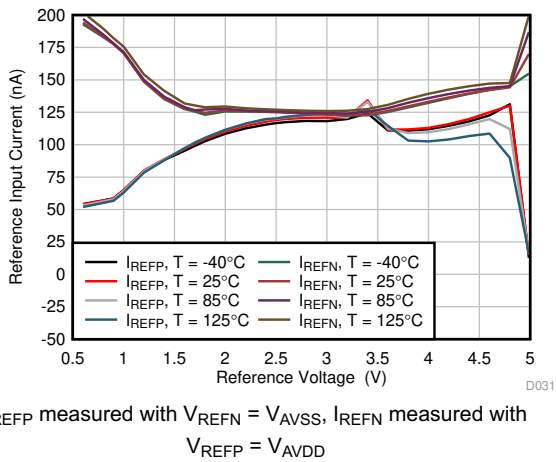


Figure 7-34. ADC1 Reference Input Current

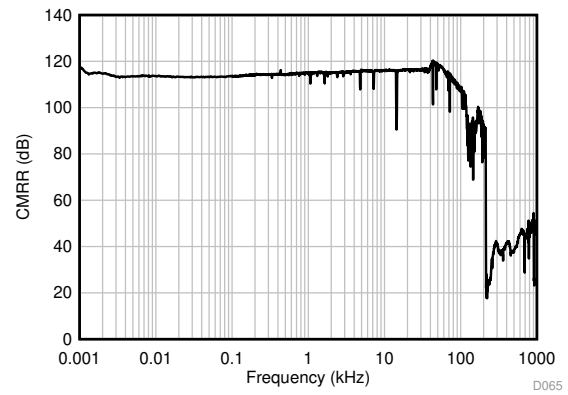


Figure 7-35. ADC1 CMRR vs Frequency

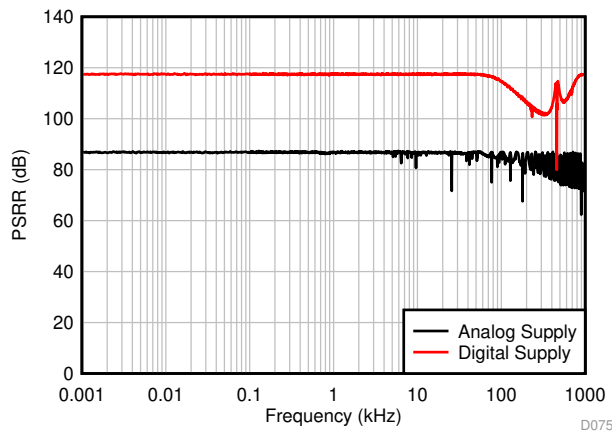


Figure 7-36. ADC1 PSRR vs Frequency

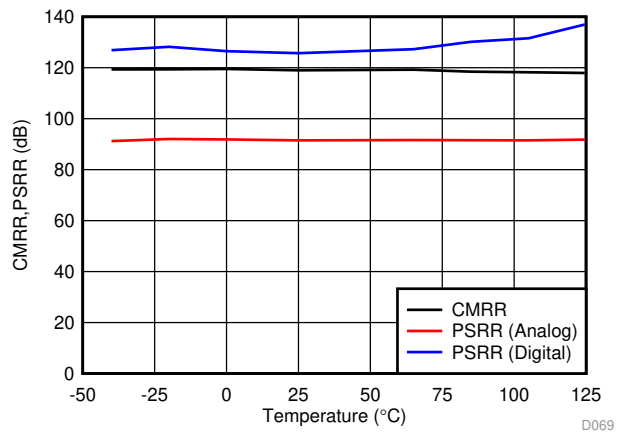


Figure 7-37. ADC1 CMRR, PSRR vs Temperature

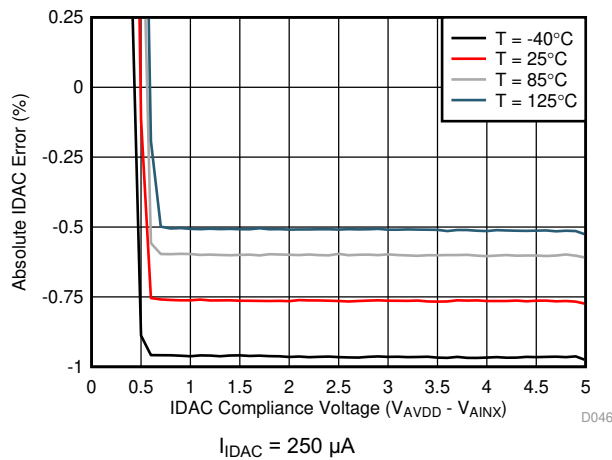


Figure 7-38. IDAC Error vs Compliance Voltage

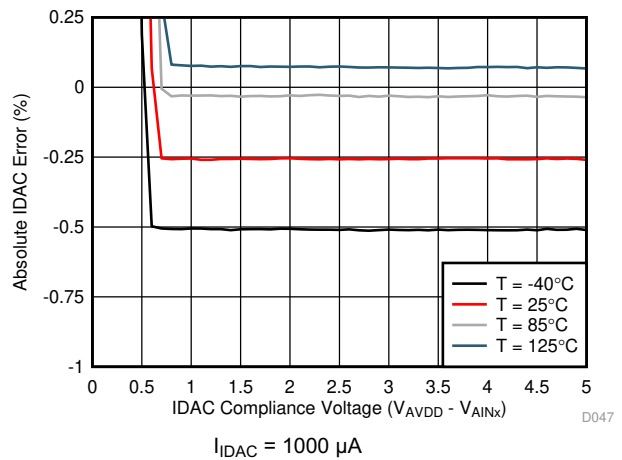


Figure 7-39. IDAC Error vs Compliance Voltage

7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1

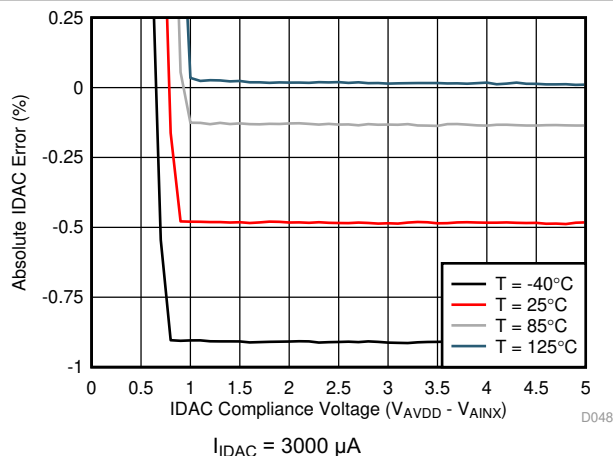


Figure 7-40. IDAC Error vs Compliance Voltage

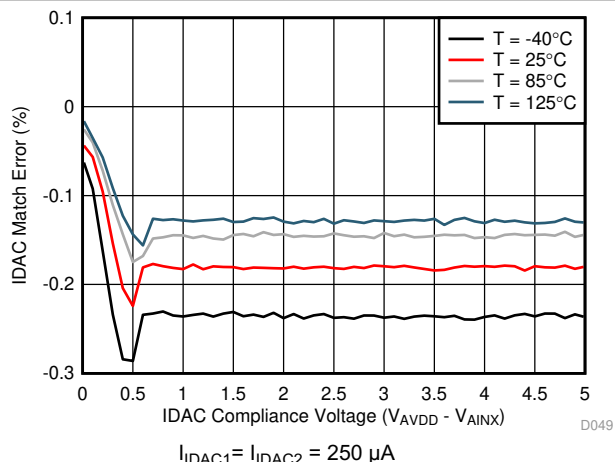


Figure 7-41. IDAC Current Error vs Compliance Voltage

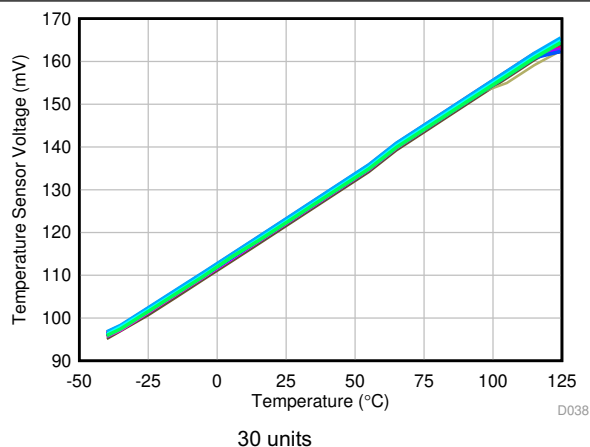


Figure 7-42. Temperature Sensor Voltage vs Temperature

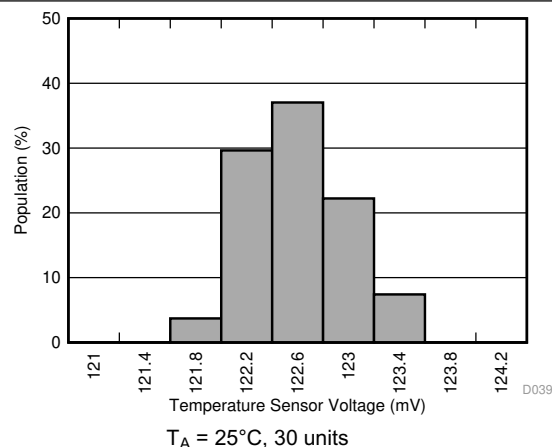


Figure 7-43. Temperature Sensor Voltage Distribution

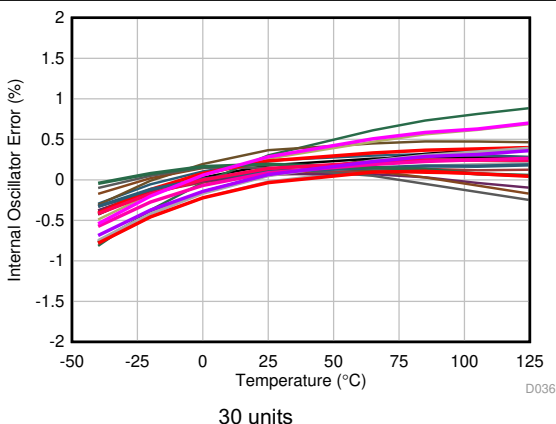


Figure 7-44. Internal Oscillator Frequency vs Temperature

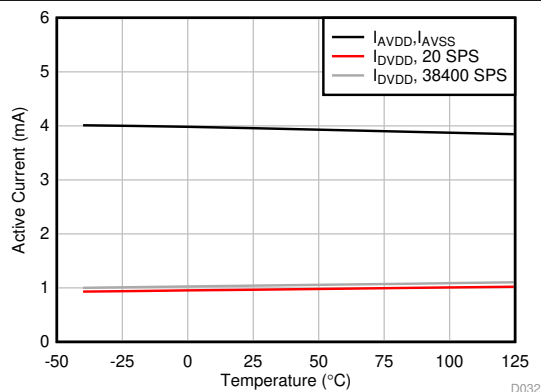


Figure 7-45. ADS1262 Active Current vs Temperature

7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1

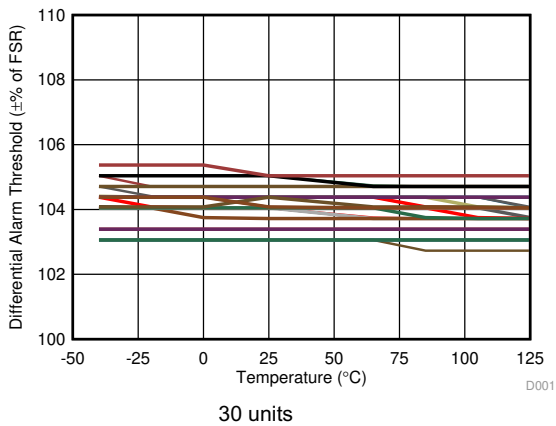


Figure 7-46. ADC1 Differential Over-range Alarm Threshold vs Temperature

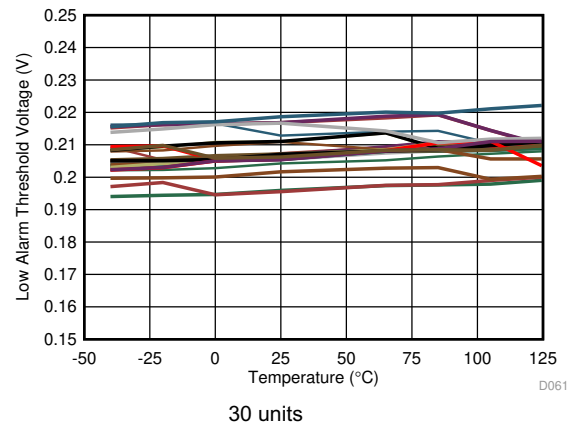


Figure 7-47. ADC1 Absolute Low Alarm Threshold vs Temperature

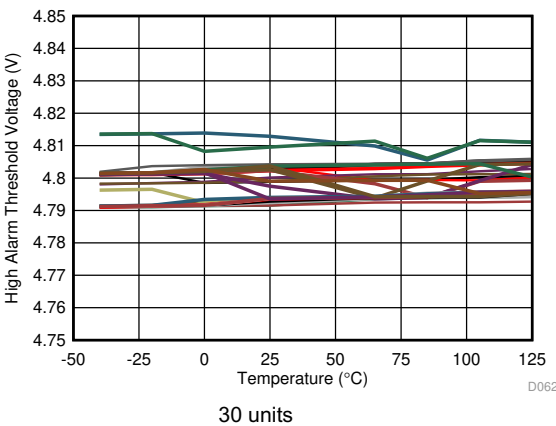


Figure 7-48. ADC1 Absolute High Alarm Threshold vs Temperature

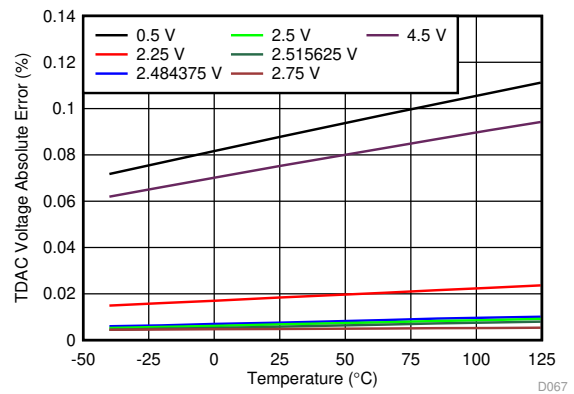


Figure 7-49. TDAC Error vs Temperature

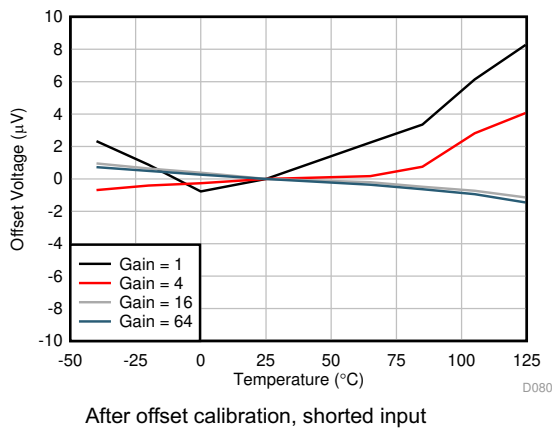


Figure 7-50. ADC2 Offset Voltage vs Temperature

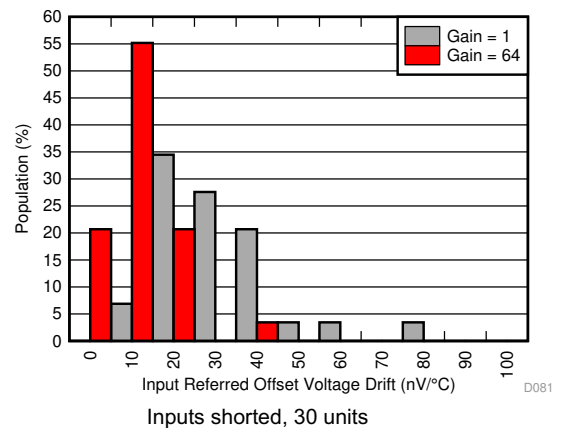


Figure 7-51. ADC2 Offset Voltage vs Temperature Distribution

7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1

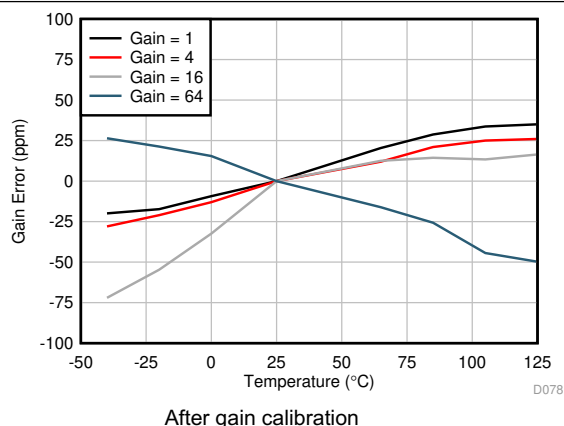


Figure 7-52. ADC2 Gain vs Temperature

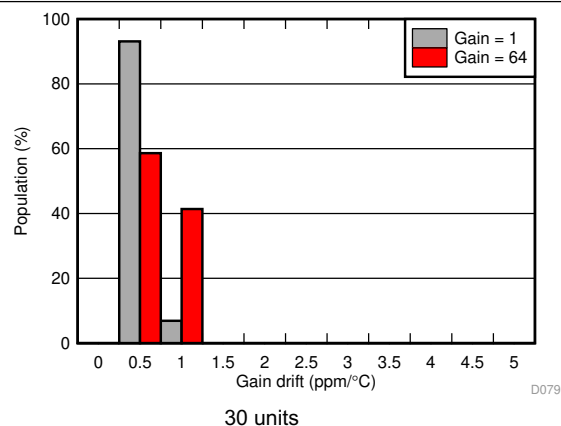
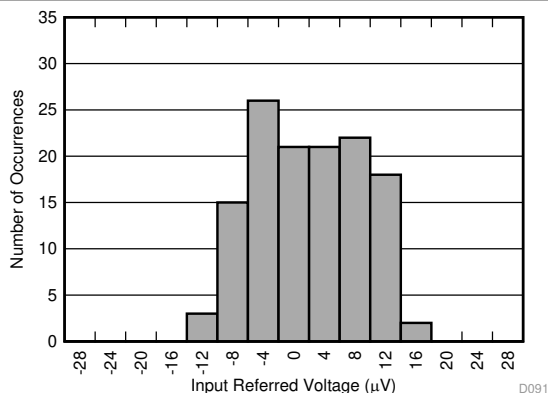
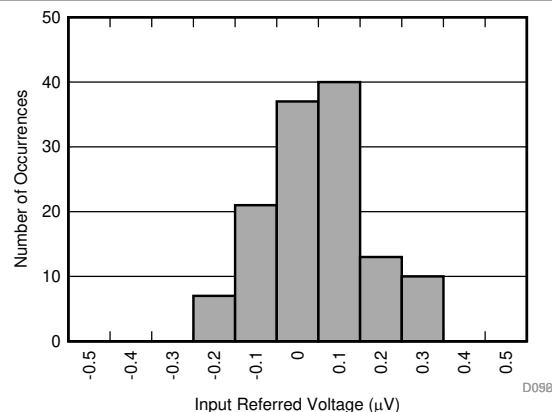


Figure 7-53. ADC2 Gain vs Temperature Distribution



Gain = 1, 10 SPS, after offset calibration, 128 samples

Figure 7-54. ADC2 Output Reading Distribution



Gain = 128, 10 SPS, after offset calibration, 128 samples

Figure 7-55. ADC2 Output Reading Distribution

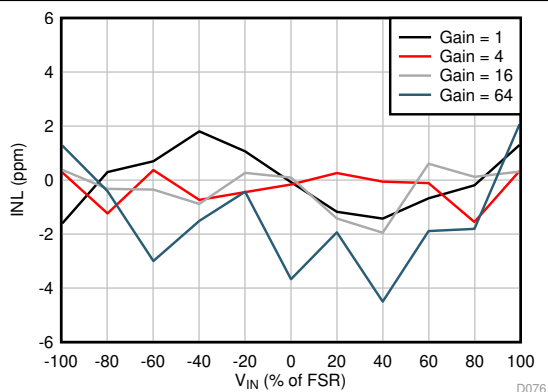


Figure 7-56. ADC2 INL vs V_{IN}

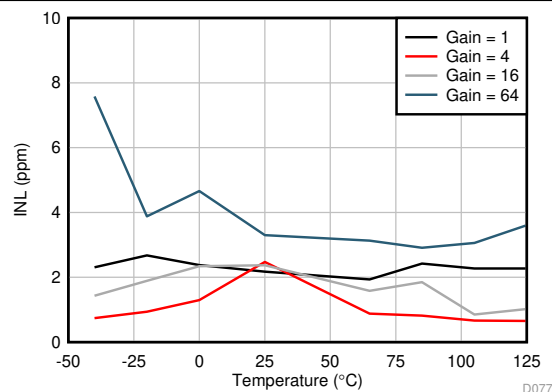


Figure 7-57. ADC2 INL vs Temperature

7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1

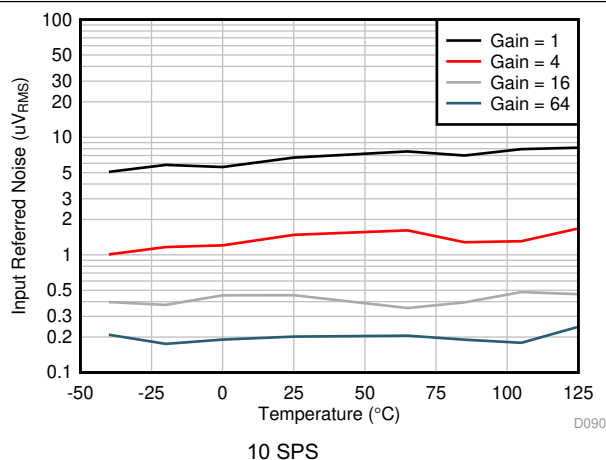


Figure 7-58. ADC2 Noise vs Temperature

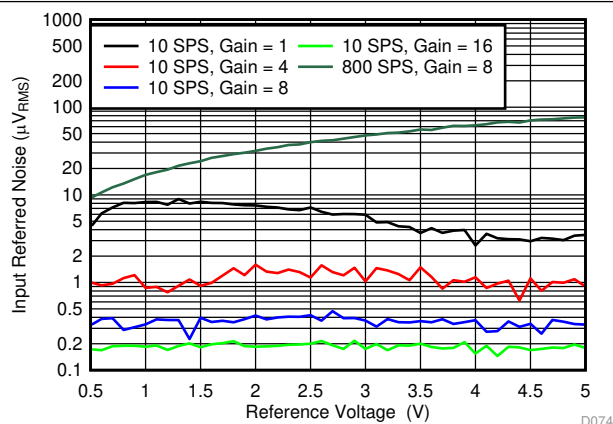


Figure 7-59. ADC2 Noise vs Reference Voltage

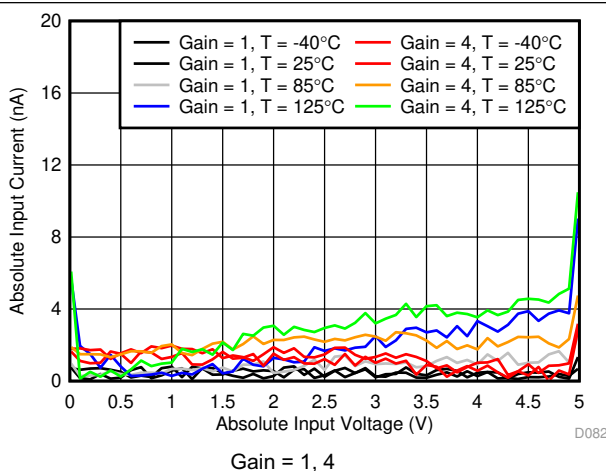


Figure 7-60. ADC2 Absolute Input Current

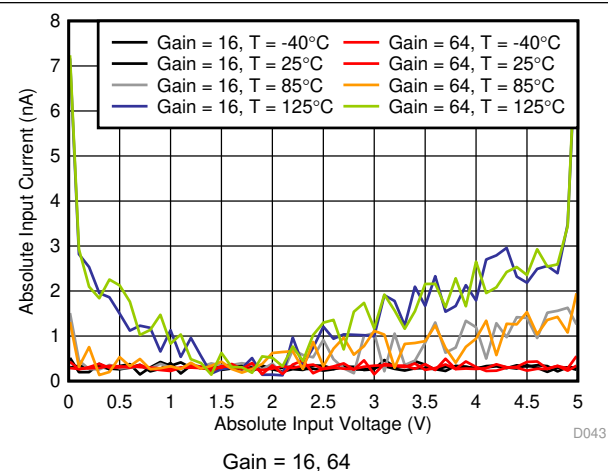


Figure 7-61. ADC2 Absolute Input Current

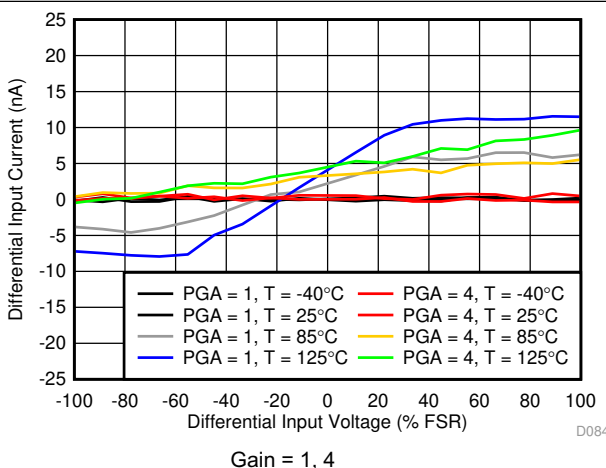


Figure 7-62. ADC2 Differential Input Current

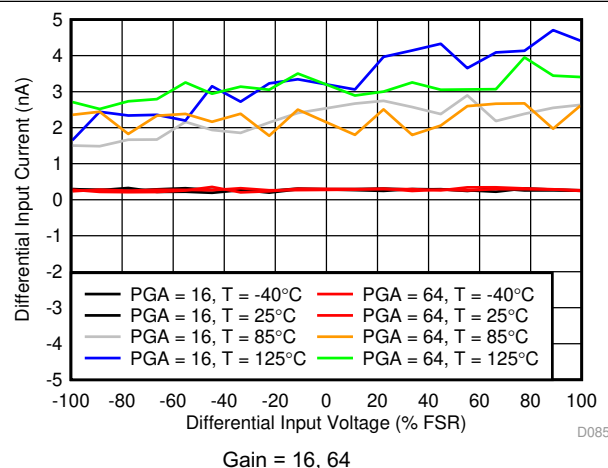


Figure 7-63. ADC2 Differential Input Current

7.9 Typical Characteristics (continued)

at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{DVDD} = 3.3\text{ V}$, $V_{REF} = 2.5\text{ V}$, and $f_{CLK} = 7.3728\text{ MHz}$ (unless otherwise noted); typical ADC1 characteristics at data rate = 20 SPS and gain = 1; typical ADC2 characteristics at data rate = 10 SPS and gain = 1

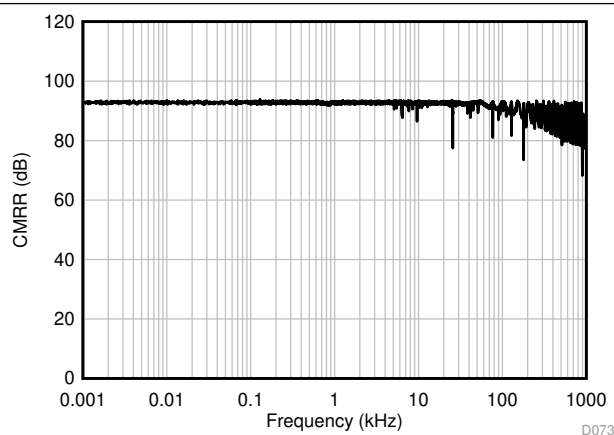
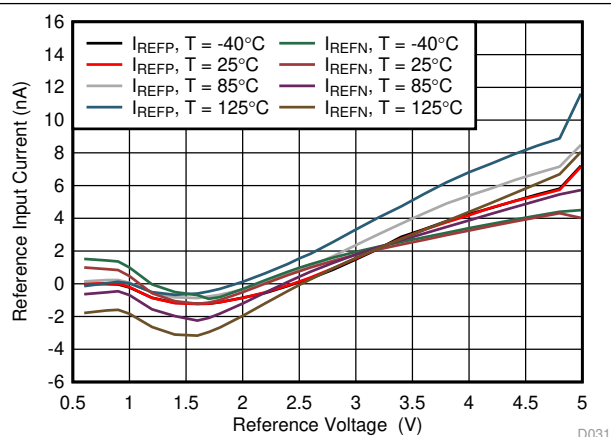


Figure 7-64. ADC2 CMRR vs Frequency



I_{REFP} measured with $V_{REFN} = V_{AVSS}$, I_{REFN} measured with $V_{REFP} = V_{AVDD}$

Figure 7-65. ADC2 Reference Input Current

8 Parameter Measurement Information

8.1 Offset Temperature Drift Measurement

Offset temperature drift is defined as the maximum change of offset voltage measured over the specified temperature range. The offset voltage drift is input referred and is calculated using the box method, as described by [Equation 1](#):

$$\text{Offset Voltage Drift} = (V_{\text{OSMAX}} - V_{\text{OSMIN}}) / (T_{\text{MAX}} - T_{\text{MIN}}) \quad (1)$$

where

- V_{OSMAX} and V_{OSMIN} are the maximum and minimum offset voltages, respectively
- T_{MAX} and T_{MIN} are the maximum and minimum temperatures, respectively, over the specified temperature range

8.2 Gain Temperature Drift Measurement

Gain temperature drift is defined as the maximum change of gain error measured over the specified temperature range. The gain error drift is calculated using the box method, as described by [Equation 2](#):

$$\text{Gain Error Drift} = (GE_{\text{MAX}} - GE_{\text{MIN}}) / (T_{\text{MAX}} - T_{\text{MIN}}) \quad (2)$$

where

- GE_{MAX} and GE_{MIN} are the maximum and minimum gain errors, respectively
- T_{MAX} and T_{MIN} are the maximum and minimum temperatures, respectively, over the specified temperature range

8.3 Common-Mode Rejection Ratio Measurement

Common-mode rejection ratio (CMRR) is defined as the rejection of the ADC output to an applied common-mode input voltage. The common-mode input is 60 Hz with a peak-to-peak amplitude equal to the specified absolute input voltage range. The standard deviation (RMS) value of the ADC output is calculated and scaled to volts. In order to measure CMRR, record two ADC readings. The first reading (V_A) is with no common-mode input signal. The first reading represents the baseline ADC noise. The second reading (V_B) is with the common-mode input applied. The second reading represents the combination of the ADC baseline noise plus the increased RMS noise caused by the common-mode input. The ADC baseline noise is extracted from the combined noise to yield the noise induced by the common-mode input voltage. The CMRR measurement is described by [Equation 3](#):

$$\text{CMRR} = 20 \cdot \log(V_{\text{IC}} / V_{\text{OC}}) \quad (3)$$

where

- V_{IC} = RMS value of the input common-mode voltage = $1.56 V_{\text{RMS}}$
- V_{OC} = Calculated RMS value of output voltage = $(V_B^2 - V_A^2)^{0.5}$
- V_A = RMS output voltage with no common-mode input
- V_B = RMS output voltage with common-mode input

For gains > 1, add 6 dB of compensation value for each binary increase of gain.

8.4 Power-Supply Rejection Ratio Measurement

Power-supply rejection ratio (PSRR) is defined as the rejection of the ADC output to the DC change of the power supply voltage referred to the input range. PSRR is calculated using two ADC mean-value readings with inputs shorted, scaled to volts. The first ADC reading (V_{OA}) is acquired at one power-supply voltage, and the second ADC reading (V_{OB}) is acquired after changing the power-supply voltage by 0.5 V. The PSRR calculation is described by [Equation 4](#):

$$\text{PSRR} = 20 \cdot \log |(V_{PSA} - V_{PSB}) / (V_{OA} - V_{OB})| - 20 \text{ dB} \quad (4)$$

where

- $V_{PSA} - V_{PSB}$ = power-supply DC voltage change = 0.5 V
- $V_{OA} - V_{OB}$ = ADC DC output voltage change (V)
- Range compensation factor = $20 \cdot \log (0.5 \text{ V} / 5 \text{ V}) = -20 \text{ dB}$ for gain = 1

For gains > 1, add an additional 6 dB of compensation value for each binary increase of gain.

8.5 Crosstalk Measurement (ADS1263)

Crosstalk is defined as the unintended coupling of signals between ADC1 and ADC2. Measure crosstalk by changing the dc input voltage of one ADC and measuring the rejection of the other ADC. The dc input voltage change is 0.3 V, and the gain of the affected ADC is 16. Acquire two mean-value readings of the affected ADC with inputs shorted. Take the first ADC reading (V_{OA}) with $V_{IN} = 0 \text{ V}$, and take the second ADC reading (V_{OB}) after changing the input voltage by 0.3 V. The crosstalk calculation is described by [Equation 5](#):

$$\text{Crosstalk} = |(V_{OA} - V_{OB}) / (V_{INA} - V_{INB})| \cdot 10^6 \text{ (}\mu\text{V/V)} \quad (5)$$

where

- $V_{OA} - V_{OB}$ = DC output voltage change of the affected ADC
- $V_{INA} - V_{INB}$ = DC input voltage change of the driven ADC = 0.3 V

8.6 Reference-Voltage Temperature-Drift Measurement

Internal reference-voltage temperature drift is defined as the maximum change in reference voltage measured over the specified temperature range. The reference voltage drift is calculated using the box method, as described by [Equation 6](#):

$$\text{Reference Drift} = (V_{\text{REFMAX}} - V_{\text{REFMIN}}) / (V_{\text{REFNOM}} \cdot (T_{\text{MAX}} - T_{\text{MIN}})) \cdot 10^6 \text{ (ppm)} \quad (6)$$

where

- V_{REFMAX} , V_{REFMIN} and V_{REFNOM} are the maximum, minimum and nominal ($T_A = 25^\circ\text{C}$) reference voltages, respectively
- T_{MAX} and T_{MIN} are the maximum and minimum temperatures, respectively, over the specified temperature range

8.7 Reference-Voltage Thermal-Hysteresis Measurement

Internal reference-voltage thermal hysteresis is defined as the change in reference voltage after operating the device at $T_A = 25^\circ\text{C}$, cycling the device through the $T_A = 0^\circ\text{C}$ to 85°C temperature range for ten minutes at each temperature and returning to $T_A = 25^\circ\text{C}$. The internal reference thermal hysteresis is defined in [Equation 7](#):

$$\text{Reference Thermal Hysteresis} = |V_{\text{REFPRE}} - V_{\text{REFPOST}}| / V_{\text{REFPRE}} \cdot 10^6 \text{ (ppm)} \quad (7)$$

where

- V_{REFPRE} and V_{REFPOST} are the reference voltages before and after the temperature cycle, respectively

8.8 Noise Performance

The ADC noise performance depends on the following ADC settings: PGA gain, data rate, digital filter mode, and chop mode. Generally, the lowest input-referred noise is achieved using the highest gain possible, consistent with the input signal range. Do not set the gain too high or the result is ADC overrange. Noise also depends on the output data rate and mode of the digital filter. As the data rate reduces, the ADC bandwidth correspondingly reduces. As the order of the digital filter mode increases, the ADC bandwidth also reduces. This reduction in total bandwidth results in lower overall noise. The ADC noise is reduced by a factor of 1.4 with chop mode enabled.

[Table 8-1](#) lists ADC1 noise performance in units of μV_{RMS} (RMS = root mean square) under the conditions shown. The values in parenthesis are peak-to-peak values. [Table 8-2](#) lists the noise performance in effective number of bits (ENOB) with an external 5-V reference voltage. The values shown in parenthesis are noise-free bits. The definition of *noise-free bits* is the resolution of the ADC with no code flicker. The noise-free bits data are based on the μV_{PP} values. Note that for data rate = 38400 SPS, noise scales with increased reference voltage. For all other data rates, noise does not scale with reference voltage.

[Table 8-3](#) lists the noise performance of ADC2 (ADS1263) in units of μV_{RMS} and (μV_{P-P}). The values in parenthesis are peak-to-peak values. [Table 8-4](#) lists the ENOB and noise-free bits of ADC2.

The ENOB and noise-free bits shown in the tables are calculated using [Equation 8](#):

$$ENOB = \ln(FSR / V_{NRMS}) / \ln(2) \quad (8)$$

where

- FSR = full scale range = $2 \cdot V_{REF}/Gain$
- V_{NRMS} = Input referred noise voltage

Achieve maximum ENOB with maximum FSR. For ADC1, achieve maximum FSR with $V_{REF} = 5\text{ V}$ and the PGA bypassed. If the PGA is enabled, the FSR is limited by the PGA input range (see the [Electrical Characteristics](#) table.) For ADC2, achieve maximum FSR with $V_{REF} = 5\text{ V}$ and gains = 1, 2, or 4. If gain = 8 to 128, then FSR is limited by the PGA input range (see the [Electrical Characteristics](#) table).

For ADC1 operation, if the reference voltage is equal to 5 V and the PGA is enabled, the available FSR is restricted because of the limited PGA range specification. For ADC2 operation, if the reference voltage is equal to 5 V, The FSR is reduced for ADC2 gains equal to or greater than eight because of the limited PGA range.

The data shown in the noise performance tables represent typical ADC performance at $T_A = 25^\circ\text{C}$. The noise-performance data are the standard deviation and peak-to-peak computations of the ADC data. Because of the statistical nature of noise, repeated noise measurements may yield higher or lower noise results. The noise data are acquired with inputs shorted, from consecutive ADC readings for a period of ten seconds or 8192 data points, whichever occurs first.

Table 8-1. ADC1 Noise in μV_{RMS} (μV_{PP}) at $T_A = 25^\circ C$, $V_{AVDD} = 5 V$, $V_{AVSS} = 0 V$, $V_{REF} = 2.5 V$

DATA RATE	FILTER MODE	GAIN					
		1	2	4	8	16	32
2.5 SPS	FIR	0.145 (0.637)	0.071 (0.279)	0.038 (0.149)	0.023 (0.089)	0.014 (0.064)	0.011 (0.051)
2.5 SPS	Sinc1	0.121 (0.510)	0.058 (0.249)	0.033 (0.143)	0.018 (0.073)	0.012 (0.054)	0.008 (0.037)
2.5 SPS	Sinc2	0.101 (0.437)	0.055 (0.225)	0.025 (0.104)	0.015 (0.064)	0.010 (0.043)	0.007 (0.031)
2.5 SPS	Sinc3	0.080 (0.307)	0.046 (0.195)	0.026 (0.116)	0.013 (0.052)	0.008 (0.034)	0.006 (0.023)
2.5 SPS	Sinc4	0.080 (0.308)	0.043 (0.180)	0.020 (0.078)	0.013 (0.049)	0.008 (0.031)	0.007 (0.027)
5 SPS	FIR	0.206 (1.007)	0.098 (0.448)	0.054 (0.252)	0.028 (0.123)	0.020 (0.098)	0.015 (0.073)
5 SPS	Sinc1	0.161 (0.726)	0.090 (0.432)	0.047 (0.246)	0.026 (0.120)	0.017 (0.083)	0.012 (0.057)
5 SPS	Sinc2	0.146 (0.661)	0.069 (0.308)	0.038 (0.195)	0.021 (0.100)	0.013 (0.061)	0.011 (0.050)
5 SPS	Sinc3	0.128 (0.611)	0.067 (0.325)	0.033 (0.153)	0.019 (0.095)	0.012 (0.054)	0.010 (0.046)
5 SPS	Sinc4	0.122 (0.587)	0.063 (0.269)	0.030 (0.144)	0.017 (0.076)	0.011 (0.048)	0.008 (0.039)
10 SPS	FIR	0.284 (1.418)	0.142 (0.753)	0.077 (0.379)	0.041 (0.197)	0.027 (0.156)	0.023 (0.118)
10 SPS	Sinc1	0.229 (1.220)	0.123 (0.662)	0.060 (0.322)	0.035 (0.177)	0.023 (0.118)	0.018 (0.103)
10 SPS	Sinc2	0.193 (1.019)	0.093 (0.488)	0.048 (0.254)	0.028 (0.149)	0.019 (0.099)	0.016 (0.079)
10 SPS	Sinc3	0.176 (0.896)	0.088 (0.452)	0.043 (0.217)	0.028 (0.137)	0.018 (0.091)	0.014 (0.067)
10 SPS	Sinc4	0.164 (0.788)	0.076 (0.389)	0.040 (0.200)	0.024 (0.119)	0.016 (0.081)	0.013 (0.065)
16.6 SPS	Sinc1	0.306 (1.708)	0.147 (0.810)	0.077 (0.436)	0.044 (0.250)	0.030 (0.176)	0.024 (0.138)
16.6 SPS	Sinc2	0.248 (1.401)	0.122 (0.729)	0.068 (0.403)	0.037 (0.213)	0.024 (0.136)	0.020 (0.111)
16.6 SPS	Sinc3	0.216 (1.221)	0.120 (0.667)	0.060 (0.332)	0.033 (0.197)	0.022 (0.130)	0.017 (0.095)
16.6 SPS	Sinc4	0.214 (1.169)	0.101 (0.544)	0.054 (0.302)	0.031 (0.175)	0.022 (0.129)	0.016 (0.092)
20 SPS	FIR	0.393 (2.467)	0.191 (1.102)	0.104 (0.603)	0.057 (0.353)	0.039 (0.222)	0.030 (0.167)
20 SPS	Sinc1	0.336 (1.861)	0.167 (0.964)	0.085 (0.486)	0.049 (0.266)	0.033 (0.191)	0.026 (0.138)
20 SPS	Sinc2	0.270 (1.560)	0.136 (0.745)	0.070 (0.376)	0.039 (0.231)	0.028 (0.149)	0.021 (0.111)
20 SPS	Sinc3	0.237 (1.415)	0.124 (0.701)	0.067 (0.399)	0.035 (0.192)	0.024 (0.130)	0.020 (0.109)
20 SPS	Sinc4	0.229 (1.285)	0.113 (0.612)	0.060 (0.325)	0.034 (0.193)	0.022 (0.123)	0.017 (0.098)
50 SPS	Sinc1	0.514 (2.925)	0.255 (1.584)	0.140 (0.940)	0.077 (0.457)	0.051 (0.315)	0.042 (0.264)
50 SPS	Sinc2	0.426 (2.400)	0.209 (1.217)	0.108 (0.666)	0.064 (0.381)	0.042 (0.265)	0.033 (0.200)
50 SPS	Sinc3	0.389 (2.324)	0.196 (1.185)	0.104 (0.624)	0.057 (0.367)	0.038 (0.228)	0.030 (0.179)
50 SPS	Sinc4	0.358 (2.319)	0.175 (1.023)	0.096 (0.597)	0.055 (0.319)	0.036 (0.217)	0.028 (0.176)
60 SPS	Sinc1	0.558 (3.574)	0.285 (1.703)	0.151 (0.913)	0.085 (0.515)	0.055 (0.335)	0.045 (0.271)
60 SPS	Sinc2	0.465 (2.753)	0.235 (1.424)	0.121 (0.760)	0.068 (0.417)	0.046 (0.276)	0.036 (0.208)
60 SPS	Sinc3	0.414 (2.704)	0.208 (1.187)	0.112 (0.655)	0.064 (0.396)	0.042 (0.276)	0.034 (0.197)
60 SPS	Sinc4	0.383 (2.288)	0.195 (1.174)	0.105 (0.623)	0.059 (0.347)	0.040 (0.242)	0.031 (0.188)
100 SPS	Sinc1	0.734 (4.715)	0.361 (2.276)	0.192 (1.209)	0.108 (0.679)	0.071 (0.473)	0.058 (0.362)
100 SPS	Sinc2	0.604 (3.662)	0.305 (1.934)	0.156 (1.072)	0.088 (0.579)	0.059 (0.371)	0.048 (0.321)
100 SPS	Sinc3	0.531 (3.431)	0.277 (1.780)	0.143 (0.935)	0.081 (0.545)	0.054 (0.343)	0.043 (0.288)
100 SPS	Sinc4	0.511 (3.340)	0.255 (1.632)	0.134 (0.861)	0.076 (0.479)	0.050 (0.322)	0.041 (0.271)
400 SPS	Sinc1	1.438 (10.374)	0.734 (5.410)	0.380 (2.657)	0.215 (1.469)	0.143 (1.066)	0.116 (0.843)
400 SPS	Sinc2	1.186 (8.523)	0.607 (4.333)	0.313 (2.280)	0.178 (1.313)	0.119 (0.884)	0.095 (0.676)
400 SPS	Sinc3	1.072 (7.923)	0.550 (3.999)	0.285 (1.991)	0.161 (1.132)	0.107 (0.781)	0.087 (0.630)
400 SPS	Sinc4	0.995 (7.107)	0.508 (3.664)	0.266 (1.947)	0.151 (1.061)	0.101 (0.708)	0.081 (0.583)
1200 SPS	Sinc1	2.451 (17.755)	1.254 (9.305)	0.651 (5.044)	0.368 (2.807)	0.244 (1.846)	0.197 (1.519)
1200 SPS	Sinc2	2.038 (15.480)	1.037 (8.128)	0.545 (4.107)	0.309 (2.315)	0.205 (1.586)	0.165 (1.283)
1200 SPS	Sinc3	1.858 (14.005)	0.960 (7.223)	0.494 (3.833)	0.281 (2.145)	0.186 (1.374)	0.148 (1.094)
1200 SPS	Sinc4	1.743 (13.428)	0.890 (6.585)	0.459 (3.405)	0.261 (2.018)	0.174 (1.337)	0.139 (1.032)

Table 8-1. ADC1 Noise in μV_{RMS} (μV_{PP}) at $T_A = 25^\circ C$, $V_{AVDD} = 5 V$, $V_{AVSS} = 0 V$, $V_{REF} = 2.5 V$ (continued)

DATA RATE	FILTER MODE	GAIN					
		1	2	4	8	16	32
2400 SPS	Sinc1	3.411 (26.095)	1.724 (13.528)	0.903 (6.609)	0.510 (3.920)	0.335 (2.626)	0.270 (2.107)
2400 SPS	Sinc2	2.870 (21.677)	1.468 (11.032)	0.770 (5.932)	0.435 (3.379)	0.286 (2.123)	0.230 (1.758)
2400 SPS	Sinc3	2.656 (20.100)	1.337 (9.936)	0.705 (5.355)	0.395 (3.035)	0.262 (1.951)	0.211 (1.533)
2400 SPS	Sinc4	2.475 (19.447)	1.262 (9.452)	0.657 (4.966)	0.371 (2.869)	0.245 (1.885)	0.198 (1.576)
4800 SPS	Sinc1	4.590 (34.155)	2.329 (17.298)	1.221 (8.943)	0.682 (5.252)	0.446 (3.239)	0.361 (2.957)
4800 SPS	Sinc2	4.091 (30.903)	2.070 (15.168)	1.077 (8.141)	0.606 (4.777)	0.398 (2.986)	0.321 (2.397)
4800 SPS	Sinc3	3.720 (28.423)	1.894 (14.842)	0.998 (7.626)	0.560 (4.176)	0.367 (2.890)	0.297 (2.211)
4800 SPS	Sinc4	3.535 (27.437)	1.784 (13.760)	0.926 (7.273)	0.527 (4.004)	0.349 (2.626)	0.277 (2.184)
7200 SPS	Sinc1	5.326 (42.076)	2.709 (19.749)	1.407 (11.126)	0.792 (5.784)	0.516 (3.881)	0.409 (3.189)
7200 SPS	Sinc2	4.867 (36.820)	2.467 (18.627)	1.280 (9.874)	0.726 (5.612)	0.472 (3.531)	0.379 (2.792)
7200 SPS	Sinc3	4.567 (35.194)	2.310 (17.516)	1.209 (9.036)	0.682 (5.181)	0.445 (3.590)	0.359 (2.666)
7200 SPS	Sinc4	4.365 (34.008)	2.211 (17.432)	1.143 (8.804)	0.642 (5.075)	0.426 (3.261)	0.341 (2.467)
14400 SPS	Sinc5	6.377 (48.242)	3.235 (25.178)	1.675 (12.508)	0.929 (7.280)	0.596 (4.430)	0.466 (3.524)
19200 SPS	Sinc5	8.720 (65.389)	4.432 (32.931)	2.285 (17.055)	1.227 (9.870)	0.747 (5.725)	0.555 (4.058)
38400 SPS	Sinc5	103.55 (759.91)	51.76 (371.46)	25.95 (192.20)	13.02 (99.09)	6.493 (46.060)	3.276 (24.435)

Table 8-2. ADC1 ENOB (Noise Free Bits) at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{REF} = 5\text{ V}$

DATA RATE	FILTER MODE	GAIN					
		1 (BYPASS)	2	4	8	16	32
2.5 SPS	FIR	26.0 (23.9)	25.9 (23.9)	25.8 (23.8)	25.5 (23.6)	25.4 (23.0)	24.6 (22.4)
2.5 SPS	Sinc1	26.3 (24.2)	26.2 (24.1)	26.0 (23.9)	25.9 (23.8)	25.6 (23.3)	25.0 (22.8)
2.5 SPS	Sinc2	26.6 (24.4)	26.3 (24.2)	26.4 (24.3)	26.1 (24.0)	25.8 (23.6)	25.2 (23.1)
2.5 SPS	Sinc3	26.9 (25.0)	26.5 (24.4)	26.3 (24.2)	26.3 (24.3)	26.1 (23.9)	25.6 (23.5)
2.5 SPS	Sinc4	26.9 (25.0)	26.6 (24.5)	26.7 (24.7)	26.3 (24.4)	26.2 (24.1)	25.2 (23.3)
5 SPS	FIR	25.5 (23.2)	25.4 (23.2)	25.3 (23.1)	25.2 (23.1)	24.8 (22.4)	24.1 (21.9)
5 SPS	Sinc1	25.9 (23.7)	25.5 (23.3)	25.5 (23.1)	25.4 (23.1)	25.0 (22.7)	24.4 (22.2)
5 SPS	Sinc2	26.0 (23.9)	25.9 (23.8)	25.8 (23.4)	25.7 (23.4)	25.4 (23.1)	24.6 (22.4)
5 SPS	Sinc3	26.2 (24.0)	26.0 (23.7)	26.0 (23.8)	25.8 (23.5)	25.6 (23.3)	24.7 (22.5)
5 SPS	Sinc4	26.3 (24.0)	26.1 (24.0)	26.1 (23.9)	25.9 (23.8)	25.7 (23.5)	25.0 (22.8)
10 SPS	FIR	25.1 (22.7)	24.9 (22.5)	24.8 (22.5)	24.7 (22.4)	24.4 (21.8)	23.5 (21.2)
10 SPS	Sinc1	25.4 (23.0)	25.1 (22.7)	25.1 (22.7)	24.9 (22.6)	24.6 (22.2)	23.8 (21.4)
10 SPS	Sinc2	25.6 (23.2)	25.5 (23.1)	25.4 (23.0)	25.2 (22.8)	24.9 (22.4)	24.1 (21.7)
10 SPS	Sinc3	25.8 (23.4)	25.6 (23.2)	25.6 (23.3)	25.2 (22.9)	25.0 (22.5)	24.2 (22.0)
10 SPS	Sinc4	25.9 (23.6)	25.8 (23.4)	25.7 (23.4)	25.5 (23.1)	25.1 (22.7)	24.4 (22.0)
16.6 SPS	Sinc1	25.0 (22.5)	24.8 (22.4)	24.8 (22.3)	24.6 (22.1)	24.2 (21.6)	23.5 (20.9)
16.6 SPS	Sinc2	25.3 (22.8)	25.1 (22.5)	24.9 (22.4)	24.8 (22.3)	24.6 (21.9)	23.7 (21.2)
16.6 SPS	Sinc3	25.5 (23.0)	25.1 (22.7)	25.1 (22.7)	25.0 (22.4)	24.6 (22.0)	23.9 (21.5)
16.6 SPS	Sinc4	25.5 (23.0)	25.4 (22.9)	25.3 (22.8)	25.1 (22.6)	24.7 (22.0)	24.0 (21.5)
20 SPS	FIR	24.6 (22.0)	24.5 (21.9)	24.3 (21.8)	24.2 (21.6)	23.9 (21.2)	23.1 (20.7)
20 SPS	Sinc1	24.8 (22.4)	24.7 (22.1)	24.6 (22.1)	24.4 (22.0)	24.1 (21.5)	23.3 (20.9)
20 SPS	Sinc2	25.1 (22.6)	24.9 (22.5)	24.9 (22.5)	24.7 (22.2)	24.3 (21.8)	23.6 (21.2)
20 SPS	Sinc3	25.3 (22.8)	25.1 (22.6)	25.0 (22.4)	24.9 (22.4)	24.5 (22.0)	23.7 (21.3)
20 SPS	Sinc4	25.4 (22.9)	25.2 (22.8)	25.1 (22.7)	25.0 (22.4)	24.6 (22.1)	23.9 (21.4)
50 SPS	Sinc1	24.2 (21.7)	24.0 (21.4)	23.9 (21.2)	23.8 (21.2)	23.5 (20.7)	22.6 (20.0)
50 SPS	Sinc2	24.5 (22.0)	24.3 (21.8)	24.3 (21.7)	24.0 (21.5)	23.7 (21.0)	23.0 (20.4)
50 SPS	Sinc3	24.6 (22.0)	24.4 (21.8)	24.3 (21.8)	24.2 (21.5)	23.9 (21.2)	23.1 (20.6)
50 SPS	Sinc4	24.7 (22.0)	24.6 (22.0)	24.4 (21.8)	24.3 (21.7)	24.0 (21.3)	23.2 (20.6)
60 SPS	Sinc1	24.1 (21.4)	23.9 (21.3)	23.8 (21.2)	23.6 (21.0)	23.4 (20.6)	22.5 (20.0)
60 SPS	Sinc2	24.4 (21.8)	24.2 (21.6)	24.1 (21.5)	24.0 (21.3)	23.6 (20.9)	22.9 (20.3)
60 SPS	Sinc3	24.5 (21.8)	24.3 (21.8)	24.2 (21.7)	24.0 (21.4)	23.7 (20.9)	23.0 (20.4)
60 SPS	Sinc4	24.6 (22.1)	24.4 (21.8)	24.3 (21.8)	24.1 (21.6)	23.8 (21.1)	23.1 (20.5)
100 SPS	Sinc1	23.7 (21.0)	23.5 (20.9)	23.5 (20.8)	23.3 (20.6)	23.0 (20.1)	22.2 (19.5)
100 SPS	Sinc2	24.0 (21.4)	23.8 (21.1)	23.8 (21.0)	23.6 (20.9)	23.2 (20.5)	22.4 (19.7)
100 SPS	Sinc3	24.2 (21.5)	23.9 (21.2)	23.9 (21.2)	23.7 (20.9)	23.4 (20.6)	22.6 (19.9)
100 SPS	Sinc4	24.2 (21.5)	24.0 (21.4)	24.0 (21.3)	23.8 (21.1)	23.5 (20.7)	22.7 (20.0)
400 SPS	Sinc1	22.7 (19.9)	22.5 (19.6)	22.5 (19.7)	22.3 (19.5)	22.0 (19.0)	21.2 (18.3)
400 SPS	Sinc2	23.0 (20.2)	22.8 (20.0)	22.7 (19.9)	22.6 (19.7)	22.2 (19.2)	21.5 (18.6)
400 SPS	Sinc3	23.2 (20.3)	22.9 (20.1)	22.9 (20.1)	22.7 (19.9)	22.4 (19.4)	21.6 (18.7)
400 SPS	Sinc4	23.3 (20.4)	23.0 (20.2)	23.0 (20.1)	22.8 (20.0)	22.5 (19.6)	21.7 (18.8)
1200 SPS	Sinc1	22.0 (19.1)	21.7 (18.9)	21.7 (18.7)	21.5 (18.6)	21.2 (18.2)	20.4 (17.5)
1200 SPS	Sinc2	22.2 (19.3)	22.0 (19.0)	21.9 (19.0)	21.8 (18.9)	21.5 (18.4)	20.7 (17.7)
1200 SPS	Sinc3	22.4 (19.4)	22.1 (19.2)	22.1 (19.1)	21.9 (19.0)	21.6 (18.6)	20.8 (17.9)
1200 SPS	Sinc4	22.5 (19.5)	22.2 (19.3)	22.2 (19.3)	22.0 (19.1)	21.7 (18.6)	20.9 (18.0)

Table 8-2. ADC1 ENOB (Noise Free Bits) at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{REF} = 5\text{ V}$ (continued)

DATA RATE	FILTER MODE	GAIN					
		1 (BYPASS)	2	4	8	16	32
2400 SPS	Sinc1	21.5 (18.5)	21.3 (18.3)	21.2 (18.3)	21.0 (18.1)	20.7 (17.7)	20.0 (17.0)
2400 SPS	Sinc2	21.7 (18.8)	21.5 (18.6)	21.4 (18.5)	21.3 (18.3)	21.0 (18.0)	20.2 (17.3)
2400 SPS	Sinc3	21.8 (18.9)	21.7 (18.8)	21.6 (18.6)	21.4 (18.5)	21.1 (18.1)	20.3 (17.5)
2400 SPS	Sinc4	21.9 (19.0)	21.7 (18.8)	21.7 (18.8)	21.5 (18.5)	21.2 (18.2)	20.4 (17.4)
4800 SPS	Sinc1	21.1 (18.2)	20.8 (18.0)	20.8 (17.9)	20.6 (17.7)	20.3 (17.4)	19.5 (16.5)
4800 SPS	Sinc2	21.2 (18.3)	21.0 (18.1)	21.0 (18.0)	20.8 (17.8)	20.5 (17.5)	19.7 (16.8)
4800 SPS	Sinc3	21.4 (18.4)	21.1 (18.2)	21.1 (18.1)	20.9 (18.0)	20.6 (17.5)	19.8 (16.9)
4800 SPS	Sinc4	21.4 (18.5)	21.2 (18.3)	21.2 (18.2)	21.0 (18.1)	20.7 (17.7)	19.9 (16.9)
7200 SPS	Sinc1	20.8 (17.9)	20.6 (17.8)	20.6 (17.6)	20.4 (17.5)	20.1 (17.1)	19.4 (16.4)
7200 SPS	Sinc2	21.0 (18.1)	20.8 (17.8)	20.7 (17.8)	20.5 (17.6)	20.2 (17.2)	19.5 (16.6)
7200 SPS	Sinc3	21.1 (18.1)	20.9 (17.9)	20.8 (17.9)	20.6 (17.7)	20.3 (17.2)	19.5 (16.7)
7200 SPS	Sinc4	21.1 (18.2)	20.9 (17.9)	20.9 (17.9)	20.7 (17.7)	20.4 (17.4)	19.6 (16.8)
14400 SPS	Sinc5	20.6 (17.7)	20.4 (17.4)	20.3 (17.4)	20.2 (17.2)	19.9 (16.9)	19.2 (16.3)
19200 SPS	Sinc5	20.1 (17.2)	19.9 (17.0)	19.9 (17.0)	19.8 (16.8)	19.6 (16.6)	18.9 (16.0)
38400 SPS	Sinc5	15.6 (12.6)	15.4 (12.6)	15.4 (12.5)	15.3 (12.5)	15.5 (12.6)	15.4 (12.5)

Table 8-3. ADC2 (ADS1263) Noise in μV_{RMS} (μV_{PP}) at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{REF} = 2.5\text{ V}$

DATA RATE	FILTER	GAIN							
		1	2	4	8	16	32	64	128
10 SPS	Sinc1	7.34 (32.6)	3.54 (16.5)	1.52 (7.57)	0.87 (4.22)	0.47 (2.42)	0.28 (1.43)	0.20 (1.08)	0.14 (0.70)
100 SPS	Sinc3	10.3 (65.2)	5.58 (36.0)	3.13 (20.4)	1.80 (11.5)	0.96 (6.30)	0.62 (4.03)	0.48 (3.08)	0.32 (2.04)
400 SPS	Sinc3	56.8 (827)	29.2 (345)	15.3 (158)	7.88 (76.9)	4.02 (36.2)	2.18 (17.9)	1.32 (9.94)	0.80 (5.56)
800 SPS	Sinc3	299 (3195)	151 (1756)	76.8 (875)	38.9 (417)	19.8 (199)	10.0 (90.0)	5.21 (43.6)	2.71 (21.9)

Table 8-4. ADC2 (ADS1263) ENOB (Noise Free Bits) at $T_A = 25^\circ\text{C}$, $V_{AVDD} = 5\text{ V}$, $V_{AVSS} = 0\text{ V}$, $V_{REF} = 5\text{ V}$

DATA RATE	FILTER	GAIN							
		1	2	4	8	16	32	64	128
10 SPS	Sinc1	21.4 (18.8)	21.3 (18.8)	21.1 (18.6)	20.6 (18.2)	20.6 (18.1)	20.2 (17.8)	19.4 (17.0)	19.1 (16.7)
100 SPS	Sinc3	20.3 (17.5)	20.1 (17.3)	19.8 (17.2)	19.4 (16.7)	19.3 (16.5)	18.9 (16.2)	18.2 (15.6)	17.8 (15.0)
400 SPS	Sinc3	16.5 (12.5)	16.5 (12.5)	16.4 (12.7)	16.2 (12.8)	16.2 (12.6)	16.2 (13.0)	16.1 (13.0)	15.9 (13.0)
800 SPS	Sinc3	14.0 (10.7)	14.0 (10.7)	14.0 (10.4)	13.8 (10.4)	13.8 (10.4)	13.8 (10.4)	13.7 (10.6)	13.7 (10.7)

9 Detailed Description

9.1 Overview

The ADS1262 and ADS1263 are precision 32-bit, delta-sigma ($\Delta\Sigma$) ADCs with an integrated analog front end (AFE) to simplify connection to sensors. A 32-bit ADC (ADC1) provides output data rates from 2.5 SPS to 38400 SPS for flexibility in resolution and data rates over a wide range of applications. The ADC low noise and low drift architecture make these devices suitable for precise digitization of low-level transducers, such as load cell bridges and temperature sensors. The ADS1263 includes an auxiliary 24-bit delta-sigma ADC (ADC2).

The ADS1262 and the ADS1263 incorporate several functions that provide increased utility. The key integrated functions include:

- Low-drift voltage reference
- Dual, matched, sensor-excitation current sources (IDAC)
- Input-level-shift voltage
- Eight GPIOs
- Dual-sensor, bias current sources
- Low-noise, CMOS PGA with integrated signal fault detection
- Internal test signal source (TDAC)
- Temperature sensor
- Internal oscillator
- Three sets of buffered external reference inputs with low reference voltage alarm

As shown in the [Functional Block Diagram](#), these devices feature 11 analog inputs that are configurable as either ten single-ended inputs, five differential inputs, or any combination, to either ADC1 or ADC2. Many of the analog inputs are multifunction as programmed by the user. The analog inputs can be programmed to the following extended functions:

- Three external reference inputs: pins AIN0, AIN1, AIN2, AIN3, AIN4 and AIN5
- Two sensor excitation current source: all analog input pins
- Level shift (VBIAS): AINCOM pin
- Eight GPIO: pins AIN3, AIN4, AIN5, AIN6, AIN7, AIN8, AIN9, AINCOM
- Sensor break current source: all analog input pins
- Two test signal output: pins AIN6, AIN7

Following the input multiplexer (mux), ADC1 features a high-impedance, CMOS, programmable gain amplifier (PGA). The PGA provides very low voltage and current noise, enabling direct connection to low-level transducers, and in many cases, eliminating the need for an external amplifier. The PGA gain is programmable from 1 V/V to 32 V/V in binary steps. The PGA can be bypassed to allow the input range to extend below ground. The PGA has voltage overrange monitors to improve the integrity of the conversion result. The PGA overrange alarm is latched during the conversion phase and appended to the conversion data. The programmable sensor bias uses a test current to help detect a failed sensor or sensor connection.

An inherently stable delta-sigma modulator measures the ratio of the input voltage to the reference voltage to provide the ADC result. The ADC operates with the internal 2.5-V reference, or with up to three external reference inputs. The external reference inputs are continuously monitored for low (or missing) voltage. The reference alarm status is latched during the conversion phase and appended to the conversion data. The REFOUT pin is the buffered 2.5-V internal voltage reference output.

Dual excitation current sources (IDAC) provide bias to resistance sensors (such as 3-wire RTD). The ADC integrates several system monitors for readback, such as temperature sensor and supply monitor. The ADC features an internal test signal voltage (TDAC) that is used to verify the ADC operation across all gains. The TDAC has two outputs to provide test voltages for single-ended and differential input configurations. Eight GPIO ports are available on the analog input pins.

The digital filter provides two functional modes, sinc and FIR, allowing optimization of settling time and line-cycle rejection. The sinc/x (sinc) filter is programmable to sinc orders one through four to tradeoff filter settling time and 50-Hz and 60-Hz line-cycle rejection. The finite impulse response (FIR) filter mode provides single-cycle settled data with 50-Hz and 60-Hz line cycle rejection at data rates up to 20 SPS.

The ADS1263 includes an auxiliary 24-bit delta-sigma ADC (ADC2) featuring buffered PGA inputs, gains from 1 V/V to 128 V/V, and data rates up to 800 SPS. All analog inputs and reference inputs are available to ADC2. ADC2 can be used to provide redundant measurements or system measurements such as sensor temperature compensation and thermocouple cold junction compensation (CJC). The ADS1263 is pin and functionally compatible to the ADS1262.

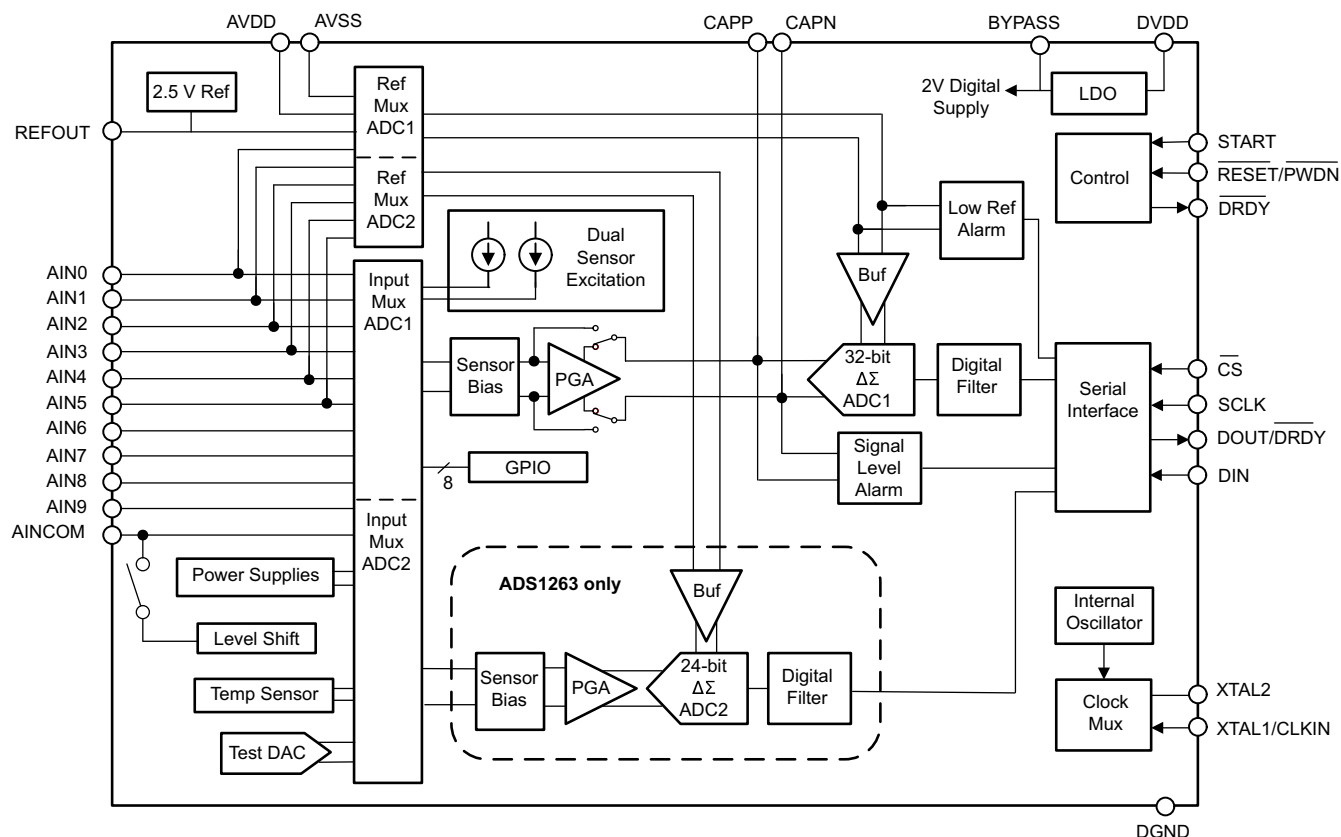
The SPI™-compatible serial interface is used to read the conversion data and also to configure and control the ADC. The serial interface consists of four signals: $\overline{\text{CS}}$, SCLK, DIN and DOUT/DRDY. The conversion data are provided with a CRC code for improved data integrity. The dual function DOUT/DRDY output indicates when conversion data are ready and also provides the data output. The serial interface can be implemented with as little as three connections by tying $\overline{\text{CS}}$ low.

The ADC has three clock options: internal oscillator, external crystal, and external clock. The ADC detects the clock mode automatically. The nominal clock frequency is 7.3728 MHz.

ADC conversions are started by a control pin or by commands. The ADC can be programmed to free-run mode or perform one-shot conversions. The DRDY and DOUT/DRDY pins are driven low when the conversion data are ready. The RESET/PWDN digital input resets the ADC when momentarily pulsed low, and when held low, enables the ADC power-down mode.

The ADC operates with bipolar (± 2.5 V) supplies, or with a single 5-V supply. For single-supply operation, use the internal level-shift voltage to level-shift isolated (floating) sensors. The digital power-supply range is 2.7 V to 5.25 V. The BYPASS pin is the subregulator output (2 V) that is used for internal digital supply.

9.2 Functional Block Diagram



9.3 Feature Description

9.3.1 Multifunction Analog Inputs

The ADS1262 and ADS1263 have 11 multifunction analog inputs configurable in a variety of extended functions. Figure 9-1 shows the internal analog signal routing to the circuit blocks. Table 9-1 summarizes the input pin functions. The devices have two cross-point multiplexers; one multiplexer for ADC1, and one multiplexer for ADC2. The multiplexers select any analog input for the positive PGA input and any input for the negative PGA input. The ADCs are also configurable for a number of internal monitor functions. The internal monitors are temperature sensor, TDAC test voltage, analog power-supply voltage, and digital power-supply voltage. The dual excitation-current sources (IDAC1 and IDAC2) are independently connected to any analog input pin. Eight analog inputs are configurable as GPIO. The GPIOs are programmable as inputs or outputs, and are referenced to the analog power-supply voltages (V_{AVDD} and V_{AVSS}). The level-shift function (VBIAS) is available on AINCOM and is used to provide an input level-shift voltage for isolated sensors. The internal TDAC test voltage is available on output pins AIN6 and AIN7. The ADC has two voltage-reference multiplexers; one reference multiplexer for ADC1, and one reference multiplexer for ADC2. Through the reference multiplexers, select the internal reference, three external reference sources, or the analog power-supply voltage ($V_{AVDD} - V_{AVSS}$).

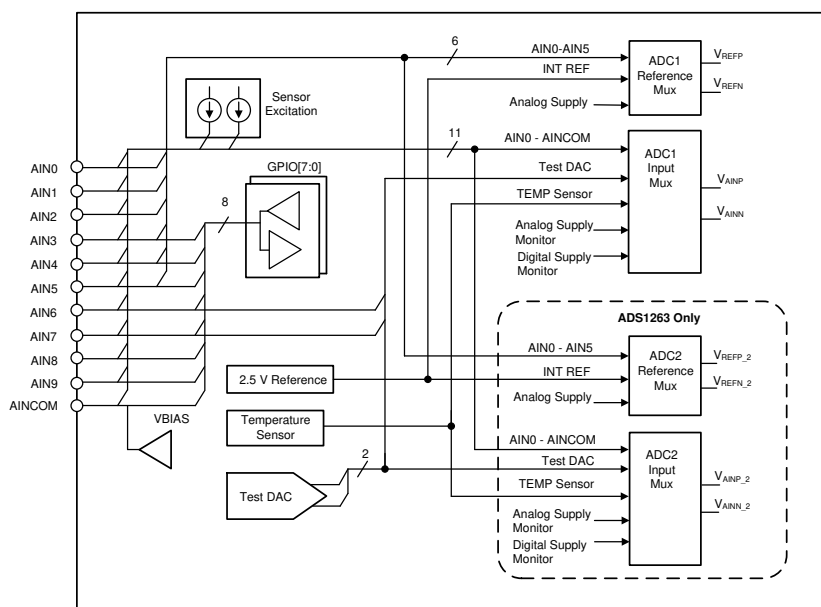


Figure 9-1. Analog Input Routing Overview

Table 9-1. Analog Input Pin Functions

PIN	ADC1 INPUT	ADC2 INPUT	ADC1 REF INPUT ⁽¹⁾	ADC2 REF INPUT	IDAC1 OUTPUT	IDAC2 OUTPUT	GPIO	TDAC OUTPUT	LEVEL SHIFT OUTPUT
AIN0	Yes	Yes	REFP1, REFN1	REFP1	Yes	Yes	—	—	—
AIN1	Yes	Yes	REFP1, REFN1	REFN1	Yes	Yes	—	—	—
AIN2	Yes	Yes	REFP2, REFN2	REFP2	Yes	Yes	—	—	—
AIN3	Yes	Yes	REFP2, REFN2	REFN2	Yes	Yes	GPIO[0]	—	—
AIN4	Yes	Yes	REFP3, REFN3	REFP3	Yes	Yes	GPIO[1]	—	—
AIN5	Yes	Yes	REFP3, REFN3	REFN3	Yes	Yes	GPIO[2]	—	—
AIN6	Yes	Yes	—	—	Yes	Yes	GPIO[3]	TDACP	—
AIN7	Yes	Yes	—	—	Yes	Yes	GPIO[4]	TDACN	—
AIN8	Yes	Yes	—	—	Yes	Yes	GPIO[5]	—	—
AIN9	Yes	Yes	—	—	Yes	Yes	GPIO[6]	—	—
AINCOM	Yes	Yes	—	—	Yes	Yes	GPIO[7]	—	Yes

(1) The reference voltage of ADC1 can be either polarity and reversed by programming.

9.3.2 Analog Input Description

As shown in Figure 9-2, the analog inputs of the device consist of ESD protection diodes, an ADC1 and ADC2 cross-point input multiplexer, the sensor bias circuit, and individual PGAs for each ADC. The ADC has 11 external inputs, four internal monitor signals, and one no-connection (float). Note that in figures throughout this document, italic text shows the associated register and register settings.

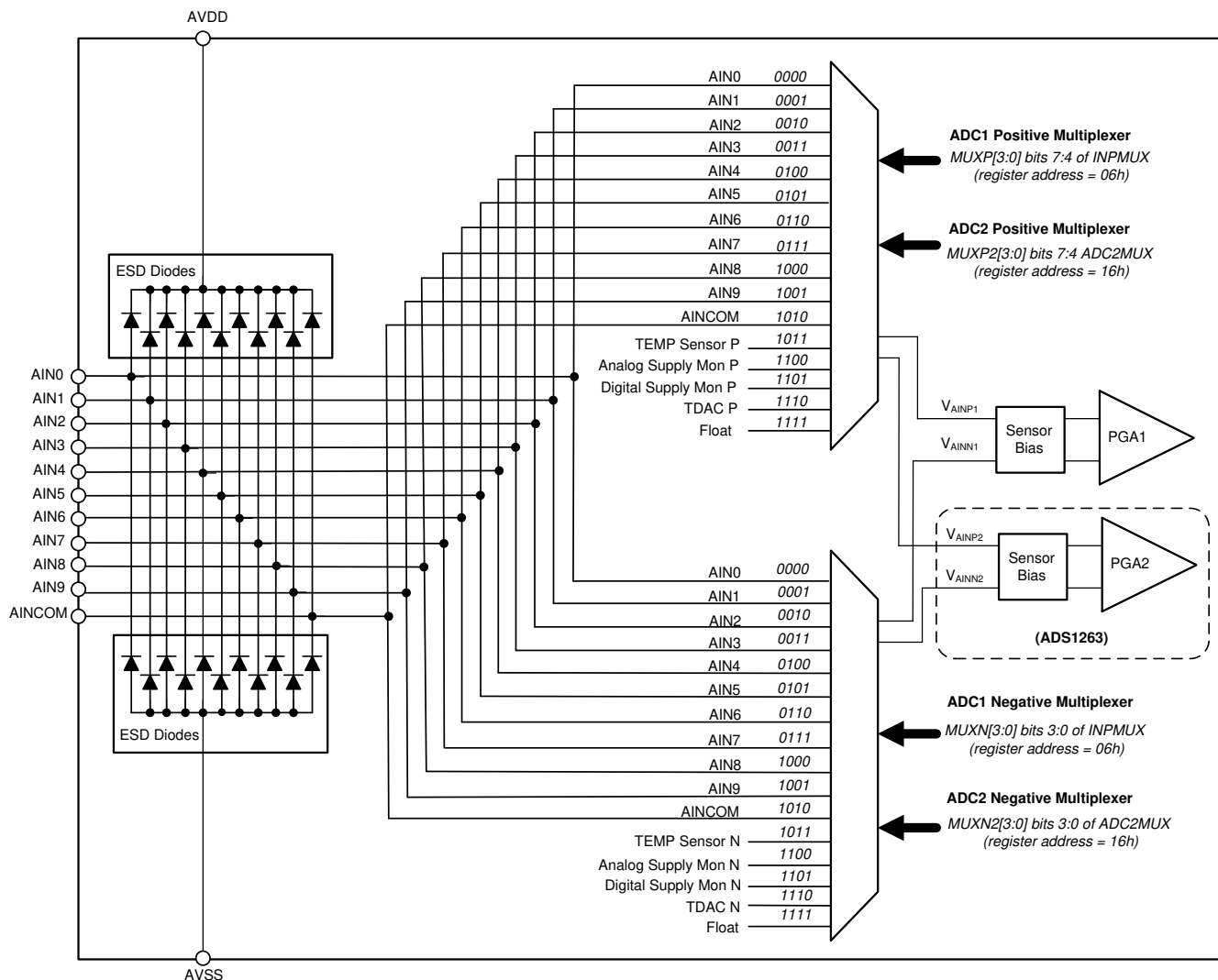


Figure 9-2. ADC1 and ADC2 Input Block Diagram

9.3.2.1 ESD Diode

The analog inputs have internal ESD diodes that are connected to the analog supplies (AVDD and AVSS). The function of the diodes is to protect the ADC inputs from ESD events. If the input signal exceeds V_{AVDD} by more than 0.3 V or goes below V_{AVSS} by more than -0.3 V, the diodes may conduct. When the diodes conduct, input current flows into the analog inputs through the AVDD or AVSS pins. If an input overvoltage is possible, limit the input current to less than $|\pm 10 \text{ mA}|$. In many applications, a resistor in series with the input is sufficient to limit the current. Depending on the application requirements, be aware of the thermal noise of the current limit resistor.

9.3.2.2 Input Multiplexer

Use the dual, cross-point input multiplexers to select from one of the 11 external inputs, one of the four internal monitors, and a floating connection, in any combination, to either ADC. One input is selected by the positive multiplexer, and one input is selected by the negative multiplexer. The ADC1 positive and negative multiplexers are programmed by bits MUXP[3:0] and bits MUXN[3:0] in the INPMUX register (address = 06h). The ADC2

positive and negative multiplexers have identical functionality and are programmed by bits MUXP2[3:0] bits and bits MUXN2[3:0] in the ADC2MUX register (address = 16h).

9.3.3 Sensor Bias

The ADC incorporates a sensor bias current source that can be used to apply a small test current to diagnose broken sensor leads or problems existing in the sensor. Figure 9-3 shows the sensor bias block diagram. The sensor bias circuit consists of programmable current sources and bias resistors. The sensor bias circuit connects to the outputs of either the ADC1 or ADC2 multiplexers. Program the sensor bias to either pull-up or pull-down mode. In pull-up mode, the current flows into the positive input and flows out of the negative input. In pull-down mode, the polarities are reversed. Configure the sensor bias either to a 10-M Ω bias resistor, or to current with magnitudes of ± 0.5 , ± 2 , ± 10 , ± 50 , or ± 200 μ A.

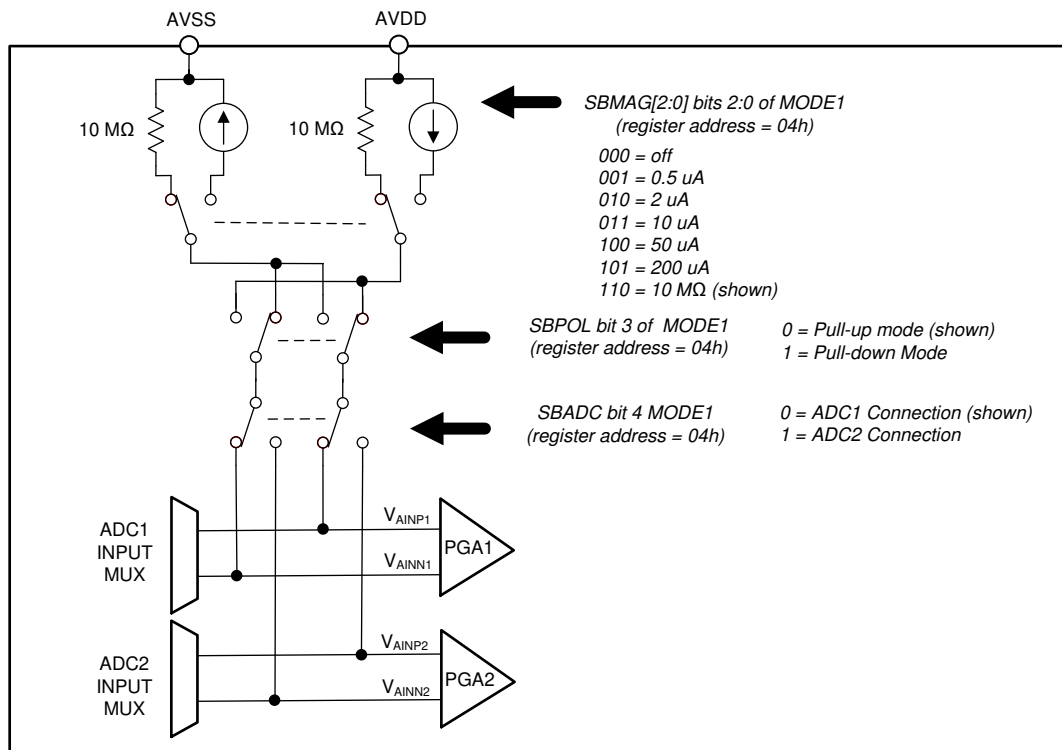


Figure 9-3. Sensor Bias Block Diagram

In pull-up mode, an open sensor results in the positive input pulled to V_{AVDD} , and the negative input pulled to V_{AVSS} . An open sensor in pull-up mode results in a positive full-scale reading. A full-scale reading can also be an indication of sensor overload or that the reference voltage is lower than expected. The sensor bias can remain on while actively converting, or pulsed on periodically to test the sensor. When pulsed on, allow time for settling because external capacitance loads the sensor bias when first enabled. Be aware of offset error as a result of sensor bias current flowing through the multiplexer switch resistance.

9.3.4 Temperature Sensor

The ADC incorporates an integrated temperature sensor. The temperature sensor is comprised of two internal diodes with one diode having 16 times the current density of the other, as shown in Figure 9-4. The difference in current density of the diodes yields a differential output voltage that is proportional to absolute temperature. Measure the temperature sensor voltage with either ADC1 or ADC2. For ADC1 measurement, set the INPMUX register (address 06h) to BBh. For ADC2 measurement, set the ADC2MUX register (address 16h) to BBh. Equation 9 shows how to convert the temperature sensor reading to degrees Celcius (°C):

$$\text{Temperature (}^{\circ}\text{C)} = [(\text{Temperature Reading (}\mu\text{V)} - 122,400) / 420 \mu\text{V/}^{\circ}\text{C}] + 25^{\circ}\text{C} \quad (9)$$

where

- Temperature reading units are in μV

Before temperature sensor measurement, enable the PGA, set gain = 1, disable chop mode, and make sure the internal voltage reference is powered on. As a result of the low package-to-PCB thermal resistance, the internal device temperature closely tracks the PCB temperature. Note that ADC self-heating results in an increase of 0.7°C relative to the temperature of the surrounding PCB.

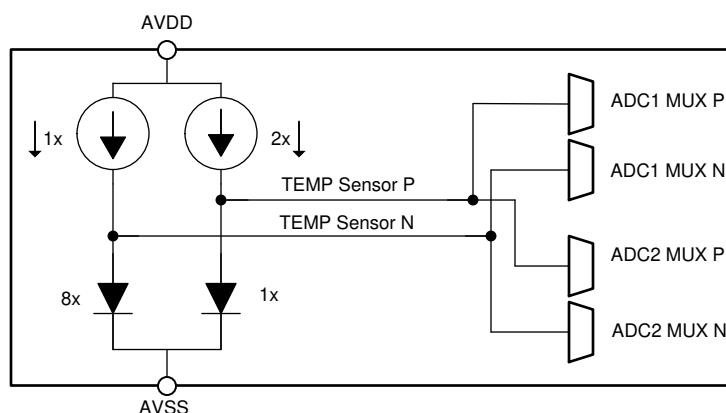


Figure 9-4. Temperature Sensor

9.3.5 Power-Supply Monitor

To internally monitor the ADC power supplies, use either ADC1 or ADC2. As shown in [Figure 9-5](#), the power supply voltages are divided by a resistor network to reduce the voltages within the ADC input range. The reduced power-supply voltage is routed to the ADC input multiplexers. The analog (V_{ANLMON}) and digital (V_{DIGMON}) power supply readings are scaled by [Equation 10](#) and [Equation 11](#), respectively:

$$V_{ANLMON} = (V_{AVDD} - V_{AVSS}) / 4 \quad (10)$$

$$V_{DIGMON} = (V_{DVDD} - V_{DGND}) / 4 \quad (11)$$

Measure the supply monitor readings using either the internal or an external reference. For an external reference, the minimum reference voltage is 1.5 V.

Before measurement, enable the PGA, set gain = 1, and disable chop mode.

For analog supply monitor ADC1 measurement, set the INPMUX register (address 06h) to CCh.

For digital supply monitor ADC1 measurement, set the INPMUX register to DDh.

For analog supply monitor ADC2 measurement, set the ADC2MUX register (address 16h) to CCh.

For digital supply monitor ADC2 measurement, set the ADC2MUX register to DDh.

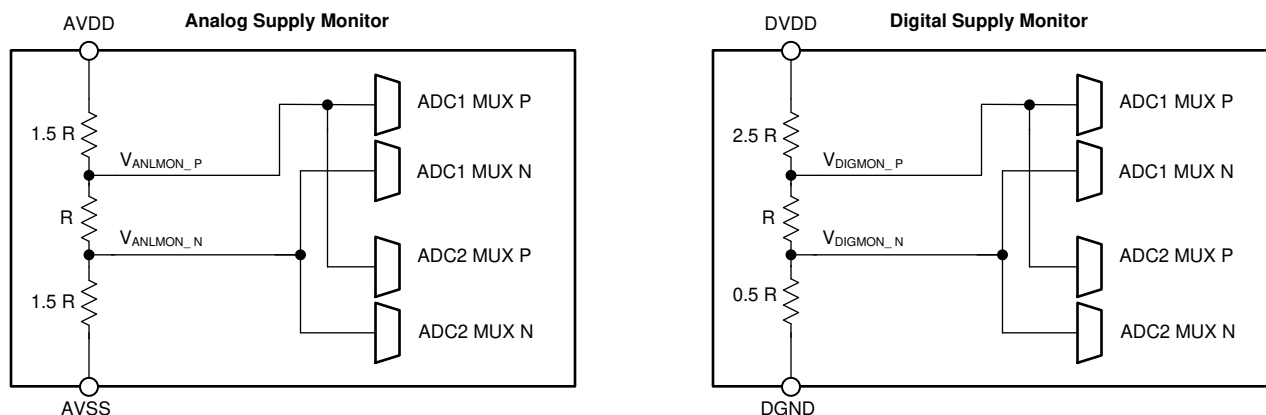


Figure 9-5. Power-Supply Monitors

9.3.6 PGA

The ADC1 PGA is a low-noise, programmable gain, CMOS differential-input, differential-output amplifier. The PGA extends the ADC dynamic range of sensors with low input-signal levels. The PGA provides gains of 1, 2, 4, 8, 16, and 32. Bypass the PGA to extend the analog input range to below ground (if the AVSS pin is grounded).

Figure 9-6 shows the PGA block diagram. The PGA consists of two chopper-stabilized amplifiers (A1 and A2), and a resistor network that is programmed to set the PGA gain. The PGA input is equipped with a high-frequency, electromagnetic-interference (EMI) input filter consisting of two 350-Ω input resistors, and several filter capacitors, as shown in the figure. Bypass the PGA to directly connect the inputs to the ADC. The PGA output is monitored by an overrange voltage monitor. The voltage monitor triggers an alarm when the absolute or differential PGA output voltage exceeds the linear range of operation. Pins CAPP and CAPN are the PGA positive and negative outputs, respectively. Connect a 4.7-nF (C0G) capacitor as shown in the figure. The capacitor provides an analog antialias filter, as well as the deglitch filter for the modulator sample pulses. Place the capacitor close to the pins using short, direct traces. Avoid running clock traces or other digital traces close to the pins.

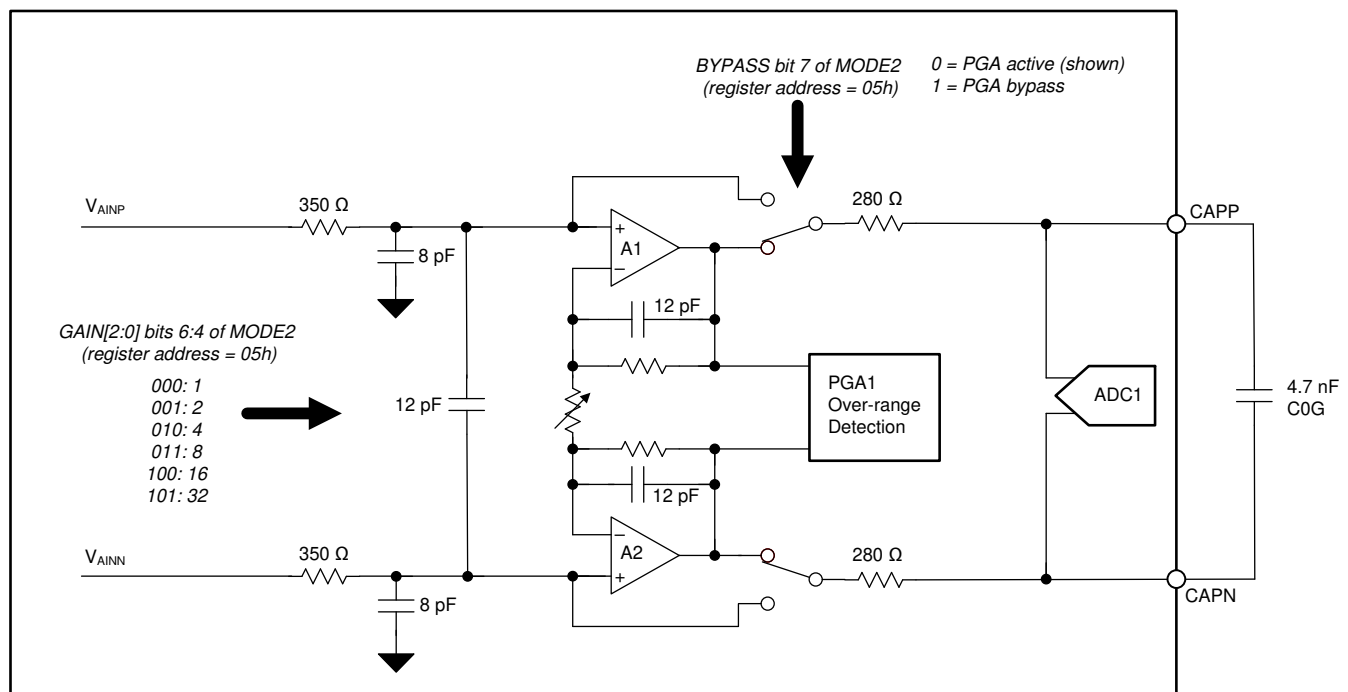


Figure 9-6. ADC1 PGA Block Diagram

The ADC1 full-scale voltage range is determined by the reference voltage and the PGA gain. Table 9-2 shows the full-scale voltage range versus gain for reference voltage = 2.5 V. The full-scale voltage range scales with the reference voltage and is increased or decreased by changing the reference voltage.

Table 9-2. ADC1 Full-Scale Voltage Range

GAIN[2:0] BITS OF REGISTER MODE2	GAIN (V/V)	FULL SCALE RANGE (V) ⁽¹⁾
000	1	±2.500 V
001	2	±1.250 V
010	4	±0.625 V
011	8	±0.312 V
100	16	±0.156 V
101	32	±0.078 V

(1) $V_{REF} = 2.5$ V. The full-scale input range is proportional to V_{REF}

As with many amplifiers, the PGA has an absolute input voltage range requirement that cannot be exceeded. The maximum and minimum absolute input voltages are limited by the voltage swing capability of the PGA output. The specified minimum and maximum absolute input voltages (V_{INP} and V_{INN}) depend on the PGA gain, the input differential voltage (V_{IN}), and the tolerance of the analog power-supply voltages (V_{AVDD} and V_{AVSS}). The absolute positive and negative input voltages must be within the specified range, as shown in Equation 12:

$$V_{AVSS} + 0.3 + |V_{IN}| \cdot (\text{Gain} - 1) / 2 < V_{INP} \text{ and } V_{INN} < V_{AVDD} - 0.3 - |V_{IN}| \cdot (\text{Gain} - 1) / 2 \quad (12)$$

where

- V_{INP} , V_{INN} = absolute input voltage
- V_{IN} = differential input voltage = $V_{INP} - V_{INN}$

The relationship between the PGA input to the PGA output is shown graphically in Figure 9-7. The PGA output voltages (V_{OUTP} , V_{OUTN}) depend on the PGA gain and the input voltage magnitudes. For linear operation, the PGA output voltages must not exceed $V_{AVDD} - 0.3$ or $V_{AVSS} + 0.3$. Note the diagram depicts a positive differential input voltage that results in a positive differential output voltage.

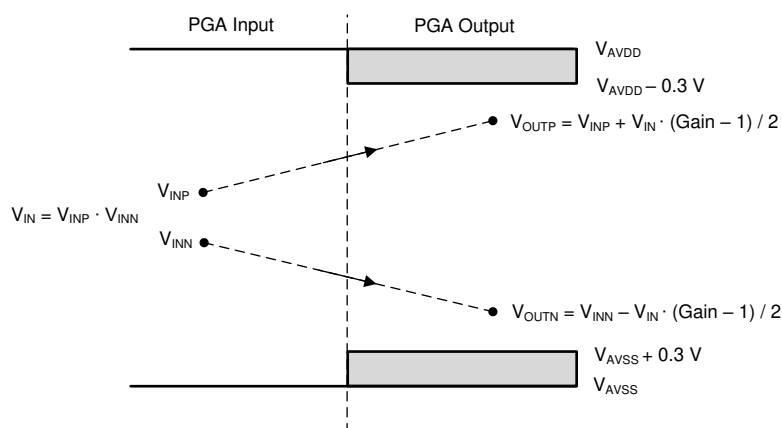


Figure 9-7. PGA Input/Output Range

If the PGA is bypassed, the ADC absolute input voltage range extends beyond the V_{AVDD} and V_{AVSS} power supplies allowing input voltages at or below ground. The absolute input voltage range when the PGA is bypassed is shown in Equation 13:

$$V_{AVSS} - 0.1 < V_{INP} \text{ and } V_{INN} < V_{AVDD} + 0.1 \quad (13)$$

9.3.7 PGA Voltage Overrange Monitors

ADC1 incorporates two PGA output-voltage monitors. The monitors trigger an alarm if the PGA output is driven into overrange. The corresponding bits are set (= 1) in the data output status byte when an alarm is triggered. The PGA output voltage is monitored in two ways:

- 1) Differential: If the PGA differential output voltage exceeds either +105% or -105% FSR.
- 2) Absolute: If either PGA absolute output voltage is higher than $V_{AVDD} - 0.2$ V or lower than $V_{AVSS} + 0.2$ V.

The alarms automatically reset when the PGA is no longer in voltage overload. The monitors are fast-responding, analog, voltage-level comparators. Therefore, these monitors detect short-duration voltage overrange events that are not necessarily evident in the output as clipped codes because of averaging of the digital filter that may span one or more conversion cycles. Use the monitor function to detect certain type of faults (such as signal overranges, incorrect gain settings, sensor faults, input miswiring, and so on) without the need to change input configuration or interrupt readings.

9.3.7.1 PGA Differential Output Monitor

ADC1 incorporates a differential PGA output voltage monitor. This voltage monitor triggers an alarm when the magnitude of the *differential* PGA output voltage is more positive than +105% or more negative than –105% of full scale, but only during a conversion cycle. The alarm event, corresponding to the conversion cycle when the alarm occurred, is set in the status byte (PGAD_ALM). For the next conversion, the alarm resets. If the magnitude of differential output voltage is within the range of $\pm 105\%$ of full-scale range, the alarm remains reset. The PGA differential monitor block diagram is shown in [Figure 9-8](#).

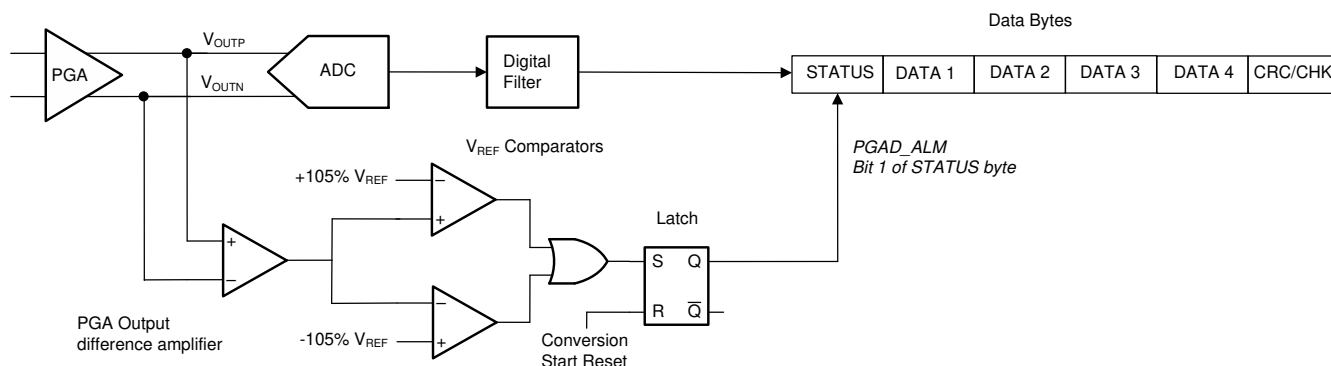


Figure 9-8. PGA Differential Overload Monitor

[Figure 9-9](#) shows an example of the differential overrange monitor event.

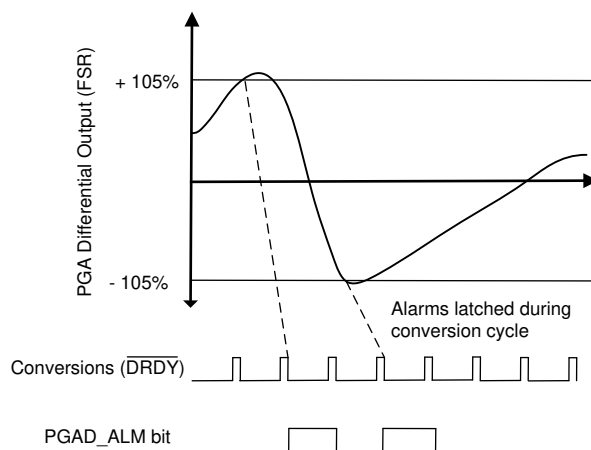


Figure 9-9. PGA Differential Alarm

9.3.7.2 PGA Absolute Output-Voltage Monitor

ADC1 contains an integrated a PGA absolute output-voltage monitor. If the absolute level of the PGA positive or negative output exceeds $V_{AVDD} - 0.2$ V, the PGA high alarm triggers (PGAH_ALM). If the absolute level of the PGA positive or negative output voltage is less than $V_{AVSS} + 0.2$ V, the PGA low alarm triggers (PGAL_ALM). The alarms are set in the status byte corresponding to the conversion cycle when the alarms occurred. For the next conversion cycle, the alarms reset. If the magnitude of PGA output voltages remains within the range ($V_{AVDD} - 0.2$ V and $V_{AVSS} + 0.2$ V), the alarms remain reset. The PGA absolute output-voltage monitor block diagram is shown in Figure 9-10.

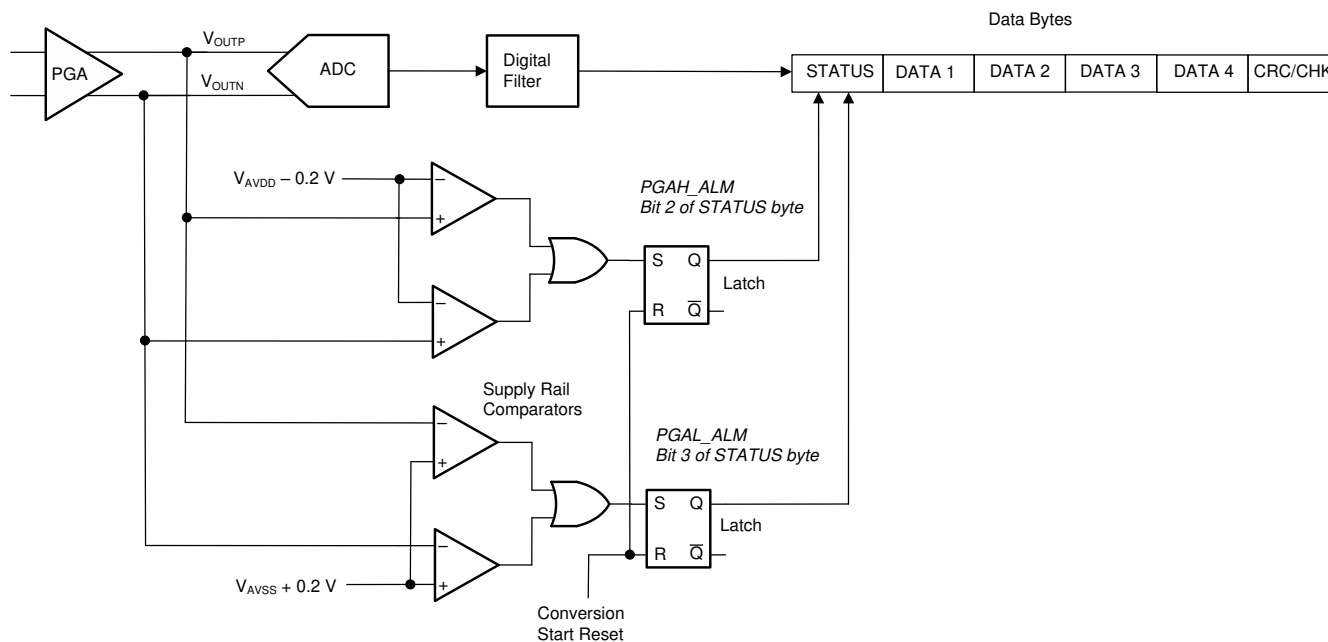


Figure 9-10. PGA Absolute Output-Voltage Monitor

Figure 9-11 shows an example of the PGA absolute output-voltage monitor overrange event.

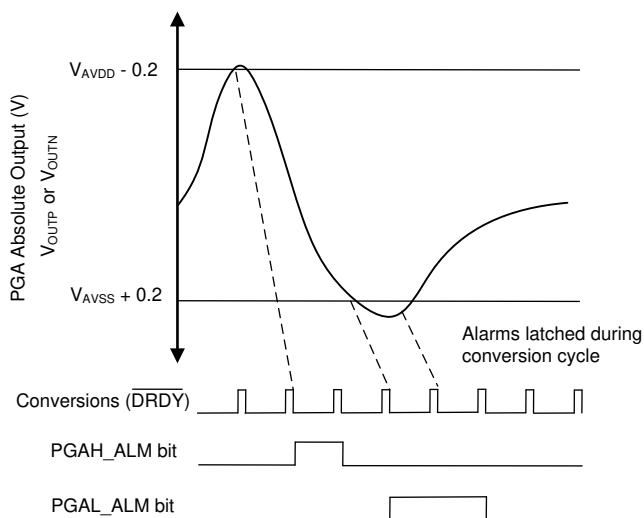
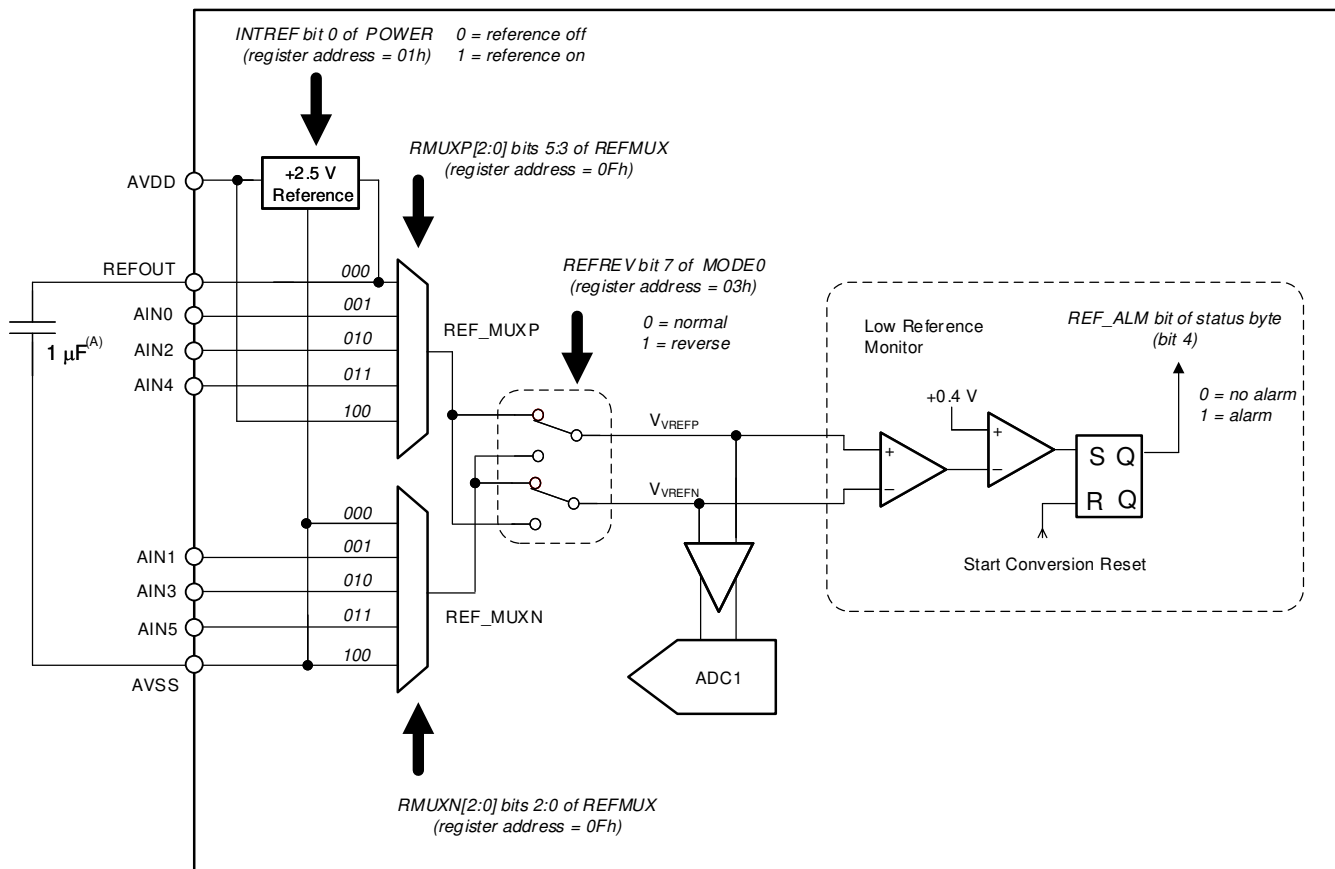


Figure 9-11. PGA Absolute Alarm

9.3.8 ADC Reference Voltage

These devices require a reference voltage for operation. Both ADCs default to the same internal reference, however, the reference voltage of ADC1 is independent of the ADC2 reference voltage. The reference voltage is provided internally by the internal 2.5-V reference, or externally, by one of the three external reference inputs. The specified external reference voltage range is 0.9 V to 5 V. The reference voltage is defined as $V_{REF} = V_{REFP} - V_{REFN}$, where V_{REFP} and V_{REFN} are the absolute positive and absolute negative reference voltages, respectively. The polarity of the reference voltage internal to the ADC must be always positive. The magnitude of the reference voltage together with the PGA gain establishes the ADC full-scale differential input range as defined by $V_{IN} = \pm V_{REF} / \text{gain}$. Figure 9-12 shows the block diagram of the ADC1 reference multiplexer. Use the reference multiplexer to select the internal reference, one of three external reference inputs, or the analog power supply.



A. The internal reference requires a 1-μF capacitor connected to pins REFOUT and AVSS.

Figure 9-12. ADC1 Reference Multiplexer Block Diagram

The ADC1 reference multiplexer consists of a positive multiplexer and a negative multiplexer. The positive and negative multiplexers are programmed by the RMUXP[2:0] and RMUXN[2:0] bits, respectively, of the REFMUX register. The positive reference input is either internal (2.5 V), external (pins AIN0, AIN2, AIN4), or the analog power-supply voltage (V_{AVDD}). The negative reference input is either internal (2.5 V), external (pins AIN1, AIN3, AIN5), or the analog power-supply voltage (V_{AVSS}). A reference polarity-reversal switch changes the reference polarity from negative to positive. The polarity switch allows either positive or negative external reference polarity. Set the reversal switch to the normal position ($REFREV = 0$) when using the internal reference or analog power supplies.

The ADC also contains an integrated low-reference voltage monitor. This monitor provides continuous detection of a low or missing reference during the conversion cycle. The low reference alarm is appended to the data output status byte (REF_ALM , bit 4 of the status byte).

9.3.8.1 Internal Reference

The ADC incorporates an integrated, precision, 2.5-V reference featuring very low drift. The internal reference is enabled by setting INTREF equal to 1 (default is on). To select the internal reference for use with ADC1, set the RMUXP and RMUXN bits of register REFMUX to 0h. The REFOUT pin provides a buffered reference output voltage. The negative reference (return) is the AVSS pin, as shown in [Figure 9-12](#). Be careful when laying out the REFOUT return to the AVSS pin. Connect a 1- μ F capacitor from the REFOUT pin to the AVSS pin. The capacitor can be increased up to 10 μ F (maximum) to decrease the reference noise, but results in increased reference start-up time. The capacitor is not required if the internal reference is not used. The internal reference must be powered if using the IDACs or the internal temperature sensor. After internal reference start-up, the reference requires start-up time before beginning the first conversion; see [Figure 7-33](#).

9.3.8.2 External Reference

The ADC provides three external reference inputs. The reference inputs are differential with independent positive and negative inputs. The reference inputs are the analog pins, AIN0 to AIN5. Typically, the positive reference is applied to pin AIN0, AIN2, or AIN4, and the negative reference is applied to pin AIN1, AIN2, or AIN3. The reference polarity can be negative, but the ADC requires a positive voltage reference. In this case, reverse the polarity using the internal polarity-reversal switch (ADC1 reference only). The reference polarity-reversal switch changes the reference polarity from negative to positive, and is controlled by REFREV (bit 7 of MODE0).

The reference inputs are high impedance. A reference input current flowing through a reference-voltage source impedance leads to possible loading errors (see [Figure 7-34](#)). To reduce the input current, use an external reference buffer; however, in most applications, an external reference buffer is not necessary.

Connect a 100-nF bypass capacitor across the external reference input pins. Follow the specified absolute and differential reference voltage requirements.

9.3.8.3 Power-Supply Reference

A third option for ADC reference is the internal analog power supply. However, an increase of linearity error results with this connection, and therefore, use this option only for less-critical applications, such as ADC self-diagnostics.

For critical applications, do not use the power-supply reference option. For applications that use the power-supply voltage as the reference voltage, connect the power-supply voltage to the external reference inputs, and select the appropriate external reference bits in the REFMUX register. For example, to measure a 6-wire load-cell, connect the bridge excitation voltages to the external reference inputs, and select the appropriate REFMUX bits.

9.3.8.4 Low-Reference Monitor

ADC1 incorporates a low-reference monitor to detect a low or missing reference. If the differential reference voltage ($V_{REF} = V_{REFP} - V_{REFN}$) falls below 0.4 V (typical), the low reference alarm triggers (REF_ALM). The low-reference monitor sets the corresponding alarm bit in the conversion data status byte. The alarm resets at the start of each new conversion. Use the low-reference monitor to detect a missing or failed reference voltage connection. Connect a 100-k Ω resistor across the reference inputs to provide the necessary bias. If either reference input is missing or unconnected, this external resistor biases the reference inputs to each other. The low-reference monitor is a fast-responding analog comparator; therefore, transients in the reference voltage may trigger the alarm.

9.3.9 ADC1 Modulator

The ADC1 modulator is an inherently stable, fourth-order, 2 + 2 pipelined $\Delta\Sigma$ modulator. The modulator samples the analog input voltage at a high sample rate ($f_{MOD} = f_{CLK} / 8 = 921.6$ kHz) and converts the analog input to a ones density bit stream. The digital filter receives the ones density bit stream output, and then filters and decimates the data to yield the final conversion result.

9.3.10 Digital Filter

The digital filter of ADC1 receives the modulator output data and produces a high-resolution conversion result. The digital filter low-pass filters and decimates the modulator data (rate reduction), yielding the final data output. By adjusting the type of filtering, tradeoffs are made between resolution, data rate, line cycle rejection, and conversion latency.

The digital filter has two selectable modes: $\sin(x)/x$ (sinc) mode and finite impulse response (FIR) mode (see Figure 9-13). The sinc mode provides data rates of 2.5 SPS through 38400 SPS with selectable sinc orders of 1 through 5. The FIR filter provides simultaneous rejection of 50-Hz and 60-Hz power-line frequencies with data rates 2.5 SPS through 20 SPS with single-cycle settled conversions.

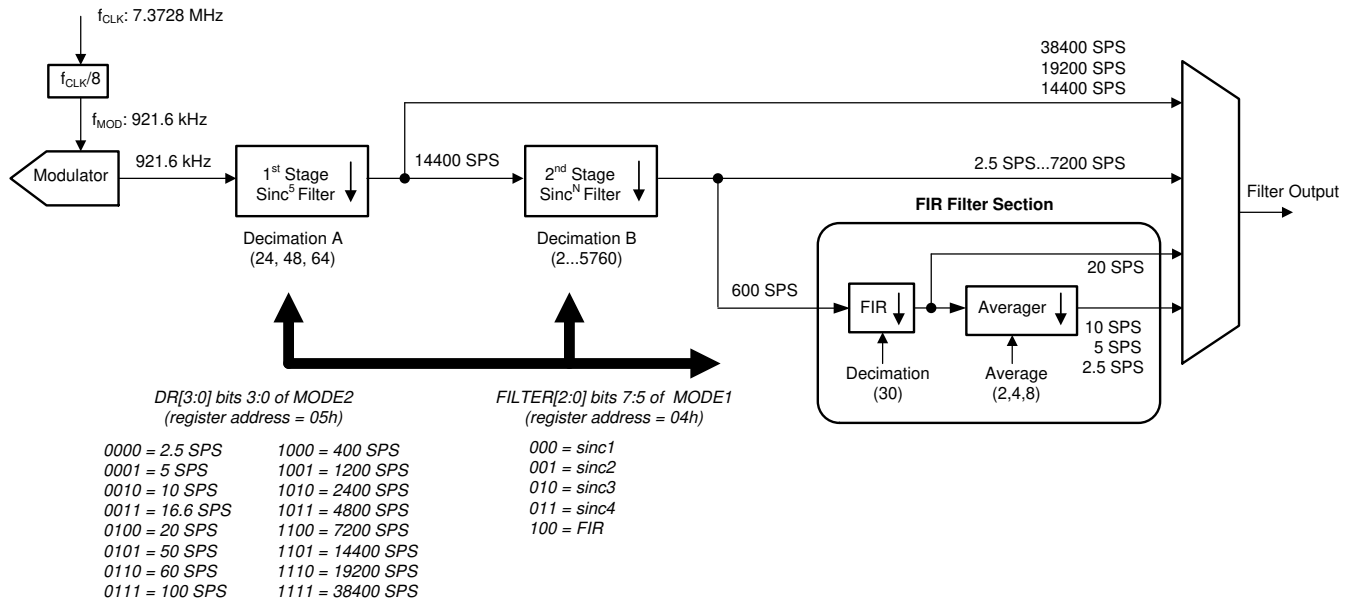


Figure 9-13. Digital Filter Block Diagram

9.3.10.1 Sinc Filter Mode

The sinc filter consists of two stages: a variable-decimation, fixed-order sinc5 filter, followed by a variable-decimation, variable-order sinc filter. The first-stage filter is sinc5. The sinc5 stage filters and down-samples the modulator data ($f_{CLK} / 8 = 921.6$ kHz) to 38400 SPS, 19200 SPS, and 14400 SPS by decimating to 24, 48, and 64, respectively. These data rates bypass the second filter stage and as a result have a sinc5 frequency response profile. The second filter stage receives the data from the first stage at 14400 SPS. The second stage reduces the data rate to produce output data of 7200 SPS to 2.5 SPS. The second stage is a variable-order sinc filter that is programmable.

The combined decimation ratio of the first and second stages determine the output data rate as follows: data rate = $921.6 \text{ kHz} / (A \cdot B)$. The filter order of the second stage affects the 50-Hz and 60-Hz rejection together with conversion latency. The high-order sinc filter yields the widest 50-Hz and 60-Hz response null widths, but correspondingly increases the conversion latency. The sinc order is programmed by the FILTER[2:0] bits of register MODE1. Table 9-3 lists the decimation ratio corresponding to the first and second filter stages (A and B, respectively) for each data rate. The data rate is programmed by the DR[3:0] bits of register MODE2.

Table 9-3. Sinc Filter Mode Data Rates and Decimation Ratio

DATA RATE (SPS) ⁽¹⁾	DR[3:0] BITS OF REGISTER MODE2	FIRST-STAGE DECIMATION RATIO A	SECOND-STAGE DECIMATION RATIO B
2.5	0000	64	5760
5	0001	64	2880
10	0010	64	1440
16.6	0011	64	864
20	0100	64	720
50	0101	64	288
60	0110	64	240
100	0111	64	144
400	1000	64	36
1200	1001	64	12
2400	1010	64	6
4800	1011	64	3
7200	1100	64	2
14400	1101	64	1
19200	1110	48	1
38400	1111	24	1

(1) $f_{CLK} = 7.3728$ MHz. Data rate scales with f_{CLK} **9.3.10.1.1 Sinc Filter Frequency Response**

The low-pass filtering effect of the sinc filter sets the overall frequency response of the ADC. The frequency response of data rates 14400 SPS, 19200 SPS and 38400 SPS is that of the first filter stage. The frequency response of data rates 2.5 SPS ranging to 7200 SPS is the product of the first and second stage individual frequency responses. The overall filter response is given in Equation 14:

$$|H(f)| = |H_{\text{sinc}^5}(f)| \times |H_{\text{sinc}^N}(f)| = \left| \frac{\sin\left[\frac{8\pi f A}{f_{CLK}}\right]}{A \times \sin\left[\frac{8\pi f}{f_{CLK}}\right]} \right|^5 \times \left| \frac{\sin\left[\frac{512\pi f B}{f_{CLK}}\right]}{B \times \sin\left[\frac{512\pi f}{f_{CLK}}\right]} \right|^N \quad (14)$$

where

- f = signal frequency
- f_{CLK} = ADC clock frequency
- A = First-stage decimation ratio (see Table 9-3)
- B = Second-stage decimation ratio (see Table 9-3)
- N = Second-stage filter order where $N = 1$ (sinc1), 2 (sinc2), 3 (sinc3), or 4 (sinc4)

The digital filter attenuates out-of-band noise that is present in the signal, and noise within the PGA and ADC modulator. Adjusting the filter by changing the decimation ratio and sinc order changes the filter bandwidth. Tradeoffs are made between signal bandwidth, noise, and filter latency.

As shown in [Figure 9-14](#) and [Figure 9-15](#), the first-stage sinc5 filter has frequency response nulls occurring at the data rate (f_{MOD} / A) and at data rate multiples. At the null frequencies, the filter has zero gain.

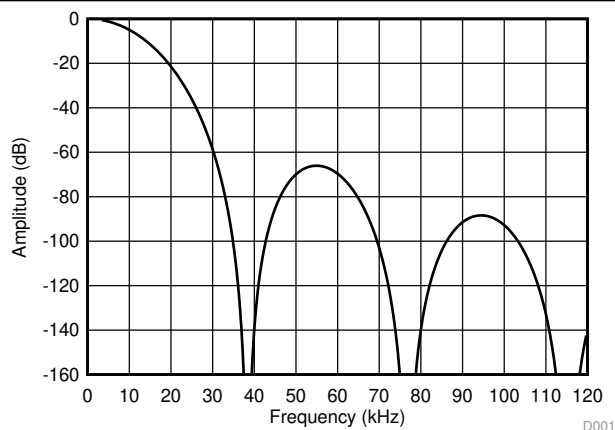


Figure 9-14. Sinc Frequency Response (38400 SPS)

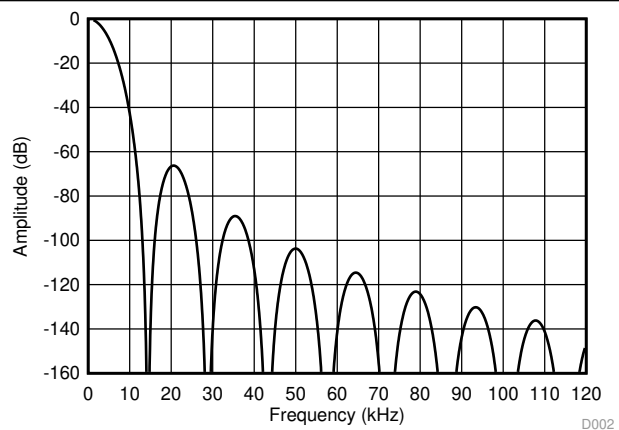


Figure 9-15. Sinc Frequency Response (14400 SPS)

The second stage superimposes new nulls in the frequency response over the nulls produced by the first stage. The first of the superimposed frequency response nulls occur at the output data rate, followed by nulls occurring at data rate multiples.

[Figure 9-16](#) illustrates the frequency response of data rate 2400 SPS produced by the combined filter stages. This data rate has five equally-spaced nulls between the larger nulls produced by the first stage. The frequency response is also characteristic of data rates 2.5 SPS to 7200 SPS that are also produced by the second-stage filter. [Figure 9-17](#) shows the frequency response nulls for 10 SPS.

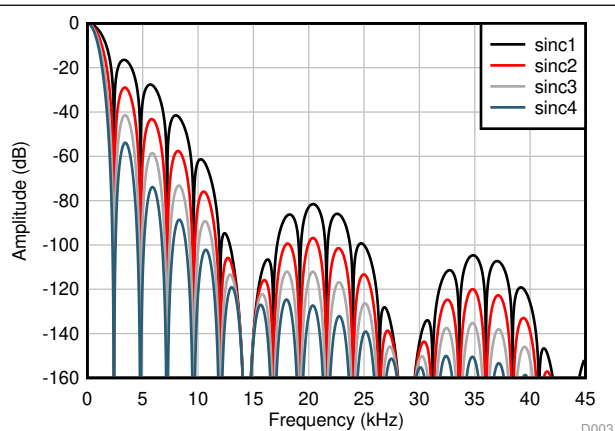


Figure 9-16. Sinc Frequency Response (2400 SPS)

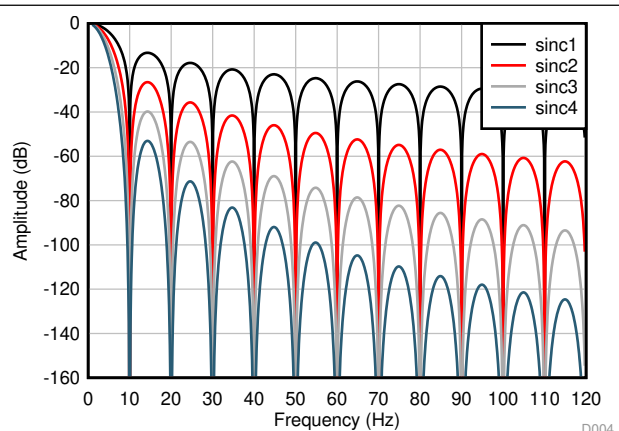


Figure 9-17. Sinc Frequency Response (10 SPS)

Figure 9-18 and Figure 9-19 illustrate the frequency response of data rates 50 SPS and 60 SPS. The frequency response is plotted out to the 50-Hz 12th harmonic (10th harmonic for 60 Hz). The 50-Hz or 60-Hz fundamental frequency and harmonics are suppressed by increasing the second-stage filter order, as shown in the figures.

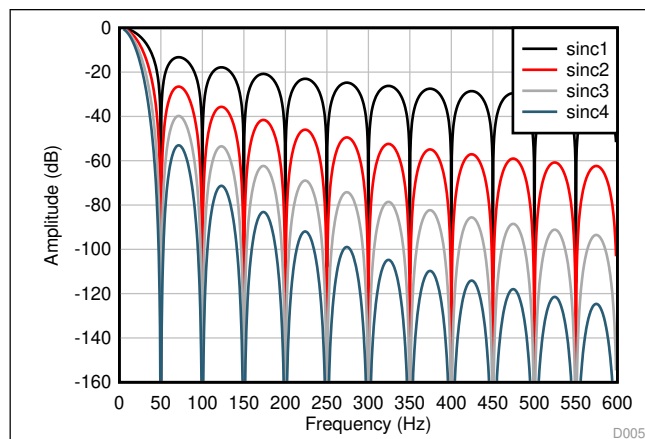


Figure 9-18. Sinc Frequency Response (50 SPS)

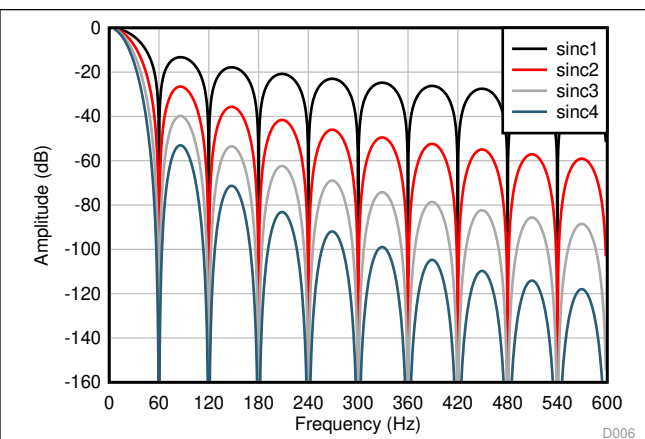


Figure 9-19. Sinc Frequency Response (60 SPS)

Figure 9-20 and Figure 9-21 plot the detailed frequency response of 50-SPS and 60-SPS data rates of different sinc-filter orders. Note that the high-order sinc filter increases the width of the null and improves line cycle rejection. The high-order filter decreases the sensitivity of the ratio tolerance between the ADC clock frequency and the line frequency that can otherwise degrade line cycle rejection. As shown in the plots, the best 50-Hz or 60-Hz rejection is provided by the sinc4 order, but has longer filter latency compared to the sinc1 order.

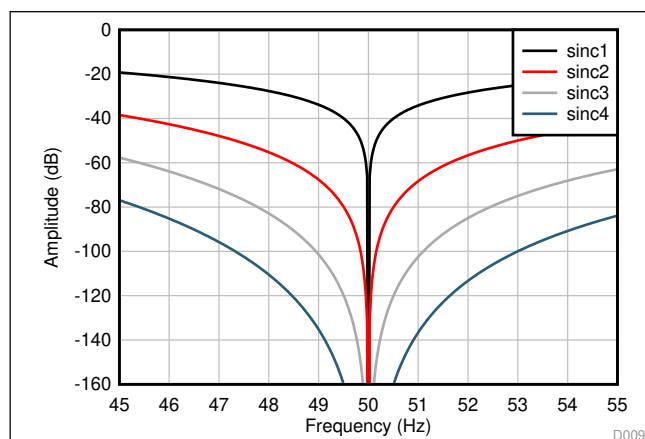


Figure 9-20. Sinc Frequency Response Zoom (50 SPS)

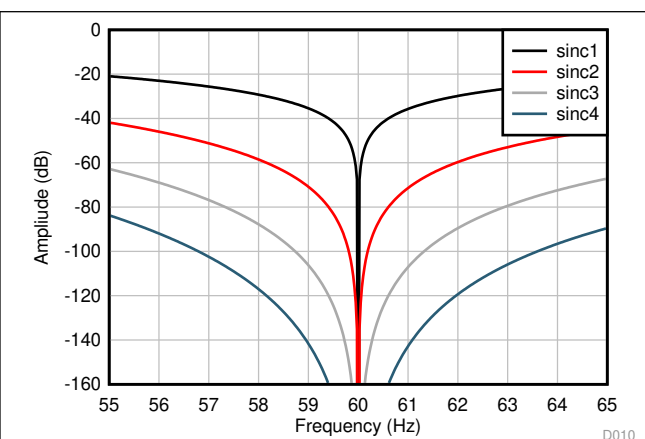


Figure 9-21. Sinc Frequency Response Zoom (60 SPS)

The overall sinc filter frequency has a low-pass response that rolls off high-frequency components in the signal. The signal bandwidth depends on the output data rate and the order of the sinc filter. Note the overall system bandwidth is the combination of the digital filter, the antialias filter, and external filter components. [Table 9-4](#) lists the –3-dB filter bandwidth of the sinc filter. Note the bandwidth reduction of the higher-order sinc filters.

Table 9-4. Sinc Filter Bandwidth

DATA RATE (SPS)	-3-dB BANDWIDTH (Hz)				
	SINC1	SINC2	SINC3	SINC4	SINC5
2.5	1.10	0.80	0.65	0.58	—
5	2.23	1.60	1.33	1.15	—
10	4.43	3.20	2.62	2.28	—
16.6	7.38	5.33	4.37	3.80	—
20	8.85	6.38	5.25	4.63	—
50	22.1	16.0	13.1	11.4	—
60	26.6	19.1	15.7	13.7	—
100	44.3	31.9	26.2	22.8	—
400	177	128	105	91.0	—
1200	525	381	314	273	—
2400	1015	751	623	544	—
4800	1798	1421	1214	1077	—
7200	2310	1972	1750	1590	—
14400	—	—	—	—	2940
19200	—	—	—	—	3920
38400	—	—	—	—	7740

9.3.10.2 FIR Filter

The finite impulse response (FIR) filter of ADC1 is a coefficient-based filter that provides simultaneous rejection of 50-Hz and 60-Hz line cycle frequencies and harmonics. The FIR filter data rates are 2.5, 5, 10 and 20 SPS. All of the FIR data rates settle within a single conversion cycle. As shown in [Figure 9-13](#), the FIR filter section receives data from the second-stage sinc filter at 600 Hz. The FIR filter section decimates by 30 to yield the output data rate of 20 SPS. A first-order averager (sinc1) with variable decimation provides the data rates of 10 SPS, 5 SPS, and 2.5 SPS.

As shown in [Figure 9-22](#) and [Figure 9-23](#), the FIR filter frequency response has a series of response nulls close to 50 Hz and 60 Hz. The response nulls repeat close to the 50-Hz and 60-Hz harmonics. The FIR frequency response superimposes with the response of the 600-SPS pre-stage filter.

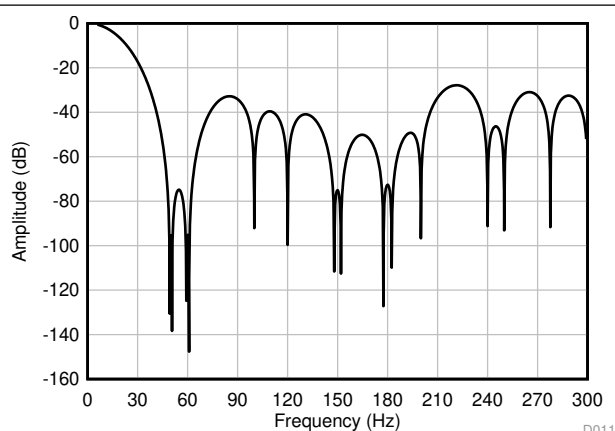


Figure 9-22. FIR Frequency Response (20 SPS)

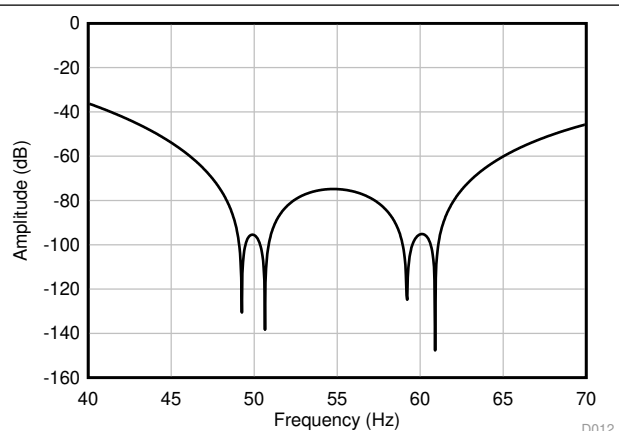


Figure 9-23. FIR Frequency Response Detail (20 SPS)

Figure 9-24 is the FIR filter response at 10 SPS. As a result of the sinc1 averager in the FIR filter block, new frequency-response nulls are superimposed to the response in Figure 9-22. The first of the added response nulls occur at 10 Hz. Additional nulls occur at folded frequencies around 20-Hz multiples. These additional nulls are seen at 10 Hz and 30 Hz.

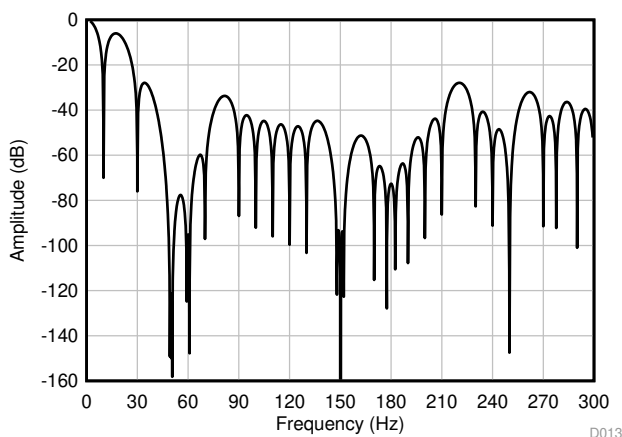


Figure 9-24. FIR Frequency Response (10 SPS)

Similar to the response of the sinc filter, the overall FIR filter frequency has a low-pass response that rolls off high frequencies of the signal. The response is such that the FIR filter limits the bandwidth of the input signal. The FIR filter signal bandwidth depends on the output data rate. Table 9-5 lists the –3-dB filter bandwidth of the FIR filter. The total system bandwidth is the combined individual responses of the digital filter, the ADC antialias filter, and external filter components.

Table 9-5. FIR Filter Bandwidth

DATA RATE (SPS)	–3-dB BANDWIDTH (Hz)
2.5	1.2
5	2.4
10	4.7
20	13

9.3.10.3 50-Hz and 60-Hz Line Cycle Rejection

If the ADC connection leads are in close proximity to industrial motors and conductors, coupling of 50-Hz and 60-Hz power line frequencies can occur. The coupled noise interferes with the signal voltage, and may lead to inaccurate or unstable conversions. The digital filter provides enhanced rejection of power-line coupled noise for data rates of 60 SPS and less. Program the filter to tradeoff data rate and conversion latency versus the desired level of line cycle rejection. [Table 9-6](#) summarizes the ADC1 50-Hz and 60-Hz line-cycle rejection based on 2% and 6% ratio tolerance of power-line to ADC clock frequency. Best possible power line rejection is provided by the high-order sinc filter and by using an accurate ADC clock.

Table 9-6. 50-Hz and 60-Hz Line Cycle Rejection

DATA RATE (SPS)	FILTER TYPE	DIGITAL FILTER Response (dB)			
		50 Hz $\pm 2\%$	60 Hz $\pm 2\%$	50 Hz $\pm 6\%$	60 Hz $\pm 6\%$
2.5	FIR	-113	-99	-88	-80
2.5	Sinc1	-36	-37	-40	-37
2.5	Sinc2	-72	-74	-80	-74
2.5	Sinc3	-108	-111	-120	-111
2.5	Sinc4	-144	-148	-160	-148
5	FIR	-111	-95	-77	-76
5	Sinc1	-34	-34	-30	-30
5	Sinc2	-68	-68	-60	-60
5	Sinc3	-102	-102	-90	-90
5	Sinc4	-136	-136	-120	-120
10	FIR	-111	-94	-73	-68
10	Sinc1	-34	-34	-25	-25
10	Sinc2	-68	-68	-50	-50
10	Sinc3	-102	-102	-75	-75
10	Sinc4	-136	-136	-100	-100
16.6	Sinc1	-34	-21	-24	-21
16.6	Sinc2	-68	-42	-48	-42
16.6	Sinc3	-102	-63	-72	-63
16.6	Sinc4	-136	-84	-96	-84
20	FIR	-95	-94	-66	-66
20	Sinc1	-18	-34	-18	-24
20	Sinc2	-36	-68	-36	-48
20	Sinc3	-54	-102	-54	-72
20	Sinc4	-72	-136	-72	-96
50	Sinc1	-34	-15	-24	-15
50	Sinc2	-68	-30	-48	-30
50	Sinc3	-102	-45	-72	-45
50	Sinc4	-136	-60	-96	-60
60	Sinc1	-13	-34	-12	-24
60	Sinc2	-27	-68	-24	-48
60	Sinc3	-40	-102	-36	-72
60	Sinc4	-53	-136	-48	-96

9.3.11 Sensor-Excitation Current Sources (IDAC1 and IDAC2)

The ADS1262 and ADS1263 incorporate two, integrated, matched current sources (IDAC1, IDAC2). The current sources provide excitation current to resistive temperature devices (RTDs), thermistors, diodes, and other sensors that require constant current biasing. These devices also contain an internal IDAC multiplexer that provides connection of IDAC1 or IDAC2 to one of the 11 analog pins (AIN0 to AINCOM). The IDACs can be programmed over these current ranges: 50 μ A, 100 μ A, 250 μ A, 500 μ A, 750 μ A, 1000 μ A, 1500 μ A, 2000 μ A, 2500 μ A, and 3000 μ A. Figure 9-25 details the IDAC connection. The IDAC switches shown in the diagram are used in the IDAC rotation mode.

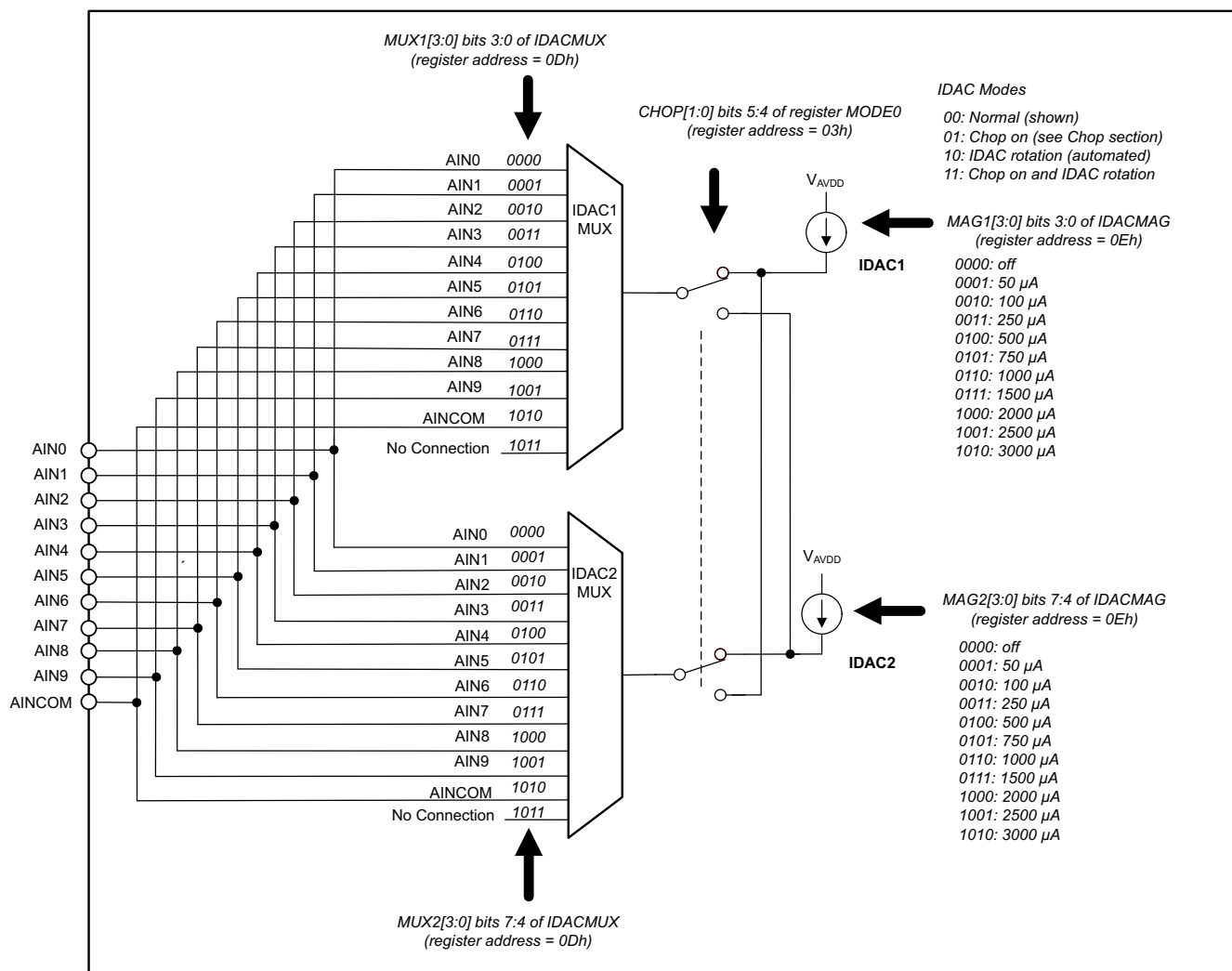


Figure 9-25. IDAC Block Diagram

The internal reference must be enabled for IDAC operation. Take care not to exceed the compliance voltage of the IDACs. In other words, the voltage on the input pin must not exceed $V_{AVDD} - 1.1$ V; otherwise, the specified accuracy of the IDAC current is not met.

The IDAC currents track the internal reference voltage. As a result of using the same reference voltage for IDAC1 and IDAC2, the current sources are matched. Matched performance is important for applications such as hardware compensated, 3-wire RTDs. IDAC to IDAC mismatch can be improved further by use of the IDAC rotation mode. The rotation mode automatically swaps the IDAC1 and IDAC2 connections of alternate conversions. The ADC averages the alternate conversions to eliminate IDAC mismatch. IDAC rotation can be performed manually by the user (by alternating the IDAC pin connections) or by the IDAC automatic rotation mode. The IDAC rotation sequence is shown as follows:

- Conversion 1: IDAC1, IDAC2 normal → first output result withheld
- Conversion 2: IDAC1, IDAC2 rotated positions → Output result 1 = (Conversion 1 + Conversion 2) / 2
- Conversion 3: IDAC1, IDAC2 normal → Output result 2 = (Conversion 3 + Conversion 2) / 2
- Conversion 4: IDAC1, IDAC2 rotated positions → Output result 3 = (Conversion 4 + Conversion 3) / 2

The sequence repeats for all succeeding conversions.

In rotation mode, the ADC provides a time delay to allow for settling after the IDAC pin connections are alternated. Note IDAC switching transients may interact with external components that may require additional time to settle. Additional settling time are provided by bits DELAY[3:0] in the MODE0 register. The total delay time results in a reduction of the nominal data rate (see the [Conversion Latency](#) section). Nevertheless, the existing frequency response nulls provided by the digital filter remain unchanged.

9.3.12 Level-Shift Voltage

The ADC integrates an optional level-shift voltage on the AINCOM pin. As shown in [Figure 9-26](#), the level-shift voltage is the mid-voltage of the analog power supply. The level-shift voltage shifts floating sensors (that is, sensors isolated from the ADC ground) to within the ADC specified input range. Thermocouple and 4-mA to 20-mA transmitters (isolated supply) are examples of floating signals.

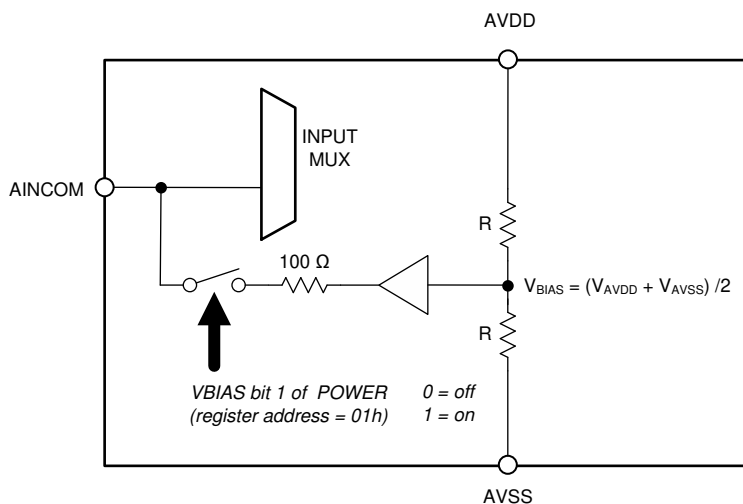


Figure 9-26. Level-Shift Voltage Diagram

When operating the ADC with ± 2.5 -V analog supplies, either ground the AINCOM pin or use the level-shift voltage. Level shift other inputs by connecting the input pins to the REFOUT pin (2.5 V). The turn-on time of the level-shift voltage depends on the pin load capacitance. The total capacitance includes those connected to AVDD, AVSS and ground. [Table 9-7](#) lists the level-shift voltage settling times for various external load capacitances. Be certain the level-shift voltage is fully settled before starting a conversion.

Table 9-7. Level-Shift Enable Time

LOAD CAPACITANCE	LEVEL-SHIFT VOLTAGE SETTLING TIME
0.1 μ F	0.22 ms
1 μ F	2.2 ms
10 μ F	22 ms

9.3.13 General-Purpose Input/Output (GPIO)

Eight analog inputs can be programmed as GPIO functions (GPIO[0] through GPIO[7]). The GPIO function is a digital input/output with a logic value that is read and written by the GPIODAT data register. The GPIO voltage levels are referenced to the ADC analog power supply voltages, V_{AVDD} and V_{AVSS} . The GPIO input voltage threshold for logic 1 is $(V_{AVDD} + V_{AVSS}) / 2$. As shown in Figure 9-27, analog inputs, AIN3 through AINCOM, can be programmed for GPIO function. Register GPIOCON programs the GPIO connection for each pin (1 = connect). Register GPIODIR programs the direction of each pin, either as input or output (0 = output). Register GPIODAT is the GPIO data value register. Note if a GPIO pin is programmed as an output, the readback data value of the corresponding GPIODAT register bit is zero.

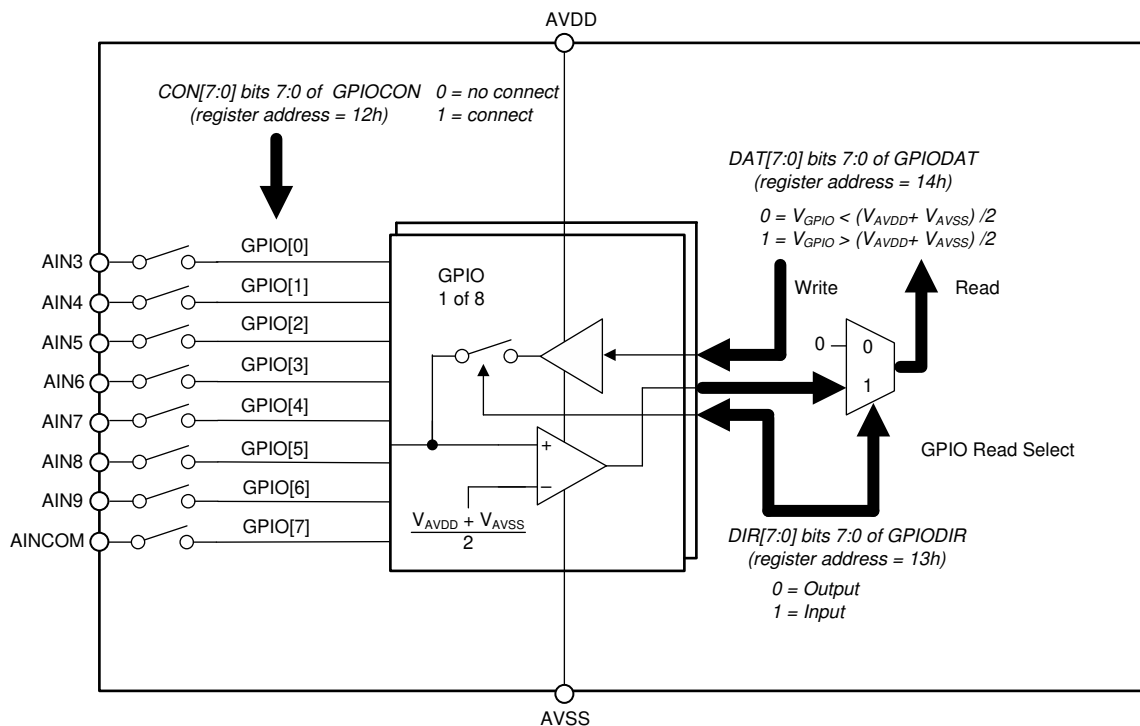


Figure 9-27. GPIO Block Diagram

9.3.14 Test DAC (TDAC)

The ADC includes a test voltage digital-to-analog converter (TDAC) intended for ADC self-testing and verification. The TDAC is capable of providing single-ended, differential and common mode test voltages. The voltages are suitable to test the ADCs under all gains and input configurations.

As shown in Figure 9-28, the TDAC consists of two independent DACs, TDACP, and TDACN. The DACs have independent control registers to program the output voltage. TDACP is programmed by register TDACP and TDACN is programmed by register TDACN. The TDACP output connects to the ADC1 and ADC2 positive input multiplexer input and TDACN connects to the ADC1 and ADC2 negative input multiplexer. The OUT1 and OUT2 bits can be programmed to connect the TDAC outputs to pins AIN6 and AIN7. The TDAC outputs are unbuffered and should not be loaded. The TDAC reference voltage is the analog supply ($V_{AVDD} - V_{AVSS}$); therefore, the output levels refer to, and scale with, the analog power supply. Note that chop mode must be disabled to test the ADC with the TDAC.

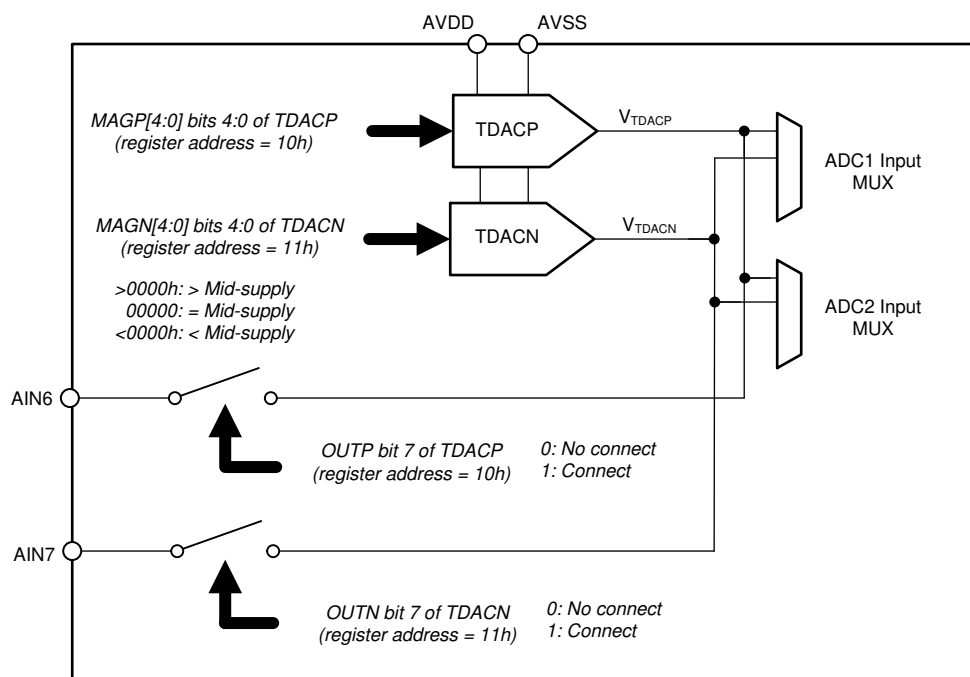


Figure 9-28. Test DAC Block Diagram

Table 9-8 shows the TDAC output voltages and the corresponding output impedance. The TDAC settings are binary-weighted and correspond to the binary-weighted ADC gains. To generate a single ended test voltage, set TDACN = 00h. This value sets the TDACN voltage to mid-supply. Set TDACP above or below the TDACN voltage to generate positive or negative test voltages, respectively. Differential test voltages are generated by setting TDACP and TDACN to symmetric values centered around a common-code value (typical common value = 00h). For example, use code values equal to 01h and 11h, 02h and 12h, and so forth, to generate a differential voltage. To generate common-mode test voltages, set the TDACs to equal values.

Table 9-8. TDAC Output Voltage

TDACP, TDACN REGISTER VALUES	DIVIDER RATIO (V/V)	OUTPUT VOLTAGE (V), 5-V SUPPLY ⁽¹⁾	OUTPUT VOLTAGE (V), ±2.5-V SUPPLY ⁽¹⁾	OUTPUT IMPEDANCE (kΩ)
09h	0.9	4.5	2	2.9
08h	0.7	3.5	1	6.4
07h	0.6	3	0.5	8.7
06h	0.55	2.75	0.25	10
05h	0.525	2.625	0.125	10.7
04h	0.5125	2.5625	0.0625	9.6
03h	0.50625	2.53125	0.03125	8.7
02h	0.503125	2.515625	0.015625	8.1
01h	0.5015625	2.5078125	0.0078125	7.8
00h	0.5	2.5	0	7.5
11h	0.4984375	2.4921875	–0.0078125	7.8
12h	0.496875	2.484375	–0.015625	8.1
13h	0.49375	2.46875	–0.03125	8.7
14h	0.4875	2.4375	–0.0625	9.6
15h	0.475	2.375	–0.125	10.7
16h	0.45	2.25	–0.25	10
17h	0.4	2	–0.5	8.7
18h	0.3	1.5	–1	6.4
19h	0.1	0.5	–2	2.9

(1) Output voltages relative to V_{DGND} .

9.3.15 ADC2 (ADS1263)

The ADS1263 includes an auxiliary, 24-bit, delta-sigma ADC (ADC2). ADC2 operation is independent of ADC1, with independent selections of input channel, reference voltage, sample rate, and channel gain. All input configurations (channel select, IDAC, level shift, sensor bias) are available to ADC2. Use ADC2 to perform main channel (ADC1) cross-checking measurements (for example, diagnostics purposes and redundant channel measurements), system background measurements, or temperature compensation of the primary sensor (such as thermocouple cold junction compensation). Using data rates of 10, 100, and 400 SPS for both ADCs, ADC2 performs virtual parallel conversions with ADC1 on the same input channel.

As shown in [Figure 9-29](#), the ADC2 consists of an input signal multiplexer followed by a high-impedance PGA. The input multiplexer has the same functionality as the ADC1 input multiplexer. The sensor bias current source or a 10-M Ω bias resistor can be connected to the multiplexer output. Connect the sensor bias to either of the ADCs. ADC2 provides gains of 1, 2, 4, 8, 16, 32, 64, and 128. Depending on the gain settings, ADC2 gains are either implemented in the PGA or in the modulator. For gains of 1, 2, and 4, the PGA is bypassed and the gain is performed in the modulator. For gains of 8, 16, 32, 64, and 128, the modulator gain is fixed at gain = 4, and the additional gains are performed in the PGA.

The PGA drives an inherently-stable, second-order, delta-sigma modulator. The modulator output data are filtered and down-sampled by a programmable decimation digital filter. The digital filter provides data rates of 10, 100, 400, or 800 SPS with 24-bit resolution. A calibration block follows the digital filter. The calibration block consists of 16-bit offset correction and 16-bit, full-scale correction registers. The ADC2 reference multiplexer selects from one of three external reference input pairs, the analog power supply, or the internal reference. The reference input is buffered to minimize errors caused by external circuit loading.

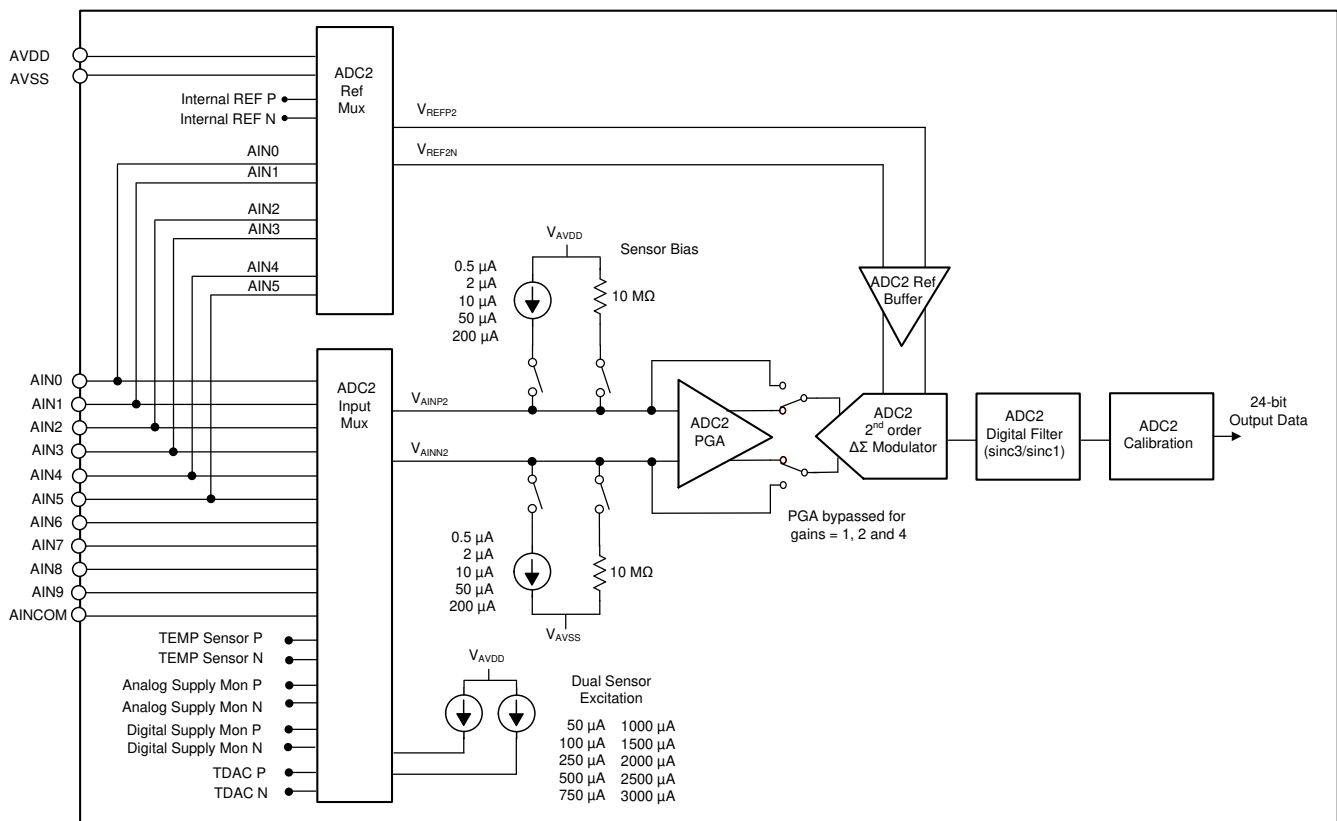


Figure 9-29. ADC2 Block Diagram

9.3.15.1 ADC2 Inputs

ADC2 features an independent input multiplexer with identical channel selections of ADC1. As shown in [Figure 9-2](#), all the external and internal inputs are available to ADC2. The ADC2 positive input is programmed by the value of MUXP2[3:0] bits (register ADC2MUX) and the negative input is programmed by the value of the MUXN2[3:0] bits of same ADC2MUX register. The ADC2MUX register address is 16h.

9.3.15.2 ADC2 PGA

ADC2 features a low-drift, low-noise CMOS PGA. The ADC2 PGA is bypassed for gains = 1, 2 and 4. Therefore, for these gains the input signal is connected directly to the buffered modulator input.

The full-scale voltage range of ADC2 is determined by the reference voltage and gain. [Table 9-9](#) shows the ADC2 full-scale voltage range versus gain using reference voltage = 2.5 V. The full-scale voltage range scales with the reference voltage and is increased or decreased by changing the reference voltage.

Table 9-9. ADC2 Full-Scale Voltage Range

GAIN2[2:0] BITS OF ADC2CFG REGISTER	GAIN (V/V)	FULL-SCALE INPUT RANGE (V) ⁽¹⁾
000	1	±2.500 V
001	2	±1.250 V
010	4	±0.625 V
011	8	±0.312 V
100	16	±0.156 V
101	32	±0.078 V
110	64	±0.039 V
111	128	±0.0195 V

(1) $V_{REF} = 2.5$ V. The full-scale voltage range is proportional to V_{REF} .

As with many amplifiers, do not exceed the PGA absolute input voltage requirement. For gains ≥ 8 (PGA active), the absolute input voltage is limited by the PGA output voltage swing range. The specified minimum and maximum absolute input voltages (V_{INP2} and V_{INN2}) depend on the PGA gain, the input differential voltage (V_{IN2}), and the tolerance of the analog power supply voltages (V_{AVDD} , V_{AVSS}). If using ADC2 in an overall gain ≥ 8 , the absolute positive and negative input voltage must be within the specified range, as shown in [Equation 15](#):

$$V_{AVSS} + 0.3 + |V_{IN2}| \cdot (\text{Gain} - 1) / 2 < V_{INP2} \text{ and } V_{INN2} < V_{AVDD} - 0.3 - |V_{IN2}| \cdot (\text{Gain} - 1) / 2 \quad (15)$$

where

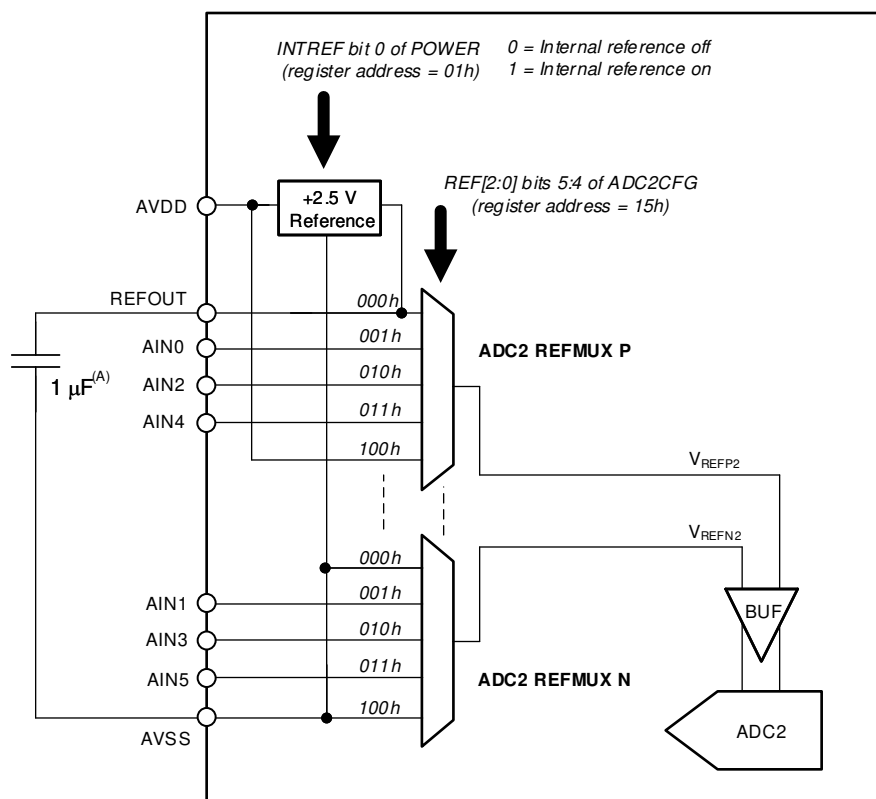
- V_{INP2} , V_{INN2} = ADC2 absolute input voltage
- V_{IN2} = ADC2 differential input voltage = $V_{INP2} - V_{INN2}$
- Gain = 8, 16, 32, 64 or 128

For gains 1, 2, or 4, the ADC2 absolute input voltage range extends beyond the V_{AVDD} and V_{AVSS} supply voltages, allowing voltage inputs at or below ground. The absolute input voltage range corresponding to gains 1, 2 and 4 is shown in [Equation 16](#):

$$V_{AVSS} - 0.1 < V_{INP2} \text{ and } V_{INN2} < V_{AVDD} + 0.1 \quad (16)$$

9.3.15.3 ADC2 Reference

ADC2 requires a reference voltage for operation. Use the ADC2 reference multiplexer to select from one of the external reference sources on pins AIN0 to AIN5, the internal 2.5-V internal reference, or the analog power supply, as shown in Figure 9-30. The external reference uses positive and negative pairs for the positive and negative references, respectively. The external reference input pairs are pins AIN0-AIN1, AIN2-AIN3, and AIN4-AIN5, for the positive and negative references, respectively.



A. The internal reference requires a 1- μ F capacitor connected to pins REFOUT and AVSS.

Figure 9-30. ADC2 Reference Multiplexer

9.3.15.4 ADC2 Modulator

ADC2 is an inherently stable, second-order, $\Delta\Sigma$ modulator. The modulator samples the analog input voltage at $f_{MOD2} = f_{CLK} / 144 = 51.2$ kHz and converts the analog input to a ones density bit-stream output. The digital filter receives the ones density bit stream output, and then filters and decimates the data to yield the final conversion result.

9.3.15.5 ADC2 Digital Filter

The ADC2 digital filter receives the modulator output and produces a 24-bit digital output. The digital filter low-pass filters and down-samples the modulator data to yield the final data rate. The ADC2 digital filter is a cascade of two stages. The first stage is a sinc3 filter that decimates by 64, 128, or 512, to derive data rates of 800 SPS, 400 SPS, or 100 SPS, respectively. The second stage receives the output of the first stage at 100 SPS. The second stage is a sinc1 filter with decimation equal to ten that derives the data rate of 10 SPS, as illustrated in Figure 9-31. The ADC bypasses the second stage for data rates of 800 SPS, 400 SPS, and 100 SPS. Table 9-10 shows the sinc filter data rates and decimation ratios (A and B) that correspond to each filter stage. The overall filter decimation ratio is the product of A and B decimation ratios. The data rate is programmed by the DR2[1:0] bits of register ADC2CFG.

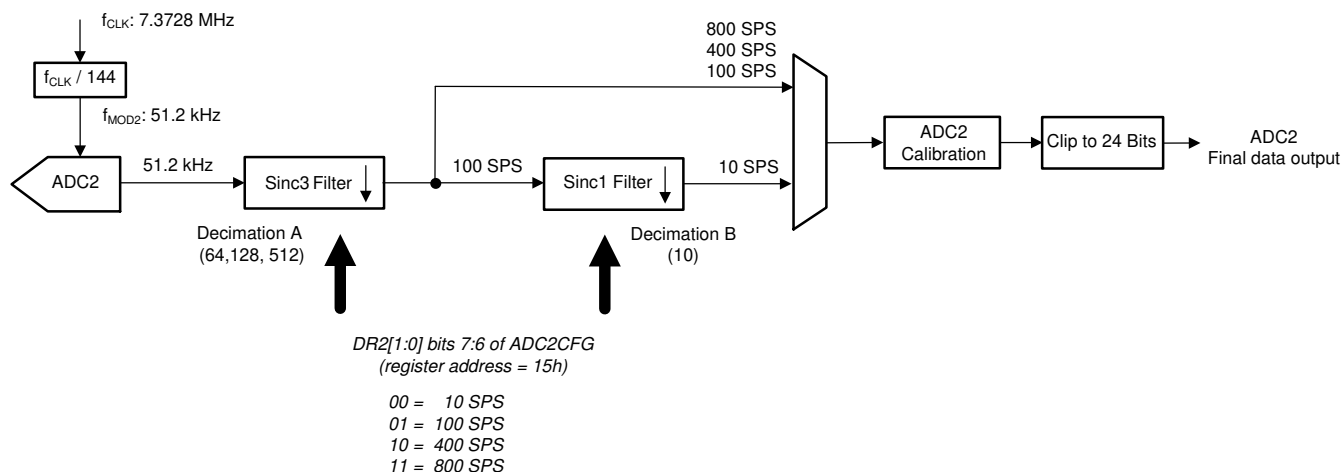


Figure 9-31. ADC2 Digital Filter Block Diagram

Table 9-10. ADC2 Data Rates and Filter Decimation Ratios

DATA RATE (SPS) ⁽¹⁾	DR2[1:0] BITS OF REGISTER ADC2CFG	1st STAGE DECIMATION RATIO A	2nd STAGE DECIMATION RATIO B
10 (default)	00	512	10
100	01	512	-
400	10	128	-
800	11	64	-

(1) $f_{CLK} = 7.3728$ MHz. The data rate scales with f_{CLK} .

The low pass nature of the ADC2 sinc filter establishes the overall frequency response. The frequency response is given by Equation 17:

$$|H(f)| = |H_{\text{sinc}^3}(f)| \times |H_{\text{sinc}}(f)| = \left| \frac{\sin\left[\frac{144\pi f A}{f_{CLK}}\right]}{A \times \sin\left[\frac{144\pi f}{f_{CLK}}\right]} \right|^3 \times \left| \frac{\sin\left[\frac{73728\pi f B}{f_{CLK}}\right]}{B \times \sin\left[\frac{73728\pi f}{f_{CLK}}\right]} \right| \quad (17)$$

where

- f = Input frequency
- f_{CLK} = ADC clock (7.3728 MHz)
- A = First stage decimation ratio
- B = Second stage decimation ratio

Figures [Figure 9-32](#) through [Figure 9-36](#) show the frequency response of different ADC2 data rates. Nulls are located in the frequency response at the data rate and at data rate multiples. [Figure 9-32](#) (data rate = 10 SPS) has frequency response nulls at 50 Hz and 60 Hz and their multiples. Therefore, the rate of 10 SPS provides rejection of power line cycle frequencies. [Figure 9-33](#) shows filter response detail of frequencies centered around 50 Hz and 60 Hz.

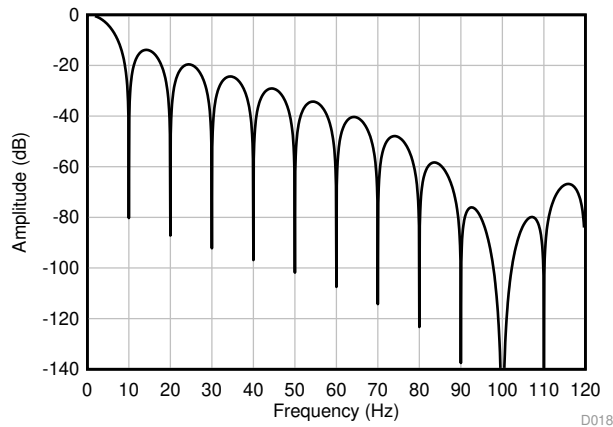


Figure 9-32. ADC2 10 SPS Frequency Response

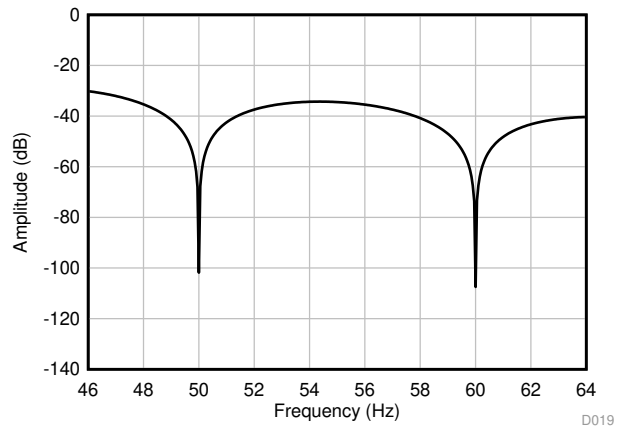


Figure 9-33. ADC2 10 SPS Frequency Response 50-Hz and 60-Hz Detail

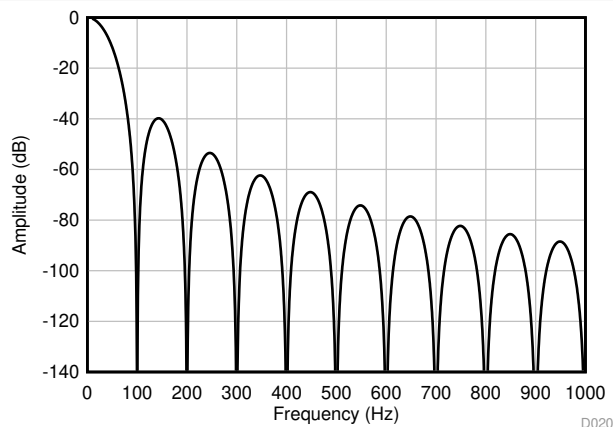


Figure 9-34. ADC2 100 SPS Frequency Response

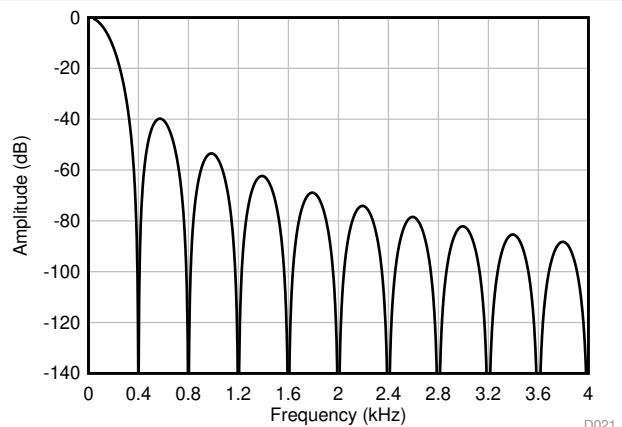


Figure 9-35. ADC2 400 SPS Frequency Response

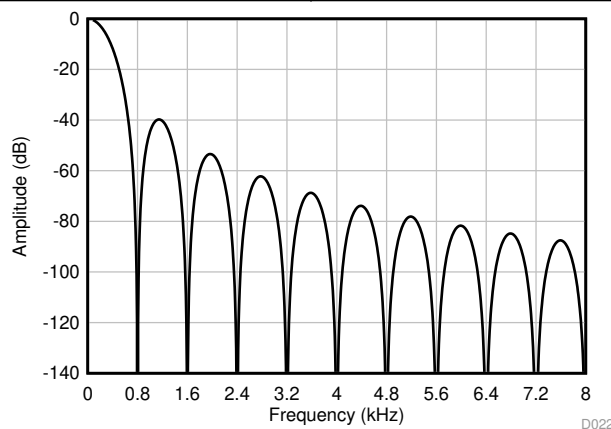


Figure 9-36. ADC2 800 SPS Frequency Response

Table 9-11 summarizes the ADC2 digital filter –3-dB bandwidth and 50-Hz and 60-Hz line-cycle rejection based on 2% and 6% ratio tolerance of power-line frequency to ADC clock frequency. The sample rate of 10 SPS has frequency response nulls at 50 Hz and 60 Hz; therefore, this data rate provides the best possible rejection of power-line interference.

Table 9-11. ADC2 –3-dB Bandwidth, 50-Hz and 60-Hz Line Cycle Rejection

DATA RATE (SPS)	-3-dB BANDWIDTH (Hz)	DIGITAL FILTER RESPONSE (dB)			
		50-Hz REJECTION $\pm 2\%$	60-Hz REJECTION $\pm 2\%$	50-Hz REJECTION $\pm 6\%$	60-Hz REJECTION $\pm 6\%$
10	4.4	–41	–47	–32	–36
100	26	–12	–17	–10	–16
400	104	–0.5	–0.9	–0.5	–0.9
800	208	–0.2	–0.2	–0.1	–0.2

The ADC digital filter provides attenuation of frequencies greater than $\frac{1}{2}$ of the data rate (Nyquist frequency) to minimize out-of-band frequencies folding back to the bandwidth of interest. As with all digital filters, response images appear at frequency multiples of the filter input frequency ($f_{MOD2} = f_{CLK} / 144 = 51.2$ kHz). [Figure 9-37](#) shows the frequency response to 175 kHz for DR = 800 SPS. The response near dc is the desired signal bandwidth. Note how the filter response repeats at multiplies of 51.2 kHz. The filter response repeats at frequencies shown in [Equation 18](#):

$$\text{Aliased frequency bands} = N \cdot f_{MOD2} \pm f_{DR2} \quad (18)$$

where

- $N = 1, 2, 3...$
- $f_{DR2} = \text{ADC2 data-rate frequency}$

The digital filter attenuates signal or noise up to the frequency where the response repeats. However, any signal or noise present within the frequency bands where the response repeats aliases into the passband, unless attenuated by an analog filter. Often, using a simple RC analog filter is sufficient to reject these frequencies.

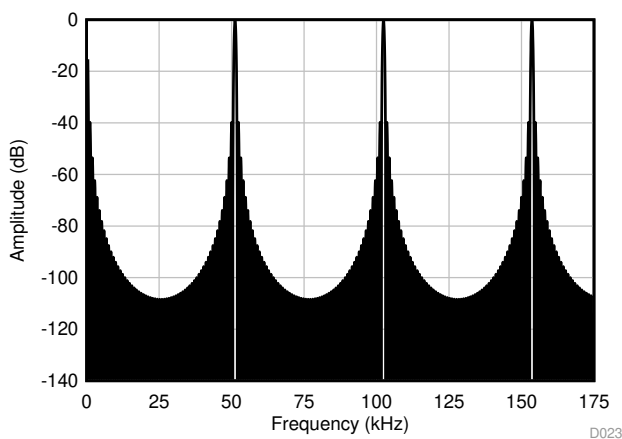


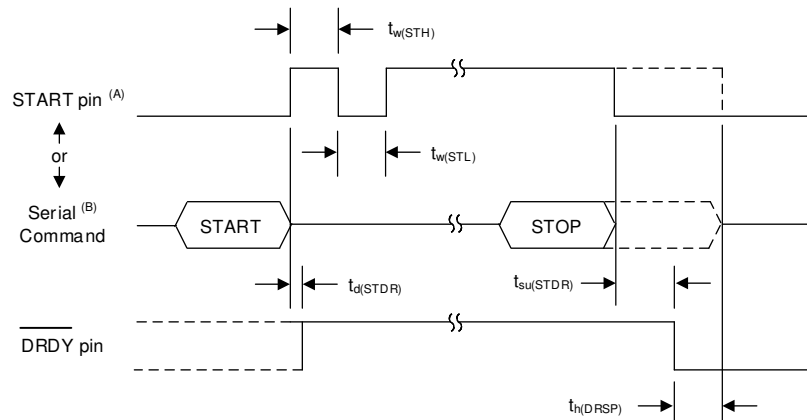
Figure 9-37. ADC2 Frequency Response to 175 kHz

9.4 Device Functional Modes

9.4.1 Conversion Control

ADC1 conversions are controlled by the START pin or by serial commands. If using commands to control ADC1 conversions, keep the START pin low to avoid possible contentions between the START pin and commands.

ADC1 has two conversion modes: continuous or pulse. Continuous-conversion mode converts indefinitely until stopped by the user. Pulse-conversion mode performs one conversion after the START pin is taken high or after the start command is sent. Use RUNMODE (bit 6, MODE0) to program the conversion mode. Figure 9-38 shows the start and stop timing to control ADC conversions.



- A. START and $\overline{\text{DRDY}}$ pins apply only to ADC1 operation.
 B. Start and stop opcodes take effect on the 7th SCLK falling edge.
 C. Start and stop opcodes:
- START1 for ADC1: 08h or 09h
 - START2 for ADC2: 0Ch or 0Dh
 - STOP1 for ADC1: 0Ah or 0Bh
 - STOP2 for ADC2: 0Eh or 0Fh

Figure 9-38. ADC1 Start and Stop Conversion Timing

Table 9-12. ADC1 START and STOP Conversion Timing Requirements

PARAMETER	TEST CONDITIONS	MIN	MAX	UNIT
$t_{w(\text{STH})}$	START pin high: pulse duration	4		t_{CLK} ⁽¹⁾
$t_{w(\text{STL})}$	START pin low to <i>re-start</i> conversion: pulse duration	4		t_{CLK}
$t_{d(\text{STDR})}$	Start condition to $\overline{\text{DRDY}}$ high: delay time		2	t_{CLK}
$t_{su(\text{STDR})}$	Stop condition to $\overline{\text{DRDY}} \downarrow$ stopping additional conversions: set-up time	16		t_{CLK}
$t_{h(\text{DRSP})}$	$\overline{\text{DRDY}} \downarrow$ to stop condition to continue current conversion: hold time	16		t_{CLK}

(1) $t_{\text{CLK}} = 1 / f_{\text{CLK}}$

9.4.1.1 Continuous Conversion Mode

To start ADC1 conversions, take the START pin high or send the START1 command. In this mode, ADC1 continuously converts until stopped by taking the START pin low or by sending the STOP1 command. To restart a conversion in progress, toggle the START pin or send a STOP1 and START1 command sequence. $\overline{\text{DRDY}}$ is driven high if conversions are restarted. $\overline{\text{DRDY}}$ is driven low when the conversion data are ready. See Figure 9-38 and Table 9-12 for stop-to- $\overline{\text{DRDY}}$ timing requirements in order to stop further conversions.

9.4.1.2 Pulse Conversion Mode

In pulse conversion mode, ADC1 performs one conversion each time the START pin is taken high or the START1 command is sent. After the first conversion completes, further conversions are automatically stopped. To restart a conversion in progress, toggle the START pin or send a STOP1 and START1 command sequence. The $\overline{\text{DRDY}}$ output is driven high to indicate conversion start, and is driven low when the conversion data are ready. If a stop command is sent during an ongoing conversion, the command has no effect because the ADC completes the conversion.

The pulse conversion mode cannot be used in conjunction with chop mode. Instead, use the continuous-conversion mode when chop is enabled by briefly pulsing the START pin (or sequentially send the START1 and STOP1 commands) to achieve the same single-conversion result provided by the pulse conversion mode.

9.4.1.3 ADC2 Conversion Control (ADS1263)

ADC2 conversions are independent of ADC1 conversions, and are controlled by commands only. The ADC2 conversion mode is similar to the ADC1 continuous conversion mode. To start an ADC2 conversion, send the START2 command. Conversions continue until the STOP2 command is sent. To restart a conversion in progress, send a STOP2 and START2 command sequence.

9.4.2 Conversion Latency

The digital filter averages and down-samples data from the modulator to provide the final data rate (rate reduction). The order of the digital filter affects the amount of data averaging and in turn, the time delay of the conversion (or filter latency). The FIR and sinc1 filter modes are zero latency providing the conversion result in single cycle. The higher order sinc filters (sinc2, 3, 4, 5) have more than one conversion latency and therefore require more conversion cycles to provide fully settled data. Tradeoffs can be made between 50-Hz and 60-Hz line cycle rejection verses conversion latency by selection of the sinc filter order. A higher order sinc filter increases the rejection of the 50-Hz and 60-Hz line cycles, but also increases the filter latency. Filter latency is an important consideration when multiplexing (scanning) through input channels. To make sure that conversions are settled after changing channels, start a new conversion for each channel using the START pin or start command. Note if the multiplexer is changed during ongoing conversions, the conversion is stopped and restarted at the time multiplexer register is changed.

Table 9-13 lists the filter latency after starting the *first* conversion. Note the conversion latency depends on the filter setting. The conversion latency is illustrated in Figure 9-39. Parameter $t_{d(\text{STDR})}$ shows the latency from start to conversion data ready ($\overline{\text{DRDY}}$ low). Note that settled data are provided, assuming the analog input is settled before the start condition. After the first conversion is completed (in continuous conversion mode), subsequent conversions occur at the nominal data rate. The latency values are for the programmable time-delay parameter set to off (DELAY[3:0] = 000).

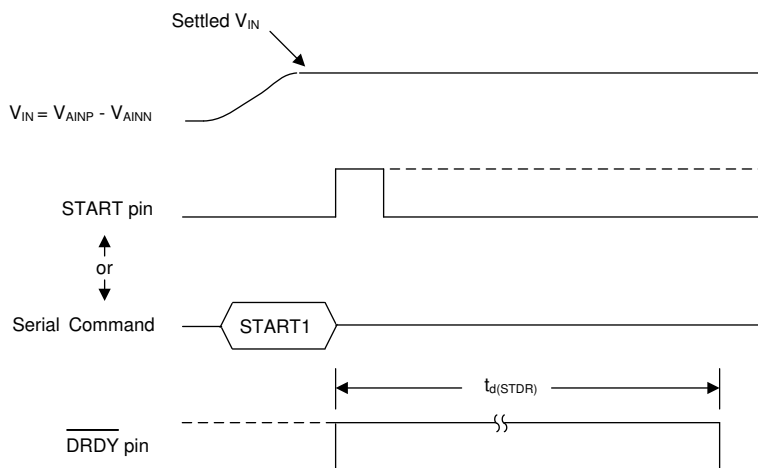


Figure 9-39. Conversion Latency After Start Condition

Table 9-13. ADC1 Conversion Latency, t_d (STDR)

DATA RATE (SPS)	CONVERSION LATENCY ⁽¹⁾ (ms)					
	SINC1	SINC2	SINC3	SINC4	SINC5	FIR
2.5	400.4	800.4	1,200	1,600	—	402.2
5	200.4	400.4	600.4	800.4	—	202.2
10	100.4	200.4	300.4	400.4	—	102.2
16.6	60.35	120.4	180.4	240.4	—	—
20	50.35	100.4	150.4	200.4	—	52.22
50	20.35	40.42	60.42	80.42	—	—
60	17.02	33.76	50.42	67.09	—	—
100	10.35	20.42	30.42	40.42	—	—
400	2.855	5.424	7.924	10.42	—	—
1200	1.188	2.091	2.924	3.758	—	—
2400	0.771	1.258	1.674	2.091	—	—
4800	0.563	0.8409	1.049	1.258	—	—
7200	0.494	0.702	0.841	0.980	—	—
14400	—	—	—	—	0.424	—
19200	—	—	—	—	0.337	—
38400	—	—	—	—	0.207	—

(1) Chop and IDAC rotation off, DELAY[3:0] = 0000.

If using chop or IDAC rotation modes, the latency of the first conversion increases. The latency of chop and IDAC rotation modes is shown in [Equation 19](#) and [Equation 20](#).

$$\text{Chop or IDAC rotation mode: latency} = 2 \cdot (t_{d(\text{STDR})} + \text{DELAY}[3:0] \text{ value}) \quad (19)$$

$$\text{Chop and IDAC rotation modes: latency} = 4 \cdot (t_{d(\text{STDR})} + \text{DELAY}[3:0] \text{ value}) \quad (20)$$

In addition, chop or IDAC rotation mode can reduce the conversion data rate depending on the time-delay parameter. The 50-Hz and 60-Hz filter response nulls are not altered by chop or IDAC rotation modes. [Equation 21](#) shows the effective data rate with the DELAY parameter.

$$\text{Chop or IDAC rotation mode data rate} = 1 / (t_{d(\text{STDR})} + \text{DELAY}[3:0] \text{ value}) \quad (21)$$

[Table 9-14](#) shows the first conversion latency of ADC2. The filter latency is the elapsed time after sending the START2 command before the first conversion is ready.

Table 9-14. ADC2 Conversion Latency, t_d (STDR)

DATA RATE (SPS)	CONVERSION LATENCY (ms)
10	121
100	31.2
400	8.71
800	4.97

If the input signal changes while the ADC is continuously converting, the output data are a mix of old and new data, as shown in Figure 9-40. The filter latency values for *settled data* ($t_{d(STDR)}$) with an input step change while continuously converting is shown in Table 9-15. The filter latency values listed in the table ($t_{d(STDR)}$) assume the analog input is settled before the start of the first whole conversion period.

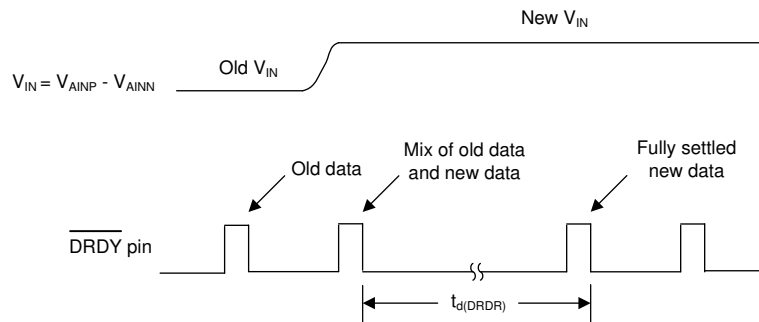


Figure 9-40. ADC1 Latency Timing While Continuously Converting

**Table 9-15. Fully-Settled Conversion Values for
Figure 9-40**

DIGITAL FILTER	FULLY SETTLED CONVERSION $t_{d(DRDR)}$ (1 / DR) ⁽¹⁾
FIR	1
Sinc1	1
Sinc2	2
Sinc3	3
Sinc4	4
Sinc5	5

(1) Chop and IDAC rotation modes off.

9.4.3 Programmable Time Delay

When a new conversion is started, the ADC provides an internal delay of 52 μ s before the actual start of the conversion. This timed delay is provided to allow for the integrated, analog, antialias filter to settle. In some cases, more delay is required to allow for external settling effects. Program additional time by using bits DELAY[3:0] of the MODE register. The programmable range is 8.7 μ s to 8.8 ms in binary steps. As an alternative to using the programmable time delay, the initiation of the start condition can also be delayed as needed after an ADC configuration change. For CHOP or IDAC rotation modes, additional time delay may be necessary to allow for external settling effects, and can only be provided by the DELAY bits; see Table 9-38 for the delay settings.

9.4.4 Serial Interface

The ADC has an SPI-compatible, bidirectional serial interface that is used to read the conversion data as well as configure and control the ADC. The serial interface consists of four control lines: $\overline{\text{CS}}$, SCLK, DIN, and DOUT/ $\overline{\text{DRDY}}$. If the ADS1262 or ADS1263 is the only device connected to the SPI bus, the $\overline{\text{CS}}$ input can be tied low, resulting in a minimum of three control signals for communications: SCLK, DIN, and DOUT/ $\overline{\text{DRDY}}$. The ADC has a data ready output signal ($\overline{\text{DRDY}}$) that asserts regardless of interface selection. The $\overline{\text{DRDY}}$ functionality is also integrated with the DOUT/ $\overline{\text{DRDY}}$ pin.

9.4.4.1 Chip Select ($\overline{\text{CS}}$)

The $\overline{\text{CS}}$ pin is an active low input that enables the ADC serial interface for communication. $\overline{\text{CS}}$ must be low during the entire data transaction. When $\overline{\text{CS}}$ is high, the serial interface is reset, SCLK input activity is ignored (blocking input commands), and the DOUT/ $\overline{\text{DRDY}}$ output pin enters a high-impedance state. ADC conversions are not affected by the state of $\overline{\text{CS}}$. If the serial bus is dedicated to the ADC, the $\overline{\text{CS}}$ pin can be optionally tied low to reduce the serial interface from four I/Os to three I/Os. Tying the $\overline{\text{CS}}$ pin low permanently enables the ADC serial interface. The $\overline{\text{DRDY}}$ output asserts low when conversion data are ready and is not affected by $\overline{\text{CS}}$.

9.4.4.2 Serial Clock (SCLK)

The serial interface clock is a noise-filtered, Schmidt-triggered input used to clock data into and out of the ADC. Input data to the ADC is latched on the falling SCLK edge and data output from the ADC is updated on the rising SCLK edge. Return SCLK low after the data sequence is complete. Even though the SCLK is a noise-immune, keep SCLK as clean as possible to prevent unintentional SCLK transitions. Avoid ringing and voltage overshoot on the SCLK input. Place a series termination resistor at the SCLK drive pin to help reduce ringing.

9.4.4.3 Data Input (DIN)

The DIN pin is the serial data input to the ADC. DIN is used to input commands and register data to the ADC. The ADC latches input data on the falling edge of SCLK. During direct-mode data readback, when no command is intended, keep DIN low.

9.4.4.4 Data Output/Data Ready (DOUT/ $\overline{\text{DRDY}}$)

The DOUT/ $\overline{\text{DRDY}}$ pin is a dual-function output. The pin functions as the digital data output and the ADC1 data-ready indication. When $\overline{\text{CS}}$ is high, the DOUT/ $\overline{\text{DRDY}}$ pin is in high-impedance mode (tri-state). Output data are updated on the rising edge of SCLK. As a data-ready indicator, the DOUT/ $\overline{\text{DRDY}}$ pin transitions low at the same time as the $\overline{\text{DRDY}}$ pin. Therefore, monitor either the DOUT/ $\overline{\text{DRDY}}$ pin or the $\overline{\text{DRDY}}$ pin to determine when ADC1 data are ready. $\overline{\text{CS}}$ must be low to monitor DOUT/ $\overline{\text{DRDY}}$ as a data-ready indicator.

9.4.4.5 Serial Interface Autoreset

The $\overline{\text{CS}}$ input resets the serial interface when taken high. However, applications that tie the $\overline{\text{CS}}$ pin low do not have the ability to reset the serial interface by using this pin. If a false SCLK occurs (for example, caused by a noise pulse or clocking glitch), the serial interface may inadvertently advance one or more bit positions, resulting in loss of synchronization to the external microcontroller. If loss of synchronization occurs, the interface does not respond correctly until the interface is reset.

For applications that tie $\overline{\text{CS}}$ low, the ADC provides a feature that automatically resets the serial interface in the event of a glitch. As shown in Figure 9-41, after the first SCLK low-to-high transition is detected by the ADC (either caused by a glitch or a normal SCLK input), if the ADC does not detect seven additional SCLK transitions within $65536 f_{\text{CLK}}$ cycles (approximately 8.9 ms), the serial interface resets. After reset, the interface is ready for the next transaction four f_{CLK} cycles later.

If the seven SCLK transitions are detected within the $65536 f_{\text{CLK}}$ cycles, the serial interface is not reset, and the SCLK detection cycle restarts at the next SCLK transition.

If the serial interface loses synchronization to an external controller, reset the serial interface by holding SCLK low for $65536 f_{\text{CLK}}$ cycles.

The serial interface autoreset function is enabled by the setting TIMEOUT = 1 (bit 3 of the INTERFACE register). The default mode is off.

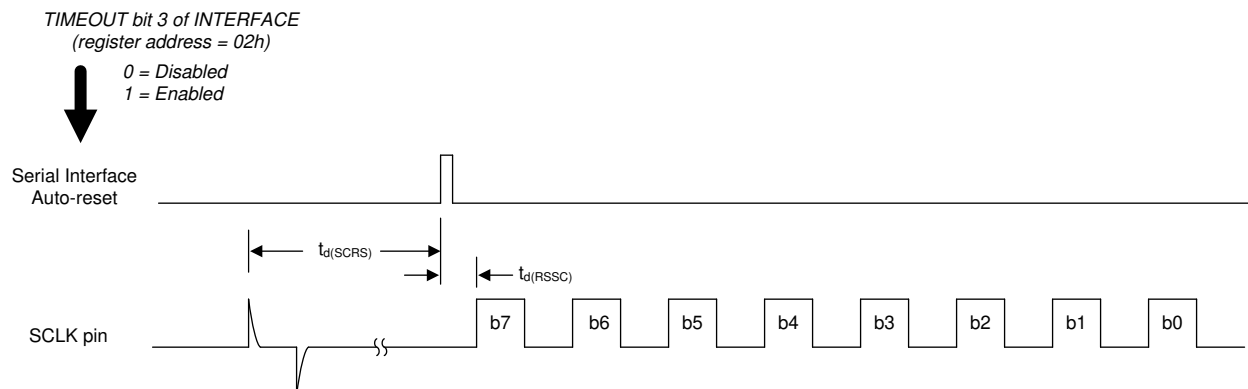


Figure 9-41. Serial Interface Autoreset

Table 9-16. Autoreset Timing Requirement

PARAMETER	TEST CONDITIONS	MIN	UNIT
$t_{d(\text{SCRS})}$	SCLK $\uparrow$ transition to interface reset : delay time	TIMEOUT bit =1	t_{CLK} ⁽¹⁾
$t_{d(\text{RSSC})}$	Serial interface reset to first SCLK $\uparrow$: delay time	TIMEOUT bit =1	4

(1) $t_{\text{CLK}} = 1 / f_{\text{CLK}}$.

9.4.5 Data Ready Pin ($\overline{\text{DRDY}}$)

The $\overline{\text{DRDY}}$ pin is an output that transitions low to indicate when ADC1 conversion data are ready for retrieval. Figure 9-42 depicts the $\overline{\text{DRDY}}$ operation. Initially, $\overline{\text{DRDY}}$ is high at power-on. When converting, the state of $\overline{\text{DRDY}}$ depends on the conversion mode (continuous or pulse) and whether the conversion data are retrieved or not. In Continuous conversion mode, after $\overline{\text{DRDY}}$ goes low, $\overline{\text{DRDY}}$ is driven high on the first SCLK falling edge. If data are not read, $\overline{\text{DRDY}}$ remains low and then pulses high 16 f_{CLK} cycles before the next $\overline{\text{DRDY}}$ falling edge. The data must be retrieved before the next $\overline{\text{DRDY}}$ falling edge otherwise the data are overwritten by new data and previous data are lost. In Pulse mode, $\overline{\text{DRDY}}$ is driven high when a conversion is started and goes low when the conversion data are ready. $\overline{\text{DRDY}}$ remains low until the next conversion is started.

The DOUT/ $\overline{\text{DRDY}}$ output operates similarly to $\overline{\text{DRDY}}$. DOUT/ $\overline{\text{DRDY}}$ transitions low when ADC1 conversion data are ready. If data are not retrieved, the DOUT/ $\overline{\text{DRDY}}$ pin stays low and is pulsed high for 16 f_{CLK} cycles at the next data ready. Note that $\overline{\text{CS}}$ must be low to enable the DOUT/ $\overline{\text{DRDY}}$ pin.

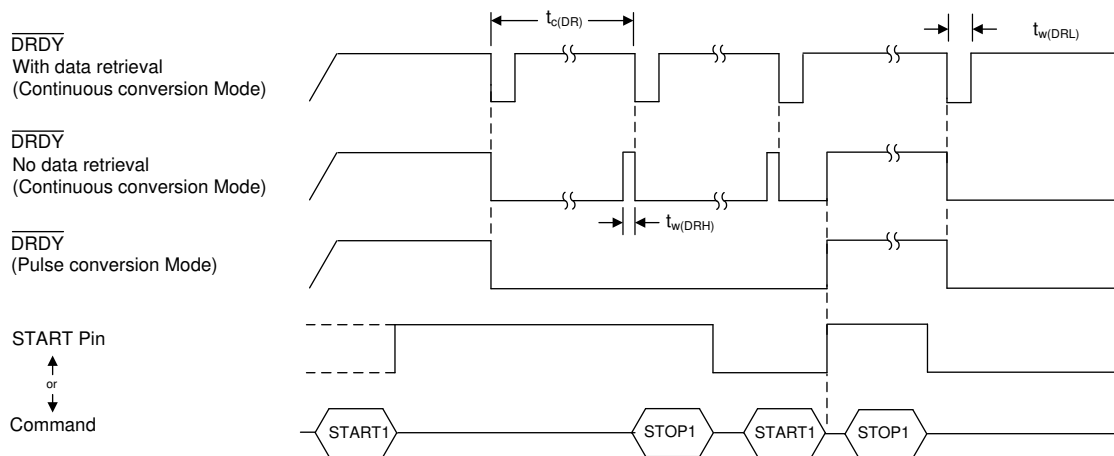


Figure 9-42. $\overline{\text{DRDY}}$ Operation

Table 9-17. $\overline{\text{DRDY}}$ Timing Characteristics

PARAMETER	TEST CONDITIONS	TYP	UNIT
$t_{c(\text{DR})}$	$\overline{\text{DRDY}} \downarrow$ to $\overline{\text{DRDY}} \downarrow$ conversion time: $\overline{\text{DRDY}}$ period	After first conversion	1 1/data rate
$t_{w(\text{DRL})}$	$\overline{\text{DRDY}} \downarrow$ to $\overline{\text{DRDY}} \uparrow$: delay time	With data retrieval, Continuous conversion mode	$\overline{\text{DRDY}}$ drives high on first falling SCLK edge
$t_{w(\text{DRH})}$	$\overline{\text{DRDY}}$ pulse high: pulse duration	No data retrieval, Continuous conversion mode,	16 t_{CLK} ⁽¹⁾

(1) $t_{\text{CLK}} = 1 / f_{\text{CLK}}$.

9.4.6 Conversion Data Software Polling

In addition to hardware polling using $\overline{\text{DRDY}}$, new conversion data are also detected by software polling. In software polling, read either ADC1 or ADC2 conversion data and poll the STATUS byte ADC1 or ADC2 data-ready bits. The data ready bits are set if the corresponding ADC1 or ADC2 conversion data are new since the last ADC1 or ADC2 data read; otherwise, the bits are cleared. If the bits are cleared, the corresponding conversion data are the previous data. To avoid missing data when using software polling, poll the status bits at a rate faster than the corresponding ADC1 or ADC2 conversion rate. The ADC2 status bit is valid only when the data are read by command (RDATA1 or RDATA2).

9.4.7 Read Conversion Data

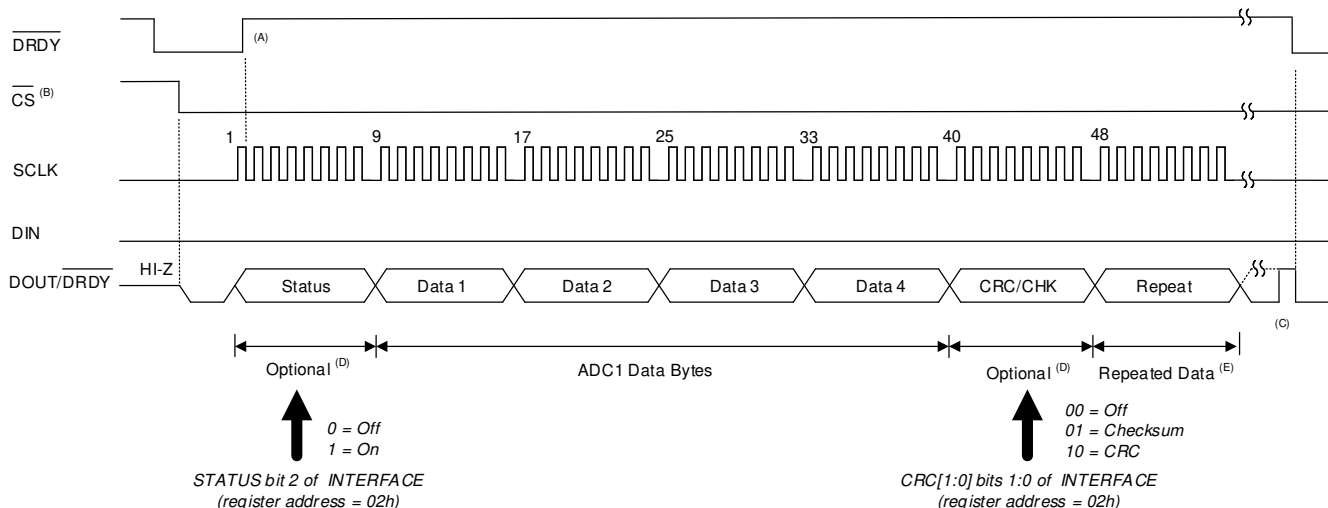
ADC1 data are read by two methods: read data direct or read data by command. ADC1 writes new conversion data to the output shift register and the internal data-holding register. Two registers hold the conversion data; therefore, data are read either from the output shift register (in direct mode) or read from the data-holding register (in command mode). Reading data from the data-holding register (command mode) does not require synchronizing the start of data readback to $\overline{\text{DRDY}}$. ADC2 data are read only from the ADC2 data-holding register (command mode).

9.4.7.1 Read Data Direct (ADC1 Only)

In this method of data retrieval, ADC1 conversion data are shifted out directly from the output shift register. No opcode is necessary. Read data direct requires that no serial activity occur from the time of $\overline{\text{DRDY}}$ low to the readback, or the data are invalid. The serial interface is full duplex; therefore, commands are decoded during the data readback. If no command is intended, keep the DIN pin low during readback. If an input command is sent during readback, the ADC executes the command, and data interruption may result. The data readback operation must be completed 16 f_{CLK} cycles before the next $\overline{\text{DRDY}}$, or the old data are overwritten with new data. Synchronize the data readback to $\overline{\text{DRDY}}$ or to DOUT/ $\overline{\text{DRDY}}$ to make sure the data are read before the next $\overline{\text{DRDY}}$ falling edge.

If new ADC1 conversion data are ready during an ongoing data or register read or write operation, data are not loaded to the output register but are written only to the data holding register. Retrieve the conversion data later from the holding register by sending a read command. However, writing new data to certain registers results in a conversion-cycle restart. Conversion restart clears the contents of the conversion data-holding register; therefore, the previous conversion data are not available. Read the conversion data before the register write operation.

As shown in Figure 9-43, the ADC1 data field is 4, 5, or 6 bytes long, depending on programming. The data field consists of an optional status byte, four bytes of conversion data, and an optional checksum byte. After all the bytes are read, the data-byte sequence is repeated by continuing SCLK. The byte sequence repeats starting with the first byte. In order to help verify error-free communication, read the same data multiple times in each conversion interval and compare.



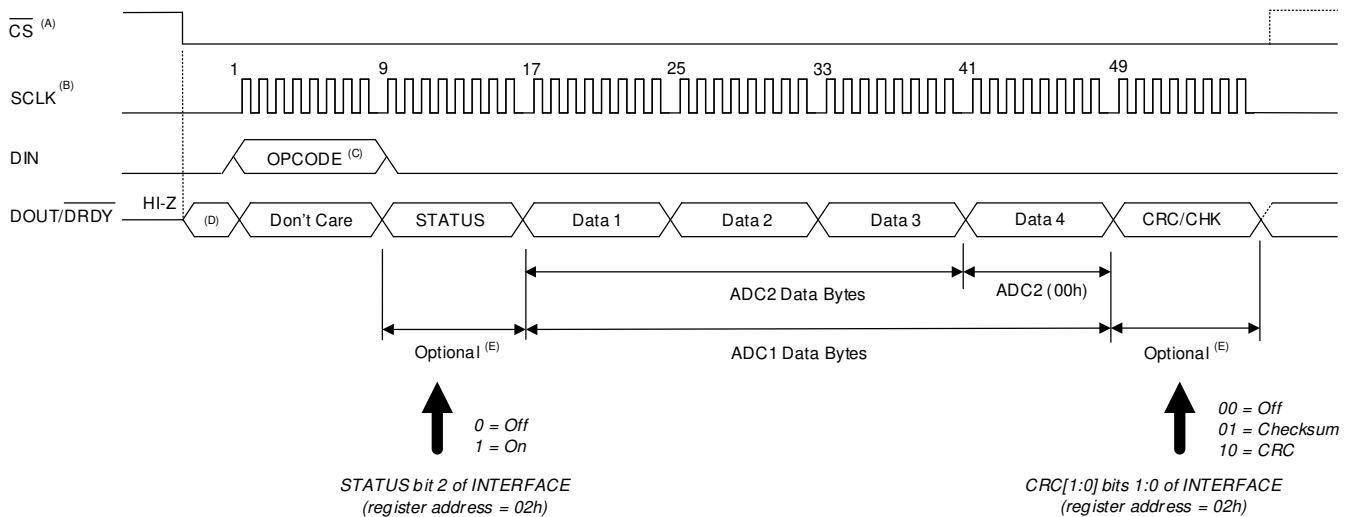
- In Continuous conversion mode, $\overline{\text{DRDY}}$ returns high on the first SCLK falling edge. In Pulse Conversion mode, $\overline{\text{DRDY}}$ stays low until the next conversion is started.
- $\overline{\text{CS}}$ can be tied low. If $\overline{\text{CS}}$ is low, DOUT/ $\overline{\text{DRDY}}$ asserts low at the same time as $\overline{\text{DRDY}}$.
- Data read must be completed before DOUT/ $\overline{\text{DRDY}}$ and $\overline{\text{DRDY}}$ goes high (16 t_{CLK} before new data ready).
- The STATUS and CRC/CHK bytes are optional.
- The byte sequence repeats by continuing SCLK.

Figure 9-43. Data Read Direct

9.4.7.2 Read Data by Command

In this method of data retrieval, a command is used to read ADC1 or ADC2 data. When the command is sent, the data are retrieved from the respective ADC data-holding register. Read the data at any time because the command method does not require synchronizing to $\overline{\text{DRDY}}$. In addition to hardware polling of $\overline{\text{DRDY}}$ or $\text{DOUT}/\overline{\text{DRDY}}$ to determine when ADC1 data are ready, the ADC can be software polled by reading bits ADC1 and ADC2 of the status byte. If the ADC1 or ADC2 status bit is 1, the ADC1 or ADC2 data are new since the last ADC1 or ADC2 read operation. If data are read again before the new data are ready, the status bit is 0 and the previous data are returned. ADC2 data can only be read by the command method.

Figure 9-44 shows the read data by command sequence. The output data MSB begins on the first rising edge of SCLK after the command. The output data field can be 4, 5, or 6 bytes long, depending on programming. The data field consists of an optional status byte, four bytes of conversion data, and an optional checksum byte. The ADC2 data block consists of the optional status byte, 3 bytes of data, a fixed-value byte equal to 00h (zero pad byte), and the optional checksum byte. A read data command must be sent for each read operation. The ADC does not respond to commands until the read operation is complete, or terminated by taking $\overline{\text{CS}}$ high.



- A. $\overline{\text{CS}}$ can be tied low. If $\overline{\text{CS}}$ is low, $\text{DOUT}/\overline{\text{DRDY}}$ asserts low with $\overline{\text{DRDY}}$.
- B. In continuous conversion mode, $\overline{\text{DRDY}}$ returns high on the first SCLK falling edge of sending the opcode. For pulse conversion mode, $\overline{\text{DRDY}}$ stays low until the next conversion is started.
- C. Read ADC1 command byte = 12h or 13h, Read ADC2 command byte = 14h or 15h
- D. $\text{DOUT}/\overline{\text{DRDY}}$ is driven low with $\overline{\text{DRDY}}$. If a read operation occurs after $\overline{\text{DRDY}}$ falling edge, then DOUT can be high or low.
- E. The STATUS and CRC/CHK bytes are optional.

Figure 9-44. Read Data by Command

9.4.7.3 Data-Byte Sequence

The ADC1 data sequence can be 4, 5, or 6 bytes long, depending on whether the optional status and checksum bytes are enabled. The entire data sequence consists of the status byte, four bytes of the 32-bit conversion word, and the checksum byte. The ADC2 data sequence is the same, except the conversion data are three bytes long (24-bit word), followed by a zero-value pad byte. If the status byte is not enabled, the remaining bytes are left shifted.

9.4.7.3.1 Status Byte

The status byte is the first byte in the sequence. The status byte indicates new ADC1 and ADC2 data, the state of the ADC1 PGA alarms, the low-reference alarm state, the clock mode, and the reset state. The status byte is enabled by the STATUS bit of the INTERFACE register (bit 2 of register 02h). Figure 9-45 and Table 9-18 shows the status-byte field description.

Figure 9-45. Status Byte

7	6	5	4	3	2	1	0
ADC2	ADC1	EXTCLK	REF_ALM	PGAL_ALM	PGAH_ALM	PGAD_ALM	RESET

Table 9-18. Status Byte Field Descriptions

Bit	Field	Type	Description
7	ADC2	Read Only	ADC2 Data⁽¹⁾ This bit indicates the status of ADC2 conversion data 0: ADC2 data not new since the last ADC2 read operation 1: ADC2 data new since the last ADC2 read operation
6	ADC1	Read Only	ADC1 Data This bit indicates the status of ADC1 conversion data 0: ADC1 data not new since the last ADC1 read operation 1: ADC1 data new since the last ADC1 read option
5	EXTCLK	Read Only	ADC Clock This bit indicates the ADC clock source 0: ADC clock is internal 1: ADC clock is external
4	REF_ALM	Read Only	ADC1 Low Reference Alarm⁽²⁾ This bit is the low reference voltage alarm of ADC1. The alarm bit is set if $V_{REF} \leq 0.4$ V, typical. 0: No alarm 1: Low reference alarm
3	PGAL_ALM	Read Only	ADC1 PGA Output Low Alarm⁽²⁾ This bit is the ADC1 PGA absolute low voltage alarm. The bit is set if the absolute voltage of either PGA output is less than $V_{AVSS} + 0.2$ V. See the PGA Absolute Output-Voltage Monitor section. 0: No alarm 1: PGA low voltage alarm
2	PGAH_ALM	Read Only	ADC1 PGA Output High Alarm⁽²⁾ This bit is the ADC1 PGA absolute high voltage alarm. The bit is set if the absolute voltage of either PGA output is greater than $V_{AVDD} - 0.2$ V. See the PGA Absolute Output-Voltage Monitor section. 0: No alarm 1: PGA high voltage alarm
1	PGAD_ALM	Read Only	ADC1 PGA Differential Output Alarm⁽²⁾ This bit is the ADC1 PGA differential output range alarm. The bit is set if the PGA differential output voltage exceeds +105% FS or -105% FS. See the PGA Differential Output Monitor section. 0: No alarm 1: PGA differential range alarm
0	RESET	Read Only	RESET Indicates device reset. Device reset occurs at power-on, by the RESET / PWDN pin or by the reset command. This bit is the same as the RESET bit of the POWER register (see Table 9-36). 0: No reset occurred since the RESET bit in power register last cleared by the user 1: Device reset occurred

(1) The ADC2 status bit is valid only with use of RDATA1 or RDATA2 read data commands.

(2) These bits are valid during the readback of ADC1 data only. All other bits are valid during the readback of either ADC1 or ADC2.

9.4.7.3.2 Data Byte Format

ADC1 data are 32 bits in a two's complement format that represents positive and negative values, and are output starting with the most significant bit first (ADC1[31]). The data are scaled so that $V_{IN} = 0\text{ V}$ results in ideal code value of 00000000h; see Table 9-19 for other ideal code values. Some applications require reduction of the 32-bit data to 24-bit data in order to provide compatibility to 24-bit systems. This reduction is done by simple truncation (or rounding) of the 32-bit data to 24 bits. See Figure 9-46 for the ADC1 data byte field.

Table 9-19. ADC1 and ADC2 Output Codes

INPUT SIGNAL (V) ⁽¹⁾	ADC1 OUTPUT CODE (32 Bits) ⁽²⁾	ADC2 OUTPUT CODE (24 Bits) ⁽²⁾
$\geq V_{REF} / \text{Gain} \cdot (2^{N-1} - 1) / 2^{N-1}$	7FFFFFFFh	7FFFFFFh
$V_{REF} / (\text{Gain} \cdot 2^{N-1})$	00000001h	000001h
0	00000000h	000000h
$-V_{REF} / (\text{Gain} \cdot 2^{N-1})$	FFFFFFFh	FFFFFFh
$\leq -V_{REF} / \text{Gain}$	80000000h	800000h

(1) N = 32 (ADC1), N = 24 (ADC2)

(2) Ideal output code, excluding effects of ADC noise, offset, gain and linearity errors.

Figure 9-46. ADC1 Data Field, Four Bytes - 32 Bits

31	30	29	28	27	26	25	24
ADC1[31:24]							
R-0							
23	22	21	20	19	18	17	16
ADC1[23:16]							
R-0							
15	14	13	12	11	10	9	8
ADC1[15:8]							
R-0							
7	6	5	4	3	2	1	0
ADC1[7:0]							
R-0							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

ADC2 data are 24 bits in a two's complement data format that represents positive and negative code values, and are output starting with the most significant bit first (ADC2[23]). The data are scaled so that a zero-voltage input results in an ideal code value of 000000h; see Table 9-19 for other ideal code values. See Figure 9-47 for the ADC2 data-byte field.

Figure 9-47. ADC2 Data Field, Three Bytes - 24 Bits

23	22	21	20	19	18	17	16
ADC2[23:16]							
R-0							
15	14	13	12	11	10	9	8
ADC2[15:8]							
R-0							
7	6	5	4	3	2	1	0
ADC2[7:0]							
R-0							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

9.4.7.3.3 Checksum Byte (CRC/CHK)

The checksum byte is the last byte in the data-byte sequence. The checksum byte can be programmed to checksum mode or to cyclic redundancy check (CRC) mode. The checksum byte is optional and is enabled by the CRC[1:0] bits of the INTERFACE register. Use the checksum byte to detect transmission errors during data read-back. Figure 9-48 and Table 9-20 shows the checksum byte description. The checksum byte is not provided when register data are read.

Figure 9-48. Checksum Byte (CRC/CHK)

7	6	5	4	3	2	1	0
SUM[7:0]							

Table 9-20. Checksum Byte (CRC/CHK) Field Descriptions

Bit	Field	Type	Description
7:0	SUM[7:0]	R	CRC or Checksum value This byte is the CRC or checksum of four ADC1 data bytes, or three ADC2 data bytes.

9.4.7.3.3.1 Checksum Mode (CRC[1:0] = 01h)

In checksum mode, the checksum byte is the lower 8-bit sum of the data conversion bytes plus an offset value 9Bh. The offset value is added to help detect whether the DOUT/DRDY has failed and is in a permanent low state. ADC1 sums four data bytes, and ADC2 sums three data bytes. To verify the correct checksum, sum the data bytes plus 9Bh and compare the value read from the ADC. If the checksum values do not match, a data transmission error occurred. In the event of a data transmission error, read the data again for verification. The checksum provides basic levels of error detection caused by single-bit errors, and limited combinations of multiple-bit errors.

Example computation of ADC1 four-data-byte checksum calculation:

- Data byte 1: 12h
- Data byte 2: 34h
- Data byte 3: 56h
- Data byte 4: 78h
- Constant: 9Bh
- Checksum value = AFh

9.4.7.3.4 CRC Mode (CRC[1:0] = 10h)

In CRC mode, the checksum byte is the 8-bit remainder of the bitwise exclusive-OR (XOR) of the data bytes by a CRC polynomial. For ADC1, use four conversion data bytes in the calculation; for ADC2, use three conversion data bytes. The CRC is based on the CRC-8-ATM (HEC) polynomial: $X^8 + X^2 + X + 1$.

The nine binary coefficients of the polynomial are: 100000111. Calculate the CRC by dividing (XOR operation) the data bytes (excluding the CRC) with the polynomial and compare the calculated CRC values to the ADC CRC value. If the values do not match, a data transmission error has occurred. In the event of a data transmission error, read the data again. The CRC provides a higher level of detection of multiple-bit errors.

The following list shows a general procedure to compute the CRC value:

1. Left shift the initial ADC1 32-bit data value by 8 bits, with zeros padded to the right, creating a new 40-bit data value (the starting data value). For ADC2, left shift the 24-bit value to create a new 32-bit starting data value.
2. Align the MSB of the CRC polynomial (100000111) to the left-most, logic-one value of the data.
3. Perform an XOR operation on the data value with the aligned CRC polynomial. The XOR operation creates a new, shorter-length value. The bits of the data values that are not in alignment with the CRC polynomial drop down and append to the right of the new XOR result.
4. When the XOR result is less than 100h, the procedure ends, yielding the 8-bit CRC value. Otherwise, continue with the XOR operation shown in step 2, using the current data value. The number of loop iterations depends on the value of the initial data.

9.4.8 ADC Clock Modes

The ADC conversion process requires a clock for operation. These devices have three clock operating modes:

1. Internal oscillator
2. External clock
3. External crystal

The nominal clock frequency is 7.3728 MHz. The output data rate and the corresponding 50-Hz and 60-Hz filter response nulls scale with clock frequency. Good line-cycle rejection requires an accurate clock frequency that is best provided by a crystal oscillator.

As depicted in Figure 9-49, the ADC contains an integrated clock generator and automatic detection circuit. If no external clock is detected, the ADC automatically selects the internal oscillator. If an external clock is detected, the ADC automatically selects the external clock. The clock mode can be verified by reading the EXTCLK bit, bit 5 of the status byte (0 = internal clock).

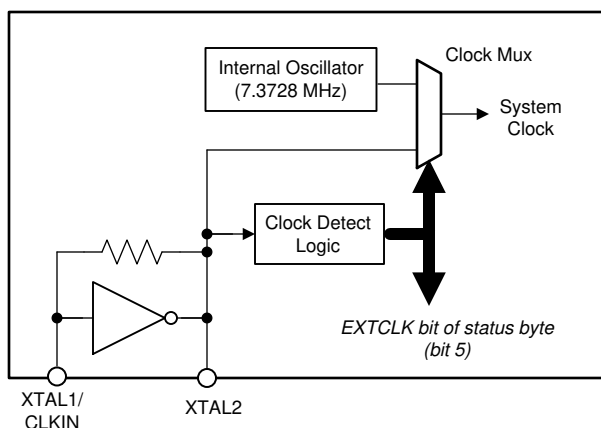


Figure 9-49. ADC Clock Block Diagram

Figure 9-50 illustrates the configuration for the three clock modes.

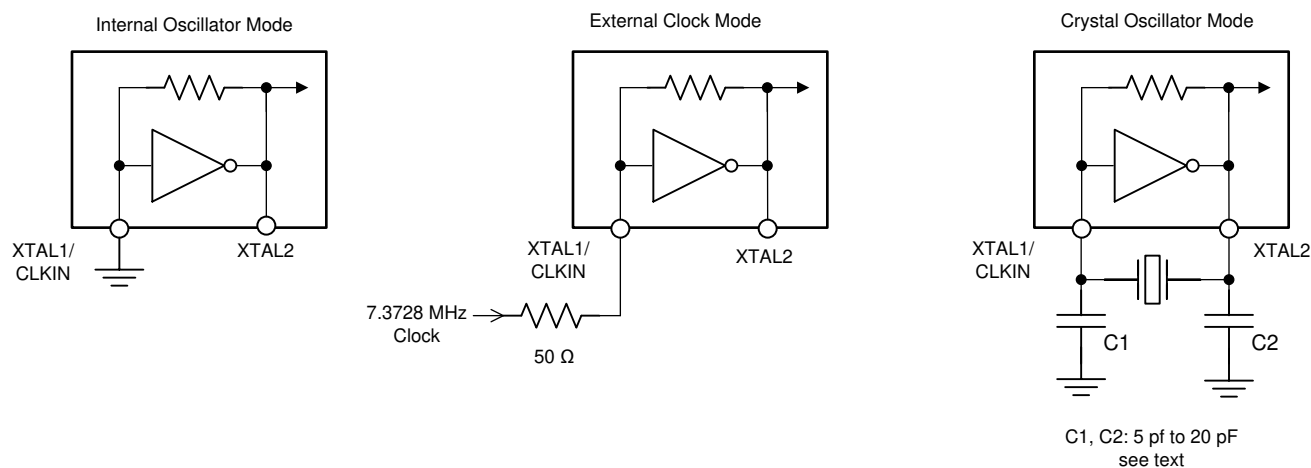


Figure 9-50. Clock Mode Configurations

9.4.8.1 Internal Oscillator

The ADC contains an integrated 7.3728-MHz internal oscillator. After ADC power-on, the internal oscillator immediately starts. To select the internal oscillator, ground the XTAL1/CLKIN pin and float the XTAL2 pin; see [Figure 9-50](#) (internal oscillator mode).

9.4.8.2 External Clock

Drive the ADC with an external clock by applying the clock input to the XTAL1/CLKIN pin and floating the XTAL2 pin; see [Figure 9-50](#) (external clock mode). The ADC automatically detects the external clock. Be sure the external clock is free of overshoot and glitches. A source-terminating resistor placed at the external clock buffer often helps to reduce overshoot.

9.4.8.3 Crystal Oscillator

The ADC contains an integrated oscillator circuit for use with an external crystal. Connect the crystal and load capacitors to the XTAL1/CLKIN and XTAL2 pins; see [Figure 9-50](#) (crystal oscillator mode). Place the crystal and crystal load capacitors close to the ADC pins using short direct traces. Connect the load capacitors to digital ground. Do not connect any other external circuit to the crystal oscillator. [Table 9-21](#) shows approved crystals for use with the ADS1262 and ADS1263. The crystal oscillator start-up time is characterized at 10 ms (typical), and can be longer depending on the crystal characteristics.

Table 9-21. Recommended Crystals

MANUFACTURER	FREQUENCY	LOAD CAPACITORS	OPERATING TEMPERATURE RANGE	PART NUMBER
Citizen	7.3728 MHz	18 pF	-40°C to +85°C	HCM497372800ABJT
CTS	7.3728 MHz	18 pF	-40°C to +85°C	ATS073BSM-1E
Abracon	7.3728 MHz	18 pF	-40°C to +125°C	ABLS-7.3728MHZ-K4T

9.4.9 Calibration

The ADC incorporates offset and gain calibration commands, as well as user-offset and full-scale calibration registers to calibrate the ADC. The ADC1 calibration registers are 24-bits wide and the ADC2 calibration registers are 16-bits wide. Use calibration to correct internal ADC errors or overall system errors. Calibrate by sending calibration commands to the ADC, or by direct user calibration. In user calibration, the user calculates and writes the correction values to the calibration registers. The ADC performs self or system-offset calibration, or for full-scale calibration, system calibration. Perform offset calibration before full-scale calibration. After power-on, but before calibrating, wait for the power supplies and reference voltage to fully settle.

9.4.9.1 Offset and Full-Scale Calibration

Use the offset and full-scale (gain) registers to correct ADC error. As shown in Figure 9-51, the value of the offset calibration register is subtracted from the filter output and then multiplied by the full-scale register value divided by 400000h. The data are then clipped to a 32-bit value to provide the final output.

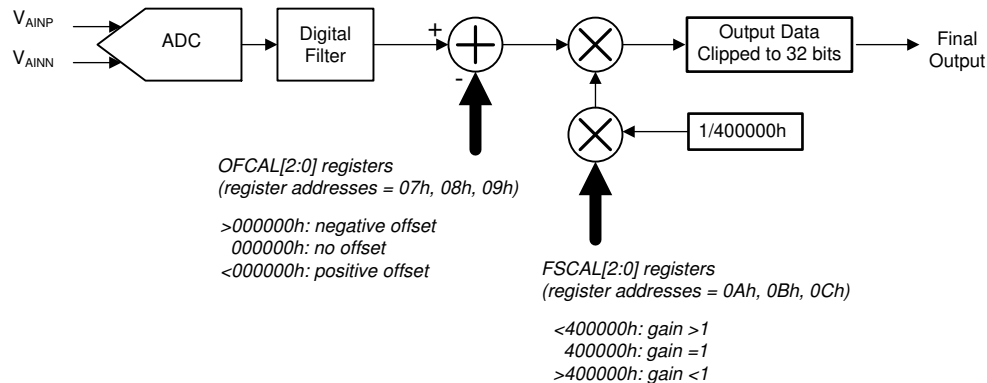


Figure 9-51. ADC1 Calibration Block Diagram

Equation 22 shows the internal calibration. ADC2 calibration registers are 16 bit. For ADC2, the ADC2FSC[1:0] registers are for full-scale calibration and the ADC2OFC[1:0] registers are for offset calibration.

$$\text{ADC1 Final Output Data} = (\text{Filter Output} - \text{OFCAL}[2:0]) \cdot \text{FSCAL}[2:0] / 400000h \quad (22)$$

9.4.9.1.1 Offset Calibration Registers

The ADC1 offset calibration word is 24 bits, consisting of three 8-bit registers, as shown in [Table 9-22](#). The offset value is two's complement format with a maximum positive value equal to 7FFFFFFh (for negative offset), and a maximum negative value equal to 800000h (for positive offset). The 24-bit register is internally left-shifted to align with the 32-bit data before subtraction occurs. A register value equal to 000000h has no offset correction. Although the offset calibration register allows a wide range of offset values, the input signal cannot exceed $\pm 106\%$ of the precalibrated range in order to prevent ADC overrange. If chop mode is enabled, the offset calibration register is disabled. [Table 9-23](#) shows example settings of the offset register.

Table 9-22. ADC1 Offset Calibration Registers

REGISTER	BYTE ORDER	ADDRESS	BIT ORDER							
OFCAL0	LSB	07h	B7	B6	B5	B4	B3	B2	B1	B0 (LSB)
OFCAL1	MID	08h	B15	B14	B13	B12	B11	B10	B9	B8
OFCAL2	MSB	09h	B23 (MSB)	B22	B21	B20	B19	B18	B17	B16

Table 9-23. ADC1 Offset Calibration Register Values

OFCAL[2:0] REGISTER VALUE	OFFSET CALIBRATED 32-BIT OUTPUT CODE ⁽¹⁾
000001h	FFFFFF00h
000000h	00000000h
FFFFFFh	00000100h

(1) Ideal output code with shorted input, excluding ADC noise and offset voltage error.

The ADC2 offset calibration word is 16 bits, consisting of two 8-bit registers, as shown in [Table 9-24](#). The 16-bit calibration value is internally aligned with the 24-bit ADC2 conversion result. The offset calibration value is subtracted from the conversion data.

Table 9-24. ADC2 Offset Calibration Registers

REGISTER	BYTE ORDER	ADDRESS	BIT ORDER							
ADC2OFC0	LSB	17h	B7	B6	B5	B4	B3	B2	B1	B0 (LSB)
ADC2OFC1	MSB	18h	B15	B14	B13	B12	B11	B10	B9	B8

9.4.9.1.2 Full-Scale Calibration Registers

The ADC1 full-scale calibration word is 24 bits consisting of three 8-bit registers, as shown in [Table 9-25](#). The full-scale calibration value is straight binary, normalized to a unity-gain correction factor at a register value equal to 400000h. [Table 9-26](#) shows register values for selected gain factors. Correct ADC gain for gain errors greater than one (resulting in full-scale register values less than 400000h, or less than 4000h for ADC2). However, to prevent ADC overrange, the input signal must not exceed $\pm 106\%$ of the precalibrated input range. Do not exceed the PGA input range limits during full-scale calibration.

Table 9-25. ADC1 Full-Scale Calibration Registers

REGISTER	BYTE ORDER	ADDRESS	BIT ORDER							
FSCAL0	LSB	0Ah	B7	B6	B5	B4	B3	B2	B1	B0 (LSB)
FSCAL1	MID	0Bh	B15	B14	B13	B12	B11	B10	B9	B8
FSCAL2	MSB	0Ch	B23 (MSB)	B22	B21	B20	B19	B18	B17	B16

Table 9-26. ADC1 Full-Scale Calibration Register Values

FSCAL[2:0] REGISTER VALUE	GAIN FACTOR
433333h	1.05
400000h	1.00
3CCCCCh	0.95

The ADC2 full-scale calibration word is 16 bits consisting of two 8-bit registers, as shown in [Table 9-27](#). The full-scale calibration value is straight binary, normalized to a unity correction factor at a register value equal to 4000h. A full-scale register value greater than 4000h increases the ADC2 gain factor.

Table 9-27. ADC2 Full-Scale Calibration Registers

REGISTER	BYTE ORDER	ADDRESS	BIT ORDER							
ADC2FSC0	LSB	19h	B7	B6	B5	B4	B3	B2	B1	B0 (LSB)
ADC2FSC1	MSB	1Ah	B15	B14	B13	B12	B11	B10	B9	B8

9.4.9.2 ADC1 Offset Self-Calibration (SFOCAL1)

The offset self-calibration command corrects internal ADC1 offset error. Program the ADC1 input multiplexer register (INPMUX) to FFh to force open all input connections before sending the command. When the SFOCAL1 self-calibration command is sent, the ADC shorts together the internal PGA inputs, then averages 16 readings to reduce the conversion noise for an accurate calibration. When calibration is complete, the calibration result is written to the 24-bit offset calibration register (OFCAL[2:0]). After calibration, set the input multiplexer to the desired measurement channel. The offset calibration register is disabled in chop mode.

9.4.9.3 ADC1 Offset System Calibration (SYOCAL1)

The offset system-calibration command corrects ADC1 system offset error. For this type of calibration, externally short the system inputs before the command. When the SYOCAL1 command is sent, the ADC averages 16 readings to reduce conversion noise for an accurate calibration. When calibration is complete, the offset calibration result is written to the 24-bit offset calibration register (OFCAL[2:0]). The offset calibration register is disabled in chop mode.

9.4.9.4 ADC2 Offset Self-Calibration ADC2 (SFOCAL2)

The offset self-calibration command corrects internal ADC2 offset error. Program the ADC2 input multiplexer register (ADC2MUX) to FFh to force open all input connections before sending the command. When the SFOCAL2 self-calibration command is sent, the ADC shorts together the internal PGA inputs of ADC2, then averages 16 readings to reduce conversion noise for an accurate calibration. When calibration is complete, the offset calibration result is written to the 16-bit ADC2 offset calibration register ADC2OFC[1:0]. After calibration, set the input multiplexer to the desired channel.

9.4.9.5 ADC2 Offset System Calibration ADC2 (SYOCAL2)

The offset system-calibration command corrects ADC2 system offset error. For this offset calibration, externally short the ADC2 inputs before sending the command. When the SYOCAL2 command is sent, the ADC averages 16 readings to reduce conversion noise for an accurate calibration. When calibration is complete, the calibration result is written to the 16-bit ADC2 offset calibration register (ADC2OFC[1:0]).

9.4.9.6 ADC1 Full-Scale System Calibration (SYGCAL1)

The full-scale calibration command corrects ADC1 system gain error. To calibrate, apply a positive full-scale dc signal to the ADC, wait until the signal is fully settled, and then send the command. When the SYGCAL1 command is sent, the ADC averages 16 readings to reduce conversion noise for an accurate calibration. When calibration completes, the ADC computes a full-scale calibration where the applied voltage calibrates to a positive full-scale code value. The computed result is written to the 24-bit offset calibration register (FSCAL[2:0]). The precalibrated ADC overrange limitation is 106% FSR.

9.4.9.7 ADC2 Full-Scale System Calibration ADC2 (SYGCAL2)

The full-scale system calibration command corrects ADC2 system gain error. To calibrate, apply a positive full-scale dc signal to ADC2, wait until the signal is fully settled, and then send the command. When the SYGCAL2 command is sent, the ADC averages 16 readings to reduce conversion noise for an accurate calibration. When calibration is complete, the full-scale calibration is computed so that applied voltage calibrates to positive full scale. The computed result is written to the ADC2 16-bit, full-scale calibration register (ADC2FSC[1:0]). The precalibrated ADC overrange limitation is 106% FSR.

9.4.9.8 Calibration Command Procedure

The following steps show the ADC calibration procedure using commands. Make sure that the reference voltage is stable before calibrating the ADC. Perform offset calibration before full-scale calibration.

1. Enable continuous-conversion mode (ADC1 only).
2. Select the desired gain and reference voltage of the ADC.
3. Choose calibration type:
 - a. For offset self-calibration, program the ADC1 or ADC2 input multiplexer register to FFh to open all inputs before sending the calibration command.
 - b. For *system* calibration, select the input channel and short the external inputs (offset calibration); or apply positive full-scale input (full-scale calibration). If performing full-scale calibration, the analog inputs cannot exceed 106% FSR. Do not exceed the specified absolute or differential PGA input range when calibrating.
4. Start conversions:
 - a. If calibrating ADC1, take the START pin high (or send the ADC1 start command).
 - b. If calibrating ADC2, send the ADC2 start command.
5. Send the desired calibration command. When the calibration command is received, calibration is started, and for ADC1, $\overline{\text{DRDY}}$ is driven high. Keep CS and SCLK low during the calibration time. The calibration time depends on the data rate and digital filter mode, as shown in [Table 9-28](#) for ADC1 and [Table 9-29](#) for ADC2.

For ADC1, $\overline{\text{DRDY}}$ is driven low when calibration is complete. The new offset or full-scale calibration values are written to the calibration registers. New conversion data are now ready using the new calibration coefficients. For ADC2, the end of calibration is not indicated by $\overline{\text{DRDY}}$. Instead, wait for the time shown in [Table 9-29](#) before reading ADC2 data.

Table 9-28. ADC1 Calibration Time (ms)

DATA RATE (SPS)	FILTER MODE ⁽¹⁾					
	SINC1	SINC2	SINC3	SINC4	SINC5	FIR
2.5	6801	7601	8401	9201	—	6805
5	3401	3801	4201	4601	—	3405
10	1701	1901	2101	2300	—	1705
16.6	1021	1141	1261	1381	—	—
20	850.7	951.0	1051	1151	—	854.5
50	340.9	380.9	421.0	460.9	—	—
60	284.1	317.7	350.9	384.4	—	—
100	170.8	190.9	210.9	230.8	—	—
400	43.27	48.43	53.42	58.41	—	—
1200	14.93	16.72	18.40	20.07	—	—
2400	7.845	8.816	9.643	10.48	—	—
4800	4.302	4.858	5.276	5.692	—	—
7200	3.123	3.534	3.815	4.095	—	—
14400	—	—	—	—	1.941	—
19200	—	—	—	—	1.490	—
38400	—	—	—	—	0.812	—

(1) $f_{\text{CLK}} = 7.3728 \text{ MHz}$. CHOP disabled.

Table 9-29. ADC2 Calibration Time

DATA RATE (SPS)	ADC2 CALIBRATION TIME (ms) ⁽¹⁾
10	1742
100	212
400	54.6
800	28.3

(1) $f_{\text{CLK}} = 7.3728 \text{ MHz}$.

9.4.9.9 User Calibration Procedure

The user calibration procedure is similar to the calibration command procedure, except the user computes the calibration coefficients and writes the corresponding values to the calibration registers. Before starting user calibration, preset the ADC1 offset and full-scale registers to 000000h and 400000h, respectively.

For ADC2, preset the offset and full-scale registers to 0000h and 4000h, respectively.

For offset calibration, short the ADC inputs or system inputs and average the conversions (averaging reduces noise for a more accurate calibration). Write the average value to the offset calibration registers. The ADC subtracts the value from the conversion result.

For full-scale calibration, apply a dc calibration voltage that is less than positive full scale to avoid clipped codes ($V_{IN} < +FSR$), and average the conversions to reduce noise for a more accurate calibration. Full-scale calibration is computed as shown in [Equation 23](#):

$$\text{Full-Scale Calibration} = (\text{Expected Code Value} / \text{Actual Code Value}) \times \text{NF} \quad (23)$$

where

- NF = Normalization Factor = 400000h for ADC1, or 4000h for ADC2

If the actual code is higher than the expected value, the calculated calibration value is less than 400000h (4000h) and the ADC gain is subsequently reduced. Write the calibration value to the full-scale register.

9.4.10 Reset

The ADC is reset in one of three ways:

- 1) Power-on reset
- 2) $\overline{\text{RESET}}/\text{PWDN}$ pin
- 3) RESET command

When the ADC is reset, the device registers reset to default values and the analog-to-digital conversion cycles restart. After reset, the RESET bit of the status byte (bit 0) and of the power register (bit 4) are set to 1 to indicate reset has occurred. Set the RESET register bit to 0 to clear the reset flag. If the RESET bit is then set after clearing the bit, a new reset has occurred.

9.4.10.1 Power-On Reset (POR)

After the power supplies are turned on, the ADC remains in reset until V_{DVDD} , the internal LDO output (BYPASS pin voltage), and the combined $(V_{\text{AVDD}} - V_{\text{AVSS}})$ power supply voltage have exceeded their respective POR voltage thresholds. Figure 9-52 shows the POR sequence. When the power supplies have crossed the voltage thresholds, the ADC is operational 65536 f_{CLK} cycles later (9 ms, typical). Note the 1- μF capacitor connected to the BYPASS pin requires charging at power-on, and as a result, can delay when the ADC is operational. Wait at least 9 ms after the power supplies have fully stabilized before beginning ADC communication.

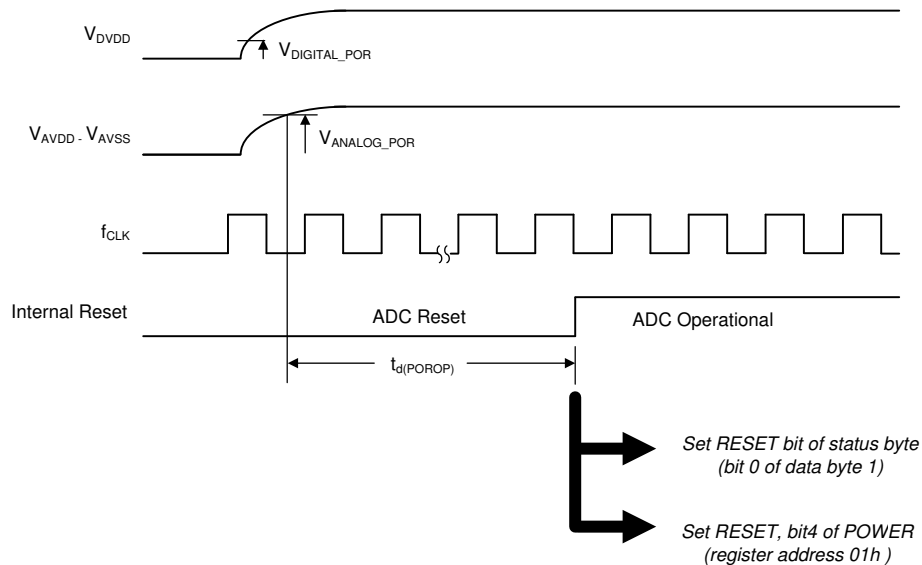


Figure 9-52. Power-On Reset

Table 9-30. POR Characteristics

PARAMETER	TEST CONDITIONS	TYP	UNIT
$V_{\text{DIGITAL_POR}}$	Digital power supply POR threshold	1	V
$V_{\text{ANALOG_POR}}$	Analog power supply POR threshold	3.5	V
$t_{\text{d(POROP)}}$	Propagation delay from last POR supply threshold to ADC operational	65536	t_{CLK} ⁽¹⁾

(1) $t_{\text{CLK}} = 1 / f_{\text{CLK}}$.

9.4.10.2 RESET/PWDN Pin

Reset the ADC by taking the $\overline{\text{RESET/PWDN}}$ pin low for a minimum four f_{CLK} cycles, and then returning the pin high, as shown in Figure 9-53. Holding the $\overline{\text{RESET/PWDN}}$ pin low for longer than 65536 f_{CLK} cycles (9 ms) engages power-down mode. As depicted in the diagram, after the $\overline{\text{RESET/PWDN}}$ pin is taken high, the delay time shown in Table 9-31 is required before sending the first serial interface command.

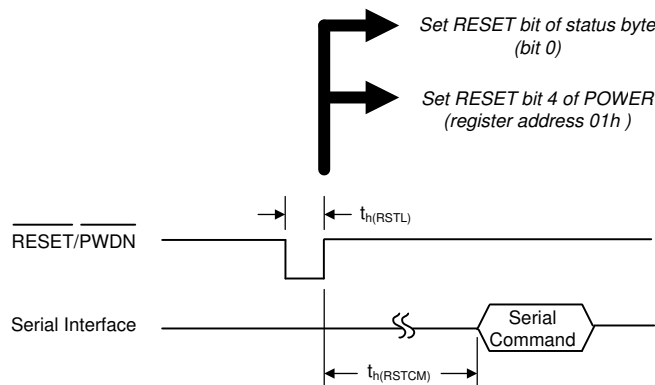


Figure 9-53. RESET/PWDN Pin Timing

Table 9-31. RESET/PWDN Pin Timing Requirements

PARAMETER		TEST CONDITIONS	MIN	UNIT
$t_{h(RSTL)}$	RESET/PWDN low for <i>reset</i> : hold time		4	t_{CLK} ⁽¹⁾
	RESET/PWDN low for <i>power down</i> : hold time		65536	
$t_{h(RSTCM)}$	RESET/PWDN high to serial command: hold time	After reset	8	t_{CLK}
	RESET/PWDN high to serial command: hold time	After exiting power down	65536	

(1) $t_{\text{CLK}} = 1 / f_{\text{CLK}}$.

9.4.10.3 Reset by Command

Reset the ADC by using the reset command (opcode = 06h or 07h). Toggle $\overline{CS}$ high first to make sure the serial interface resets before sending the command. For applications that tie $\overline{CS}$ low, see the [Serial Interface Autoreset](#) section for information on how to reset the serial interface. After sending the reset command, provide an $8 \cdot f_{CLK}$ -cycle delay before sending the next command, as shown in [Figure 9-54](#) and [Table 9-32](#).

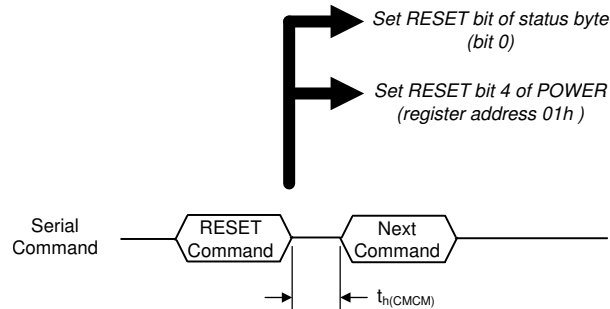


Figure 9-54. RESET Command Timing

Table 9-32. RESET Command Timing Requirements

PARAMETER		MIN	UNIT ⁽¹⁾
$t_{h(MCM)}$	RESET command to next command: hold time	8	t_{CLK}

(1) $t_{CLK} = 1 / f_{CLK}$.

9.4.11 Power-Down Mode

Power down the ADC by holding the $\overline{RESET/PWDN}$ pin low. To reset the ADC without engaging power-down mode, pulse the pin low for less than 65536 clock cycles. In power-down mode, the ADC (including the internal reference) is shutdown. The internal low-dropout regulator (LDO) output to the BYPASS pin remains on, typically drawing 25- μ A idle current from the DVDD power supply. To exit power-down mode, take the $\overline{RESET/PWDN}$ pin high.

While in power-down mode, the ADC digital outputs remain driven and the analog inputs and reference inputs are high impedance. Maintain the digital inputs at V_{IH} or V_{IL} levels (do not float the digital inputs). When power-down mode is exited, the ADC resets, resulting in the registers resetting to default values. Wait the required $65536 f_{CLK}$ cycles (9 ms) before first communication to the ADC. Make sure to allow time for the internal reference to settle before starting the first conversion.

9.4.12 Chop Mode

The device uses a chopper-stabilized PGA and modulator in order to provide very low input voltage offset drift (V_{OS}/dT). However, because of nonidealities arising from chopper stabilization, a small amount of offset voltage drift sometimes remains. ADC1 incorporates a global chop option to reduce the offset voltage and offset voltage drift to very low levels. When Chop is enabled, the ADC performs two internal conversions to cancel the input offset voltage. The first conversion is taken with normal input polarity. The ADC reverses the internal input polarity for the second conversion. The difference of the two conversions is computed to yield the final corrected result with the offset voltage removed. See Figure 9-55. The ADC internal offset voltage is modeled as V_{OFS} .

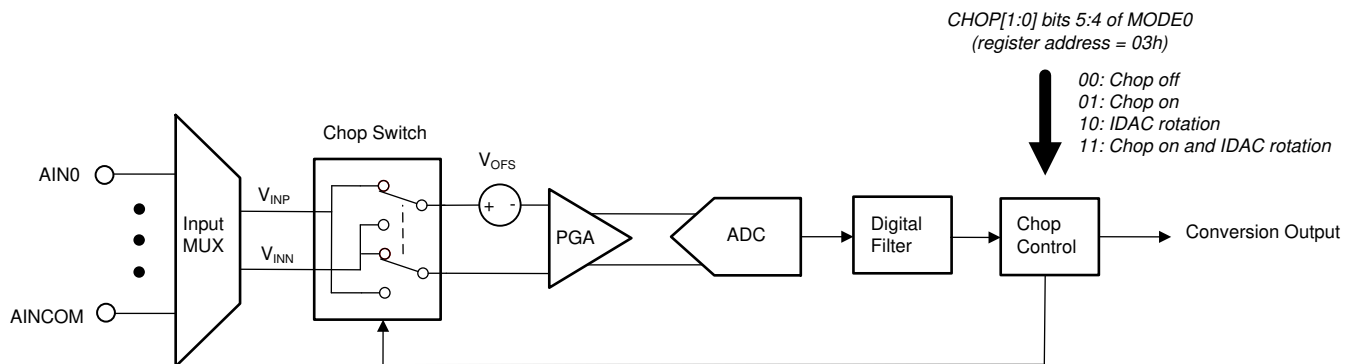


Figure 9-55. ADC1 Chop Block Diagram

The following is the internal Chop mode sequence.

Internal Conversion 1: $V_{AINP} - V_{AINN} - V_{OFS} \Rightarrow$ First conversion withheld

Internal Conversion 2: $V_{AINN} - V_{AINP} - V_{OFS} \Rightarrow$ Output result 1 = (Conversion 1 - Conversion 2) / 2 = $V_{AINP} - V_{AINN}$

Internal Conversion 3: $V_{AINP} - V_{AINN} - V_{OFS} \Rightarrow$ Output result 2 = (Conversion 3 - Conversion 2) / 2 = $V_{AINP} - V_{AINN}$

Internal Conversion 4: $V_{AINN} - V_{AINP} - V_{OFS} \Rightarrow$ Output result 3 = (Conversion 3 - Conversion 4) / 2 = $V_{AINP} - V_{AINN}$

The internal chop sequence repeats for all successive conversions.

As a result of the delay required by the digital filter to settle after reversing the inputs, the chop-mode data rate is less than the nominal data rate, depending on the digital filter order and programmed settling delay. Nevertheless, if the data rate currently in use has 50-Hz and 60-Hz frequency response nulls, the null frequencies remain unchanged. Chop mode also reduces the ADC noise by a factor of 1.4 because of the averaging of two conversions. In some cases, it is necessary to increase the time delay parameter, DELAY[3:0], to allow for settling of external components.

9.5 Programming

Commands are used to access the configuration and data registers and also to control the ADC. Many of the ADC commands are stand-alone (that is, single-byte). The register write and register read commands, however, are multibyte, consisting of two opcode bytes plus the register data byte or bytes. The commands are listed in [Table 9-33](#).

Commands can be sent at any time, either during a conversion or while conversions are stopped. However, if register read/write commands are in progress when conversion data are ready, the ADC blocks loading of conversion data to the output shift register. The $\overline{CS}$ input pin can be taken high between commands; or held low between consecutive commands. $\overline{CS}$ must stay low for the entirety of the command sequence. Complete the command, or terminate before command completion by taking $\overline{CS}$ high. Only send the commands that are listed in [Table 9-33](#).

Table 9-33. ADC Commands

COMMAND MNEMONIC	COMMAND TYPE	DESCRIPTION	OPCODE 1 BYTE	OPCODE 2 BYTE
NOP	NOP	No operation	0000 0000 (00h)	
RESET	Control	Reset the ADC	0000 011x (06h or 07h) ⁽¹⁾	
START1		Start ADC1 conversions	0000 100x (08h or 09h) ⁽¹⁾	
STOP1		Stop ADC1 conversions	0000 101x (0Ah or 0Bh) ⁽¹⁾	
START2		Start ADC2 conversions	0000 110x (0Ch or 0Dh) ⁽¹⁾	
STOP2		Stop ADC2 conversions	0000 111x (0Eh or 0Fh) ⁽¹⁾	
RDATA1	Conversion data read	Read ADC1 data	0001 001x (12h or 13h) ⁽¹⁾	
RDATA2		Read ADC2 data	0001 010x (14h or 15h) ⁽¹⁾	
SYOCAL1	Calibration	ADC1 system offset calibration	0001 0110 (16h)	
SYGCAL1		ADC1 system gain calibration	0001 0111 (17h)	
SFOCAL1		ADC1 self offset calibration	0001 1001 (19h)	
SYOCAL2		ADC2 system offset calibration	0001 1011 (1Bh)	
SYGCAL2		ADC2 system gain calibration	0001 1100 (1Ch)	
SFOCAL2		ADC2 self offset calibration	0001 1110 (1Eh)	
RREG	Register data read and write	Read registers	001r rrrr (20h+000r rrrr) ⁽²⁾	000n nnnn ⁽³⁾
WREG		Write registers	010r rrrr (40h+000r rrrr) ⁽²⁾	000n nnnn ⁽³⁾

(1) x = don't care.

(2) r rrrr = register address.

(3) n nnnn = number of registers to read or write minus 1.

9.5.1 NOP Command

The NOP command sends a no operation command to the device. The NOP command opcode is 00h. Hold the DIN pin low for the NOP command.

9.5.2 RESET Command

The RESET command resets the ADC operation and resets the device registers to default. See the [Reset by Command](#) section.

9.5.3 START1, STOP1, START2, STOP2 Commands

These commands start and stop the conversions of ADC1 and ADC2. See the [Conversion Control](#) section.

9.5.4 RDATA1, RDATA2 Commands

These commands are used to read ADC1 or ADC2 conversion data from the respective data holding buffers. See the [Read Conversion Data](#) section for more details.

9.5.5 SYOCAL1, SYGCAL1, SFOCAL1, SYOCAL2, SYGCAL2, SFOCAL2 Commands

These commands are used to calibrate ADC1 or ADC2. See the [Calibration](#) section.

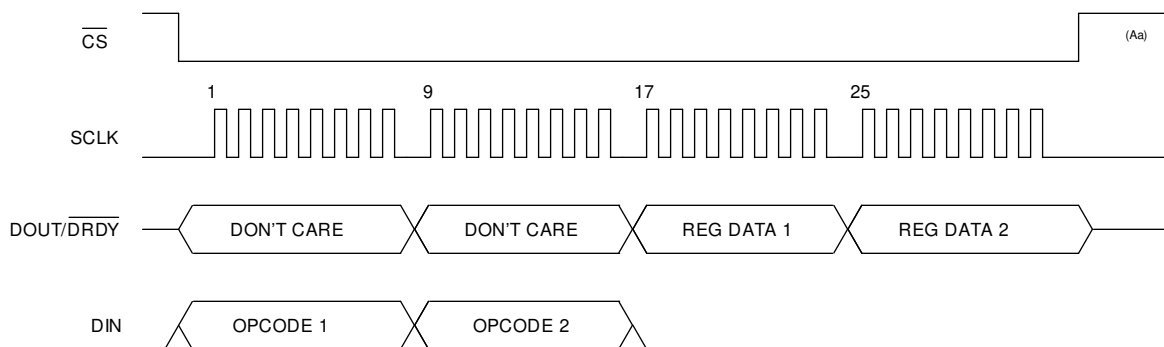
9.5.6 RREG Command

Use the RREG command to read the device register data. Read the register data one register at a time, or read as a block of register data. The starting register address is any register in the map. The RREG opcode consists of two bytes. The first byte specifies the starting register address: 001r rrr: where *r rrr* is the starting register address. The second opcode byte is the number of registers to read (minus 1): 000n nnn: where *n nnn* is the number of registers to read minus 1.

After the read command is sent, the ADC responds with one or more register data bytes, most significant bit first. If the byte count exceeds the last register address, the ADC begins to output zero data (the address pointer does not wrap). During the register read operation, if ADC1 data are ready, the conversion data are not loaded to the output shift register to avoid data contention. However, the conversion data can be retrieved later by the RDATA1 command. After the register read command has been started, further commands are disabled until one of the following conditions:

- 1) The read operation is completed.
- 2) The read operation is terminated by taking $\overline{\text{CS}}$ high.
- 3) The read operation is terminated by a serial interface autoreset.
- 4) The ADC is reset by toggling the $\overline{\text{RESET/PWDN}}$ pin.

Figure 9-56 depicts a two-register read operation example. As shown, the opcodes required to read data from two registers starting at register MODE2 (address = 05h) are: OPCODE 1 = 25h and OPCODE 2 = 01h. Keep the DIN input low after the two opcode bytes are sent.



A. $\overline{\text{CS}}$ can be set high or kept low between commands. If kept low, the command must be completed.

Figure 9-56. Read Register Sequence

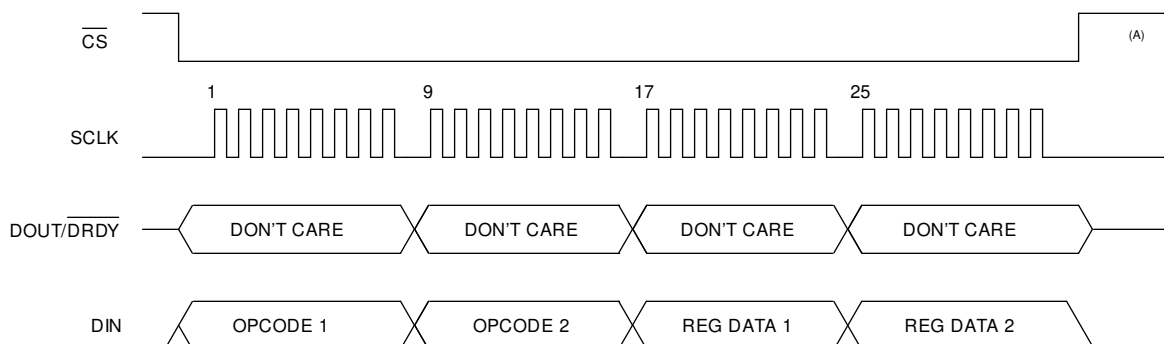
9.5.7 WREG Command

Use the WREG command to write the device register data. The register data are written one register at a time or as a block of register data. The starting register address is any register in the map.

The WREG opcode consists of two bytes. The first byte specifies the starting register address: 010r rrrr, where *r rrrr* is the starting register address. The second opcode byte is the number of registers to write (minus one): 000n nnnn, where *n nnnn* is the number of registers to write minus one. The following byte (or bytes) is the register data, most significant bit first. If the byte count exceeds the last register address, the ADC ignores the data (the address pointer does not wrap). Writing new data to certain registers results in a reset of ADC1 or ADC2 conversions, as specified in the *ADC restart* column in the [Register Maps](#). The previous conversion data are cleared at restart; therefore, read the data before the register write operation. After the register write command has been started, further commands are disabled until one of these conditions occur:

- 1) The write operation is completed.
- 2) The write operation is terminated by taking $\overline{CS}$ high.
- 3) The write operation is terminated by a serial interface auto-reset
- 4) The ADC is reset by toggling the $\overline{RESET/PWDN}$ pin.

Figure 9-57 depicts a two-register write operation example. As shown, the required opcodes to write data to two registers starting at register MODE2 (address = 05h) are: OPCODE 1 = 45h and OPCODE 2 = 01h.



A. Between commands, either set $\overline{CS}$ high or keep $\overline{CS}$ low. If $\overline{CS}$ is kept low, the command must be completed.

Figure 9-57. Write Register Sequence

The MODE2 and INPMUX registers are modified. Typically, register changes take effect immediately after the data are written. However, if the registers are part of a group, then the data are written only after all data for the grouped registers in the write block have been sent. In this example, data for MODE2 and INPMUX are written only after the data for INPMUX are sent. See the [Register Maps](#) section for those registers that are grouped when writing register data.

9.6 Register Maps

The ADS1262 register map consists of 21, 8-bit registers. The ADS1263 has six additional registers totaling 27 registers. Registers with addresses 15h through 1Ah apply exclusively to the ADC2. Collectively, these registers are used to configure and control the ADC to the desired mode of operation. Access the registers through the serial interface by using the RREG and WREG register-read and -write commands. At power-on or reset, the registers default to their initial settings, as shown in the *Default* column of [Table 9-34](#).

Writing new data to certain registers results in restart of conversions that are in progress. The registers that result in conversion restart (either ADC1 or ADC2) are shown in the *ADC Restart* column in [Table 9-34](#). The device drives the $\overline{\text{DRDY}}$ output high when ADC1 restarts. Additionally, data can be written as a block to multiple registers using a single command. If data are written as a block, the data of certain registers take effect immediately as the data are shifted in, while the data of other registers are buffered and take effect when the command is fully completed. The registers that update as a group are identified in the *Group Update* column in [Table 9-34](#). The group update registers that pertain to ADC1 operation are labeled *Group1*. The group update registers that pertain to ADC2 operation are labeled *Group2*. Update registers as a group to minimize the ADC recovery time after a configuration change. If the write command is terminated before completion, the data of group registers are not saved.

Table 9-34. Register Map

ADDR	REGISTER	DEFAULT	ADC RESTART	GROUP UPDATE	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
00h	ID	xxh			DEV_ID[2:0]			REV_ID[4:0]				
01h	POWER	11h			0	0	0	RESET	0	0	VBIAS	INTREF
02h	INTERFACE	05h			0	0	0	0	TIME OUT	STATUS	CRC[1:0]	
03h	MODE0	00h	ADC1	Group1	REFREV	RUN MODE	CHOP[1:0]		DELAY[3:0]			
04h	MODE1	80h	ADC1	Group1	FILTER[2:0]			SBADC	SBPOL	SBMAG[2:0]		
05h	MODE2	04h	ADC1	Group1	BYPASS	GAIN[2:0]			DR[3:0]			
06h	INPMUX	01h	ADC1	Group1	MUXP[3:0]				MUXN[3:0]			
07h	OFCAL0	00h			OFC[7:0]							
08h	OFCAL1	00h			OFC[15:8]							
09h	OFCAL2	00h			OFC[23:16]							
0Ah	FSCAL0	00h			FSC[7:0]							
0Bh	FSCAL1	00h			FSC[15:8]							
0Ch	FSCAL2	40h			FSC[23:16]							
0Dh	IDACMUX	BBh	ADC1	Group1	MUX2[3:0]				MUX1[3:0]			
0Eh	IDACMAG	00h	ADC1	Group1	MAG2[3:0]				MAG1[3:0]			
0Fh	REFMUX	00h	ADC1	Group1	0	0	RMUXP[2:0]			RMUXN[2:0]		
10h	TDACP	00h			OUTP	0	0	MAGP[4:0]				
11h	TDACN	00h			OUTN	0	0	MAGN[4:0]				
12h	GPIOCON	00h			CON[7:0]							
13h	GPIODIR	00h			DIR[7:0]							
14h	GPIODAT	00h			DAT[7:0]							
15h	ADC2CFG	00h	ADC2	Group2	DR2[1:0]		REF2[2:0]			GAIN2[2:0]		
16h	ADC2MUX	01h	ADC2	Group2	MUXP2[3:0]				MUXN2[3:0]			
17h	ADC2OFC0	00h			OFC2[7:0]							
18h	ADC2OFC1	00h			OFC2[15:8]							
19h	ADC2FSC0	00h			FSC2[7:0]							
1Ah	ADC2FSC1	40h			FSC2[15:8]							

9.6.1 Device Identification Register (address = 00h) [reset = x]

Figure 9-58. Device Identification Register (ID)

7	6	5	4	3	2	1	0
DEV_ID[2:0]			REV_ID[4:0]				
NOTE: Reset values are device dependent							

Table 9-35. Device Identification Register (ID) Field Descriptions

Bit	Field	Type	Reset	Description
7:5	DEV_ID[2:0]	R	x	Device ID. 000: ADS1262 001: ADS1263
4:0	REV_ID[4:0]	R	x	Revision ID <i>Note: the chip revision ID can change without notification</i>

9.6.2 Power Register (address = 01h) [reset = 11h]

Figure 9-59. Power Register (POWER)

7	6	5	4	3	2	1	0
RESERVED			RESET	RESERVED		VBIAS	INTREF
R-0h			R/W-1h	R-0h		R/W-0h	R/W-1h

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-36. Power Register (POWER) Field Descriptions

Bit	Field	Type	Reset	Description
7:5	RESERVED	R	0h	Reserved Always write 000
4	RESET	R/W	1h	Reset Indicator Indicates ADC reset has occurred. Clear this bit to detect the next device reset. 0: No new reset occurred 1: New reset has occurred (default)
3:2	RESERVED	R	0h	Reserved Always write 00
1	VBIAS	R/W	0h	Level Shift Voltage Enable Enables the internal level shift voltage to the AINCOM pin. $V_{BIAS} = (V_{AVDD} + V_{AVSS})/2$ 0: Disabled (default) 1: VBIAS enabled
0	INTREF	R/W	1h	Internal Reference Enable Enables the 2.5 V internal voltage reference. Note the IDAC and temperature sensor require the internal voltage reference. 0: Disabled 1: Internal reference enabled (default)

9.6.3 Interface Register (address = 02h) [reset = 05h]

Figure 9-60. Interface Register (INTERFACE)

7	6	5	4	3	2	1	0
RESERVED				TIMEOUT	STATUS	CRC[1:0]	
R-0h				R/W-0h	R-1h	R/W-1h	

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-37. Interface Register (INTERFACE) Field Descriptions

Bit	Field	Type	Reset	Description
7:4	RESERVED	R	0h	Reserved Always write 00h
3	TIMEOUT	R/W	0h	Serial Interface Time-Out Enable Enables the serial interface automatic time-out mode 0: Disabled (default) 1: Enable the interface automatic time-out
2	STATUS	R/W	1h	Status Byte Enable Enables the inclusion of the status byte during conversion data read-back 0: Disabled 1: Status byte included during conversion data read-back (default)
1:0	CRC[1:0]	R/W	1h	Checksum Byte Enable Enables the inclusion of the checksum byte during conversion data read-back 00: Checksum byte disabled 01: Enable Checksum byte in <i>Checksum</i> mode during conversion data read-back (default) 10: Enable Checksum byte in <i>CRC</i> mode during conversion data read-back 11: Reserved

9.6.4 Mode0 Register (address = 03h) [reset = 00h]

Figure 9-61. Mode0 Register (MODE0)

7	6	5	4	3	2	1	0
REFREV	RUNMODE	CHOP[1:0]		DELAY[3:0]			
R/W-0h	R/W-0h	R/W-0h		R/W-0h			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-38. Mode0 Register (MODE0) Field Descriptions

Bit	Field	Type	Reset	Description
7	REFREV	R/W	0h	Reference Mux Polarity Reversal Reverses the ADC1 reference multiplexer output polarity 0: Normal polarity of reference multiplexer output (default) 1: Reverse polarity of reference multiplexer output
6	RUNMODE	R/W	0h	ADC Conversion Run Mode Selects the ADC conversion (run) mode 0: Continuous conversion (default) 1: Pulse conversion (one shot conversion)
5:4	CHOP[1:0]	R/W	0h	Chop Mode Enable Enables the ADC chop and IDAC rotation options 00: Input chop and IDAC rotation disabled (default) 01: Input chop enabled 10: IDAC rotation enabled 11: Input chop <i>and</i> IDAC rotation enabled
3:0	DELAY[3:0]	R/W	0h	Conversion Delay Provides additional delay from conversion start to the beginning of the actual conversion 0000: no delay (default) 0001: 8.7 μ s 0010: 17 μ s 0011: 35 μ s 0100: 69 μ s 0101: 139 μ s 0110: 278 μ s 0111: 555 μ s 1000: 1.1 ms 1001: 2.2 ms 1010: 4.4 ms 1011: 8.8 ms 1100: Reserved 1101: Reserved 1110: Reserved 1111: Reserved

9.6.5 Mode1 Register (address = 04h) [reset = 80h]

Figure 9-62. Mode1 Register (MODE1)

7	6	5	4	3	2	1	0
FILTER[2:0]			SBADC	SBPOL	SBMAG[3:0]		
R/W-4h			R/W-0h	R/W-0h	R/W-0h		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-39. Mode1 Register (MODE1) Field Descriptions

Bit	Field	Type	Reset	Description
7:5	FILTER[2:0]	R/W	4h	Digital Filter Configures the ADC digital filter 000: Sinc1 mode 001: Sinc2 mode 010: Sinc3 mode 011: Sinc4 mode 100: FIR mode (default) 101: Reserved 110: Reserved 111: Reserved
4	SBADC	R/W	0h	Sensor Bias ADC Connection Selects the ADC to connect the sensor bias 0: Sensor bias connected to ADC1 mux out (default) 1: Sensor bias connected to ADC2 mux out
3	SBPOL	R/W	0h	Sensor Bias Polarity Selects the sensor bias for pull-up or pull-down 0: Sensor bias pull-up mode (AIN _P pulled high, AIN _N pulled low) (default) 1: Sensor bias pull-down mode (AIN _P pulled low, AIN _N pulled high)
2:0	SBMAG[2:0]	R/W	0h	Sensor Bias Magnitude Selects the sensor bias current magnitude or the bias resistor 000: No sensor bias current or resistor (default) 001: 0.5-μA sensor bias current 010: 2-μA sensor bias current 011: 10-μA sensor bias current 100: 50-μA sensor bias current 101: 200-μA sensor bias current 110: 10-MΩ resistor 111: Reserved

9.6.6 Mode2 Register (address = 05h) [reset = 04h]

Figure 9-63. Mode2 Register (MODE2)

7	6	5	4	3	2	1	0
BYPASS	GAIN[2:0]			DR[3:0]			
R/W-0h	R/W-0h			R/W-4h			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-40. Mode2 Register (MODE2) Field Descriptions

Bit	Field	Type	Reset	Description
7	BYPASS	R/W	0h	PGA Bypass Mode Selects PGA bypass mode 0: PGA enabled (default) 1: PGA bypassed
6:4	GAIN[2:0]	R/W	0h	PGA Gain Selects the PGA gain 000: 1 V/V (default) 001: 2 V/V 010: 4 V/V 011: 8 V/V 100: 16 V/V 101: 32 V/V 110: Reserved 111: Reserved
3:0	DR[3:0]	R/W	4h	Data Rate Selects the ADC data rate. In FIR filter mode, the available data rates are limited to 2.5, 5, 10 and 20 SPS. 0000: 2.5 SPS 0001: 5 SPS 0010: 10 SPS 0011: 16. $\bar{6}$ SPS 0100: 20 SPS (default) 0101: 50 SPS 0110: 60 SPS 0111: 100 SPS 1000: 400 SPS 1001: 1200 SPS 1010: 2400 SPS 1011: 4800 SPS 1100: 7200 SPS 1101: 14400 SPS 1110: 19200 SPS 1111: 38400 SPS

9.6.7 Input Multiplexer Register (address = 06h) [reset = 01h]

Figure 9-64. Input Multiplexer Register (INPMUX)

7	6	5	4	3	2	1	0
MUXP[3:0]				MUXN[3:0]			
R/W-0h				R/W-1h			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-41. Input Multiplexer Register (INPMUX) Field Descriptions

Bit	Field	Type	Reset	Description
7:4	MUXP[3:0]	R/W	0h	Positive Input Multiplexer Selects the positive input multiplexer. 0000: AIN0 (default) 0001: AIN1 0010: AIN2 0011: AIN3 0100: AIN4 0101: AIN5 0110: AIN6 0111: AIN7 1000: AIN8 1001: AIN9 1010: AINCOM 1011: Temperature sensor monitor positive 1100: Analog power supply monitor positive 1101: Digital power supply monitor positive 1110: TDAC test signal positive 1111: Float (open connection)
3:0	MUXN[3:0]	R/W	1h	Negative Input Multiplexer Selects the negative input multiplexer. 0000: AIN0 0001: AIN1 (default) 0010: AIN2 0011: AIN3 0100: AIN4 0101: AIN5 0110: AIN6 0111: AIN7 1000: AIN8 1001: AIN9 1010: AINCOM 1011: Temperature sensor monitor negative 1100: Analog power supply monitor negative 1101: Digital power supply monitor negative 1110: TDAC test signal negative 1111: Float (open connection)

9.6.8 Offset Calibration Registers (address = 07h, 08h, 09h) [reset = 00h, 00h, 00h]

Figure 9-65. Offset Calibration Registers (OFCAL0, OFCAL1, OFCAL2) 24-bit, 3 Rows

7	6	5	4	3	2	1	0
OFC[7:0]							
R/W-00h							
15	14	13	12	11	10	9	8
OFC[15:8]							
R/W-00h							
23	22	21	20	19	18	17	16
OFC[23:16]							
R/W-00h							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-42. Offset Calibration Registers (OFCAL0, OFCAL1, OFCAL2) Field Descriptions

Bit	Field	Type	Reset	Description
23:0	OFC[23:0]	R/W	000000h	Offset Calibration Three registers compose the 24-bit offset calibration word. The 24-bit word is two's complement format, and is internally left-shifted to align with the 32-bit conversion result. The ADC subtracts the register value from the 32-bit conversion result before the full-scale operation.

9.6.9 Full-Scale Calibration Registers (address = 0Ah, 0Bh, 0Ch) [reset = 40h, 00h, 00h]

Figure 9-66. Full-Scale Calibration Registers (FSCAL0, FSCAL1, FSCAL2) 24-bit, 3 Rows

7	6	5	4	3	2	1	0
FSCAL[7:0]							
R/W-00h							
15	14	13	12	11	10	9	8
FSCAL[15:8]							
R/W-00h							
23	22	21	20	19	18	17	16
FSCAL[23:16]							
R/W-40h							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-43. Full-Scale Calibration Registers (FSCAL0, FSCAL1, FSCAL2) Field Descriptions

Bit	Field	Type	Reset	Description
23:0	FSCAL[23:0]	R/W	400000h	Full-Scale Calibration Three 8-bit registers compose the 24-bit full scale calibration word. The 24-bit word format is straight binary. The ADC divides the 24-bit value by 400000h to derive the gain coefficient. The ADC multiplies the gain coefficient by the 32-bit conversion result after the offset operation.

9.6.10 IDACMUX Register (address = 0Dh) [reset = BBh]

Figure 9-67. IDAC Multiplexer Register (IDACMUX)

7	6	5	4	3	2	1	0
MUX2[3:0]				MUX1[3:0]			
R/W-Bh				R/W-Bh			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-44. IDAC Multiplexer Register (IDACMUX) Field Descriptions

Bit	Field	Type	Reset	Description
7:4	MUX2[3:0]	R/W	Bh	IDAC2 Output Multiplexer Selects the analog input pin to connect IDAC2 0000: AIN0 0001: AIN1 0010: AIN2 0011: AIN3 0100: AIN4 0101: AIN5 0110: AIN6 0111: AIN7 1000: AIN8 1001: AIN9 1010: AINCOM 1011: No Connection (default) 1100: Reserved 1101: Reserved 1110: Reserved 1111: Reserved
3:0	MUX1[3:0]	R/W	Bh	IDAC1 Output Multiplexer Selects the analog input pin to connect IDAC1 0000: AIN0 0001: AIN1 0010: AIN2 0011: AIN3 0100: AIN4 0101: AIN5 0110: AIN6 0111: AIN7 1000: AIN8 1001: AIN9 1010: AINCOM 1011: No Connection (default) 1100: Reserved 1101: Reserved 1110: Reserved 1111: Reserved

9.6.11 IDACMAG Register (address = 0Eh) [reset = 00h]

Figure 9-68. IDAC Magnitude Register (IDACMAG)

7	6	5	4	3	2	1	0
MAG2[3:0]				MAG1[3:0]			
R/W-0h				R/W-0h			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-45. IDAC Magnitude (IDACMAG) Field Descriptions

Bit	Field	Type	Reset	Description
7:4	MAG2[3:0]	R/W	0h	IDAC2 Current Magnitude Selects the current values of IDAC2 0000: off (default) 0001: 50 μ A 0010: 100 μ A 0011: 250 μ A 0100: 500 μ A 0101: 750 μ A 0110: 1000 μ A 0111: 1500 μ A 1000: 2000 μ A 1001: 2500 μ A 1010: 3000 μ A 1011: Reserved 1100: Reserved 1101: Reserved 1110: Reserved 1111: Reserved
3:0	MAG1[3:0]	R/W	0h	IDAC1 Current Magnitude Selects the current values of IDAC1 0000: off (default) 0001: 50 μ A 0010: 100 μ A 0011: 250 μ A 0100: 500 μ A 0101: 750 μ A 0110: 1000 μ A 0111: 1500 μ A 1000: 2000 μ A 1001: 2500 μ A 1010: 3000 μ A 1011: Reserved 1100: Reserved 1101: Reserved 1110: Reserved 1111: Reserved

9.6.12 REFMUX Register (address = 0Fh) [reset = 00h]

Figure 9-69. Reference Multiplexer Register (REFMUX)

7	6	5	4	3	2	1	0
RESERVED		RMUXP[2:0]			RMUXN[2:0]		
R/W-0h		R/W-0h			R/W-0h		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-46. Reference Multiplexer Register (REFMUX) Field Descriptions

Bit	Field	Type	Reset	Description
7:6	Reserved	R	0h	Reserved Always write 0h
5:3	RMUXP[2:0]	R/W	0h	Reference Positive Input Selects the positive reference input 000: Internal 2.5 V reference - P (default) 001: External AIN0 010: External AIN2 011: External AIN4 100: Internal analog supply (V_{AVDD}) 101: Reserved 110: Reserved 111: Reserved
2:0	RMUXN[2:0]	R/W	0h	Reference Negative Input Selects the negative reference input 000: Internal 2.5 V reference - N (default) 001: External AIN1 010: External AIN3 011: External AIN5 100: Internal analog supply (V_{AVSS}) 101: Reserved 110: Reserved 111: Reserved

9.6.13 TDACP Control Register (address = 10h) [reset = 00h]

Figure 9-70. TDACP Control Register (TDACP)

7	6	5	4	3	2	1	0
OUTP	RESERVED			MAGP[4:0]			
R/W-0h	R-0h			R/W-0h			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-47. TDACP Output Register (TDACP) Field Descriptions

Bit	Field	Type	Reset	Description
7	OUTP	R/W	0h	TDACP Output Connection Connects TDACP output to pin AIN6 0: No connection 1: TDACP output connected to pin AIN6
6:5	Reserved	R	0h	Reserved Always write 0
4:0	MAGP[4:0]	R/W	0h	MAGP Output Magnitude Select the TDACP output magnitude. The TDAC output voltages are ideal and are with respect to V_{AVSS} 01001: 4.5 V 01000: 3.5 V 00111: 3 V 00110: 2.75 V 00101: 2.625 V 00100: 2.5625 V 00011: 2.53125 V 00010: 2.515625 V 00001: 2.5078125 V 00000: 2.5 V 10001: 2.4921875 V 10010: 2.484375 V 10011: 2.46875 V 10100: 2.4375 V 10101: 2.375 V 10110: 2.25 V 10111: 2 V 11000: 1.5 V 11001: 0.5 V Remaining codes are reserved

9.6.14 TDACN Control Register (address = 11h) [reset = 00h]

Figure 9-71. TDACN Control Register (TDACN)

7	6	5	4	3	2	1	0
OUTN	RESERVED			MAGN[4:0]			
R/W-0h	R-0h			R/W-0h			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-48. TDAC Negative Output Register (TDACN) Field Descriptions

Bit	Field	Type	Reset	Description
7	OUTN	R/W	0h	TDACN Output Connection Connects TDACN output to pin AIN7 0: No external connection 1: TDACN output connected to pin AIN7
6:5	Reserved	R	0h	Reserved Always write 0h
4:0	MAGN[4:0]	R/W	0h	TDACN Output Magnitude Select the TDACN output magnitude. The TDAC output voltages are ideal and are with respect to V_{AVSS} 01001: 4.5 V 01000: 3.5 V 00111: 3 V 00110: 2.75 V 00101: 2.625 V 00100: 2.5625 V 00011: 2.53125 V 00010: 2.515625 V 00001: 2.5078125 V 00000: 2.5 V 10001: 2.4921875 V 10010: 2.484375 V 10011: 2.46875 V 10100: 2.4375 V 10101: 2.375 V 10110: 2.25 V 10111: 2 V 11000: 1.5 V 11001: 0.5 V Remaining codes are reserved

9.6.15 GPIO Connection Register (address = 12h) [reset = 00h]

Figure 9-72. GPIO Connection Register (GPIOCON)

7	6	5	4	3	2	1	0
CON[7:0]							
R/W-00h							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-49. GPIO Connection Register (GPIOCON) Field Descriptions

Bit	Field	Type	Reset	Description
0	CON[0]	R/W	0h	GPIO[0] Pin Connection Connects GPIO[0] to analog input pin AIN3 0: GPIO[0] not connected to AIN3 (default) 1: GPIO[0] connected to AIN3
1	CON[1]	R/W	0h	GPIO[1] Pin Connection Connects GPIO[1] to analog input pin AIN4 0: GPIO[1] not connected to AIN4 (default) 1: GPIO[1] connected to AIN4
2	CON[2]	R/W	0h	GPIO[2] Pin Connection Connects GPIO[2] to analog input pin AIN5 0: GPIO[2] not connected to AIN5 (default) 1: GPIO[2] connected to AIN5
3	CON[3]	R/W	0h	GPIO[3] Pin Connection Connects GPIO[3] to analog input pin AIN6 0: GPIO[3] not connected to AIN6 (default) 1: GPIO[3] connected to AIN6
4	CON[4]	R/W	0h	GPIO[4] Pin Connection Connects GPIO[4] to analog input pin AIN7 0: GPIO[4] not connected to AIN7 (default) 1: GPIO[4] connected to AIN7
5	CON[5]	R/W	0h	GPIO[5] Pin Connection Connects GPIO[5] to analog input pin AIN8 0: GPIO[5] not connected to AIN8 (default) 1: GPIO[5] connected to AIN8
6	CON[6]	R/W	0h	GPIO[6] Pin Connection Connects GPIO[6] to analog input pin AIN9 0: GPIO[6] not connected to AIN9 (default) 1: GPIO[6] connected to AIN9
7	CON[7]	R/W	0h	GPIO[7] Pin Connection Connects GPIO[7] to analog input pin AINCOM 0: GPIO[7] not connected to AINCOM (default) 1: GPIO[7] connected to AINCOM

9.6.16 GPIO Direction Register (address = 13h) [reset = 00h]

Figure 9-73. GPIO Direction Register (GPIODIR)

7	6	5	4	3	2	1	0
DIR[7:0]							
R/W-00h							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-50. GPIO Direction Register (GPIODIR) Field Descriptions

Bit	Field	Type	Reset	Description
0	DIR[0]	R/W	0h	GPIO[0] Pin Direction Configures GPIO[0] as a GPIO input or GPIO output 0: GPIO[0] is an output (default) 1: GPIO[0] is an input
1	DIR[1]	R/W	0h	GPIO[1] Pin Direction Configures GPIO[1] as a GPIO input or GPIO output 0: GPIO[1] is an output (default) 1: GPIO[1] is an input
2	DIR[2]	R/W	0h	GPIO[2] Pin Direction Configures GPIO[2] as a GPIO input or GPIO output 0: GPIO[2] is an output (default) 1: GPIO[2] is an input
3	DIR[3]	R/W	0h	GPIO[3] Pin Direction Configures GPIO[3] as a GPIO input or GPIO output 0: GPIO[3] is an output (default) 1: GPIO[3] is an input
4	DIR[4]	R/W	0h	GPIO[4] Pin Direction Configures GPIO[4] as a GPIO input or GPIO output 0: GPIO[4] is an output (default) 1: GPIO[4] is an input
5	DIR[5]	R/W	0h	GPIO[5] Pin Direction Configures GPIO[5] as a GPIO input or GPIO output 0: GPIO[5] is an output (default) 1: GPIO[5] is an input
6	DIR[6]	R/W	0h	GPIO[6] Pin Direction Configures GPIO[6] as a GPIO input or GPIO output 0: GPIO[6] is an output (default) 1: GPIO[6] is an input
7	DIR[7]	R/W	0h	GPIO[7] Pin Direction Configures GPIO[7] as a GPIO input or GPIO output 0: GPIO[7] is an output (default) 1: GPIO[7] is an input

9.6.17 GPIO Data Register (address = 14h) [reset = 00h]

Figure 9-74. GPIO Data Register (GPIODAT)

7	6	5	4	3	2	1	0
DAT[7:0]							
R/W-00h							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-51. GPIO Data Register (GPIODAT) Field Descriptions

Bit	Field	Type	Reset	Description
0	DAT[0]	R/W	0h	GPIO[0] Pin Data Configured as an output, read returns 0b Configured as an input, write sets the register value only 0: GPIO[0] is low 1: GPIO[0] is high
1	DAT[1]	R/W	0h	GPIO[1] Pin Data Configured as an output, read returns 0b Configured as an input, write sets the register value only 0: GPIO[1] is low 1: GPIO[1] is high
2	DAT[2]	R/W	0h	GPIO[2] Pin Data Configured as an output, read returns 0b Configured as an input, write sets the register value only 0: GPIO[2] is low 1: GPIO[2] is high
3	DAT[3]	R/W	0h	GPIO[3] Pin Data Configured as an output, read returns 0b Configured as an input, write sets the register value only 0: GPIO[3] is low 1: GPIO[3] is high
4	DAT[4]	R/W	0h	GPIO[4] Pin Data Configured as an output, read returns 0b Configured as an input, write sets the register value only 0: GPIO[4] is low 1: GPIO[4] is high
5	DAT[5]	R/W	0h	GPIO[5] Pin Data Configured as an output, read returns 0b Configured as an input, write sets the register value only 0: GPIO[5] is low 1: GPIO[5] is high
6	DAT[6]	R/W	0h	GPIO[6] Pin Data Configured as an output, read returns 0b Configured as an input, write sets the register value only 0: GPIO[6] is low 1: GPIO[6] is high
7	DAT[7]	R/W	0h	GPIO[7] Pin Data Configured as an output, read returns 0b Configured as an input, write sets the register value only 0: GPIO[7] is low 1: GPIO[7] is high

9.6.18 ADC2 Configuration Register (address = 15h) [reset = 00h]

Figure 9-75. ADC2 Configuration Register (ADC2CFG)

7	6	5	4	3	2	1	0
DR2[1:0]		REF2[2:0]			GAIN2[2:0]		
R/W-0h		R/W-0h			R/W-0h		

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-52. ADC2 Configuration Register (ADC2CFG) Field Descriptions

Bit	Field	Type	Reset	Description
7:6	DR2[1:0]	R/W	0h	ADC2 Data Rate These bits select the data rate of ADC2 00: 10 SPS (default) 01: 100 SPS 10: 400 SPS 11: 800 SPS
5:3	REF2[2:0]	R/W	0h	ADC2 Reference Input Selects the reference inputs of ADC2 as positive and negative pairs 000: Internal 2.5 V reference, positive and negative (default) 001: External AIN0 and AIN1 pin pairs as positive and negative 010: External AIN2 and AIN3 pin pairs as positive and negative 011: External AIN4 and AIN5 pin pairs as positive and negative 100: Internal V_{AVDD} and V_{AVSS} 101: Reserved 110: Reserved 111: Reserved
2:0	GAIN2[2:0]	R/W	0h	ADC2 Gain These bits configure the gain of ADC2 000: 1 V/V (default) 001: 2 V/V 010: 4 V/V 011: 8 V/V 100: 16 V/V 101: 32 V/V 110: 64 V/V 111: 128 V/V

9.6.19 ADC2 Input Multiplexer Register (address = 16h) [reset = 01h]

Figure 9-76. ADC2 Input Multiplexer Register (ADC2MUX)

7	6	5	4	3	2	1	0
MUXP2[3:0]				MUXN2[3:0]			
R/W-0h				R/W-1h			

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-53. ADC2 Input Multiplexer Register (ADC2MUX) Field Descriptions

Bit	Field	Type	Reset	Description
7:4	MUXP2[3:0]	R/W	0h	ADC2 Positive Input Multiplexer Selects the ADC2 positive input 0000: AIN0 (default) 0001: AIN1 0010: AIN2 0011: AIN3 0100: AIN4 0101: AIN5 0110: AIN6 0111: AIN7 1000: AIN8 1001: AIN9 1010: AINCOM 1011: Temperature sensor monitor positive 1100: Analog power supply monitor positive 1101: Digital power supply monitor positive 1110: TDAC test signal positive 1111: Open connection
3:0	MUXN2[3:0]	R/W	1h	ADC2 Negative Input Multiplexer Selects the ADC2 negative input 0000: AIN0 0001: AIN1 (default) 0010: AIN2 0011: AIN3 0100: AIN4 0101: AIN5 0110: AIN6 0111: AIN7 1000: AIN8 1001: AIN9 1010: AINCOM 1011: Temperature sensor monitor negative 1100: Analog power supply monitor negative 1101: Digital power supply monitor negative 1110: TDAC test signal negative 1111: Open Connection

9.6.20 ADC2 Offset Calibration Registers (address = 17h, 18h) [reset = 00h, 00h]

Figure 9-77. ADC2 Offset Calibration Registers (ADC2OFC0, ADC2OFC1) 16-bit, 2 Rows

7	6	5	4	3	2	1	0
OFC2[7:0]							
R/W-00h							
15	14	13	12	11	10	9	8
OFC2[15:8]							
R/W-00h							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-54. ADC2 Offset Calibration Registers (ADC2OFC0, ADC2OFC1) Field Descriptions

Bit	Field	Type	Reset	Description
15:0	OFC2[15:0]	R/W	0000h	ADC2 Offset Calibration Two registers compose the ADC2 16-bit offset calibration word. The 16-bit word is two's complement format and is internally left-shifted to align with the ADC2 24-bit conversion result. The ADC subtracts the register value from the conversion result before full-scale operation.

9.6.21 ADC2 Full-Scale Calibration Registers (address = 19h, 1Ah) [reset = 00h, 40h]

Figure 9-78. ADC2 Full-Scale Calibration Registers (ADC2FSC0, ADC2FSC1) 16-bit, 2 Rows

7	6	5	4	3	2	1	0
FSC2[7:0]							
R/W-00h							
15	14	13	12	11	10	9	8
FSC2[15:8]							
R/W-40h							

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 9-55. ADC2 Full-Scale Calibration Registers (ADC2FSC0, ADC2FSC1) Field Descriptions

Bit	Field	Type	Reset	Description
15:0	FSC2[15:0]	R/W	4000h	ADC2 Full-Scale Calibration Two registers compose the ADC2 16-bit full scale calibration word. The 16-bit word format is straight binary. The ADC divides the 16-bit value by 4000h to derive the scale factor for calibration. After the offset operation, the ADC multiplies the scale factor by the conversion result.

10 Application and Implementation

Note

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

10.1 Application Information

10.1.1 Isolated (or Floated) Inputs

Isolated sensors (sensors that are not referenced to the ADC ground) must have a common-mode voltage established within the specified ADC input range. Level shift the common-mode voltage by external resistor biasing, by connecting the negative lead to ground (bipolar analog supply), or by connecting to a dc voltage (unipolar analog supply). Use the level-shift voltage option on the AINCOM pin for this purpose. The 2.5-V reference output voltage is also used to provide level shifting to other floating sensor inputs.

10.1.2 Single-Ended Measurements

Single-ended measurements typically have one input connected to a fixed potential (ground or dc voltage) and the other input is the signal. Usually, the fixed connection is the negative input. The positive input is the signal and is driven above and below the negative input, as depicted in Figure 10-1. This is an example of a bipolar signal because the positive input can swing above and below the negative input. Unipolar signals are those where the positive signal is equal to or greater than the negative signal. The single-ended signal plus the level-shift voltage must be within the ADC specified operating range. In single supply configurations (5 V), the level-shift voltage is usually 2.5 V. This type of input configuration is shown in Figure 10-2. For bipolar power supplies (± 2.5 V), the negative voltage can be grounded. This type of input is shown in Figure 10-3.

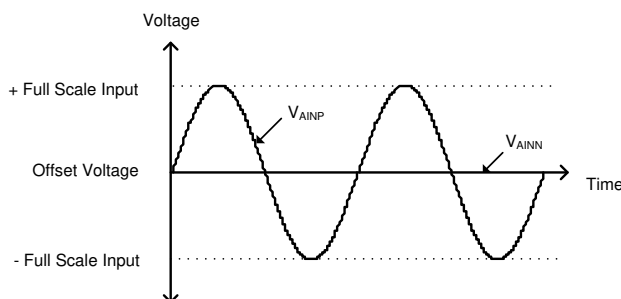


Figure 10-1. Single-Ended Input Voltage Diagram

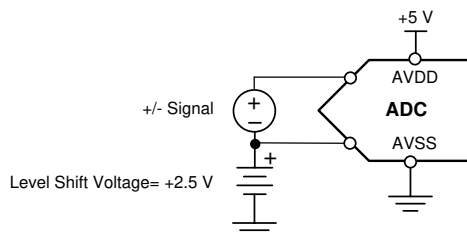


Figure 10-2. Single-Ended Input with Level-Shift Voltage

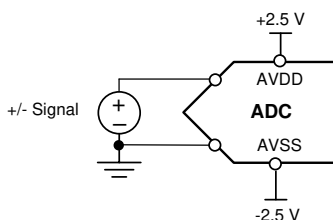


Figure 10-3. Single-Ended Input with Ground

10.1.3 Differential Measurements

A differential signal is one where both inputs are driven in symmetric and opposite polarities centered at a common-mode voltage. Optimally, the common-mode voltage is the midpoint of the ADC input range. The common-mode voltage plus the signal must always be within the ADC specified operating range to avoid signal clipping. As shown in Figure 10-4, the magnitude of each signal is maximum $\frac{1}{2}$ of the ADC full-scale range. The maximum differential signal ($V_{AINP} - V_{AINN}$) is equal to or less than the ADC FSR. For single 5-V operation, the common-mode voltage is typically equal to mid-supply (2.5 V) in order to use the full ADC input range. This type of input with single 5-V supply operation is shown in Figure 10-5. For bipolar supplies (± 2.5 V), the common-mode voltage of V_{AINP} and V_{AINN} are typically equal to ground potential. This type of input of configuration is shown in Figure 10-6. Certain types of differential signals, such as from a bridge circuits, are referenced to ADC ground; therefore, the common-mode voltage is defined.

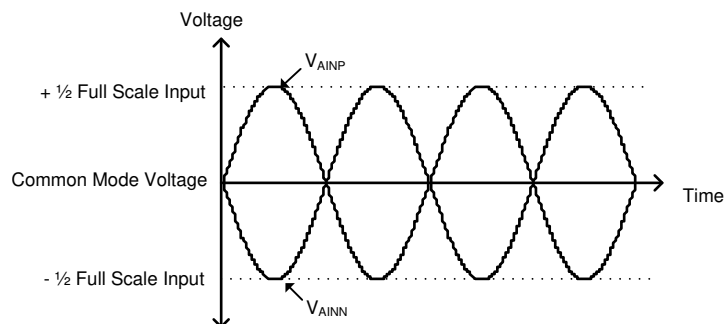


Figure 10-4. Differential Input Voltage Diagram

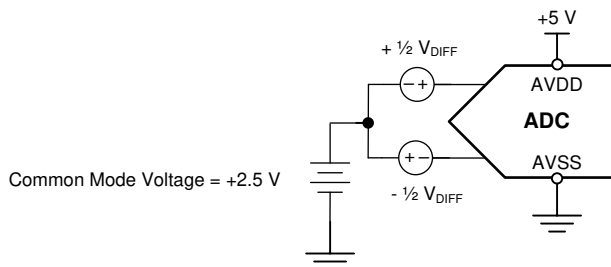


Figure 10-5. Differential Input With Common-Mode Level-Shift

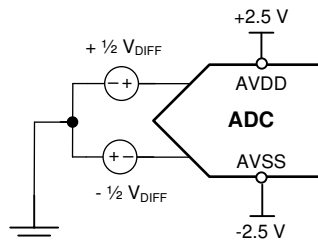


Figure 10-6. Differential Input With Common-Mode Ground

10.1.4 Input Range

For proper operation of ADC1, the PGA absolute input voltages, V_{INP} and V_{INN} , must always remain within the valid PGA input range, as shown in [Equation 12](#).

The following example uses [Equation 12](#) to determine the input-range requirement. For this example, use a thermocouple (60 mV, maximum differential output) with the negative lead connected to the internal level-shift voltage (2.5 V). Use a PGA gain of 32 and operate the ADC with a single 5-V power supply. To verify the PGA input-range requirement, the conditions are:

- V_{INN} = Negative absolute input voltage = 2.5 V
- V_{INP} = Positive absolute input voltage = 2.56 V
- V_{IN} = Differential input voltage = 0.06 V
- V_{AVDD} = 4.75 V (worst-case minimum)
- V_{AVSS} = 0 V
- Gain = 32

Filling in [Equation 12](#) with the values shown gives:

$$\begin{aligned} V_{AVSS} + 0.3 + |V_{IN}| \cdot (\text{Gain} - 1) / 2 &< V_{INP} \text{ and } V_{INN} < V_{AVDD} - 0.3 - |V_{IN}| \cdot (\text{Gain} - 1) / 2 \\ = 0 + 0.3 + 0.06 \cdot (32 - 1) / 2 &< 2.5 \text{ and } 2.56 < 4.75 - 0.3 - 0.06 \cdot (32 - 1) / 2 \\ = 1.23 \text{ V} < 2.5 \text{ V} \text{ and } 2.56 \text{ V} &< 3.52 \text{ V} \end{aligned}$$

The inequality is satisfied, therefore the V_{INN} and V_{INP} absolute input voltages are within the required PGA input range. Alternatively, measure the PGA output voltages (pins CAPP and CAPN) with a voltmeter to verify that each PGA output voltage is $< V_{AVDD} - 0.3 \text{ V}$ and $> V_{AVSS} - 0.3 \text{ V}$ under the expected minimum and maximum input conditions, respectively.

The input range requirement of ADC2 is verified in the same way as ADC1. See [Equation 15](#) for the ADC2 input range requirements.

10.1.5 Input Filtering

Analog input filtering serves two purposes: first, to limit the effect of aliasing during the sampling process; and second, to reduce external noise that affects the measurement.

10.1.5.1 Aliasing

As with all ADCs, out-of-band input signals can fold back or *alias* if not band-limited. Aliasing describes the effect of input frequencies greater than $\frac{1}{2}$ the sample rate folding back to the bandwidth of interest. An antialias filter placed at the ADC inputs reduces the magnitude of the aliased frequencies. The ADS1262 and ADS1263 incorporate analog and digital antialiasing filters to attenuate the aliased frequencies. There are two ranges of aliased frequencies: frequencies greater than $\frac{1}{2}$ of the down-sampled output data rate (Nyquist frequency) and frequencies occurring at multiples of the modulator sample rate.

Aliasing can occur at frequencies greater than $\frac{1}{2}$ the ADC output data rate. For example, at data rate of 50 SPS, aliasing occurs at frequencies greater than 25 Hz. The ADC digital filter rejects the aliased frequencies as input frequency increases. The amount of aliased frequency rejection is given by the filter type and order. [Figure 10-7](#) illustrates the frequency response of the sinc filter. Note the sinc4 filter provides the best rejection of aliased frequencies.

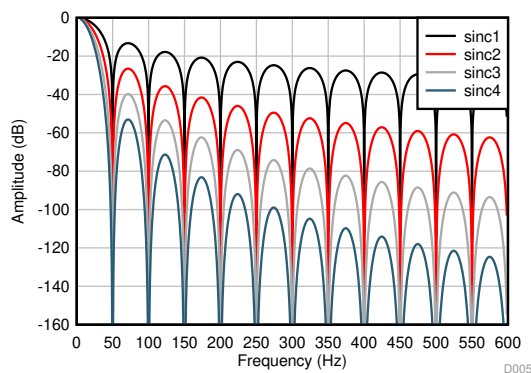


Figure 10-7. Frequency Response (50 SPS)

The second band of aliased frequencies occur at the ADC modulator sample rate multiples ($f_{\text{MOD}} = f_{\text{CLK}} / 8 = 921.6 \text{ kHz}$, multiples = 1843.2 kHz and so on). [Figure 10-8](#) shows the 38400 SPS frequency response plotted to 1.2 MHz. The response near dc is the signal bandwidth of interest. Observe how the digital filter response repeats on the sides of the modulator sample rate (921.6 kHz). [Figure 10-9](#) shows the repeated response at the modulator frequency multiples = $N \cdot f_{\text{MOD}} \pm f_{\text{DR}}$, where N = multiples of f_{MOD} starting at 1, and f_{DR} = data rate frequency. The digital filter attenuates signal or noise up to where the response repeats. However, signal or noise occurring at the modulator sample rate is not attenuated by the digital filter and therefore, is aliased to the passband.

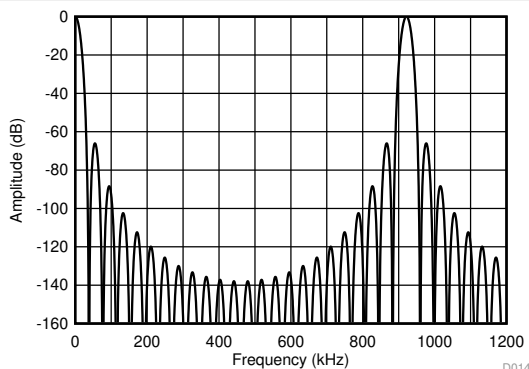


Figure 10-8. Frequency Response to 1.2 MHz (38400 SPS)

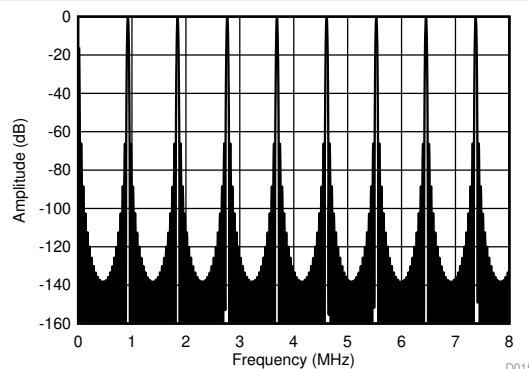


Figure 10-9. Frequency Response to 8 MHz (38400 SPS)

[Figure 10-10](#) illustrates how the frequencies alias near the modulator sample rate frequency. The final figure shows the aliased frequency rejection provided by an antialias filter. The ADC incorporates an analog antialias filter with a cutoff frequency of 60 kHz that rejects the aliased frequencies.

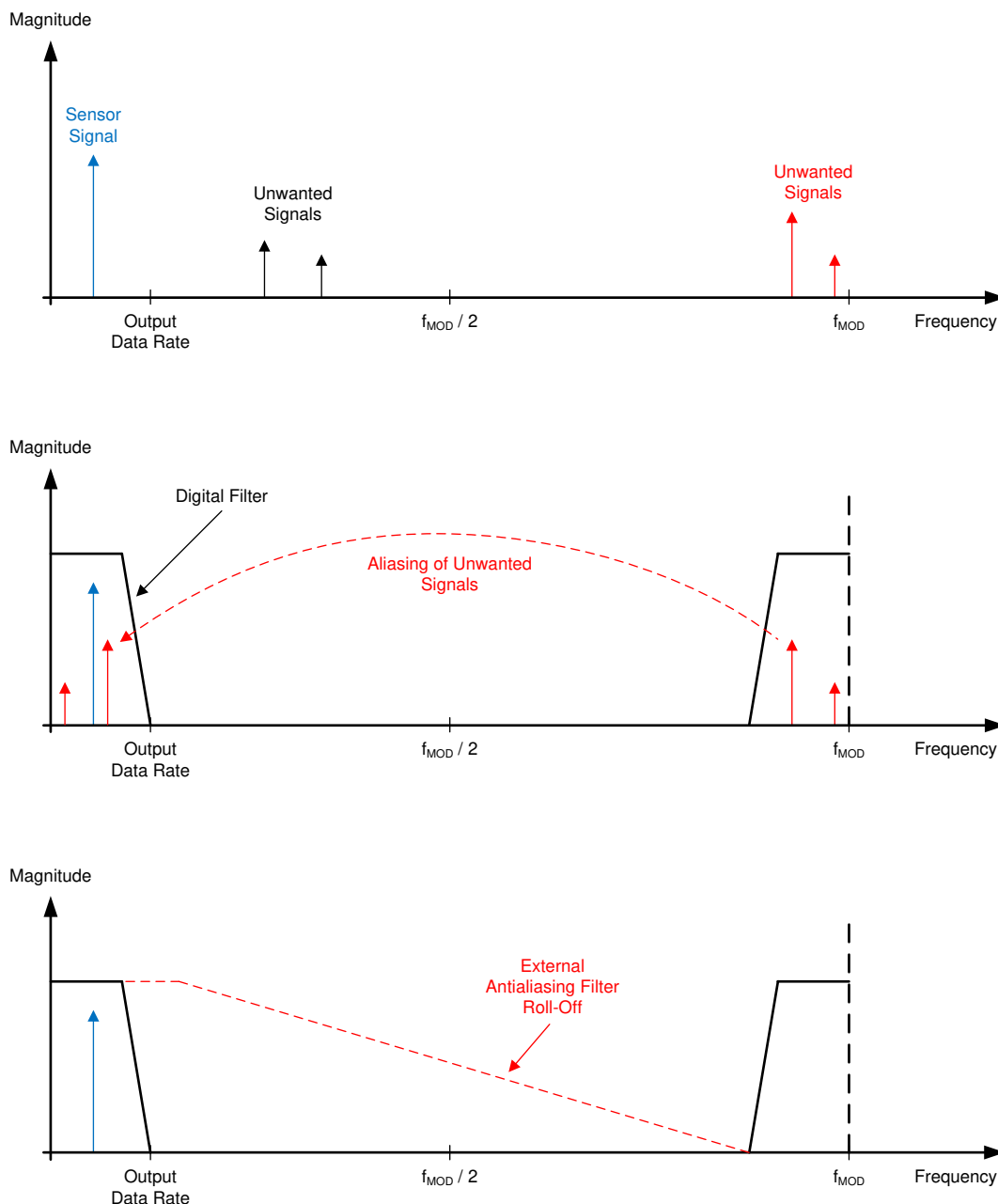


Figure 10-10. Alias Effect

Many sensor signals are inherently band-limited; for example, the output of a thermocouple has a limited rate of change. In this case, the sensor signal does not alias back into the pass band when using a $\Delta\Sigma$ ADC. However, any noise picked up along the sensor wiring or the application circuitry can potentially alias into the pass band. Power line-cycle frequency and harmonics are one common noise source. External noise is also generated from electromagnetic interference (EMI) or radio frequency interference (RFI) sources, such as nearby motors and cellular phones. Another noise source exists on the printed circuit board (PCB) in the form of clocks and other digital signals. Analog input filtering helps remove unwanted signals from affecting the measurement result. The ADC incorporates a low-pass, antialias filter with a corner frequency of 60 kHz to reduce the aliased frequencies. The filter consists of the external 4.7-nF PGA output capacitor (CAPP and CAPN pins) and internal 280- Ω resistors.

Use an input filter to provide increased rejection of aliased noise frequencies and further attenuate possible strong high-frequency interference signals. For best performance, filter strong interference frequencies at the

ADC inputs. Ideally, select a low-pass corner frequency that allows frequencies within the desired bandwidth and attenuates those frequencies outside the desired bandwidth. As a result of the stable and linear dielectric characteristics, use C0G-type MLCC capacitors in analog signal filters. In applications where high energy transients can be generated, such as caused by inductive load switching, transient voltage suppressor (TVS) diodes or external ESD diodes should be used to protect the ADC inputs.

10.1.6 Input Overload

Follow the input overvoltage precautions as outlined in the [ESD Diode](#) section. Despite external current limit provided for the input pins, if an overvoltage condition occurs on an unused channel, the overvoltage channel may crosstalk to the measurement channel. One solution is to externally clamp the inputs with low-forward voltage diodes as shown in [Figure 10-11](#). The external diodes shunt the overvoltage fault current around the ADC inputs. Be aware of the reverse leakage current in the external clamp diodes in the application.

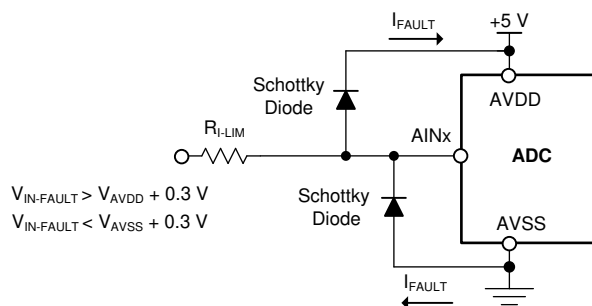


Figure 10-11. External Diode Voltage Clamp

10.1.7 Unused Inputs and Outputs

To minimize input leakage of the measurement channel, tie the unused input channels to mid-supply ($V_{AVDD} + V_{AVSS} / 2$). Use the 2.5-V reference output voltage for this purpose if operating with single 5-V supply. Do not float unused digital inputs. Tie all unused digital inputs to the appropriate levels, V_{DVDD} or V_{DGND} , including when in power-down mode. Do not float (3-state) the digital inputs to the ADC or excessive power-supply leakage current can result. If the $\overline{DRDY}$ output is unused, leave the pin unconnected or connect to an external circuit.

10.1.8 Voltage Reference

For nonratiometric (absolute) measurements where the input signal is not derived from the voltage reference, either use the internal precision voltage reference, or use an external precision reference. Examples of these types of measurements come from sensors such as thermocouples, 20-mA transmitters, and accelerometers.

For ratiometric measurements, where the input signal is derived from the voltage reference, reference noise and drift are canceled by the same ratio of noise and drift within the signal. Ratiometric operation is common with many types of bridge and RTD measurements. For best noise performance, match the reference filter and input filter time constants (see the [3-Wire RTD Measurement with Lead-Wire Compensation](#) section for more information). In general, achieve the best ADC signal-to-noise ratio by using large amplitude signals, a large reference voltage, and the highest gain setting possible.

10.1.9 Serial Interface Connections

After power up, take the $\overline{\text{CS}}$ input high to reset the ADC serial interface. $\overline{\text{CS}}$ high resets the serial interface in the event an unintentional SCLK glitch has occurred during power-on initialization. If $\overline{\text{CS}}$ is tied low, glitches at SCLK power on can interrupt synchronization to the serial interface and must be avoided. In this case, reset the ADC using the $\overline{\text{RESET/PWDN}}$ input. The SCLK input is edge sensitive, and therefore must be free of noise, glitches, and overshoot. Use a terminating resistor located at the SCLK buffer to smooth the edges and reduce overshoot.

Most microcontroller SPI peripherals can operate with the ADC. The interface operates in SPI mode 1, where CPOL = 0 and CPHA = 1. In SPI mode 1, SCLK idles low and data are updated or changed on SCLK rising edges; data are latched or read by the host and the ADC on SCLK falling edges. Details of the SPI communication protocol employed by the device is found in the [Timing Requirements: Serial Interface](#) table. Place a 47- Ω resistor in series with all digital input and output pins ($\overline{\text{CS}}$, SCLK, DIN, DOUT/ $\overline{\text{DRDY}}$, and $\overline{\text{DRDY}}$). The resistors match the characteristic impedance of the PCB trace by source termination, helping reduce overshoot and ringing.

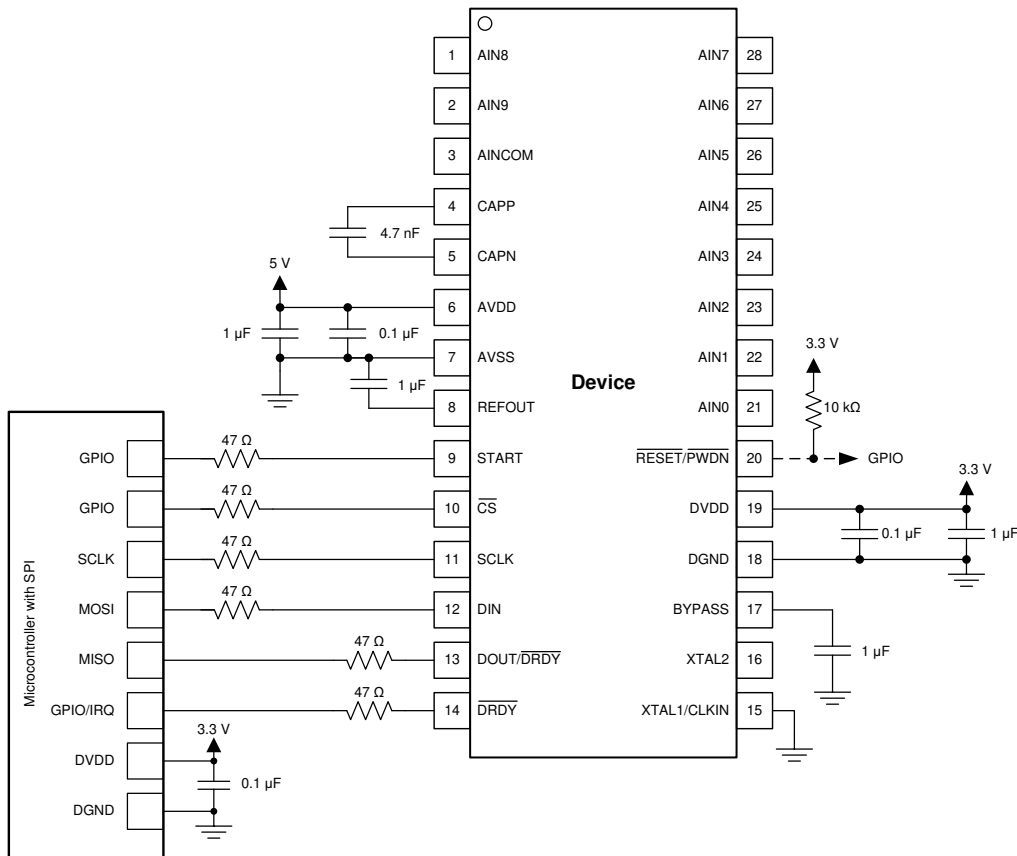


Figure 10-12. Serial Interface Connections

10.2 Typical Application

10.2.1 3-Wire RTD Measurement with Lead-Wire Compensation

Figure 10-13 is a fault-protected, filtered, 3-wire RTD application circuit with hardware-based, lead-wire compensation. Two IDAC current sources provide the lead-wire compensation. One IDAC current source (IDAC1) provides excitation to the RTD element. The ADC reference voltage input (pins AIN2 and AIN3) is derived from the same current by resistor R_{REF} , providing ratiometric cancellation of current-source drift. The other current source (IDAC2) has the same current setting, providing cancellation of lead-wire resistance by generating a voltage drop across lead-wire resistance R_{LEAD2} equal to the voltage drop of R_{LEAD1} . Because the R_{RTD} voltage is measured differentially at ADC pins AIN4 and AIN5, the voltages across the lead wire resistance cancel. Resistor R_{BIAS} level-shifts the RTD signal to within the ADC specified input range. The current sources are provided by two additional pins (AIN1 and AIN6) that connect to the RTD through blocking diodes. The additional pins are used to route the RTD excitation currents around the input resistors, avoiding the voltage drop otherwise caused by the filter resistors R_{F1} and R_{F4} . The diodes protect the ADC inputs in the event of a miswired connection. The input filter resistors limit the input fault currents flowing into the ADC.

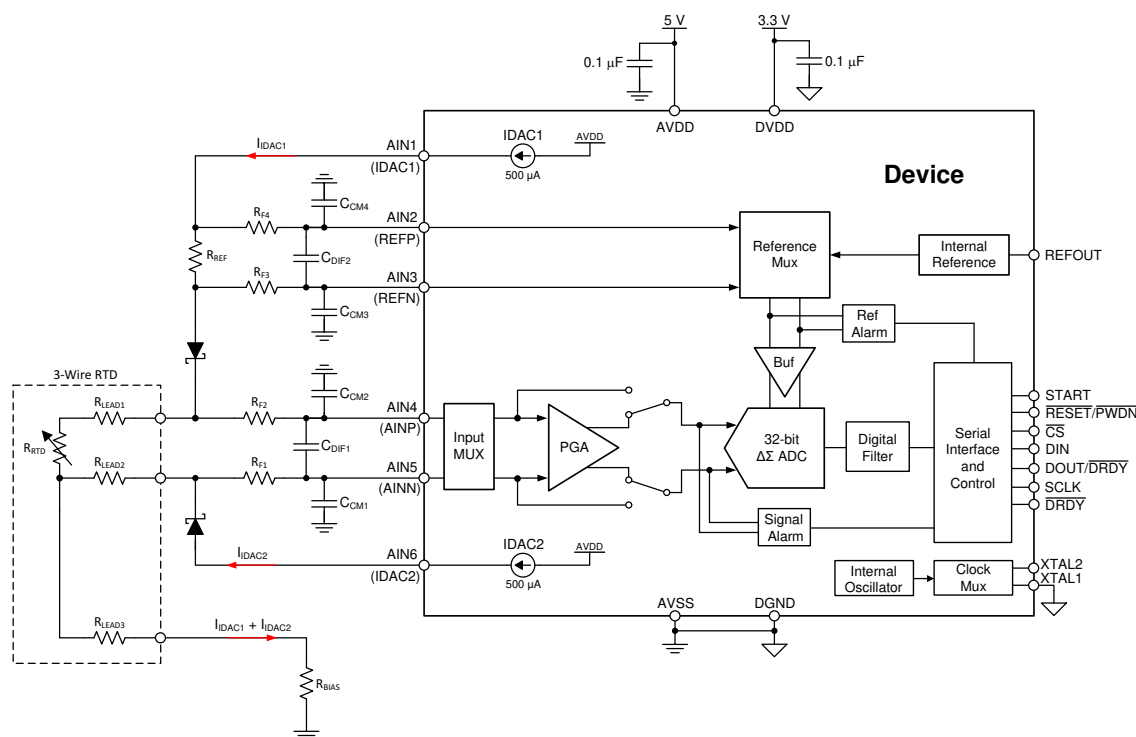


Figure 10-13. 3-Wire RTD Application

10.2.1.1 Design Requirements

Table 10-1 shows the design requirements of the 3-wire RTD application.

Table 10-1. Design Requirements

DESIGN PARAMETER	VALUE
ADC supply voltage	4.75 V (minimum)
RTD sensor type	3-wire Pt100
RTD resistance range	20 Ω to 400 Ω
RTD lead resistance range	0 Ω to 10 Ω
RTD self heating	1 mW
Accuracy ⁽¹⁾	$\pm 0.02 \Omega$

(1) $T_A = 25^\circ\text{C}$. After offset and full-scale calibration.

10.2.1.2 Detailed Design Procedure

The key considerations in the design of a 3-wire RTD circuit are the accuracy, the lead wire compensation, and the sensor self-heating. As the design values of [Table 10-2](#) illustrate, several values of excitation currents are available. The resolution is expressed in units of noise-free bits (NFR). Noise-free resolution is resolution with no code flicker. The selection of excitation currents trades off resolution against sensor self-heating. In general, measurement resolution improves with increasing excitation current. Increasing the excitation current beyond 1000 μA results in no further improvement in resolution. The design procedure is based on 500- μA excitation current, because this level of current results in very low sensor self-heating (0.4 mW).

Table 10-2. RTD Circuit Design Parameters

I _{IDAC} (μA)	NFR (bits)	P _{RTD} (mW)	V _{RTD} ⁽¹⁾ (V)	Gain ⁽²⁾ (V/V)	V _{REFMIN} ⁽³⁾ (V)	V _{REF} ⁽⁴⁾ (V)	R _{REF} ⁽⁵⁾ (k Ω)	V _{INNLIM} ⁽⁶⁾ (V)	V _{INPLIM} ⁽⁷⁾ (V)	R _{BIAS} ⁽⁸⁾ (k Ω)	V _{RTDN} ⁽⁹⁾ (V)	V _{RTDP} ⁽¹⁰⁾ (V)	V _{IDAC1} ⁽¹¹⁾ (V)
50	16.8	0.001	0.02	32	0.64	0.90	18	0.6	4.1	7.10	0.7	0.7	1.9
100	17.8	0.004	0.04	32	1.28	1.41	14.1	0.9	3.8	5.10	1.0	1.1	2.8
250	18.8	0.025	0.10	16	1.60	1.76	7.04	1.1	3.7	2.30	1.2	1.3	3.3
500	19.1	0.100	0.20	8	1.60	1.76	3.52	1.0	3.8	1.10	1.1	1.3	3.4
750	18.9	0.225	0.30	4	1.20	1.32	1.76	0.8	4.0	0.57	0.9	1.2	2.8
1000	19.3	0.400	0.40	4	1.60	1.76	1.76	0.9	3.9	0.50	1.0	1.4	3.5
1500	19.1	0.900	0.60	2	1.20	1.32	0.88	0.6	4.2	0.23	0.7	1.3	3.0
2000	18.3	1.600	0.80	1	0.80	0.90	0.45	0.3	4.5	0.10	0.4	1.2	2.4

- (1) V_{RTD} is the RTD input voltage.
- (2) Gain is the ADC gain
- (3) V_{REFMIN} is the minimum reference voltage required by the design.
- (4) V_{REF} is the design target reference voltage allowing for 10 % over-range or the minimum 0.9 V reference voltage requirement.
- (5) R_{REF} is the resistor that senses the IDAC current to generate V_{REF}.
- (6) V_{INNLIM} is the absolute minimum input voltage required by the ADC.
- (7) V_{INPLIM} is the absolute maximum input voltage required by the ADC.
- (8) R_{BIAS} establishes the level-shift voltage.
- (9) V_{RTDN} is the design target negative input voltage.
- (10) V_{RTDP} is the design target positive input voltage.
- (11) V_{IDAC1} is the design target IDAC1 loop voltage.

Initially, R_{LEAD1} and R_{LEAD2} are considered to be 0 Ω . Route the IDAC1 current through the external reference resistor, R_{REF}. IDAC1 generates the ADC reference voltage, V_{REF}, across the reference resistor. This voltage is defined by [Equation 24](#):

$$V_{REF} = I_{IDAC1} \cdot R_{REF} \quad (24)$$

Route the second current (IDAC2) to the second RTD lead.

Program both IDAC1 and IDAC2 to the same value by using the IDACMAG register; however, only the IDAC1 current flows through the reference resistor and RTD. The IDAC1 current excites the RTD to produce a voltage proportional to the RTD resistance. The RTD voltage is defined by [Equation 25](#):

$$V_{RTD} = R_{RTD} \cdot I_{IDAC1} \quad (25)$$

The ADC amplifies the RTD signal voltage (V_{RTD}) and measures the resulting voltage against the reference voltage to produce a proportional digital output code, as shown in [Equation 26](#) through [Equation 28](#).

$$\text{Code} \propto V_{RTD} \cdot \text{Gain} / V_{REF} \quad (26)$$

$$\text{Code} \propto (R_{RTD} \cdot I_{IDAC1}) \cdot \text{Gain} / (I_{IDAC1} \cdot R_{REF}) \quad (27)$$

$$\text{Code} \propto (R_{RTD} \cdot \text{Gain}) / R_{REF} \quad (28)$$

As shown in [Equation 28](#), the RTD measurement depends on the value of the RTD, the PGA gain, and the reference resistor R_{REF}, but not on the IDAC1 value. Therefore, the absolute accuracy and temperature drift of the excitation current does not matter.

The second excitation current (IDAC2) provides a second voltage drop across the second RTD lead resistance, R_{LEAD2} . The second voltage drop compensates the voltage drop caused by I_{DAC1} and R_{LEAD1} . The leads of a 3-wire RTD typically have the same length; therefore, the lead resistance is typically identical. Taking the lead resistance into account ($R_{LEADx} \neq 0$), the differential voltage (V_{IN}) across ADC inputs AIN4 and AIN5 is shown in Equation 29:

$$V_{IN} = I_{DAC1} \cdot (R_{RTD} + R_{LEAD1}) - I_{DAC2} \cdot R_{LEAD2} \quad (29)$$

If $R_{LEAD1} = R_{LEAD2}$ and $I_{DAC1} = I_{DAC2}$, the expression for V_{IN} reduces to Equation 30:

$$V_{IN} = I_{DAC1} \cdot R_{RTD} \quad (30)$$

In other words, the measurement error resulting from the voltage drop across the RTD lead resistance is compensated, as long as the lead resistance values and the IDAC values are matched.

Using Equation 25, the value of RTD resistance (400 Ω , maximum) and the excitation current (500 μ A) yields an RTD voltage of $V_{RTD} = 500 \mu\text{A} \cdot 400 \Omega = 0.2 \text{ V}$. Use the maximum gain of 8 V/V in order to limit the reference voltage requirement as well as the corresponding loop voltage of IDAC1. The total loop voltage must not exceed the maximum IDAC voltage compliance specification. Gain = 8 requires a minimum reference voltage $V_{REFMIN} = 0.2 \text{ V} \cdot 8 = 1.6 \text{ V}$. To provide a margin for the ADC operating range, increase the target reference voltage by 10% ($V_{REF} = 1.6 \text{ V} \cdot 1.1 = 1.76 \text{ V}$). Calculate the value of the reference resistor, as shown in Equation 31:

$$R_{REF} = V_{REF} / I_{DAC1} = 1.76 \text{ V} / 500 \mu\text{A} = 3.52 \text{ k}\Omega \quad (31)$$

For best results, use a precision reference resistor R_{REF} with a low temperature drift (< 10 ppm/ $^{\circ}$ C).

The next step in the design is determining the value of the R_{BIAS} resistor, in order to level shift the RTD voltage to meet the ADC absolute input-voltage specification. The required level-shift voltage is determined by calculating the minimum absolute voltage (V_{INNLIM}) as shown in Equation 32:

$$V_{AVSS} + 0.3 + V_{RTD} \cdot (\text{Gain} - 1) / 2 \leq V_{INNLIM} \quad (32)$$

where

- V_{RTD} = maximum differential RTD voltage = 0.2 V
- Gain = 8
- $V_{AVSS} = 0 \text{ V}$

The result of the equation requires a minimum absolute input voltage (V_{RTDN}) > 1.0 V. Therefore, the RTD voltage must be level shifted a minimum of 1.0 V. To meet this requirement, a target level-shift value of 1.1 V is chosen to provide 0.1 V margin. Calculate the value of R_{BIAS} as shown in Equation 33:

$$R_{BIAS} = V_{INN} / (I_{DAC1} + I_{DAC2}) = 1.1 \text{ V} / (2 \cdot 500 \mu\text{A}) = 1.1 \text{ k}\Omega. \quad (33)$$

After the level-shift voltage is determined, verify that the positive RTD voltage (V_{RTDP}) is less than the maximum absolute input voltage (V_{INPLIM}), as shown in Equation 34:

$$V_{INPLIM} \leq V_{AVDD} - 0.3 - V_{RTD} \cdot (\text{Gain} - 1) / 2 \quad (34)$$

where

- V_{RTD} = maximum differential RTD voltage = 0.2 V
- Gain = 8
- $V_{AVDD} = 4.75 \text{ V}$ (minimum)

Solving Equation 34 results in a required V_{RTDP} of less than 3.8 V. Calculate the V_{RTDP} input voltage by Equation 35:

$$V_{INP} = V_{RTDN} + I_{DAC1} \cdot (R_{RTD} + R_{LEAD1}) = 1.1 \text{ V} + 500 \mu\text{A} \cdot (400 \Omega + 10 \Omega) = 1.3 \text{ V} \quad (35)$$

Because 1.3 V is less than the 3.8-V maximum input voltage limit, the absolute positive and negative RTD voltages are within the ADC specified input range.

The next step in the design is to verify that the loop voltage of the excitation current is less than the specified IDAC compliance voltage. The IDAC compliance voltage is the maximum voltage drop developed across each IDAC current path to AVSS. In this circuit, IDAC1 has the largest voltage drop developed across its current path. The IDAC1 calculation is sufficient to satisfy IDAC2 because the IDAC2 voltage drop is always less than IDAC1 voltage drop. The sum of voltages in the IDAC1 loop is shown in [Equation 36](#):

$$V_{IDAC1} = [(I_{IDAC1} + I_{IDAC2}) \cdot (R_{LEAD3} + R_{BIAS})] + [I_{IDAC1} \cdot (R_{RTD} + R_{LEAD1} + R_{REF})] + V_D \quad (36)$$

where

- V_D = external blocking diode voltage.

The equation results in a loop voltage of $V_{IDAC1} = 3.4$ V. The worst-case current source compliance voltage is: $(V_{AVDD} - 1.1 \text{ V}) = (4.75 \text{ V} - 1.1 \text{ V}) = 3.64$ V. The V_{IDAC1} loop voltage is less than the specified current source compliance voltage ($3.4 \text{ V} < 3.64 \text{ V}$).

Many applications benefit from using an analog filter at the inputs to remove noise and interference from the signal. Filter components are placed on the ADC inputs (R_{F1} , R_{F2} , C_{DIF1} , C_{CM1} , and C_{CM2}), as well as on the reference inputs (R_{F3} , R_{F4} , C_{DIF2} , C_{CM3} , and C_{CM4}). The filters remove both differential and common-mode noise. The application shows a differential input noise filter formed by R_{F1} , R_{F2} and C_{DIF} , with additional differential mode capacitance provided by the common-mode filter capacitors, C_{M1} and C_{M2} . Calculate the differential cutoff frequency as shown in [Equation 37](#):

$$f_{DIF} = 1 / [2\pi \cdot (R_{F1} + R_{F2}) \cdot (C_{DIF1} + C_{M1} || C_{M2})] \quad (37)$$

The common-mode noise filter is formed by components R_{F1} , R_{F2} , C_{M1} and C_{M2} . Calculate the common-mode signal cutoff frequency as shown in [Equation 38](#):

$$f_{CM} = 1 / (2\pi \cdot R_{F1} \cdot C_{M1}) = 1 / (2\pi \cdot R_{F2} \cdot C_{M2}) \quad (38)$$

Mismatches in the common-mode filter components convert common-mode noise into differential noise. To reduce the effect of mismatch, use a differential mode filter with a corner frequency that is 10 times lower than the common-mode filter corner frequency. The low-frequency differential filter removes the common-mode converted noise. The filter resistors (R_{Fx}) also serve as current-limiting resistors. These resistors limit the current into the analog inputs (AINx) of the device to safe levels when an overvoltage occurs on the inputs.

Filter resistors lead to an offset voltage error due to the dc input current leakage flowing into and out of the device. Remove this voltage error by system offset calibration. Resistor values that are too large generate excess thermal noise and degrade the overall noise performance. The recommended range of the filter resistor values is 2 kΩ to 10 kΩ. The properties of the capacitors are important because the capacitors are connected to the signal; use high-quality C0G ceramics or film-type capacitors.

For consistent noise performance across the full range of RTD measurements, match the corner frequencies of the input and reference filter. Detailed information on matching the input and reference filter is found in the [RTD Ratiometric Measurements and Filtering Using the ADS1148 and ADS1248 application report](#).

10.2.1.3 Application Curve

Figure 10-14 shows the resistance measurement results. The measurements are taken at $T_A = 25^\circ\text{C}$. A system offset calibration is performed using shorted inputs. A system gain calibration is performed using a $390\text{-}\Omega$ precision resistor. The data are taken using a precision resistor simulator with a 3-wire connection in place of a 3-wire RTD. Note that the measurement data are in ohms and do not include the error of the RTD sensor itself. The measured resistance error is $< \pm 0.02\text{ }\Omega$ over the $20\text{-}\Omega$ to $400\text{-}\Omega$ range.

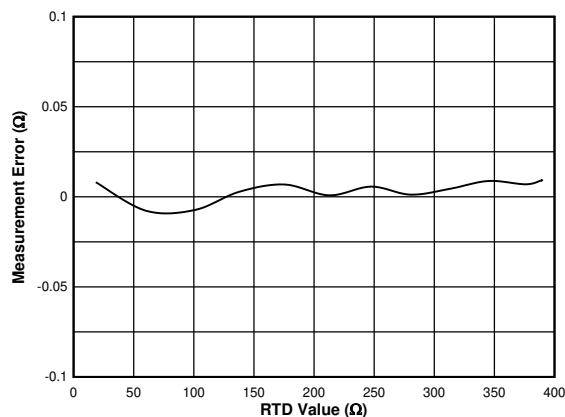


Figure 10-14. Resistance Measurement Error

10.3 What To Do and What Not To Do

- Do partition the analog, digital, and power supply circuitry into separate sections on the PCB.
- Do use a single ground plane for analog and digital grounds.
- Do place the analog components close to the ADC pins using short, direct connections.
- Do keep the SCLK pin free of glitches and noise.
- Do verify that the analog input voltages are within the specified PGA input voltage range under all input conditions.
- Do tie unused analog input pins to midsupply to minimize input leakage current.
- Do provide current limiting to the analog inputs in case overvoltage faults occur.
- Do use an LDO regulator to reduce ripple voltage generated by switch-mode power supplies.
- Don't route digital clock traces in the vicinity of the CAPP and CAPN pins.
- Don't cross digital signals over analog signals.
- Don't allow the analog and digital power supply voltages to exceed 7 V under all conditions, including during power-up and power-down.

Figure 10-15 shows Do's and Don'ts of ADC circuit connections.

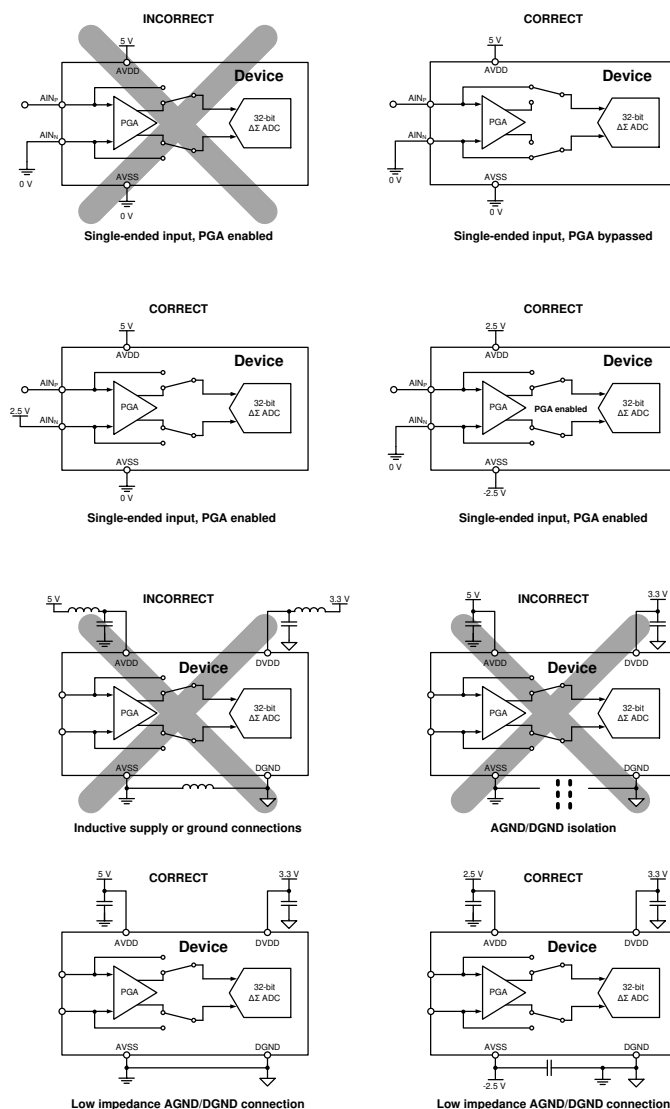


Figure 10-15. Dos and Don'ts Circuit Connections

10.4 Initialization Setup

Figure 10-16 is a general procedure that shows a typical ADS1262 configuration and measurement sequence.

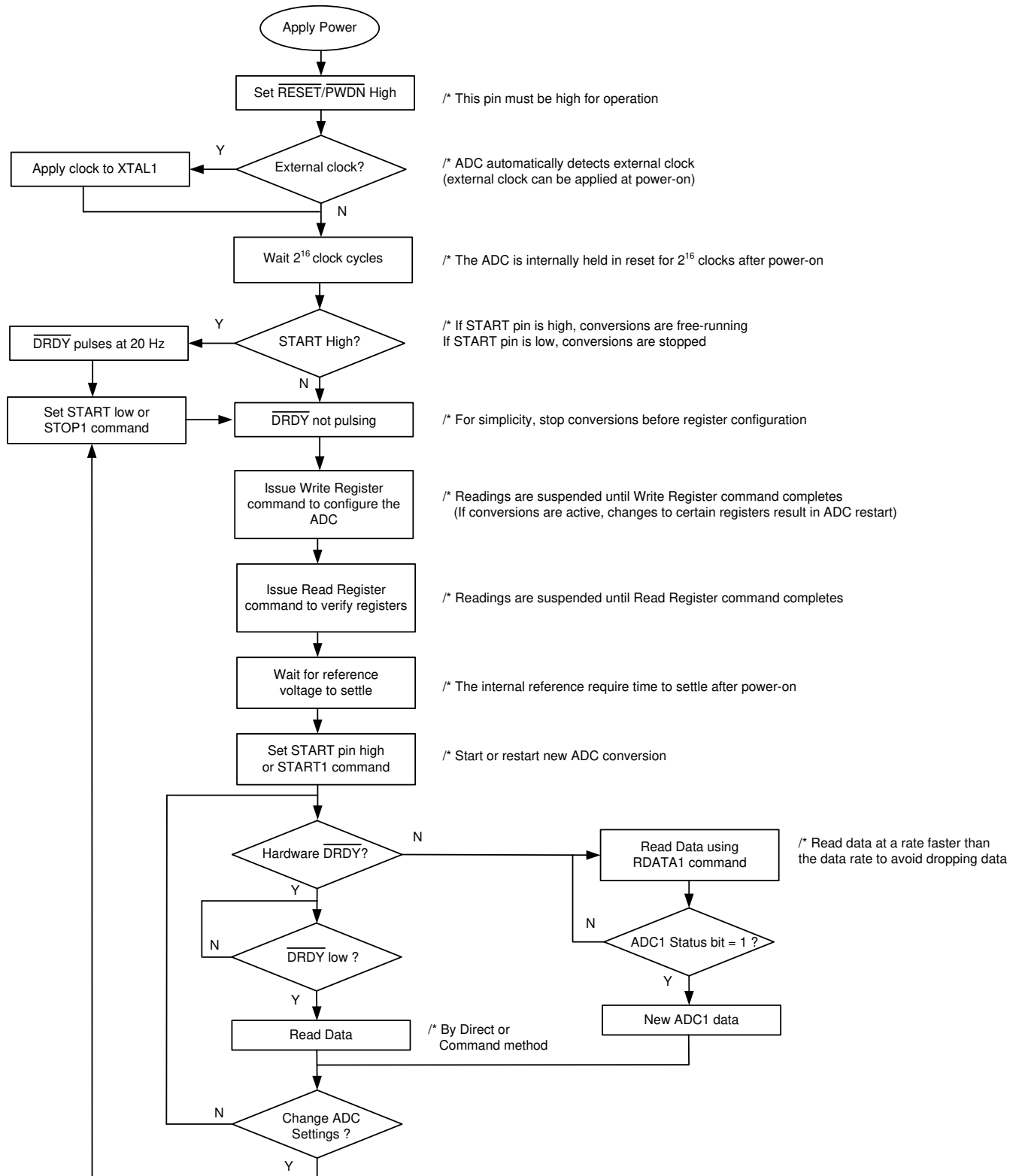


Figure 10-16. ADS1262 Configuration and Measurement Procedure

Figure 10-17 shows a general procedure to read concurrent ADC1 and ADC2 data of the ADS1263. The conversion time of ADC1 can be faster or slower than ADC2. If the conversion time of ADC1 is less than or equal to that of ADC2, and if the ADC2 status bit is equal to 1, then when ADC1 data are ready, ADC2 data are also ready. The ADC2 data can then be read by the RDATA2 command. Similarly, if the conversion time of ADC2 is less than that of ADC1, and if the ADC1 status bit is equal to 1, then when ADC2 data are ready, ADC1 data are also ready. The ADC1 data can then be read by the RDATA1 command. It is important to note an exception to the conversion time related to the data rate: the time of the *first* conversion is not always the same as (1 / data rate) because of digital filter latency. Therefore, it is possible that although the data rate of ADC1 can be faster than ADC2, the time required for the *first* conversion of ADC1 can be greater than ADC2 depending on the digital filter setting and chop mode. When checking the ADC2 status by reading ADC1 data, use the RDATA1 command.

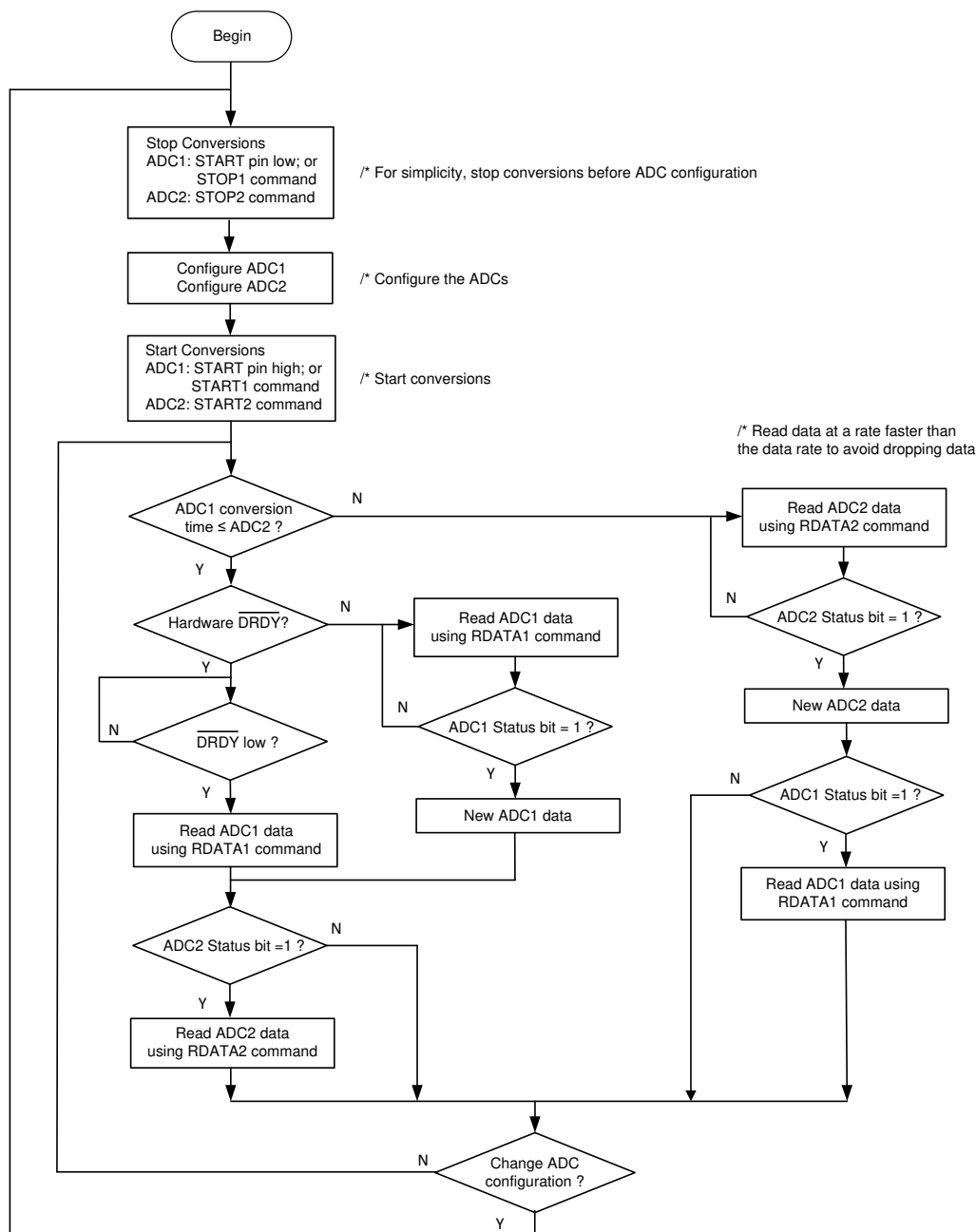


Figure 10-17. ADS1263 Concurrent Read of ADC1 and ADC2 Data

11 Power Supply Recommendations

The ADS1262 and ADS1263 require an analog power supply (V_{AVDD} , V_{AVSS}) and digital power supply (V_{DVDD}). The analog power supply can be bipolar (for example, $V_{AVDD} = +2.5$ V, $V_{AVSS} = -2.5$ V) or unipolar (for example, $V_{AVDD} = 5$ V, $V_{AVSS} = 0$ V). The digital supply (V_{DVDD}) range is 2.7 V to 5.25 V. The digital supply voltage determines the digital I/O logic levels. Keep in mind that the GPIO logic levels (AIN3-AINCOM) are referenced to the analog supply voltage and may be different from the digital I/O logic level. The analog and digital sections of the ADC are not internally isolated and the grounds for analog and digital must be connected together. Output voltage ripple produced by switch-mode power supplies may interfere with the ADC resulting in reduced performance. Use low-dropout regulators (LDOs) to reduce the power-supply ripple voltage produced by switch-mode power supplies.

11.1 Power-Supply Decoupling

Good power-supply decoupling is important in order to achieve optimum performance. Power supplies V_{AVDD} , V_{AVSS} and V_{DVDD} must be decoupled to a common ground potential. For proper power-supply decoupling, place a 0.1- μ F capacitor as close as possible to the supply with an additional 1- μ F bulk capacitor placed nearby. [Figure 11-1](#) shows decoupling for bipolar-supply (left figure) and single-supply (right figure) operation. When using bipolar supplies, bypass both AVDD and AVSS to ground separately, and include a bypass capacitor between AVDD and AVSS. Use a multilayer ceramic chip capacitors (MLCCs) that offers low equivalent series resistance (ESR) and equivalent series inductance (ESL) characteristics for power-supply decoupling purposes. The BYPASS pin is the bypass output of an internal 2-V regulator. The 2-V regulator powers the digital circuitry. Connect a ceramic or tantalum 1- μ F capacitor from this pin to DGND. Do not load this voltage by external circuits.

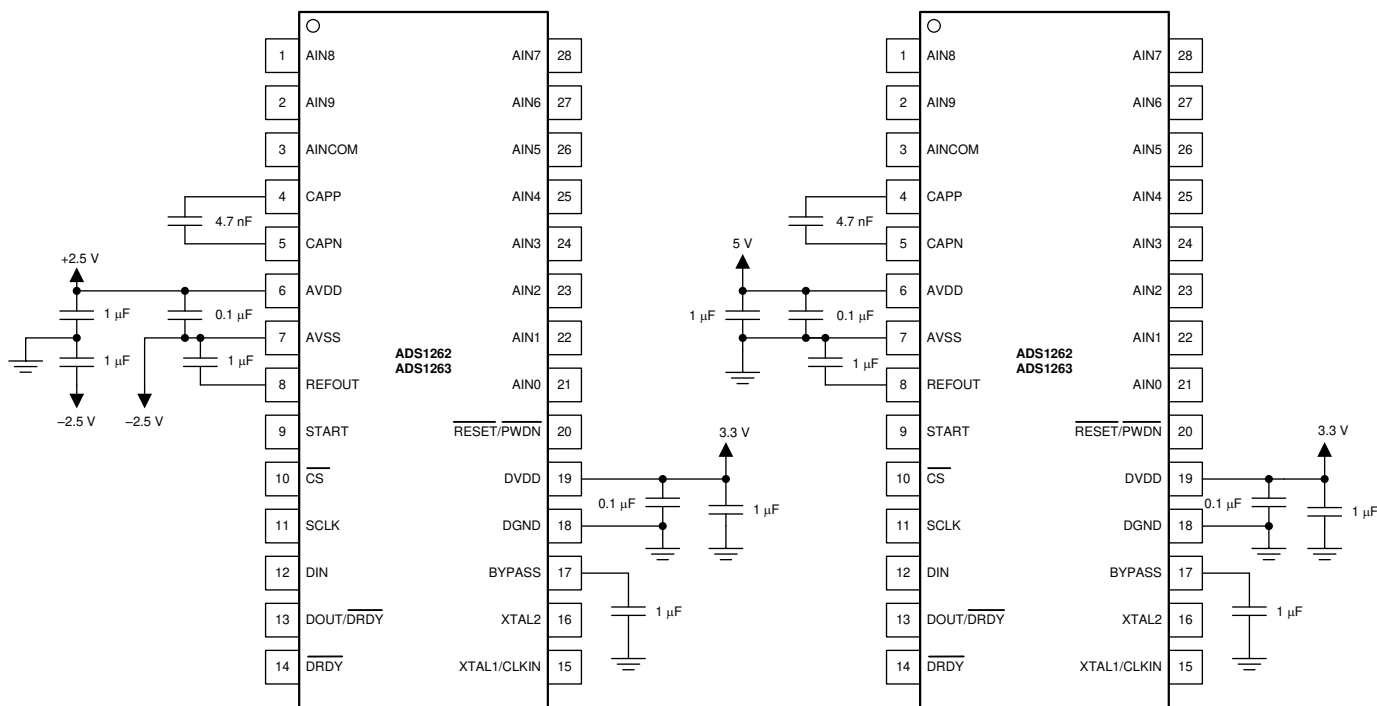


Figure 11-1. Power-Supply Decoupling for Bipolar (left) and Single-Supply (right) Operation

11.2 Analog Power-Supply Clamp

It is important to evaluate circumstances when an input signal is present while the ADC is powered and unpowered. When the input signal exceeds the power-supply voltage, it is possible to *back drive* the analog power-supply voltage with the input signal through a conduction path of the internal ESD diodes. Back driving the ADC power supply can also occur when the power-supply voltage is on. The back-drive, fault-current path is shown in [Figure 11-2](#). Depending on the external power-supply components, it is possible that the maximum rating of the ADC power-supply voltage can be exceeded if back-driven. ADC power supply overvoltage must be prevented in all cases. One solution is to clamp the AVDD to AVSS voltage with an external 6-V Zener diode.

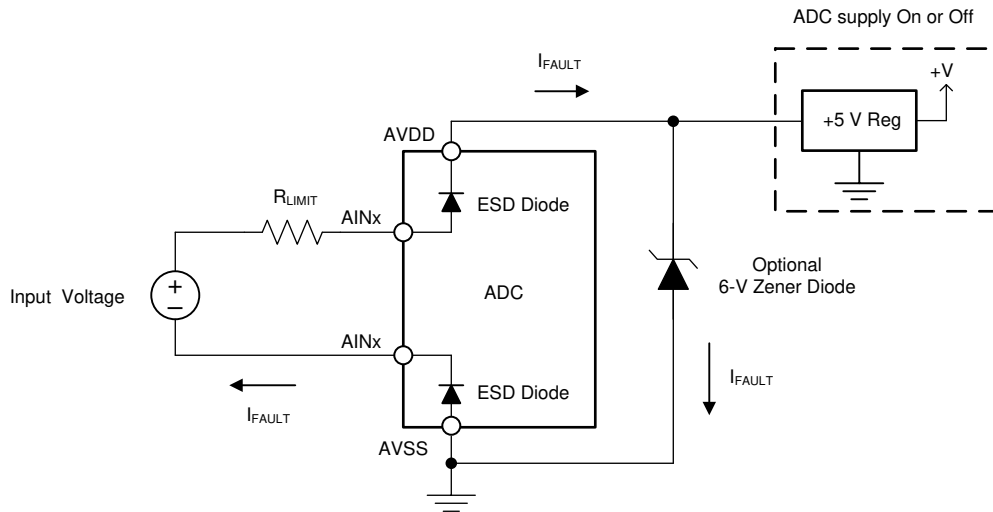


Figure 11-2. Analog Power-Supply Clamp

11.3 Power-Supply Sequencing

Sequence the power supplies in any order, but never allow an analog or digital input to exceed the respective analog or digital power-supplies without limiting the input fault current. The ADC remains in reset until both analog and digital power supplies exceed the respective power-on reset (POR) thresholds. [Figure 9-52](#) shows the power-on reset sequence. After the power supplies have crossed the reset levels (including the internal 2-V LDO), the ADC resets (POR) and is ready for communication 65536 clock periods later (nominally 9 ms).

Delay communication for 50 ms after the power supplies have stabilized within the specified range to make sure the ADC is operational. In addition to POR, make sure that the reference voltage has fully settled before starting the conversions. When using a 1-μF reference capacitor allow a minimum of 50 ms for the internal reference to settle. External references may require additional settling time.

12 Layout

Good layout practices are crucial to realize the full-performance of the ADS1262 and ADS1263. Poor grounding can quickly degrade the noise performance of the main 32-bit ADC and auxiliary 24-bit ADC. The following layout recommendations are given to help provide best results.

12.1 Layout Guidelines

Ground must be a low impedance connection for return currents to flow undisturbed back to their respective sources. Keep connections to the ground plane as short and direct as possible. When using vias to connect to the ground layer, use multiple vias in parallel to reduce impedance to ground.

A mixed-signal layout sometimes incorporates separate analog and digital ground planes that are tied together at one location; however, separating the ground planes is not necessary when analog, digital, and power supply components are properly placed. Proper placement of components partitions the analog, digital, and power supply circuitry into different PCB regions to prevent digital return currents from coupling into sensitive analog circuitry.

For best performance, dedicate an entire PCB layer to a ground plane and do not route any other signal traces on this layer. However, depending on restrictions imposed by specific form factors, single ground planes may not be possible. If ground plane separation is necessary, then make the connection at the ADC. Do not connect individual ground planes at multiple locations because this configuration creates ground loops. A single plane for analog and digital ground avoids ground loops.

If isolation is required in the application, isolate the digital signals between the ADC and controller, or provide the isolation from the controller to the remaining system. If an external crystal is used to provide the ADC clock, place the crystal and load capacitors directly to the ADC pins using short direct traces. See the [Crystal Oscillator](#) section for more details.

Supply pins must be bypassed with a low-ESR ceramic capacitor. Place the bypass capacitors as close as possible to the supply pins using short, direct traces. For optimum performance, use low-impedance connections on the ground-side connections of the bypass capacitors. Flow the supply current through the bypass capacitor pin first and then to the supply pin to make the bypassing most effective (also known as a *Kelvin connection*). If multiple ADCs are on the same PCB, use wide power supply traces or dedicated power-supply planes to minimize the potential of crosstalk between ADCs.

If external filtering is used for the analog inputs, use C0G-type ceramic capacitors when possible. C0G capacitors have stable properties and low-noise characteristics. Ideally, route the differential signals as pairs in order to minimize the loop area between the traces. For the ADC CAPP and CAPN pins, place the 4.7-nF C0G capacitor close to the pins using short direct traces. Route digital circuit traces (such as clock signals) away from all analog pins. Note the internal reference output return shares the same pin as the AVSS power supply. To minimize coupling between the power-supply trace and reference-return trace, route the two traces separately; ideally, as a star connection at the AVSS pin.

It is important the SCLK input of the serial interface is free from noise and glitches. Even with relatively slow SCLK frequencies, short digital-signal rise and fall times may cause excessive ringing and noise. For best performance, keep the digital signal traces short, use termination resistors as needed, and ensure all digital signals are routed directly above the ground plane with minimal use of vias.

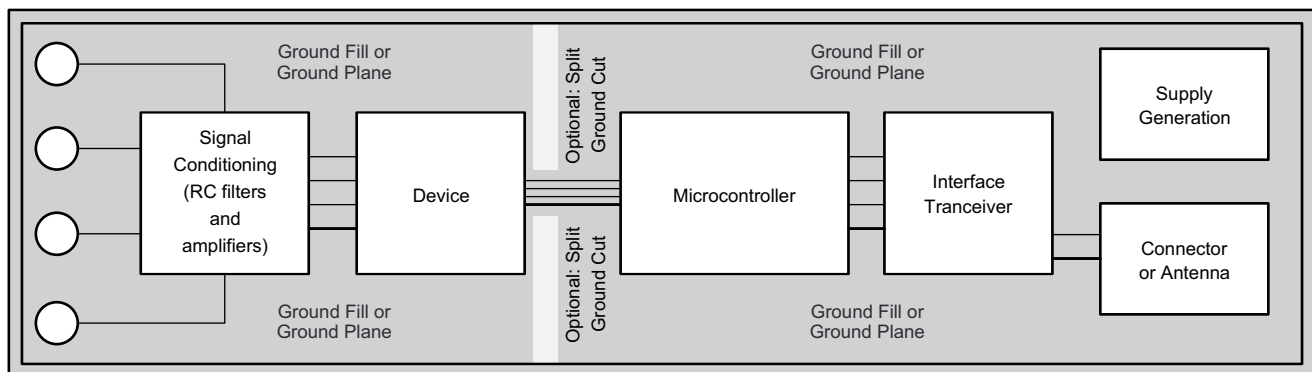


Figure 12-1. System Component Placement

12.2 Layout Example

Figure 12-2 is an example layout of the ADS1262 and ADS1263, requiring a minimum of three PCB layers. The example circuit is shown for a single analog supply (5 V) connection and an external crystal oscillator. In this example, an inner layer is dedicated to the ground plane and the outer layers are used for signal and power traces. If a four-layer PCB is used, dedicate the additional inner layers to route power traces. The ADC orientation is shown left to right to minimize crossover of the analog and digital signal traces. The PCB is partitioned with analog signals routed from the left, digital signals routed to the lower-right, and power routed from the upper-right. Analog supply bypass capacitors are placed opposite to the ADC on the bottom layer to allow the reference and PGA output capacitors to be placed closer to the ADC.

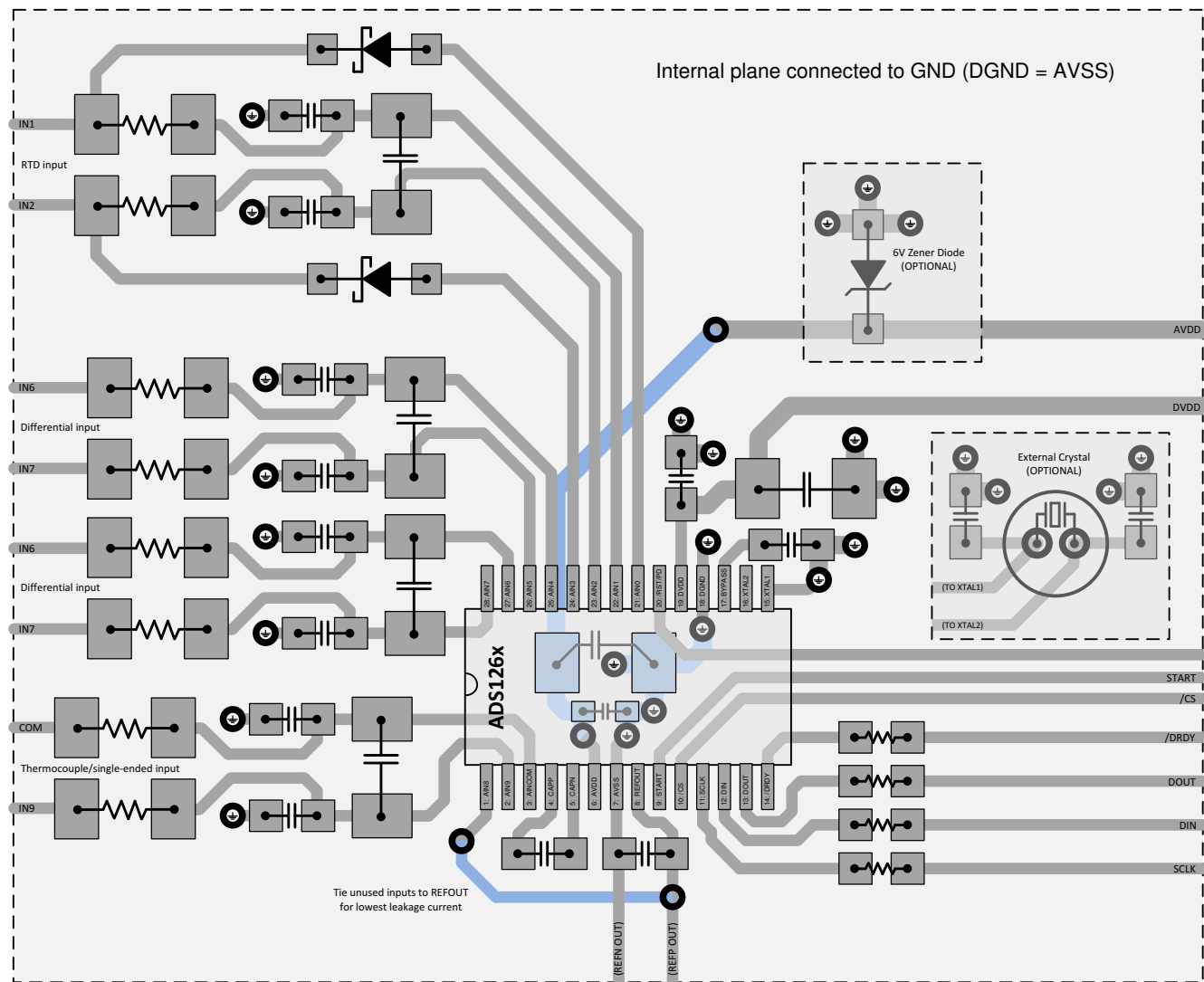


Figure 12-2. PCB Layout Example

13 Device and Documentation Support

13.1 Receiving Notification of Documentation Updates

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This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

13.5 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
ADS1262IPW	ACTIVE	TSSOP	PW	28	50	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	1262	Samples
ADS1262IPWR	ACTIVE	TSSOP	PW	28	2000	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	1262	Samples
ADS1263IPW	ACTIVE	TSSOP	PW	28	50	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	1263	Samples
ADS1263IPWR	ACTIVE	TSSOP	PW	28	2000	RoHS & Green	NIPDAU	Level-2-260C-1 YEAR	-40 to 125	1263	Samples

(1) The marketing status values are defined as follows:

LIFEBUY: Product device recommended for new designs.

ACTIVE: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSOLETE: TI has discontinued the production of the device.

(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "-" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

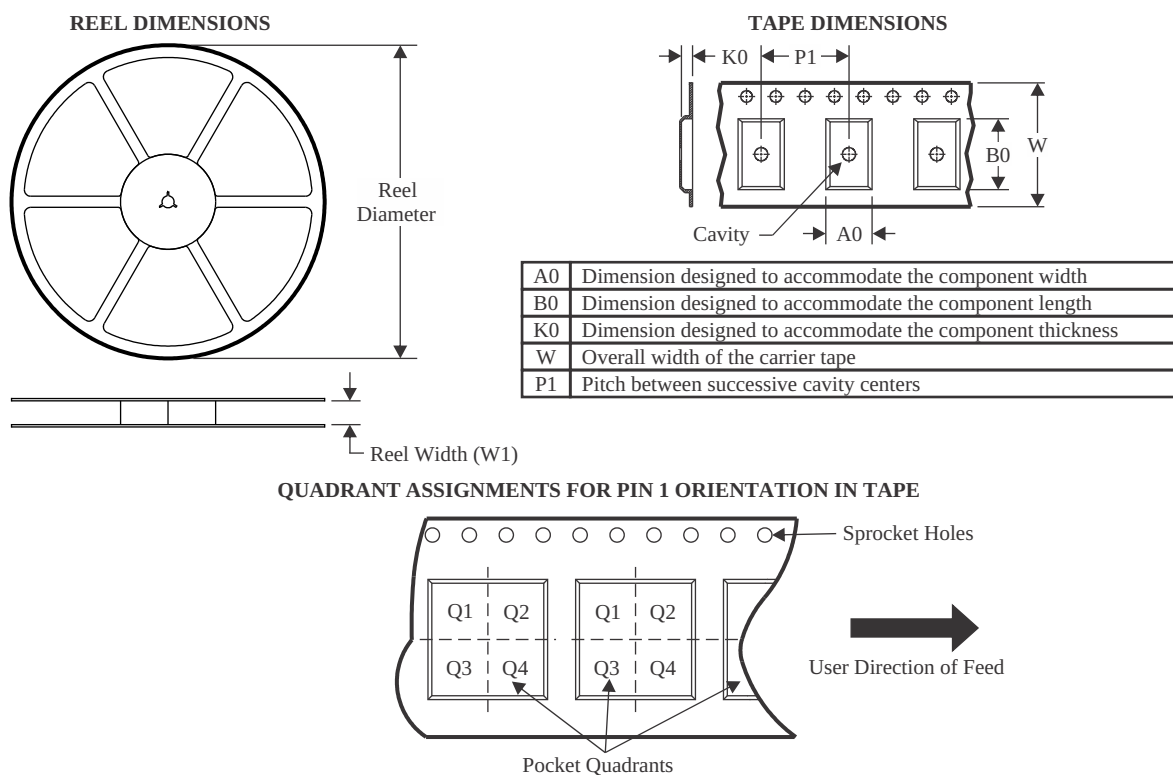
(6) Lead finish/Ball material - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

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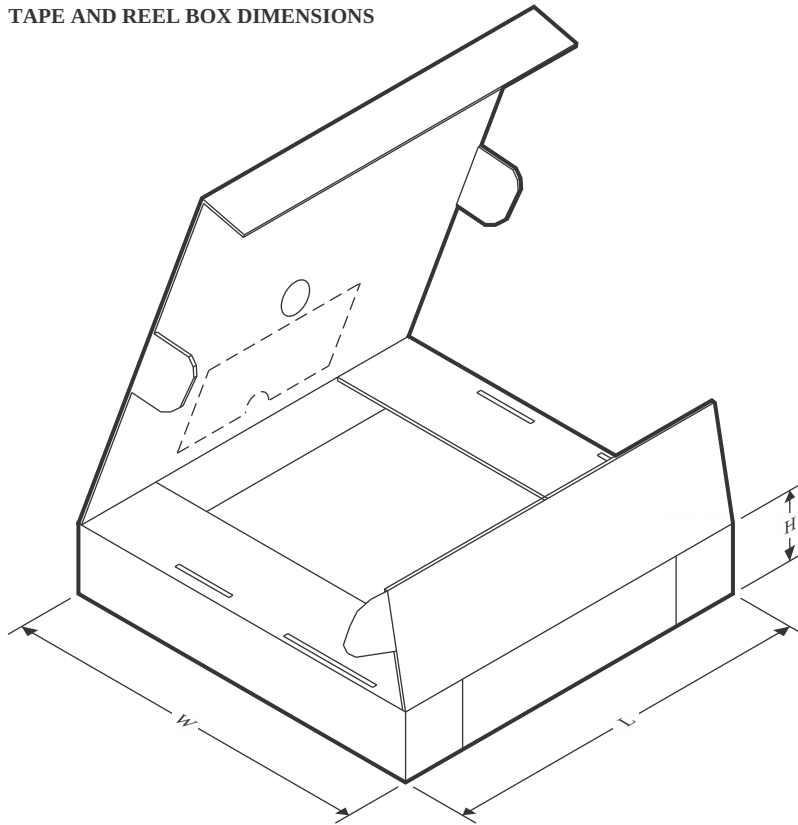
TAPE AND REEL INFORMATION



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
ADS1262IPWR	TSSOP	PW	28	2000	330.0	16.4	6.9	10.2	1.8	12.0	16.0	Q1
ADS1263IPWR	TSSOP	PW	28	2000	330.0	16.4	6.9	10.2	1.8	12.0	16.0	Q1

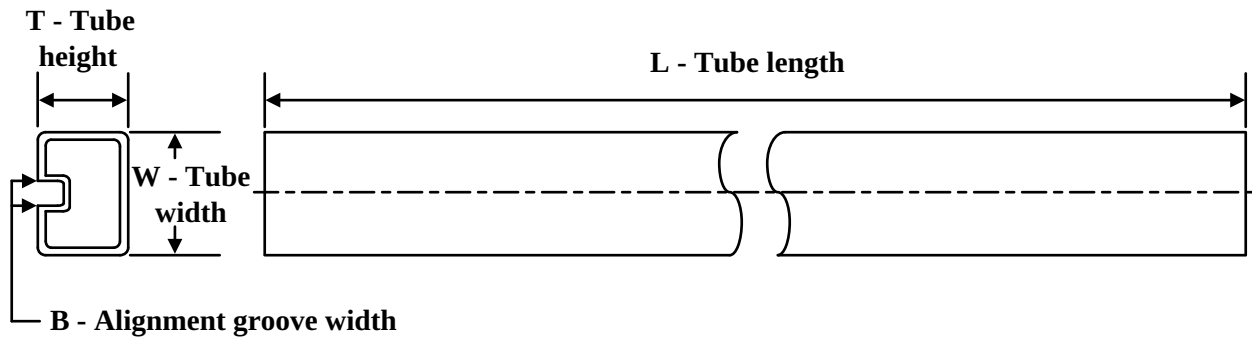
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
ADS1262IPWR	TSSOP	PW	28	2000	356.0	356.0	35.0
ADS1263IPWR	TSSOP	PW	28	2000	356.0	356.0	35.0

TUBE

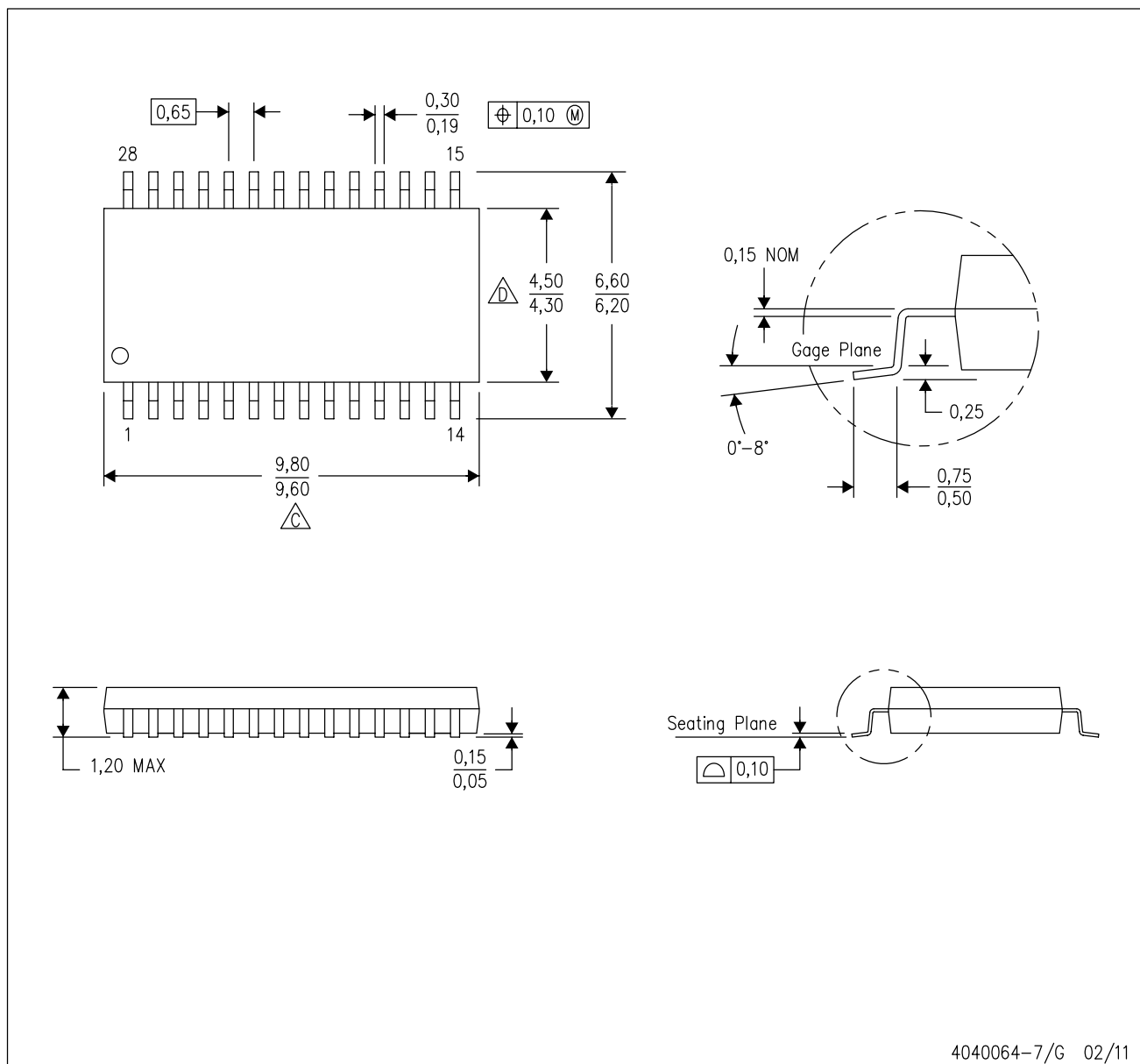


*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (μm)	B (mm)
ADS1262IPW	PW	TSSOP	28	50	530	10.2	3600	3.5
ADS1263IPW	PW	TSSOP	28	50	530	10.2	3600	3.5

PW (R-PDSO-G28)

PLASTIC SMALL OUTLINE



- NOTES:
- A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5M-1994.
 - B. This drawing is subject to change without notice.
 - C. Body length does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0,15 each side.
 - D. Body width does not include interlead flash. Interlead flash shall not exceed 0,25 each side.
 - E. Falls within JEDEC MO-153

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